

CHEST RADIOGRAPHY IN TUBERCULOSIS DETECTION

Summary of current WHO
recommendations and
guidance on programmatic
approaches



THE
END TB
STRATEGY



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Preface

The End TB Strategy puts renewed emphasis on the need to ensure early and correct diagnosis for all people with tuberculosis (TB) (1). Important progress has been made in improving laboratory services in recent decades. New bacteriological tests for TB diagnosis have become available and their use is now being scaled up (2, 3). Efforts have been made to ensure that people who seek care and have symptoms consistent with TB are correctly triaged and evaluated for TB. Systematic screening for active TB in high-risk groups is being implemented and scaled up in several places (4, 5). However, despite these efforts, many people with TB remain undiagnosed or are diagnosed only after long delays (6).

Chest radiography, or chest X-ray (CXR), is an important tool for triaging and screening for pulmonary TB, and it is also useful to aid diagnosis when pulmonary TB cannot be confirmed bacteriologically. Although recent diagnostic strategies have given specific prominence to bacteriology, CXR can be used for selecting individuals for referral for bacteriological examination, and the role of radiology remains important when bacteriological tests cannot provide a clear answer. Access to high-quality radiography is limited in many settings. Ensuring the wider and quality-assured use of CXR for TB detection in combination with laboratory-based diagnostic tests recommended by the World Health Organization (WHO), can contribute to earlier TB diagnosis and potentially to closing the TB case-detection gap when used as part of algorithms within a framework of health-system and laboratory strengthening.

This document summarizes WHO's recommendations on using CXR for TB triaging, diagnosis and screening. It also outlines a framework for the strategic planning and use of CXR within national TB programmes (NTP). Moreover, the document provides a brief overview of technical specifications, and quality assurance and safety considerations for CXR. However, because these technical aspects are generic and should be addressed as part of the general strengthening of radiography and imaging services, this document does not go into technical details. General radiography guidance is provided elsewhere (7-11).

The document focuses on CXR, with a major emphasis on detecting pulmonary TB. CXR can be useful for diagnosing other forms of TB (for example, miliary or pericardial TB, or tuberculous effusions) and other imaging techniques are also valuable for TB diagnosis, for example, for extrapulmonary TB, but these topics are not discussed in this document.

The document is mainly intended for NTPs and partners helping with the planning and implementation of national TB care and prevention efforts. It is not intended to be a clinical guide. The recommendations and principles that are summarized in this document need to be adapted to each setting's TB epidemiology and health-system capacity.

Development process

A steering group was established in January 2016, which advised WHO on the scope and content of this document. The members of the steering group were Faiz Ahmad Khan, Sevim Ahmedov, Frank Cobelens, Jacob Creswell, Claudia Denking, Christopher Gilpin, Michael Kimerling, Knut Lönnroth, Cecily Miller, YaDiul Mukadi, Ikushi Onozaki and Madhukar Pai.

After consultation with WHO's Guideline Review Committee, it was determined that the document is not a new guideline but a summary of existing WHO recommendations. Therefore, it did not need to follow WHO's guideline development process.

All major WHO publications about TB were reviewed for their relevance to the use of CXR in screening for, triaging and diagnosing TB. Recommendations across all documents were compiled and summarized. An accompanying framework for strategic planning for using CXR within NTPS was developed based on experts' opinions.

No systematic literature review was undertaken during the development of this document. The evidence base for the statements made in this document is the same as for those in the cited WHO guidelines and policy frameworks.

Scenarios for the yield of TB (true positive/true negative and false positive/false negative) for different triaging algorithms were modelled using the ScreenTB tool (12) to illustrate how the different placement of CXR in an algorithm influences yields and costs under different epidemiological scenarios. The model outputs that are included in this document should not be used for forecasting TB detection, but are included merely to demonstrate how variations in algorithms influence TB detection and costs. Readers are advised to develop setting-specific scenarios based on the local TB epidemiology and the best data about test accuracy and costs.

A first draft was completed in July 2016 and was circulated to experts (see below) for peer review. Based on comments from the peer review, a second draft was prepared ahead of a global consultation held during 28–29 September 2016. The consultation provided additional inputs on the draft document, and the documented was thereafter finalized.

Acknowledgements

The first draft was prepared by Cecily Miller and Knut Lönnroth. The following persons contributed to the development of the document or peer reviewed it, or both: Faiz Ahmad Khan, Sevim Ahmedov, Farhana Amanullah, Samiha Baghdadi, Draurio Barreira, Adriana Velazquez Berumen, Nils Billo, Annemieke Brands, Grania Brigden, Chen-Yuan Chiang, Maarten van Cleeff, Jacob Creswell, Claudia Denking, Anna-Marie Celina Garfin, Nebiat Gebreselassie, Sifrash Meseret Gelaw, Wayne van Gemert, Robert Gie, Steve Graham, Rob van Hest, Philip Hopewell, Bogomil Kohlbrenner, Alexei Korobitsyn, Devesh Gupta, Michael Kimerling, Irwin Law, Partha Pratim Mandal, Guy Marks, Giovanni Batista Migliori, Mahshid Nasehi, Nobuyuki Nishikiori, Pierre-Yves Norval, Kosuke Okada, Ikushi Onozaki, Salah-Eddine Ottmani, Madhukar Pai, Tripti Pande, Mario Raviglione, Maria del Rosario Pérez, Anna Scardigli, Eric Stern, Beat Stoll, Etienne-Leroy Terquem, Belay Tessema, Mukund Uplekar, Diana Weil, William Wells, Marieke van der Werf and Christine Whalen.

Declarations of interests

The following interests were declared by the experts consulted.

Declared interests that were deemed not significant

- Claudia Denking: took part in several clinical research projects to evaluate new diagnostic tests against the target product profiles for TB defined through consensus processes led by WHO. These studies were for diagnostic products developed by private sector companies (Cepheid, Epistem, Molbio Diagnostics, Hain Lifescience, Nipro, Becton Dickinson, Alere, YD Diagnostics, Ustar Biotechnologies and Qiagen) that provide access to know-how, equipment and reagents, and contribute through unrestricted donations as per FIND (Foundation for Innovative New Diagnostics) policy.
- Bogomil Kohlbrenner and Beat Stoll: were employed as researchers on a project to develop appropriate medical devices and appropriate training for health workers in the field of tropical medical imaging; it was a philanthropic project.

Declared interest that were deemed significant for making recommendations to WHO about whether to develop guidelines for computer-aided detection

- Faiz Ahmad Khan and Madhukar Pai: received a research grant to study the diagnostic accuracy of CAD4TB (developed by Delft Imaging Systems, Veenendaal, the Netherlands) in collaboration with Interactive Research & Development, who have purchased equipment from the makers of CAD4TB. The developers of CAD4TB are not collaborators or in any way involved in the research. Faiz Ahmad Khan and Madhukar Pai were invited to present the systematic review on CAD for TB detection, provide comments throughout the meeting, and peer-review draft documents. However, they were not part of the decision to advise WHO on the initiation of a CAD guideline development process.

Abbreviations

AFB	acid-fast bacilli
CAD	computer-aided detection
CXR	chest X-ray or chest radiography
HIV	human immunodeficiency virus
LTBI	latent tuberculosis infection
MTB	Mycobacterium tuberculosis
NTP	national tuberculosis programme
PICO	population, intervention, comparator, outcome
QUADAS	Quality Assessment of Diagnostic Accuracy Studies
SSM	sputum-smear microscopy
TB	tuberculosis
WHO	World Health Organization

Definitions

Bacteriologically confirmed TB case: A bacteriologically confirmed case of TB is one from whom a biological specimen tests positive by smear microscopy, culture or WHO-recommended rapid diagnostic (such as the Xpert MTB/RIF assay). All such cases should be notified, regardless of whether TB treatment has started (13).

Clinically diagnosed TB case: A clinically diagnosed case of TB is one who does not fulfil the criteria for bacteriological confirmation but has been diagnosed with active TB by a clinician or other medical practitioner who has decided to give the patient a full course of anti-TB treatment. This definition includes cases diagnosed on the basis of abnormalities seen on X-ray or histology suggestive of TB, and extrapulmonary cases without laboratory confirmation. Clinically diagnosed cases subsequently found to be bacteriologically positive (before or after starting treatment) should be reclassified as bacteriologically confirmed (13).

Systematic screening for active TB: is the systematic identification of people with suspected active TB in a predetermined target group, using tests, examinations or other procedures that can be applied rapidly (4). Unlike evaluations of those who actively seek care for respiratory symptoms (known as triaging), the systematic screening of individuals for TB is typically initiated by a provider and offered in a systematic way to an apparently healthy target group that has been determined to have a high risk of TB.

Triaging: For the purpose of this document, triaging is defined as the processes of deciding the diagnostic and care pathways for people seeking healthcare, based on their symptoms, signs, risk markers and test results. Triaging involves assessing the likelihood of various differential diagnoses as a basis for making clinical decisions. It can follow more- or less-standardized protocols and algorithms and may be done in multiple steps.

1. INTRODUCTION

1.1 Medical imaging

Medical imaging uses different modalities and processes to image the internal structures of the human body for diagnosis and treatment. Imaging has an important role in healthcare for all population groups. In public health and preventive medicine, as well as in both curative and palliative care, effective clinical decisions depend on correctly screening, triaging and diagnosing patients. The use of imaging services is paramount in correctly screening, confirming and documenting the course of many diseases. With the improved availability of medical equipment, global access to medical imaging has increased considerably, but is still insufficient in many settings (14). Medical imaging is a key element within many evidence-based clinical decision-support algorithms, consistent with overarching evidence-based recommendations for disease management (14). As such, medical imaging should be accessible to all and should not be exclusively a hospital service (15).

1.2 Radiography

Radiography uses X-rays to visualize the internal structures of a patient. X-rays are a form of electromagnetic radiation produced by an X-ray tube. The X-rays pass through the body and are captured behind the patient by film that is sensitive to X-rays or by a digital detector. Different tissues in the body vary in their absorption of X-rays: dense bone absorbs more radiation, but soft tissue allows more to pass through. This variance produces contrasts within the image to give a two-dimensional representation of the three-dimensional structures. As a result, the X-ray image often includes overlapping structures. A thorough knowledge of anatomy is needed to identify an abnormality on an X-ray and understand where it is in the body. Common clinical applications include imaging the chest to assess lung and intrathoracic pathologies; imaging the skeletal system to examine bone structures and diagnose fractures, dislocations or other bone pathologies; imaging the abdomen to assess obstructions or free air or fluid within the abdominal cavity; or imaging the teeth to assess common dental pathologies, such as cavities or abscesses (14).

1.3 Chest X-ray for detecting TB

Chest X-ray (CXR) is a rapid imaging technique that allows lung abnormalities to be identified. CXR is used to diagnose conditions of the thoracic cavity, including the airways, ribs, lungs, heart and diaphragm.

CXR has historically been one of the primary tools for detecting tuberculosis (TB), especially pulmonary TB. CXR has high sensitivity for pulmonary TB and thus is a valuable tool to identify TB as a differential diagnosis for patients, especially when the X-ray is read to identify any abnormality that is consistent with TB. However, CXR has poor specificity; although some CXR abnormalities are rather specific for pulmonary TB (for example, cavities), many CXR abnormalities that are consistent with pulmonary TB are seen also in several other lung pathologies and, therefore, are indicative not only of TB but also of other pathologies. Moreover, there is significant intra- and interobserver variation in the reading of CXRs. Relying only on CXR for TB diagnosis leads to overdiagnosis, as well as underdiagnosis (16). Rigorous efforts should always be made to base a TB diagnosis on bacteriological confirmation (sputum-smear microscopy, culture or a molecular test). WHO classifies TB diagnosis into bacteriologically confirmed TB, if it is based on bacteriological confirmation, or clinically diagnosed TB, if it is based on clinical assessment including CXR, but is not confirmed by bacteriological examination (13).

1.4 Overview of the use of chest X-ray in WHO's policies and guidelines

For many years, WHO has recommended CXR as a diagnostic tool to be used as a complementary part of the clinical diagnosis of bacteriologically negative TB. As such, CXR has previously been placed at the end of diagnostic algorithms. WHO's 2003 treatment guidelines for national programmes and the guideline on diagnosing smear-negative pulmonary TB from 2007 recommended that CXR be used after: (i) initial negative bacteriological testing, (ii) a course of broad-spectrum antibiotics and (iii) a second negative round of bacteriological testing (17, 18). However, CXR was recommended to be used directly after initial negative bacteriological testing to diagnose TB in people living with HIV or AIDS and in those considered to be at high risk of HIV infection (18). The 2008 handbook for national tuberculosis control programmes (19), as well as the third edition of the *International standards for tuberculosis care* in 2014 (20), suggested a more flexible approach, with the possibility of using CXR directly after an initial negative bacteriological test, and not just for people living with HIV.

None of these guidelines placed CXR as a triage test before bacteriological testing. However, none of the guidelines specifically recommend against using CXR for triaging or diagnostic evaluation of TB, and they emphasized that whenever CXR has been done and shows abnormalities consistent with TB, a bacteriological test for TB must always be performed.

All the above-mentioned documents emphasized that using CXR to diagnose TB is problematic, given that CXR has low specificity and significant interobserver variation. Moreover, poor access to high-quality radiography equipment and expert interpretation, along with the widespread use of low-quality radiography, were identified as additional barriers for promoting large-scale programmatic use.

Recently, however, CXR has been promoted as a useful tool that can be placed early in screening and triaging algorithms. An important reason for rethinking the role of CXR in screening and diagnostic algorithms is that numerous national TB prevalence surveys have demonstrated that CXR is the most sensitive screening tool for pulmonary TB and that a significant proportion of people with TB are asymptomatic, at least early in the course of the disease (see Annex 2) (21). Other factors that have contributed to CXR becoming an increasingly accepted part of programmatic approaches to TB care and prevention include:

- the increased availability of radiography, including digital radiography with its lower running costs and highly portable systems for field use, better image quality and better safety (due to decreased radiation dose) than conventional radiography, as well as possibilities for use for telemedicine;
- the documented rapidity of results and high throughput capacity, especially of digital CXR;
- a gradual shift from strictly prioritizing the diagnosis of the most infectious TB cases (that is, bacteriologically confirmed TB, especially sputum smear-positive TB, in persons with persistent cough) to programmatic targets in line with a rights-based vision of universal access to high-quality diagnosis for all people with all forms of TB as well as concern with diagnosis of other lung diseases;
- the increasing availability of rapid molecular tests with higher sensitivity and specificity than sputum-smear microscopy which allows for higher diagnostic accuracy among people with CXR abnormalities consistent with TB (and, thus, reduces the risk of overdiagnosis). Moreover, available molecular tests have significantly higher costs than sputum smear microscopy, which often necessitates a method of triaging of patients for evaluation for TB.

The limitations of, and recent advances in, CXR are summarized in Box 1.

BOX 1. Limitations of and advances in chest X-ray (21)

The main limitations associated with using chest X-ray include:

- it produces two-dimensional representations of a three-dimensional structure;
- there is intrareader and interreader variability;
- no abnormalities are definitive of TB, therefore the specificity is low;
- a universally accepted reporting system is lacking;
- patients are exposed to ionizing radiation;
- special equipment (with adequate input power) is needed;
- trained personnel are required to operate the machine and interpret the results;
- there is often limited access in rural areas; access is often limited to district or regional levels;
- there is limited archiving of hard copies;
- out-of-pocket costs for patients are often high.

Recent advances in digital chest X-ray technology include:

- lower operating costs;
- improved and more reproducible image quality with enlargement capability;
- a decreased radiation dose;
- improved portable systems that can be used for mobile units;
- efforts to harmonize interpretation and reporting;
- the potential of objective tools for interpretation of digital images, such as computer-aided detection;
- better (digital) archiving facilities;
- film processing equipment and hard copies no longer required;
- the possibility of electronically transmitting images (for example, for telemedicine or quality assurance).

The importance of CXR is reflected by the recommendations about using CXR for screening, triaging and assisting in the diagnosis of TB that have been included in recent policies developed or endorsed by WHO, including:

- *Systematic screening for active tuberculosis: principles and recommendations* (4)
- *Guidance for national tuberculosis programmes on the management of tuberculosis in children* (22)
- *Tuberculosis prevalence surveys: a handbook* (21)
- *Consolidated guidelines on the use of antiretroviral drugs for treating and preventing HIV infection* (23)
- the implementation manual for the Xpert MTB/RIF assay (Cepheid, Sunnyvale, CA, United States) (3)
- International standards for tuberculosis care, third edition (20)
- *Guidelines on the management of latent tuberculosis infection* (24).

Box 2 summarizes the recent recommendations from WHO.

However, despite the demonstrated utility of CXR and multiple WHO recommendations about when and how to use it, the programmatic and rational use of CXR for TB detection remains limited. The lack of consolidated programmatic guidance is one possible reason, hence the need for this document. Also contributing to its restricted use are the limited availability of radiography in some regions (including a lack of systems and incentives to keep X-ray machines operational), constraints on human resources, insufficient training (including qualification programmes and post-graduate training), a lack of quality assurance programmes and, often, high out-of-pocket costs for patients.

In the following chapters, the options for using CXR for different elements of TB detection are outlined in more detail. This is followed by a brief summary of technical specifications, and quality control and safety issues. Finally, guidance on the programmatic planning and implementation of CXR is discussed.

BOX 2. Summary of recommendations on using chest X-ray (CXR) for TB in recent WHO guidelines and policies

CHEST X-RAY: AN ESSENTIAL TOOL TO END TB

CXR IS A SENSITIVE TOOL FOR SCREENING FOR ACTIVE TB

Reference: *Systematic screening for active tuberculosis: principles and recommendations* (4)

- CXR has higher sensitivity for pulmonary TB than screening for TB symptoms.

AN ABNORMAL CXR IS AN INDICATION FOR FULL DIAGNOSTIC EVALUATION

Reference: *International standards for tuberculosis care* (20)

- All patients with unexplained findings suggestive of TB on CXR should be evaluated for TB with a bacteriological diagnostic test.
- CXR can be used as a supplementary diagnostic aid, although the specificity is low.
- A bacteriologically confirmed diagnosis is always preferred.

CXR IS AN IMPORTANT TOOL FOR DIAGNOSING CHILDHOOD TB

Reference: *Guidance for national tuberculosis programmes on the management of tuberculosis in children* (22)

- CXR is useful in diagnosing pulmonary and extrapulmonary TB in children in combination with history, evidence of TB infection and microbiological testing.

CXR CAN IMPROVE THE EFFICIENCY OF USING THE XPRT MTB/RIF ASSAY

Reference: *implementation manual for the Xpert MTB/RIF assay* (3)

- CXR and further clinical assessment can be used to triage who should be tested with the Xpert MTB/RIF assay to reduce the number of individuals tested and the associated costs, as well as to improve the pre-test probability for TB and, thus, the predictive value of the Xpert MTB/RIF assay.

CXR CAN ASSIST IN DIAGNOSING TB AMONG PEOPLE LIVING WITH HIV

Reference: *Consolidated guidelines on the use of antiretroviral drugs for treating and preventing HIV infection* (23)

- CXR can assist in diagnosing TB among people living with HIV. It is particularly useful for ruling out TB disease before providing treatment for latent TB infection.

CXR HELPS RULE OUT ACTIVE TB BEFORE TREATING LATENT TB INFECTION

Reference: *Guidelines on the management of latent tuberculosis infection* (24)

- CXR used in combination with symptom screening has the highest sensitivity for detecting TB and, thus, should be used to exclude active TB before initiating treatment of latent TB infection.
- Individuals with any radiological abnormality or TB symptoms should be investigated further for active TB and other conditions.

CXR IS AN ESSENTIAL TECHNOLOGY FOR PREVALENCE SURVEYS

Reference: *Tuberculosis prevalence surveys: a handbook* (21)

- CXR is a necessary screening tool to identify survey participants eligible for bacteriological examination; in recent surveys, CXR has proven essential for detecting a large proportion of prevalent TB cases.

More information can be found in the following resources:

- WHO's diagnostic imaging website (14)
- *The WHO manual of diagnostic imaging: radiographic anatomy and interpretation of the chest and the pulmonary system* (7)
- *International standards for tuberculosis care, third edition* (20)
 - *Standards for TB care in India* (25)
 - *European Union standards for tuberculosis care* (26)
 - *Canadian tuberculosis standards, seventh edition* (27)
- *Systematic screening for active tuberculosis: principles and recommendations* (4)
- *Systematic screening for active tuberculosis: an operational guide* (5)
- *Guidelines on the management of latent tuberculosis infection* (24)
- *Guidance for national tuberculosis programmes on the management of tuberculosis in children, second edition* (22)
- *Tuberculosis prevalence surveys: a handbook* (21)
- Xpert MTB/RIF implementation manual (3)
- *Consolidated guidelines on the use of antiretroviral drugs for treating and preventing HIV infection* (23)
- *Pocket book of hospital care for children* (28)
- the website for the Practical approach to Lung Health (known as PAL) (29).

2. CHEST X-RAY AS A TRIAGE TOOL

2.1 Definition of triaging

For the purpose of this document, triaging is defined as the processes of deciding the diagnostic and care pathways for people seeking healthcare, based on their symptoms, signs, risk markers and test results. Triaging involves assessing the likelihood of various differential diagnoses as a basis for making clinical decisions. It can follow more- or less-standardized protocols and algorithms and may be done in multiple steps. Effective triaging that helps to rapidly identify TB is important both for optimizing care for the individual and for ensuring good infection control (30).

Triaging protocols should be adapted to the disease's epidemiology in a given setting because the prevalence of different diseases determines the predictive values of symptoms, signs, risk markers and test results.

Triaging is different from systematic screening in that it focuses on the clinical management of a person seeking healthcare for one or several unexplained complaints or concerns, while systematic screening normally is initiated by a provider and targets apparently healthy individuals with or without risk markers for a given disease; for more information on systematic screening see *Systematic screening for active tuberculosis: principles and recommendations* (4) and the associated *Systematic screening for active tuberculosis: an operational guide* (5).

2.2 Triaging for TB among people with respiratory complaints

Proper triaging of people seeking healthcare with respiratory complaints is essential for diagnosing TB correctly and early, as well as for the early diagnosis of other conditions. Unfortunately, not all people seeking care with symptoms consistent with TB receive an adequate evaluation for TB. These failures result in missed opportunities for detecting TB early and lead to increased disease severity, more complications and a higher risk of poor treatment outcomes for the patients. They also can lead to a greater overall disease burden in the community because they increase the likelihood of transmission of *Mycobacterium tuberculosis* in health facilities and to family members and others in the community. For this reason, effective and efficient clinical triaging algorithms are of utmost importance; for more information see the *International standards for tuberculosis care* (20).

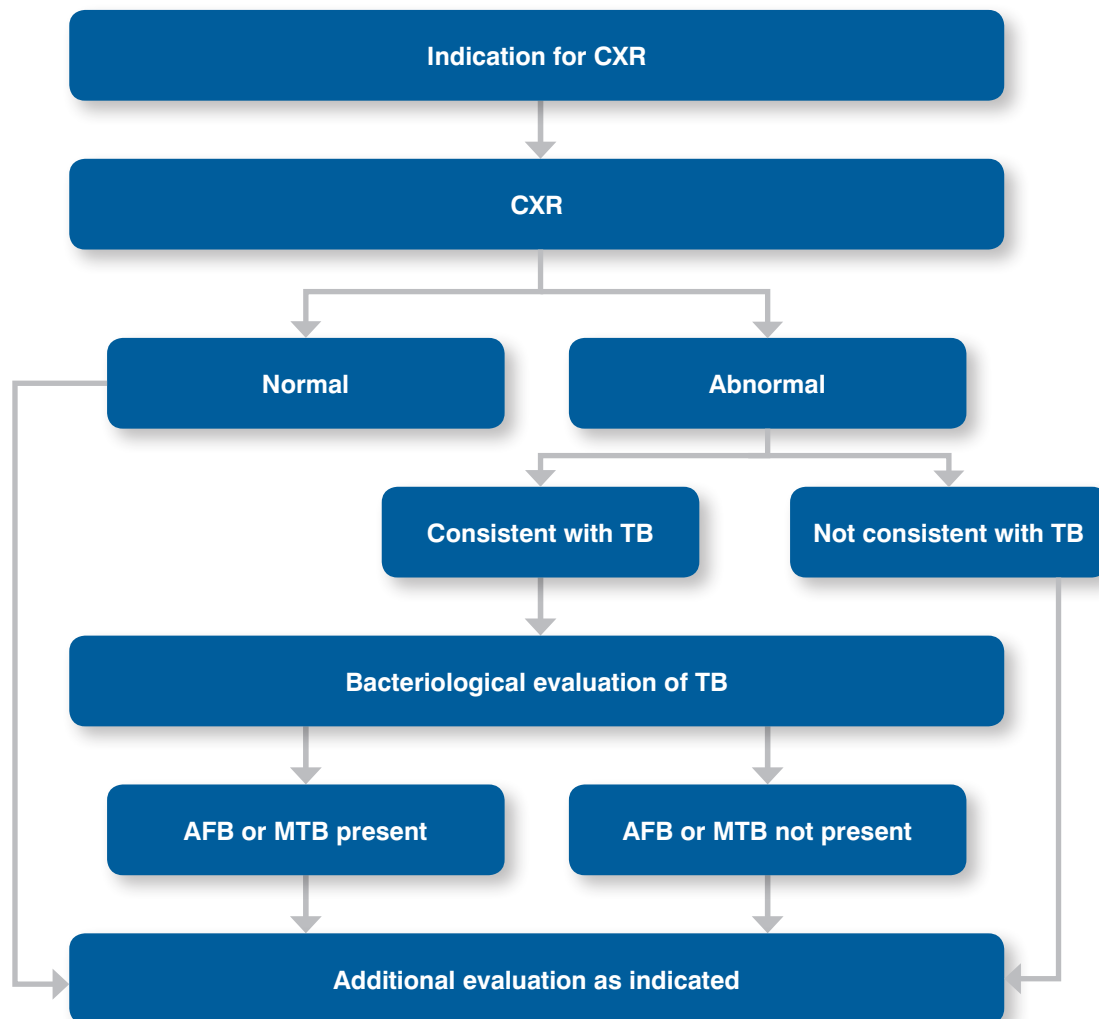
People with pulmonary TB who are seeking care often initially present with non-specific respiratory symptoms that need to be evaluated. Respiratory conditions are among the most common acute and chronic diseases worldwide; they occur in all societies and in all age groups. The heavy burden of respiratory diseases means that their symptoms are some of the most common reasons why patients seek primary healthcare: respiratory complaints (including cough, sputum production, and shortness of breath) often constitute around 20% of the symptoms that prompt a visit to a primary health centre (31); for more information see the website for the Practical approach to Lung Health (known as PAL) (29).

Thus, respiratory symptoms are both common and non-specific. Most people with respiratory symptoms consistent with TB do not have TB, even in setting where TB is highly endemic. Therefore, it is important to identify in a sensitive and efficient manner those who have a high likelihood of TB among those with respiratory symptoms and to determine the underlying cause of disease for those who are not ultimately diagnosed with TB. Triage algorithms that are appropriate to specific patient populations, the epidemiology of respiratory conditions, and healthcare-system capacity are essential for providing high-quality care.

Where it is available and feasible in the outpatient primary care setting, CXR can be used as an effective triage test for those seeking care for respiratory complaints. CXR is a sensitive tool for identifying TB, meaning that it identifies most people with a high likelihood of having the disease, while correctly ruling out TB in most persons when the X-ray is read to look for any abnormality consistent with TB. In addition, CXR can help identify other pulmonary conditions, such as lung cancer and occupational lung diseases like silicosis, as well as other intrathoracic diseases that require further diagnostic evaluation. Therefore, CXR is a useful general triage test for pulmonary conditions because it helps identify which type of further diagnostic evaluation patients require to correctly diagnose the cause of their illness. A normal CXR helps

rule out a number of pulmonary conditions and prompts diagnostic evaluation for conditions consistent with no radiological findings, while an abnormal CXR prompts evaluation for conditions consistent with radiographic changes, including but not limited to bacteriological evaluation for TB (Fig. 1). In any case, when used as a triage test, CXR should be followed by further diagnostic evaluation to establish a diagnosis. Generating differential diagnoses for conditions other than TB may be the primary objective of ordering a CXR. Regardless of the reason for obtaining a CXR, it is important that any CXR abnormality consistent with TB be further evaluated with a bacteriological test (20).

FIG. 1. Using chest radiography as a triage tool



AFB: acid-fast bacilli; CXR: chest X-ray; MTB: *Mycobacterium tuberculosis*; TB: tuberculosis.

CXR may have higher specificity for pulmonary TB than assessing symptoms alone, depending on how the X-ray is read. Therefore, triaging using CXR can help reduce the number of persons who undergo bacteriological TB testing without decreasing the detection of true TB cases. CXR also improves the positive predictive value of subsequent bacteriological tests by increasing the pre-test probability of TB (4).

Beyond identifying active TB disease, CXR also identifies one of the populations at highest risk of developing TB disease: those who have inactive TB or fibrotic lesions without a history of TB treatment. Once active TB has been excluded, patients with fibrotic lesions should be followed-up, given their high risk for developing active disease (4).

There is no comprehensive WHO guidance on using CXR in triaging individuals with respiratory symptoms. In the absence of such guidance, this chapter presents options for approaches to triage, and the contribution of CXR to each approach is described.

2.3 TB triage algorithm options

In this section, different triage algorithms for patients with respiratory complaints are discussed (up to the point of ordering an initial bacteriological test for TB) and compared to help guide the choice of an appropriate algorithm for different situations. At the end of the section, the algorithms are displayed schematically, together with indications of the yield of true positives and false positives for TB and the cost per true TB case detected, based on modelled yields and costs in a hypothetical scenario in which the prevalence of TB is 0.5% (500 cases/100 000 population) among persons entering the triage algorithm (Fig. 2). Details and additional scenarios are provided in Annex 1. The estimated yield shown concerns only bacteriologically confirmed TB, and the yield of false-positive TB corresponds to a false positive only on initial bacteriological testing. Actions to be taken after a positive or negative initial bacteriological test – including further evaluation for TB, drug resistance and for other underlying conditions, as well as the yield of true-positive and false-positive TB based on clinical diagnosis – are discussed in Section 2.3.3. and Chapter 3.

In line with the progressive realization of universal health coverage, out-of-pocket expenditures for CXR, as well as for bacteriological and other tests, should be minimized (32). Costs should be covered through fair third-party financing. If CXR is used for triaging and the subsequent diagnostic evaluation of patients with respiratory complaints, CXR and bacteriological tests should be free of charge for patients. The algorithm scenarios discussed here and displayed in Fig. 2 and in Annex 1 assume there are no direct costs for patients and estimate the potential magnitude of costs from a healthcare perspective of various approaches to triaging for TB, including using CXR and other tools. When tools are available only on the referral level, additional indirect costs for patients, as well as transport, feasibility and time factors, need to be considered when choosing an appropriate algorithm. Further discussion of CXR infrastructure, planning and financing can be found in Chapter 6.

2.3.1 Optimizing TB triaging when sputum-smear microscopy is used as bacteriological test

Traditionally, chronic and productive cough (with a duration of longer than 2–3 weeks) has been used by national tuberculosis programmes (NTPs) as a triaging criterion for determining who should undergo sputum-smear microscopy (Algorithm 1). This approach identifies people with an advanced stage of TB, and it is a rational public health approach when the priority is to detect highly infectious pulmonary TB or when CXR is not available. If available, CXR can be used as an additional triage test after initially triaging for chronic cough (Algorithm 2). Because CXR is sensitive, Algorithm 2 has a similar yield to Algorithm 1 of true-positive cases detected with sputum-smear microscopy while reducing the number of persons who need to undergo sputum-smear microscopy. However, the cost can be higher, depending on the cost of CXR as compared with the cost of sputum-smear microscopy. The number of false-positive sputum-smear microscopy results is low for both of these algorithms when the TB prevalence is moderate to high, although it is lower in Algorithm 2, which includes CXR. However, introducing CXR before a bacteriological test can increase the total number of clinically diagnosed cases and, thus, also the total number of false-positive cases, depending on what further diagnostic evaluation and treatment decisions are made for patients with abnormal CXR and negative bacteriological tests (see Section 2.3.3. and Chapter 3).

Triage algorithms based on chronic cough have a low sensitivity for TB, and, thus, many cases will be missed with this approach, especially in the early stages of disease. Recent TB prevalence surveys have demonstrated that a large proportion of people with bacteriologically confirmed TB do not experience chronic cough (21). Annex 2 shows the proportion detectable through screening with CXR and screening for chronic cough among persons with bacteriologically confirmed TB detected in recent TB prevalence surveys.

Earlier and more complete detection of TB among people seeking healthcare may, therefore, require bacteriological testing for TB using broader indications, especially in TB-endemic settings. This may include testing all people with any symptom or sign consistent with TB (that is, any one of cough, haemoptysis, fever, night sweats or weight loss) as in Algorithm 3, or testing those with a predefined constellation of symptoms, signs and clinical or population-based risk markers for TB (for example, HIV, other immune-compromising conditions, diabetes, renal failure, smoking, alcohol or substance abuse, undernutrition, poverty, homelessness, history of imprisonment, migration from a TB-endemic setting). The appropriate threshold or indication for applying a bacteriological test depends on the local TB epidemiology. As a guiding principle, the higher the TB prevalence, the broader the indication for TB testing should be.

Testing using a broad indication, such as any symptom consistent with TB, increases the total yield of TB cases detected. However, it can lead to high demands on resources and overloaded laboratories. It also leads to a lower positive predictive value for the bacteriological test result and, thus, a higher risk of false-positive test results. Narrowing symptom criteria will, in most situations, lower the sensitivity while increasing the specificity, and reduce the number who need to undergo bacteriological testing. Using CXR as an additional triage test is especially useful in the context of more inclusive initial symptom triaging, and it can lead to a high total yield with fewer bacteriological tests per detected case and fewer false-positive sputum-smear microscopy results as compared with symptom screening alone (see Algorithm 4). However, introducing CXR into an algorithm that uses smear microscopy as the bacteriological test can increase the cost per true case detected, depending on the relative costs of CXR and microscopy. It can also increase the number of clinically diagnosed cases, of which a substantial proportion may be false positives if proper quality control of clinical diagnosis is not in place (see Section 2.3.3. and Chapter 3).

Algorithm 5 uses CXR as an initial triage test (regardless of symptoms, signs and other risk markers), which has been suggested as a rigorous triaging approach for healthcare facilities in some hyperendemic settings). It improves the sensitivity of triaging as compared with initial symptom-based triaging and, thus, improves case detection, but it can increase resource demands considerably, including for laboratories. Such an approach is equivalent to systematic TB screening in health facilities, which is further discussed in Chapter 3.

2.3.2 Chest X-ray triaging to optimize use of the Xpert MTB/RIF assay

As NTPs adopt and roll out the Xpert MTB/RIF assay into routine practice for TB diagnosis, it becomes important to determine how the test can be most efficiently used in evaluating patients with suspected TB; for more information see the Xpert MTB/RIF implementation manual (3).

In Algorithm 6, replacing sputum-smear microscopy with the Xpert MTB/RIF assay as the primary bacteriological test after initial triaging for chronic cough increases the yield of bacteriologically confirmed TB. However, the cost per true case detected is considerably higher than with an algorithm using sputum-smear microscopy due to the higher costs for the Xpert MTB/RIF assay. One approach to reducing the number of individuals who undergo Xpert MTB/RIF testing without significantly reducing the yield of TB cases detected is to use Algorithm 7; in this algorithm CXR is used as a second triage test after initial triaging for chronic cough, after which patients with abnormal CXR results are referred for Xpert MTB/RIF testing for confirmatory diagnostic evaluation. If CXR is readily available and considerably cheaper than Xpert MTB/RIF testing, which is often the case, especially with digital CXR, this could greatly reduce the cost per true case detected.

CXR triaging also increases the positive predictive value of Xpert MTB/RIF testing. The specificity of the Xpert MTB/RIF assay for detecting TB is high (99%, with liquid culture as reference standard) (2). However, given that it is not 100%, the positive predictive value of Xpert MTB/RIF testing may be low when applied to groups with a relatively low TB prevalence. Using CXR as a triage test for those who have chronic cough and then referring those with abnormal results for confirmatory testing with the Xpert MTB/RIF assay results in a higher prevalence of TB in the group tested and, thus, increases the positive predictive value and minimizes false-positive results. Using CXR as a triage test is further discussed in the Xpert MTB/RIF implementation manual (3).

However, the sensitivity of Algorithms 6 and 7 remains limited due to the low sensitivity of the initial triaging for chronic cough. In order for Xpert MTB/RIF testing to significantly improve early TB detection it may need to be used with a broad indication. Using Xpert MTB/RIF testing as the primary diagnostic test after triaging for any TB symptoms is a highly sensitive approach (see Algorithm 8). However, this algorithm is expensive and requires high-throughput capacity for Xpert MTB/RIF testing. Using CXR as a second triage test is especially valuable when the initial triaging is for any TB symptom (see Algorithm 9). Algorithm 9 can significantly reduce the laboratory burden, cost per true case detected and false-positive bacteriological test results.

As discussed in Section 2.3.1, introducing CXR before a bacteriological test can increase the total number of clinically diagnosed cases and, thus, also the total number of false-positive cases, depending on what further evaluation and treatment decisions are made for patients with abnormal CXR and negative bacteriological tests. This is more likely when sputum-smear microscopy is used than when Xpert MTB/

RIF testing is used because the Xpert MTB/RIF assay is more sensitive and, therefore, has a much higher negative predictive value than sputum-smear microscopy. This means that the likelihood is low that a person with a negative Xpert MTB/RIF test has TB. It is important that clinicians interpret a negative Xpert MTB/RIF result differently than a negative sputum-smear microscopy result. Although no bacteriological test can completely rule out TB, a clinical diagnosis needs to be considered for a much smaller proportion of persons who are negative by Xpert MTB/RIF testing than for persons who are negative by sputum-smear microscopy (see Chapter 3).

The most sensitive algorithm is to use CXR for initial triaging regardless of symptoms, followed by Xpert MTB/RIF testing (see Algorithm 10). However, this algorithm is more expensive and results in a higher number of false-positive results than sequential triaging with symptoms and CXR.

FIG. 2. Algorithm options for triaging patients with respiratory complaints consistent with TB^a

Triage algorithm	\$ Cost per true case detected + Yield of true-positive results + Yield of false-positive results
1. Cough followed by microscopy 	\$ + + +
2. Cough followed by CXR followed by microscopy 	\$ \$ + +
3. Any TB symptom followed by microscopy 	\$ \$ + + + + + + + +
4. Any TB symptom followed by CXR followed by microscopy 	\$ \$ \$ + + + +
5. CXR followed by microscopy 	\$ \$ \$ \$ \$ + + + + + + +
6. Cough followed by Xpert MTB/RIF testing 	\$ \$ + + +
7. Cough followed by CXR followed by Xpert MTB/RIF testing 	\$ \$ + + +
8. Any TB symptom followed by Xpert MTB/RIF testing 	\$ \$ \$ \$ + + + + + + +
9. Any TB symptom followed by CXR, followed by Xpert MTB/RIF testing 	\$ \$ \$ + + + + + +
10. CXR followed by Xpert MTB/RIF testing 	\$ \$ \$ \$ \$ + + + + + + + +

CXR: chest X-ray.

^a The figure shows the potential yield of triage and the subsequent diagnostic evaluation for TB based on using the indicated tools. The cost and yield indicator values are based on a hypothetical scenario of a triage population of 100 000 with a TB prevalence of 0.5% (500 cases/100 000 population). (Details are available in Annex 1.) The Cost indicator (\$) corresponds to the cost per true case detected for each algorithm; \$ corresponds to the algorithm with the lowest relative cost per true case detected and \$\$\$\$ corresponds to the algorithm with the highest cost per true case detected. The True-positives indicator (green +) corresponds to the yield of true TB cases (with liquid culture as the gold standard); one green + corresponds to the algorithm with the lowest yield of true-positive cases and five green + + + + + signs correspond to the algorithm with the highest yield of true-positive cases. The False-positives indicator (red +) corresponds to the number of false-positive bacteriological test results; one red + corresponds to the algorithm with the lowest number of false-positive test results and five red + + + + + signs correspond to the algorithm with the highest number of false-positive results. Note that true and false positive indicators within each algorithm are not proportional and can therefore not be directly compared to assess positive predictive values. Also note that predictive values change with prevalence. See Annex 1 for actual numbers with different prevalence.

2.3.3 Using chest X-ray after a negative bacteriological test result

A negative initial bacteriological test result may require additional bacteriological testing (see Chapter 3). This can be done in parallel with a CXR if CXR was not done previously. For the diagnosis of non-bacteriologically confirmed TB, CXR has previously been recommended to be performed after a negative bacteriological test result and a trial treatment with broad spectrum antibiotics other than those used to treat TB (17, 18). However, trial treatment with broad spectrum antibiotics is not in line with general principles on the use of antibiotics, which should be reserved for treating a clear indication and not primarily used for diagnostic purposes. Trial treatment with antibiotics is particularly discouraged in children (22).

Putting CXR at the end of a diagnostic algorithm may sometimes be the only option – for example in primary healthcare facilities where X-ray is not available and patients need to be referred to a secondary care facility. In this case, often patients with symptoms highly suggestive of TB and negative smear microscopy results are then referred to secondary care facilities for further evaluation with CXR. This approach is rational when infectious TB is the priority or if radiography access is limited. However, the approach can lead to delayed TB diagnosis and loss to follow-up during the diagnostic work-up, especially when a bacteriological test with low sensitivity is used, such as sputum-smear microscopy (which misses many true cases).

One rationale for not doing a CXR before a bacteriological test is to avoid identifying a large group of people as possible TB patients via an early CXR, which can lead, in turn, to false-positive CXR-based clinical diagnosis of TB (see Sections 2.3.1 and 2.3.2, and Chapter 3). However, the risk of a false-positive clinical diagnosis can be reduced by using proper quality control as well as by improving access to sensitive and rapid molecular tests, or culture, or both, which can more effectively rule out TB than sputum-smear microscopy and can, therefore, reduce the number of cases diagnosed based on CXR findings.

2.3.4 Estimating the yield and cost to decide where to place chest X-ray in triaging for TB

The judgement of whether to expand CXR use as part of triaging for TB and the decision of where to put CXR in a triaging algorithm need to be based on an assessment of the risks of overdiagnosis versus underdiagnosis, and in the context of resource demands, the availability of human resources, the accessibility of CXR, and the feasibility and cost effectiveness. It is also important to consider that if all tools included in a diagnostic algorithm are not located in the same physical location, patients risk being lost in the diagnostic pathway. A case study on CXR in TB detection algorithms in India is provided in box 3.

It may be helpful to estimate for different triaging protocols the number of CXR investigations and number of bacteriological tests required (and related resource demands), as well as the potential yield (including true positives and false positives, and true negatives and false negatives; see Annex 1 for an example). Unfortunately, the scientific evidence about the performance and cost effectiveness of different triaging algorithms using CXR is limited. The ScreenTB tool (12), initially developed for systematic screening, can be used for preliminary modelling of the theoretical expected yield and related costs required for some triage algorithms and the epidemiological scenarios in a given setting. However, the assumptions (originally put in place for screening) may need to be changed, and the outputs of the tool are only indicative.

The introduction of any new algorithm should be coupled with careful monitoring and evaluation of its performance, in terms of the yield and the proportion of detected cases that are bacteriologically confirmed (5).

BOX 3. Case study on chest radiography in the TB detection algorithms in India

India has the highest TB burden of any country, representing 23% of the total global burden. Although policies requiring mandatory case notification, a new web-based information system and greater engagement with the private sector have led to significant recent increases in case notification, there is still a large case-detection gap. In response, the country is looking for ways to increase case detection, including developing more sensitive approaches for identifying persons who need to be evaluated for TB. CXR has, therefore, returned to the agenda after several decades of marginal programmatic use.

CXR has long featured in TB detection in India, but its use has varied. In the 1960s, radiography was used for mass TB screening and clinical diagnosis of TB. These uses were accompanied by high rates of clinically diagnosed TB. Beginning in the 1990s, with the adoption of DOTS, CXR was still used to diagnose TB but less often because it was placed at the end of the recommended diagnostic algorithms, to be used only after multiple negative smear examinations and a trial course of antibiotics. Accordingly, the amount of training for providers dedicated to the use of CXR decreased substantially over the same period. When India launched its first national TB programme in 1962, District TB Officers were trained at the national level for 3 months. For more than 30 years, the main component of training aimed to help the officers develop skills in the CXR-based diagnosis of TB. After DOTS was introduced in the 1990s, under the Revised National TB Control Programme, training was gradually reduced to 2 weeks. The use of CXR was discouraged because previously TB had been overdiagnosed in the NTP when diagnosis was based on CXR. Consequently, the focus of training shifted to ensuring high-quality smear microscopy, and training in CXR vanished from the human resources development plan. Overall, this led to a decrease in the skills needed to evaluate CXRs, an inadequate infrastructure and availability of radiology, and minimal CXR monitoring and quality assurance, especially in peripheral areas. Throughout the period from the 1960s to the present, however, CXR has been widely used in the private sector, where it has been the preferred initial tool for TB detection, and it has also been used extensively for clinical diagnosis. No formal quality assurance procedures for CXR and clinical diagnosis in the private sector have been undertaken.

In 2016, after programme evaluations suggested that only 5% of patients symptomatic for TB were screened completely using the existing diagnostic algorithms, and data from prevalence surveys suggested there would be an additional 30–40% diagnostic yield using CXR as a screening tool, the Ministry of Health and Family Welfare of India recommended revised diagnostic algorithms that feature CXR as an early triage tool for evaluating patients with suspected TB, to be used with bacteriological confirmatory evaluation and continued clinical assessment of bacteriologically negative patients.

However, the implementation of the revised triage and diagnostic algorithms will likely be hampered by the limited availability of high-quality CXR. Currently, CXR technology is available only in the private sector and at the public sector referral level. Along with limited equipment for CXR, only limited personnel have been trained to deliver and interpret CXRs. Additionally, the financing for radiology services varies by state, and although many states have provisions for free-of-charge CXR in the general health budget for patients with suspected TB, some require fees from patients. Technological innovations may provide future opportunities to address financial and human-resource capacity gaps in CXR access, including by using digital CXR in combination with telemedicine.

More information can be found in the following resources:

- *International standards for tuberculosis care*, third edition (20)
- the website for the Practical approach to Lung Health (known as PAL) (29)
- Xpert MTB/RIF policy update and implementation manual (2, 3)
- *Consolidated guidelines on the use of antiretroviral drugs for treating and preventing HIV infection* (23)
- *Systematic screening for active tuberculosis: principles and recommendations* (4)
- *Systematic screening for active tuberculosis: an operational guide* (5)
- ScreenTB tool: target prioritization and strategy selection for tuberculosis screening (12).

3. CHEST X-RAY AS A DIAGNOSTIC AID

3.1 Chest X-ray as a diagnostic aid for respiratory and other intrathoracic diseases

As discussed in Chapter 2, people with respiratory symptoms who are seeking care need to be evaluated not only for TB but for all relevant respiratory diseases, which may include, for example, non-TB infections, diseases of airflow obstruction (such as chronic obstructive pulmonary disease or asthma), neoplasms (such as lung cancer or pulmonary metastasis), occupational lung diseases (such as silicosis) or bronchiectasis. CXR is a useful diagnostic aid for several of these conditions, as well as for diseases that affect extrapulmonary intrathoracic structures (such as mediastinal lymph nodes, the pleura or pericardium). Good quality chest radiographs are essential for proper evaluation.

CXR should be seen as a tool that can be used in comprehensive pathways for diagnosing respiratory and other intrathoracic diseases. Beyond algorithms for TB diagnosis and treatment, protocols are needed for the proper referral and management of other diseases, in line with the principles of the practical approach to lung health (29).

CXR is an essential health technology that should be accessible to all. However, because access remains limited in many settings, and it may be available only in tertiary care, the choice to use CXR as a diagnostic aid clearly depends on the availability of and access to CXR (see Chapter 6).

3.2 Chest X-ray as a complement to bacteriological TB tests

It must be underscored that although CXR is a useful adjunct in diagnosing TB, CXR alone cannot establish a diagnosis. Bacteriological confirmation of TB should always be attempted. The triage algorithms described in the previous chapter show only the steps leading to an initial bacteriological test, but not the subsequent bacteriological testing and clinical assessment that may be required.

Additional bacteriological testing may be required after a negative initial bacteriological result. This is especially relevant if the initial test is sputum-smear microscopy, which has low sensitivity. For such patients, subsequent Xpert MTB/RIF testing or culture should be done if the clinical suspicion of TB is moderate to high – for example, due to CXR findings consistent with TB.

Although Xpert MTB/RIF testing is more sensitive than smear microscopy, the sensitivity of the test in a given setting depends on the prevalence of smear-negative TB in the patient population; this is because the Xpert MTB/RIF assay is highly sensitive for smear-positive TB but only moderately sensitive for TB that is smear negative (2). Hence, repeated testing should be considered using the Xpert MTB/RIF assay or culture, or both, after an initially negative Xpert MTB/RIF assay or culture. Culture can improve the sensitivity of bacteriological confirmation, although it typically takes 6–8 weeks before results are available. Considerations for sequential bacteriological testing and suggested bacteriological test algorithms, including for diagnosing drug-resistant TB, are included in WHO's framework on implementing TB diagnostics as well as in the *International standards for tuberculosis care* (20, 33).

Making a clinical diagnosis based on medical history (symptoms, TB exposure, risk markers), signs and CXR findings is sometimes reasonable in persons in whom TB cannot be ruled out despite negative bacteriological tests. In combination with clinical assessment, CXR may provide important circumstantial evidence for clinical diagnosis (20). Repeated CXR examination after a period of time could be considered, as interpretation of image changes could aid in clinical evaluation for TB and other diagnoses.

Clinical diagnosis is particularly relevant in certain groups for whom it can be difficult to confirm a TB diagnosis with a bacteriological test. This includes patients for whom bacteriological tests tend to have lower sensitivity, such as people living with HIV or people with other immune-compromising conditions. It also includes patients from whom it is difficult to collect samples for bacteriological confirmation, such as young children. Moreover, for seriously ill patients (particularly persons with HIV infection), a clinical decision to start treatment often must be made without waiting for test results. Such patients may die if appropriate treatment is not begun promptly (4, 20, 23). In such patients, CXR can be particularly useful

as a diagnostic aid given the rapidity with which it delivers results. The risks associated with a delayed or missed TB diagnosis in these groups can be higher than the risks associated with a possible false-positive clinical diagnosis, which means that a clinical diagnosis made with a relatively low positive predictive value may be acceptable. However, it should be noted that the accuracy of CXR in these groups may be lower than in other groups. If a patient is not critically ill, follow-up bacteriological testing, repeat CXR and re-evaluation of TB and differential diagnoses should be considered (19, 20).

Patients in whom a clinical diagnosis of TB has been made should be followed closely to ensure that a non-TB disease has not been misdiagnosed and left untreated. Follow-up should include repeat clinical and radiological assessment. For those who deteriorate or fail to improve, repeat bacteriological testing for TB should be considered (that is, to ensure the initial results were not falsely negative) and also diagnostic testing for non-TB diseases.

Quality assurance for clinical TB diagnoses is important (see Chapter 5), both to not miss TB cases and to avoid overdiagnosis. Unnecessary TB treatment should be avoided to protect patients from harm and avoid wasting resources, both for patients and society. Moreover, making a false-positive clinical TB diagnosis when the illness has another cause can delay proper diagnosis and treatment of the true illness – for example, lung cancer (34). When an abnormality is present on CXR it is important to consider all possible differential diagnoses. Persons with confirmed or unconfirmed TB may have other concurrent lung conditions.

Quality assurance involves ensuring high-quality CXR as well as high-quality CXR reading (35), (36). Quality and standardization can be improved by conducting multidisciplinary clinical rounds, convening groups of diagnostic experts and implementing peer review procedures. Monitoring and evaluation are essential. A useful indirect indicator of quality is the proportion of TB cases that have not been bacteriologically confirmed. A high proportion of clinically diagnosed TB may indicate overdiagnosis. A low proportion may indicate underdiagnosis. However, there is no established benchmark for the appropriate proportion of cases of bacteriologically confirmed TB.

3.3 Chest X-ray as part of a comprehensive diagnostic pathway in children

CXR is useful in the diagnostic evaluation of TB as well as other intrathoracic diseases in children, especially younger children, in whom bacteriological evaluation is commonly negative. It should be part of a comprehensive diagnostic pathway that includes multiple steps, beginning with clinical assessment, the assessment of risk factors and exposure history, and CXR and bacteriological tests, as required. In most cases, children with TB have radiographic changes suggestive of TB (22). Adolescent patients with TB have radiographic changes similar to those seen in adult patients. Good-quality CXRs are essential for proper evaluation (22). A lateral CXR view may be required, especially in younger children and when bacteriological confirmation is challenging. For example, children younger than 4 years are more likely to have primary TB, and a lateral view will be important in identifying mediastinal or hilar lymphadenopathy (37). Details are provided in the *Guidance for national tuberculosis programmes on the management of tuberculosis in children* (22).

As with adults, there are no specific features on clinical examination that can confirm that the presenting symptoms are due to TB. Every effort should be made to confirm the diagnosis of TB in a child using whatever specimens and laboratory tests are available (22). The Xpert MTB/RIF assay should be used as the initial test in all children suspected of having TB, regardless of their HIV status (2).

It should be noted that children living with HIV have a substantially elevated risk of TB. The approach to diagnosing TB in children living with HIV is essentially the same as for diagnosis in HIV-negative children. However, this approach can be more challenging in children living with HIV because there is a high incidence of acute and chronic lung diseases other than TB; because children with HIV may have lung disease of more than one cause (multimorbidity), which can mask their response to therapy; and because there is an overlap of radiographic findings in TB and from other HIV-related lung diseases, making CXR less specific for detecting TB in children with HIV (22).

CXR quality assurance can be especially challenging when CXR is used in children. Radiographers and medical staff require special training on taking and reading CXRs in children, and ensuring this training

occurs is a particular challenge in low- and middle-income countries where there are few paediatric-orientated radiologists.

More information can be found in the following resources:

- *International standards for tuberculosis care, third edition (20)*
- *Guidance for national tuberculosis programmes on the management of tuberculosis in children, second edition (22)*
- Xpert MTB/RIF policy update and implementation manual (2, 3)
- *Consolidated guidelines on the use of antiretroviral drugs for treating and preventing HIV infection (23).*

4. CHEST X-RAY AS A SCREENING TOOL FOR PULMONARY TB

4.1 Chest X-ray as a sensitive tool for screening for active TB

Systematic screening for active pulmonary TB is defined as the “systematic identification of people with suspected active TB in a predetermined target group, using tests, examinations or other procedures that can be applied rapidly” (4). Unlike the evaluation of those who actively seek care for respiratory symptoms (see triaging, Chapter 2), the systematic screening of individuals for TB is typically initiated by a provider and offered in a systematic way to an apparently healthy target group that has been determined to have a high risk of TB.

Systematic screening implemented within health facilities is a special systematic screening situation in which specific risk groups at high risk of TB are targeted for screening among people seeking care in health facilities – for example, people being treated for diabetes or people living with HIV who are attending a clinic to receive antiretroviral therapy. Systematic screening of all people seeking care in an outpatient department may be considered in a setting where TB is highly endemic. Such an approach can also be seen as aggressive triaging for TB; the same principles for choosing an appropriate screening algorithm apply.

Systematic screening outside health facilities – such as in the community or in special institutions such as prisons or shelters for homeless people – is often labelled active case finding, which refers to a provider-initiated approach that actively reaches outside the health services. Such screening can help find prevalent cases of TB in the community that might otherwise go undiagnosed and untreated. It often requires screening a large number of people who do not have TB. Costs can be high, and the risk of a false-positive diagnosis is high when TB prevalence in the screened group is low or moderate (4). Because of this, choosing an accurate screening algorithm is critical, and there are specific considerations involved, including how well the screening and diagnostic tools perform in the population to be screened, the trade-off between risks and benefits to the person being screened, the ability of the screening algorithms to detect TB without risking overdiagnosis, and the feasibility and costs (4, 5).

WHO's guideline *Systematic screening for active tuberculosis: principles and recommendations* describes 10 algorithms for screening for TB (4). Eight options include a symptom screen as the initial test (screening either for cough lasting longer than 2 weeks, or screening for any symptom consistent with pulmonary TB, including cough of any duration, haemoptysis, weight loss, fever or night sweats), and two options use CXR as the initial screening test. If symptom screening is used initially, then CXR can be used as a second screen to improve the pre-test probability of the subsequent diagnostic test and to reduce the number of people who need to undergo further diagnostic evaluation. See box 4 for an example of the use of CXR as a tool for TB screening in migrants and refugees.

As discussed in Chapter 2, symptom screening has a low sensitivity, especially for detecting TB early, which is a primary objective of systematic screening. Nevertheless, symptom screening is a low-cost and feasible approach that may be the only option in some situations. CXR, however, is a good screening tool for pulmonary TB because of its high sensitivity (87% to 98%, depending on how the CXR is interpreted) (4), meaning that up to 98% of those with culture-positive TB who undergo CXR will have an abnormal result. From the perspective of the person being screened, CXR is a valuable tool because it provides rapid screening results for a range of medical conditions beyond TB (21).

Like symptom screening, CXR has low specificity in an active case-finding situation (46% to 89%, depending on how it is read) (4), meaning that a significant proportion of individuals without TB will have an abnormal test result. This is due, in part, to the fact that CXR identifies many types of lung abnormalities, whether due to TB or to other lung conditions. For this reason, CXR should be used with a bacteriological confirmatory test that has high sensitivity and specificity for TB. However, as discussed in Chapter 2, it is unavoidable that some proportion of persons who screen positive by CXR and who are negative by bacteriological tests will be diagnosed based on CXR and clinical criteria alone. Such patients should be periodically re-evaluated to assess response to treatment for TB and to exclude other diseases that might explain the

radiographic findings. A significant proportion of those with a clinical diagnosis may not, in fact, have TB and, thus, may be treated unnecessarily. At the same time, some of the true TB cases will be missed.

The proportions of false-positive and false-negative cases found through screening will depend on the TB prevalence in the screened group, on the quality of the laboratory test, and on the rigour used in making clinical diagnoses, including the quality of CXR and CXR reading, as well as the quality of clinical evaluation (4). It is essential to ensure that the diagnostic quality is high, including for clinical diagnosis. It is also important that people invited for screening are well informed about the diagnostic challenges. This is particularly important in the context of screening as compared with triaging, for two reasons: screening is normally provider-initiated rather than patient-initiated (which leads to particular ethical considerations) and the prevalence of TB in the tested group tends to be much lower (which leads to lower positive predictive values).

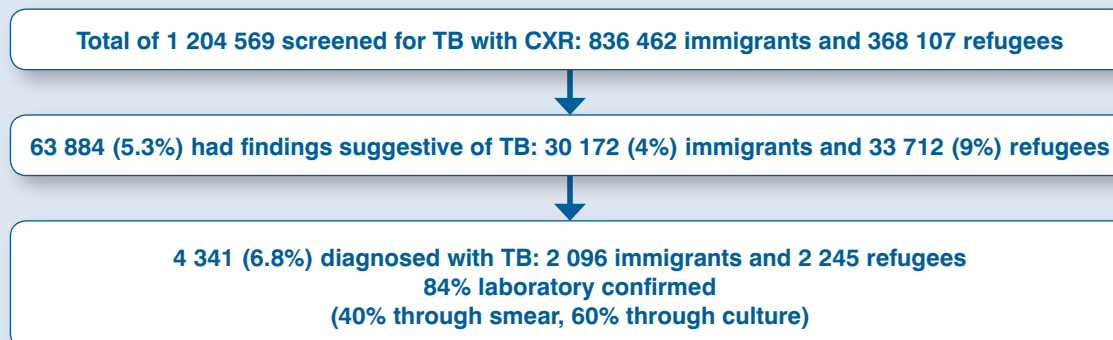
Although CXR is the preferred screening tool from the viewpoint of test accuracy, it can be expensive and logistically challenging to use, especially during active case finding when screening is done as an outreach activity outside the health services. The best choice of screening tool for any given situation depends on several factors, including the setting where screening is to be done, the populations to be screened and their epidemiology of TB and associated risk factors, the resources available (such as, human or financial), and the feasibility of various screening options. CXR is a good choice in most screening scenarios, particularly those based in the healthcare setting (see Case study 2) or where it is feasible to utilize mobile X-ray technology, but it will not be feasible in some scenarios. WHO's *Systematic screening for active tuberculosis: an operational guide* provides some guidance for the processes of planning screening and choosing an appropriate screening approach (5). The operational guide (5) and the *ScreenTB* tool (12) can be used to model the potential yield of true-positive and false-positive cases from various screening approaches, as well as costs, which can help in prioritizing the risk groups to be screened and choosing a screening algorithm. As with the discussion of optimizing triage algorithms in section 2.3, the introduction of any new algorithm for screening should be coupled with careful monitoring and evaluation of its performance, in terms of the yield and the proportion of detected cases that are bacteriologically confirmed (5).

BOX 4. Case study on the International Organization for Migration's experience of using digital chest radiography and teleradiology to screen migrants for TB.

The International Organization for Migration (IOM) conducts health assessments for selected immigrants and refugees during the process of resettlement. A key component of the health assessment is radiological screening for TB for migrants from countries with a significant burden of TB. CXR is performed on all adults and on children with indications for TB evaluation (such as a positive tuberculin skin test or interferon-gamma release assay, or being a close contact of a person with TB or HIV). For the detection of TB, an abnormal CXR suggestive of TB is the main criterion for referral for sputum-smear microscopy and culture examination; clinical suspicion is the main criterion for referral for cases with a normal CXR.

Because this health screening is conducted in an apparently healthy population, TB is usually detected during an early phase of disease, and the findings on CXR are often subtle. Skilled radiologists experienced in interpreting screening CXRs for TB are essential. The difficulty in finding appropriately skilled radiologists, especially in rural areas, prompted the IOM to develop a Global Teleradiology and Quality Control Centre in 2012. The aim was to standardize and optimize the quality of CXR used across all IOM health assessments. The centre utilizes a teleradiology system, with web-based applications and live chat for communications and to transfer digital images and provide CXR reports in real time. The centre provides primary CXR reading if necessary, with a turnaround time of 1 hour, as well as quality control and radiology-related technical support to IOM health-assessment field operations globally.

From 2011 to 2015, IOM conducted health assessments for more than 1.5 million people, of which 1.2 million immigrants and refugees were screened for TB using CXR as part of their health assessment. Among those screened, 5% had CXR findings consistent with TB, and among those almost 7% were diagnosed with TB. Among diagnosed TB cases, 84% were bacteriologically confirmed. The majority were culture-confirmed smear-negative cases.



These experiences demonstrate the potential for using CXR to screen a large group of apparently healthy individuals for TB and find a significant number of cases. The high proportion of culture-confirmed smear-negative TB indicates the low sensitivity of smear microscopy and that it is possible to maintain a high standard of bacteriological confirmation with culture. The IOM experience also shows the potential for digital and communication technologies to increase access to CXR and to skilled radiology services globally. However, nearly 400 immigrants or 164 refugees needed to be screened to detect 1 case of TB. The cost effectiveness of this type of mass screening needs to be considered. The TB detection rate was 2.5 times higher in refugees than in immigrants, which indicates that screening targeted high-risk groups will be more cost-effective than screening all migrants.

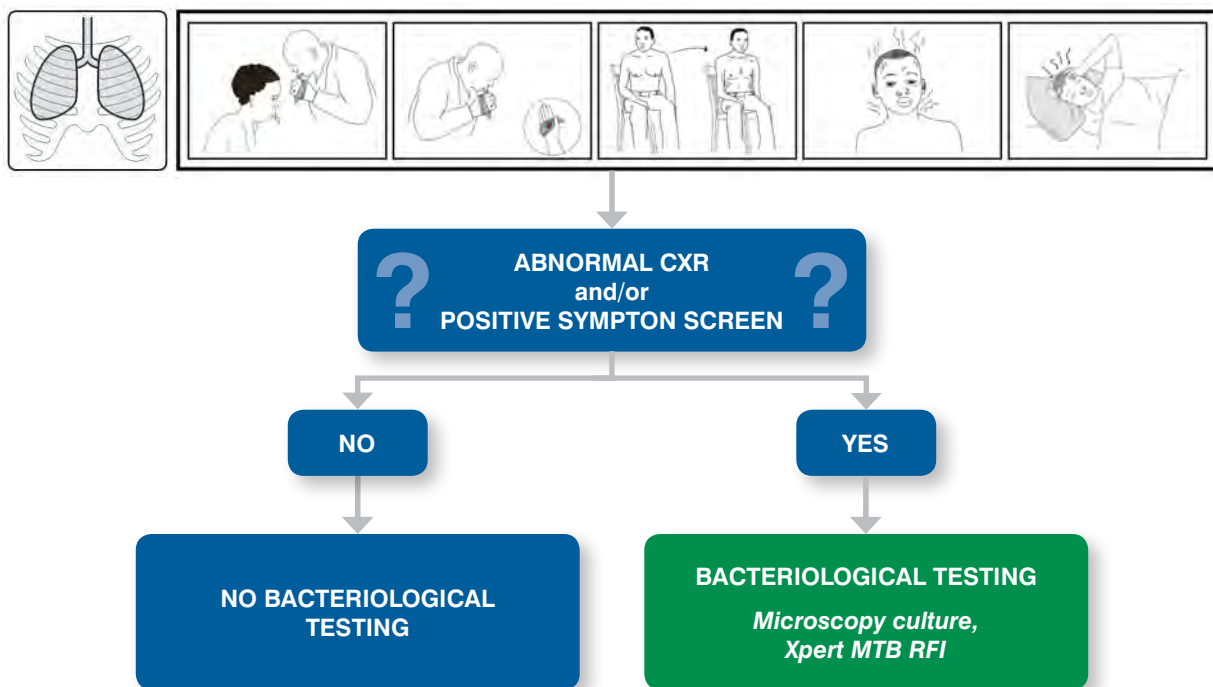
4.2 Chest X-ray screening in TB prevalence surveys

In the specific context of a national TB prevalence survey, in which a country's entire adult population is sampled and tested to determine the population prevalence of TB, screening is applied to identify individuals who should undergo bacteriological examinations. CXR is the most sensitive screening tool for identifying those survey participants with a high probability of having TB. For diagnosis, combining CXR and symptom checklists for screening (with, typically, a positive result in either category being sufficient to warrant further testing) with culture or an alternative bacteriological test with high sensitivity (such as the Xpert MTB/RIF assay), will generate the most accurate prevalence estimate for bacteriologically positive TB (which is the objective a TB prevalence survey). CXR should, therefore, be used for all participants in a survey, regardless of their symptoms or risk markers (21). It should be noted that prevalence surveys

generally do not include children (usually defined as less than 15 years of age), due to ethical concerns and because they require distinct screening and diagnostic procedures, so survey findings do not represent the entire population and cannot be extrapolated to the younger age group.

The screening strategy recommended by the WHO Global Task Force on TB Impact Measurement is shown in Fig. 3. Those persons with abnormalities on CXR and/or found to have TB-compatible symptoms during the questionnaire screening are eligible for sputum examination with the Xpert MTB/RIF assay or culture, or both. Those who do not have abnormalities or symptoms suggestive of TB during screening are not eligible for bacteriological testing and do not submit sputum samples. For more information, see Tuberculosis prevalence surveys: a handbook (21).

FIG. 3. WHO's recommended screening strategy for TB prevalence surveys (21)



CXR: chest X-ray.

A screening strategy using symptom screening without CXR screening is not recommended because it will underestimate the true prevalence of TB. Between 23% and 70% of bacteriologically positive cases detected in recent surveys had chronic cough, meaning that, on average, half of bacteriologically confirmed cases will be missed by symptom screening alone (see Annex 2) (21).

To increase sensitivity, intentional overreading of CXRs should be encouraged in the context of a prevalence survey. That is, participants with any lung abnormality (even if it may not be considered typical of TB) should be referred for bacteriological examination. Overreading should ensure that almost all potential TB cases are referred for sputum examination. Because TB in an immunocompromised person with a high risk of TB, such as an HIV-positive individual or, to a lesser extent, a person with diabetes, often shows atypical manifestations in a CXR, using CXR abnormalities suggestive of TB to identify individuals eligible for sputum examination may not be sensitive enough. Therefore, the recommended definition for screening is any CXR abnormality in the lung (21). Follow-up of participants with abnormal CXR and negative bacteriological exams is sometimes warranted.

4.3 Chest X-ray to rule out active TB before treating latent infection

Due to the crucial importance of excluding active TB before initiating treatment for latent TB infection (LTBI), one of WHO's recommendations for managing LTBI in resource-rich settings with a TB incidence of < 100 cases/100 000 population states that before initiating treatment for LTBI, individuals should both be asked about any symptoms of TB and should undergo CXR. Individuals with TB symptoms or any radiological abnormality should be investigated further for active TB and other conditions. For more information, see the Guidelines on the management of latent tuberculosis infection (24). Guidelines for managing LTBI in resource-poor high-burden settings are being developed; presently, in such settings CXR is not usually used for evaluating asymptomatic children under 5 years of age.

Using the combination of any abnormality in a CXR and/or the presence of any symptoms suggestive of TB (that is, any one of cough, haemoptysis, fever, night sweats, weight loss, chest pain, shortness of breath or fatigue) offers the highest sensitivity and negative predictive value for ruling out TB (24).

More information can be found in the following resources:

- *Systematic screening for active tuberculosis: principles and recommendations* (4)
- *Systematic screening for active tuberculosis: an operational guide* (5)
- *Tuberculosis prevalence surveys: a handbook* (21)
- *Guidelines on the management of latent tuberculosis infection* (24)
- *Screening chest X-ray interpretations and radiographic techniques* (10).

5. TECHNICAL SPECIFICATION, QUALITY ASSURANCE, QUALITY CONTROL, AND SAFETY

5.1 Technologies for chest X-ray and technical specifications

WHO first specified a basic radiological system in 1975. The concept was developed further with a series of revisions. WHO's guiding principles in designing radiological units have been:

- the radiological image must be high quality;
- the equipment must be safe for patients and personnel;
- the equipment must be easy to install and use;
- required equipment maintenance should be minimal and quality assured;
- if necessary, it must be possible to use the equipment with an unreliable electricity supply.

Two types of technology are used for CXR: analogue (that is, a system using film) or digital. It is important to highlight that both of these technologies employ the same principle of X-ray production (which is non-digital); the difference is the method of recording the result. In conventional systems, the result is recorded and displayed on an X-ray film but in digital systems, the result is recorded on a detector and displayed in a digital format on a computer screen (and it can also be printed on X-ray film or paper or sent to a digital device).

Digital systems have several advantages over conventional systems. They reduce procedure time, have very low running costs (particularly when a hard copy image is not needed), save on staff requirements because the system is more user-friendly, produce superior image quality, give a lower radiation dose, allow for easier archiving and are more environmentally friendly. Moreover, they allow for telemedicine solutions and can be used for computer-aided reading.

In 2000, WHO published the Consumer guide for the purchase of X-ray equipment (38), a comprehensive document that was aimed at small hospitals and large primary healthcare centres. The document covered not only technical specifications but also guidance on infrastructure, staffing and protection from radiation. However, because affordable digital technologies were not widely available for resource-poor settings at the time, the guide recommended only conventional X-ray technology with a manual or automatic film processing and development system.

In 2016, WHO developed technical specifications for the 61 essential medical devices needed by healthcare facilities (39). The specifications for digital imaging systems in that document should help countries procure appropriate equipment. It was compiled by WHO in collaboration with a working group of experts, and will be continually updated. Among the 61 devices, 6 diagnostic X-ray-related devices are included (as of 29 September 2016):

- the stationary basic diagnostic X-ray system, digital;
- the mobile basic diagnostic X-ray system, digital;
- the stationary basic diagnostic X-ray system, analogue;
- the mobile basic diagnostic X-ray system, analogue;
- the darkroom automated X-ray film processor; and
- the daylight automatic X-ray film processor.

Basic radiography is an essential technology in primary care, such as at health centres and district hospitals. The basic radiography unit (the stationary digital diagnostic X-ray system) has much wider functions than CXR, such as to investigate trauma or gastrointestinal diseases and to perform interventions, such as thoracic intubation, and the fixation or manipulation of fractures and dislocations. Mobile digital basic X-ray systems that can provide radiology services outside the radiology unit, such as in an operating theatre or intensive care unit, are listed in WHO's 2016 technical specifications as the second unit of radiography needed for hospitals.

In some facilities it is appropriate to have a CXR-specific stationary unit. WHO's specifications for stationary digital CXR systems are not available, but will be developed by WHO and its partners. Moreover, WHO's guidelines on radiography are limited to medical facilities. WHO is developing specifications for portable X-ray units, which are already widely used in small medical facilities, for home care, in military and disaster relief operations, for systematic TB screening and for TB prevalence surveys.

The WHO Global Task Force on TB Impact Measurement strongly recommends that countries do not use fluoroscopy or mass miniature indirect radiography for TB screening because these require a higher radiation dose (21). Since 2009, all national TB prevalence surveys, without exception, have adopted direct CXR technology.

5.2 How to choose chest X-ray technology

There are many factors to consider when choosing the appropriate X-ray equipment.

- **Settings:** For frontline hospitals and health centres, a stationary basic digital diagnostic X-ray system should be considered for the first X-ray unit, as X-ray technology has many diagnostic uses beyond CXR. An X-ray unit used only for CXR could be installed in the radiology department when the demand for CXR is high. For field use, options include a stationary X-ray unit (either in an X-ray van or container) or a portable unit. The choice depends on the accessibility of field sites, a country's regulations, the climate (temperature) and the daily demand for CXR images.
- **Costs:** Costs depend on several factors. It is useful to think about the entire lifetime of the equipment when considering costs. Different technologies have varying levels of initial investment and running costs (including requirements for consumables, operational costs and costs for maintenance and parts). Digital systems have a higher initial cost but often offer savings on consumables (particularly when a hard copy of the CXR image is not necessary) and human resources.
- **Duration of use:** Although X-ray equipment is sometimes acquired for a specific purpose or activity, such as for a systematic screening campaign or a prevalence survey, it should be assumed that it will be used for a much longer time period and for other purposes. Therefore, its utility should be considered in terms of its general use.
- **Field conditions:** If the equipment is to be used in the field, important factors to consider are portability and power requirements.
- **Personnel:** It is important to consider whether skilled personnel will be available to conduct CXR examinations, read the results and maintain the equipment in the setting in which it will be used.
- **Radiation exposure:** Although none of the options present dangerous levels of radiation, newer digital technologies provide lower exposure to radiation.
- **Throughput capacity:** Digital systems are good for tasks with a heavy workload – such as a prevalence survey, hospital-based systematic screening or active case finding in the community – because they shorten the processing time.
- **Availability of maintenance:** Particularly for digital equipment, ensuring maintenance after the initial 1-year warranty period can be difficult in countries where maintenance services are scarce.

5.3 Computer-aided detection of TB

New technologies for analyzing the results of CXR evaluations are being developed, including computer-aided detection (CAD) software that can analyze digital CXR images for abnormalities and the likelihood of TB being present. Such technology could help reduce interreader variability and delays in reading radiographs when skilled personnel are scarce.

As of 2016, WHO provides no recommendations on using CAD for TB. A systematic review of five peer reviewed articles published in 2016 concluded that the evidence of CAD's diagnostic accuracy is limited by the small number of studies of the single commercially available CAD software (CAD4TB, Delft Imaging Systems, Veenendaal, the Netherlands). There were also important methodological limitations to the studies and their findings had limited generalizability (40). To determine whether WHO should initiate a process to develop guidelines on CAD, WHO commissioned an extended systematic review to evaluate

both published and unpublished studies assessing CAD as a tool for evaluating persons with suspected TB and for TB screening (41). The following PICO questions were formulated (PICO refers to population, intervention, comparator, outcome).

1. What is the diagnostic accuracy of CXR interpreted by CAD for detecting TB confirmed with culture or a molecular test?
2. Is the diagnostic accuracy of a CXR analyzed with CAD superior, inferior or equivalent to that of a CXR interpreted by a human reader for detecting TB confirmed by culture or a molecular test?

The review stratified results by use-case: in triage use-case studies, CAD was used for evaluating persons with suspected TB (that is, persons seeking care for TB symptoms); in screening use-case studies, it was used to screen populations at risk that may not be seeking care. The review focused on CAD4TB, the only commercially available CAD program for TB. CAD4TB analyses a digital CXR and produces an abnormality score ranging from 0 to 100, with higher scores indicating greater likelihood of TB. A threshold score is the abnormality score below which TB is considered ruled out. For this to be a generalizable diagnostic test, one would predefine a threshold for each use-case. However, threshold scores are currently not preselected by the test developers.

Studies were identified through four medical databases as well as the developer of CAD4TB software, projects supported by the Stop TB partnership and researchers in the field. Of the 542 identified citations and studies with unpublished data, 13 were included in the systematic review: 5 published peer reviewed papers, 4 published conference abstracts and 4 unpublished studies. Seven studies were triage use-case, and 6 were screening use-case, of which 6 and 3, respectively, had results that addressed the PICO questions. Results were considered in light of the potential for bias and applicability concerns, and assessed using the QUADAS-2 (Quality Assessment of Diagnostic Accuracy Studies 2) approach (<http://www.quadas.org>).

A number of important methodological limitations were identified for both use-cases during quality assessment. The main sources of concern about the potential for bias and limited applicability were the inappropriate exclusion of participants (particularly in screening use-case studies) and how the threshold scores had been operationalized, in particular the use of threshold scores that were not prespecified. Concerns about generalizability also arose from the use of non-commercially available versions of CAD4TB, and from training staff and evaluating the software for CXR within the same study population. Studies were not equally affected by these methodological limitations. Overall, reviewers found that few studies used CAD4TB the way it will be used in the field – that is, with the most up-to-date commercially available version, with a prespecified threshold score and without having undertaken a pilot study to determine which threshold score to use. Furthermore, there was limited or no information about the performance of CAD4TB in several subgroups of interest, including people living with HIV and different age groups, particularly for the screening use-case studies. Finally, data disaggregated by sex or smear status were not available. No pooling of data was done due to the methodological heterogeneity of the reviewed studies, as well as to the low number of studies for each version of CAD4TB.

Across versions, the software achieved high sensitivity for both use-cases, but with variable specificity. The sensitivity and specificity of the human CXR readers whose performance was compared with that of CAD4TB were variable. In a number of studies, CAD4TB's threshold score was set to match either the sensitivity or specificity of the human readers, thus limiting the comparisons of accuracy that could be made between the software and human readers. The software's sensitivity was similar to that of human expert readers in the triage use-case, but it was less specific. Performance varied when compared with non-expert readers. For the screening use-case, data were insufficient to draw conclusions about performance compared with humans.

Based on the findings of the extended systematic review, WHO has decided not to initiate a guideline development process at the present time. CAD can be used for TB detection for research, ideally following a protocol that contributes to the required evidence base for guideline development. For this purpose, WHO plans to further specify the desirable characteristics of CAD for TB and develop advice on key research questions and appropriate study designs that can address those questions. Broad questions that should be addressed include the following:

- what is the added value of CAD in different places in the diagnostic pathway?
- what is the added value of CAD in different populations and settings?
- what should be the threshold scores for different populations or settings?
- what needs to happen if a patient is symptomatic and a CAD assessment is negative? And what needs to happen if a patient is symptomatic and the CAD assessment is positive but the microbiological test is negative?
- what is the cost effectiveness of CAD compared with human readers, considering different payment models?
- what operational or implementation challenges exist for ensuring equitable access?
- what needs to be done with preliminarily CAD-positive patients? For example, when would a human reader be engaged for final interpretation/reporting in each use case?
- how could CAD be used as an aid to human CXR readers rather than as a replacement?

Two possible research strategies have been identified to generate evidence for WHO to issue recommendations on using CAD4TB. When new evidence has been generated, WHO will assess whether a guideline process should start.

In strategy 1, data from individual patients and digital images from the higher quality studies identified in the extended review would be analysed independently of the CAD4TB developers, using the latest version of the software and pooling the results using a meta-analysis of individual patient data. Although this could provide evidence of sufficient quality for the triage use-case, it would be inadequate for the screening test use-case. Hence, a complementary undertaking in strategy 1 would be to compile a standard panel of CXRs taken from existing databases, and use this panel to evaluate CAD4TB. Potential data sources include completed or ongoing studies that utilize digital CXR in either a triage or screening use-case and also use the required reference standard as defined but that are not utilizing CAD software (in order to avoid the developers having access to the files for software training), TB prevalence surveys and other existing data sets of digital CXR files with appropriate pathology reference standards. The timeline of this strategy is estimated to be 1 year, but it needs to be further clarified.

Strategy 2 would involve identifying planned or ongoing trials of CAD, or undertaking new studies, and integrating an evaluation of CAD into that research. The main focus would be the screening use-case. Design considerations that would ensure studies are of sufficient quality would include using predefined threshold scores according to the use-case; enrolling important subgroups of patients (defined by sex, age, HIV status, smear status); using CAD for all participants (in triage studies, this would include testing all participants with the reference test; in screening studies, this would include testing at least a subset of asymptomatic persons with the reference test); and for comparison, having human readers who are blinded to CAD scores and microbiological data, and using standardized categories for reporting CXR results. The timeline for this strategy is estimated to be 2–3 years but depends on the ability to integrate an evaluation of CAD into planned studies and the availability of funding.

5.4 Quality assurance and quality control

There are two major areas of quality assurance for CXR: the quality of CXR imaging and the quality of CXR reading.

The quality of CXR imaging is associated with the quality of the hardware being used and the quality of the techniques and applications. WHO provides technical specifications for X-ray machines and the equipment related to handling X-ray films and images, as well as checklists and workbooks (http://www.who.int/diagnostic_imaging/publications/en/). The WHO manual of diagnostic imaging is available on the website of the International Society of Radiology (http://www.isradiology.org/isr/books_technique.php) (7). Also, the *Handbook for district hospitals in resource constrained settings on quality assurance of chest radiography* was developed specifically for TB programmes (35).

Quality control of CXR reading is also essential. Underreading leads to missed opportunities for diagnosing and treating TB, and overreading leads to excess burdens on TB laboratories, as well as possible false-

positive clinical diagnoses (10). Another resource developed specifically for quality assurance in TB programmes in district hospitals is the *Handbook for district hospitals in resource constrained settings for the quality improvement of chest X-ray reading in tuberculosis suspects* (36).

In addition to qualifications such as diplomas, masters' degrees and fellowships, several distance-learning or self-training opportunities are available for ensuring quality control standards. Additionally, some professional societies have developed educational materials, and some of these are listed at the end of this chapter. The International Commission on Radiology Education, part of the International Society of Radiology, provides its own educational materials as well as links to materials from other sources (11).

Standardized CXR reporting forms and peer review, which may be accomplished by holding regular conferences in hospitals or at local TB diagnostic committees, can be useful for standardizing reading and interpretations. In systematic TB screening campaigns and TB prevalence surveys involving non-specialist screening physicians or clinical officers, a specialist often assesses the primary reading (in what is known as a central reading system). Comparing laboratory results with CXR results can help to improve the quality of radiological screening and diagnosis. Monitoring the proportion of clinically diagnosed cases out of all TB cases can serve as an indicator of diagnostic quality. However, WHO has not defined formal procedures for internal or external quality assessment of CXR interpretation for TB screening and detection.

5.5 Safety

A proportion of the X-rays used in radiography are absorbed by the body. The potential effects from ionizing radiation depend on the dose. At doses much higher than those of typical diagnostic imaging exams, the damage may be extensive enough to affect tissue function, and the damage may become clinically observable (for example, as skin reddening or burns). Effects of this type are called tissue reactions or deterministic effects, and they occur only if the radiation dose exceeds a certain threshold (42). The long-term risks from ionizing radiation include an increased risk of cancer.

Direct CXR is a safe technology using a radiation dose of 0.1 mSv, which corresponds to 1/30 of the average annual radiation dose from the environment (3 mSv) and 1/10 of the annual accepted dose of ionizing radiation for the general public (1 mSv). As a point of reference, the radiation dose of one CXR is equivalent to or less than the radiation exposure received during return travel on an intercontinental flight.

Therefore, exposure to the low radiation doses delivered to patients during a CXR poses a small risk of inducing tissue reactions or cancer in the years to decades following the examination (43). However, it should be noted that a linear non-threshold relationship is assumed between radiation exposure and the risks of effects of this nature. Based on this linear model, the probability of developing cancer is presumed to increase even following exposure to low doses of radiation, although the increase in risk is extremely small. Even though the individual risk associated with radiation exposure from CXR is low, when a large number of individuals are exposed, the associated risks may still constitute a public health issue.

Children and pregnant women are especially vulnerable to ionizing radiation. Also, children have a longer life expectancy, resulting in a larger window for developing long-term radiation-induced health effects. When imaging small children and infants, exposure parameters should be adjusted from those used for adults to avoid using a higher dose than necessary. Unnecessarily high doses and their associated risks can be substantially reduced without affecting image quality by customizing exposure settings to deliver the lowest radiation dose necessary for providing an image that is fit for the clinical purpose. For pregnant women and the fetus, a CXR does not pose any significant risk, provided that good practices are observed, as the primary beam is targeted away from the pelvis (21).

Due to the potential risks, the decision to expose patients to radiation during imaging procedures must adhere to two overall principles of radiation protection in medicine: justification and optimization. Justification means that the need for medical imaging should be assessed by weighing the expected benefits against the potential radiation risk, taking into account the benefits and the risks of alternative techniques that do not involve exposure to radiation. The procedure should be judged to do more good than harm. Optimization in medical imaging refers to keeping doses as low as reasonably achievable to obtain a useful image. X-ray personnel should at all times ensure that the benefits of an examination outweigh the potential risks from radiation exposure and that the patient is exposed to as little radiation as possible (14, 44, 45).

Staff performing CXR must be familiar with protection measures for themselves, for patients and for others who may be exposed. Important concepts in radiation protection include consideration of the following areas:

Type of exposure, including:

- medical exposure (exposure of clients, such as patients and study participants);
- occupational exposure;
- public exposure.

Place of exposure, including:

- controlled areas, such as the X-ray room and areas directly connected to X-ray rooms (used by X-ray staff and assistants during the X-ray examinations);
- supervised areas, such as areas in the vicinity of the X-ray room, which are usually part of the X-ray department.

Protection measures, including:

- shielding – that is, barriers of attenuating material around the radiation source, such as concrete walls and lead-containing panels, curtains and windows;
- distance from the radiation source.

Preventing exposure for staff and the general public is particularly important when X-ray systems are taken outside of medical facilities, such as during systematic screening using mobile CXR in the community. *Tuberculosis prevalence surveys: a handbook* provides advice on how to ensure safety in the field (21). When a van with a full size X-ray system with safety and power features appropriate for a hospital is available, the safety standards applied for health facilities can be used. When portable CXR units are placed in non-shielded areas, then restricted areas should be set up, clearly marked and monitored under the guidance of the radiological unit of the survey team. Radiometers and film badge dosimeters should be used, and rules should be devised for how to direct the X-ray beam.

More information can be found in the following resources:

- *Radiation protection and safety of radiation sources: international basic safety standards*, interim edition (46)
- *Tuberculosis prevalence surveys: a handbook* (21)
- WHO's diagnostic imaging website (14)
- WHO's *Consumer guide for the purchase of X-ray equipment* (38)
- *Handbook for district hospitals in resource constrained settings for the quality improvement of chest X-ray reading in tuberculosis suspects* (36)
- *Handbook for district hospitals in resource constrained settings on quality assurance of chest radiography* (35)
- *Communicating radiation risks in paediatric imaging* (47)
- International Commission on Radiology Education (42)
- World Federation of Pediatric Imaging: TB resources (48)
- *Radiographic manifestations of tuberculosis: a primer for clinicians* (9)
- *Screening chest X-ray interpretations and radiographic techniques* (10).

6. STRATEGIC PLANNING FOR USING CHEST X-RAY IN NATIONAL TB CARE

This chapter describes the strategic planning steps needed to introduce, expand, or systematize the use of CXR in TB care and prevention. Since CXR is not a TB-specific tool, developing a strategic plan for CXR should be done as a part of making general improvements to imaging services in the healthcare sector and within the broader framework of health-system strengthening and providing universal health coverage. This will increase the relevance and acceptance of investments in the technology and related capacity strengthening, and improve the overall efficiency and cost effectiveness of radiography services.

Accordingly, the strategic planning process should involve all relevant stakeholders interested in introducing or expanding the use of CXR within the health sector. Including all stakeholders can be beneficial because different stakeholders will be able to utilize different sources of funding. However, this strategy may also be hampered by conflicting goals or expectations. If possible, the process needs to be integrated into the development of a national strategic plan for TB care, as well as into the broader national health plan. The level of integration will depend on the structure of the national health system and the level of decentralization in planning, as well as the timing of the strategic planning process at all levels.

Strategic planning involves:

1. defining the objective of using CXR in the health system;
2. performing a situation analysis;
3. defining specific TB targets to which introducing, expanding or systematizing the use of CXR will contribute;
4. developing an operational plan, with a detailed budget and financing plan.

Strategic planning for CXR should begin by **defining the objective** of using CXR within the health system and within the NTP national strategic plan – for example, it may be used to improve care for patients with respiratory diseases. Planning should then proceed by undertaking a **situation analysis** to understand the current state of and capacity for CXR in the country, the current epidemiology of TB and the possible contribution of CXR to TB care. Table 1 specifies the relevant questions that should be addressed by the situation analysis; the process should focus on those issues that are most relevant to the stated objective for current and planned use of CXR.

Area to assess	Relevant issues to be addressed	Potential information sources
CXR capacity	Determine the current state of CXR use in healthcare in the country, within both the public and the private sectors, including: - existing infrastructure, equipment (including specifications) and current workload and capacity; - existing human resources available to both conduct and read CXRs, and any available measures of their competencies; - existing internet connectivity capacity for e-health/telemedicine options, if applicable; - percentage of machines that are operational; - maintenance arrangements: how are maintenance issues reported and acted upon? What human resources and which budget (or budgets) are used for maintenance? - current financing: who pays for new equipment and for ongoing consumables, and from which budget (or budgets)? - what is the fee structure for end users, if any, in both the public and private sectors? - all stakeholders interested in using CXR and improving its use.	Service availability and readiness assessment (known as SARA) surveys; national health insurance plans or other payment structures
	Determine existing national policies and regulations affecting the use of radiology, including: - its placement on the list of essential diagnostic services for the health service; - its placement on the list of essential medical devices; - quality assurance and quality control mechanisms; - safety policies and practices; - other existing national policies regarding the use of radiography, such as policies pertaining to health screening for immigrants or for occupational health screening.	National guidelines on radiation safety; national screening policies for using CXR, such as to screen immigrants or for occupational health screening, or screening in prisons
	Map needs for human resources and capacity strengthening in the public and private sectors for introduction, scale up and quality assurance/quality control; consider: - radiologists and radiographers; - professional societies; - infrastructure for training and capacity strengthening; - capacity for e-health/telemedicine options.	Professional societies that utilize CXR (such as radiologists and pulmonologists)
Epidemiology of TB	Determine the current epidemiology of TB in the region, including: - the geographical distribution of cases; - the age distribution of cases; - current case-detection gaps; - the current epidemiology of risk factors for TB; - any results from prevalence surveys or other research indicating the proportion of bacteriologically confirmed cases reporting no or vague symptoms of TB because this indicates the contribution that CXR could make to case detection; - CXR accuracy data from prevalence surveys or other research; - the profile of persons with suspected TB and where they would likely enter the health system, or how they could be detected through active case-finding initiatives (including children), as this will determine the current gap in CXR services and the potential workload of CXR.	National TB surveillance data; WHO's global TB report; prevalence surveys; other relevant research findings from the country
TB activities involving CXR	Consider current and planned activities that require increased introduction or scale up of CXR and quality assurance in the country in terms of: - improving triaging; - improving the clinical diagnosis of TB; - improving the diagnosis of childhood TB; - implementing systematic screening in risk groups; - scaling up detection and treatment of latent TB infection.	National strategic plan for TB; concept notes from the Global Fund, or similar strategic planning for funding

CXR: chest X-ray.

Based on the findings of the situation analysis, the TB control **targets** that CXR will contribute to need to be defined for its introduction, and for expanding or systematizing its use. For example, it should be specified whether the intended target is to increase case detection, rationalize the use of Xpert MTB/RIF testing or assist with the initiation of treatment for LTBI. Similarly, stakeholders not using CXR for TB should define their targets for introducing or expanding CXR use.

The **operationalization plan** for integrating CXR into TB control should then be defined, beginning with the broader questions of CXR technology and placement in the healthcare system, and then using that information to proceed to address more detailed aspects of implementation, such as the following.

- Where in the health system will CXR will be placed – for example, at the primary healthcare level, at the referral level, in hospitals, in mobile screening units?
- Which CXR technology is best suited to the planned placement, or the desired improvement in the quality of existing CXR technology? This will be determined largely by the intended uses for CXR – for example, whether it will be used in routine care at primary healthcare centres or in a prevalence survey.
- What additional infrastructure or equipment will be required, such as vans for mobile screening units or computers and internet connectivity for digital systems?
- What is the approximate cost of such equipment, both in terms of procurement and ongoing operation and maintenance costs? Detailed budgeting comes later in the process (see below), but rough budget estimates are needed earlier to check that the proposed approaches are broadly affordable and cost effective.
- What is the place of CXR in triaging, screening and diagnostic algorithms? What possible modifications could be made to current algorithms to make better use of CXR?
- What is the expected increase in TB detection from the new algorithms or new placement of a CXR system?
- What human resources are required for CXR use?
- What are the initial training requirements and ongoing quality assurance requirements for CXR use?
- Is there a need for technical assistance during the implementation of the operational plan?
- What is the plan for monitoring and evaluating the use of CXR?
- What is the plan for any operational research to be conducted in conjunction with CXR use in TB control? How will the associated data security and patient privacy issues be addressed?
- What implementation steps are required to achieve the defined objectives and targets, and who will perform the steps and when?

Along with this operational plan, a **detailed budget** needs to specify who will pay for the costs associated with using CXR, taking into account the different levels and budgeting processes of the entire health system. The budget should show the total cost of implementing and operationalizing CXR activities, the sources of funding and the funding gaps. The NTP, other departments of the ministry of health and other concerned ministries should work together to address the gaps in funding CXR for TB care to ensure the costs are not passed on to patients. Such costs can be catastrophic for individuals already facing severe economic and other stresses as a result of their illness.

For more details, see WHO's *Toolkit to develop a national strategic plan for TB prevention, care and control* (49).

Annexes

Annex 1.

Yield and costs of triage algorithms in a hypothetical population of 100 000 with different TB prevalence levels

The yields of true-positive, false-positive, true-negative and false-negative TB diagnoses (using liquid culture as the gold standard), and the cost per true case of TB detected, were modelled for different triage algorithms and different prevalence levels in a hypothetical population of 100 000 persons seeking healthcare.

Table A1 displays the assumptions about test accuracy that are based on estimates from the systematic review done for the guideline on systematic screening for active TB (4). It should be noted that these estimates are from systematic screening studies (mainly TB prevalence surveys) and are likely to be different in a triage scenario. The reason for using these assumptions is that no data from systematic reviews are available for triage scenarios. The cost estimates are the same as the pre-set values on the ScreenTB tool. The ScreenTB tool (5, 12) can be used to model the yield and costs for these algorithms based on different assumptions.

Table A1. Assumptions used to model the yield and costs per true case of TB detected

Test	Sensitivity	Specificity	Test costs ^a	Operational costs ^a
Cough screen	0.35	0.95	0	0.05
Any symptom screen	0.77	0.68	0	0.05
Digital chest X-ray	0.98	0.75	1.00	4.00
Sputum-smear microscopy	0.61	0.98	1.50	0.50
Xpert MTB/RIF assay ^b	0.92	0.99	10.00	7.00

^a Costs are per person tested in US dollars.

^b Xpert MTB/RIF assay (Cepheid, Sunnyvale, CA, United States).

Table A2 displays the results of modelling with a TB prevalence of 250 cases per 100 000 population; Table A3 displays the results of modelling with a TB prevalence of 500 cases per 100 000 population; and Table A4 displays the results of modelling with a TB prevalence of 1 000 cases per 100 000 population.

Table A2. Yield and costs per true case of TB detected by triage algorithms in a hypothetical population of 100 000 with a TB prevalence of 250 cases/100 000^a

Algorithm	True positive	False positive	False negative	True negative	Cost per case	Number needed to screen ^b
1 (Prolonged cough → SSM)	53	100	197	99 650	286	1 887
2 (Prolonged cough → CXR → SSM)	52	25	198	99 725	635	1 924
3 (Any symptom → SSM)	117	638	133	99 112	592	855
4 (Any symptom → CXR → SSM)	115	160	135	99 590	1 582	870
5 (CXR → SSM)	149	499	101	99 251	3 694	672
6 (Prolonged cough → Xpert)	80	50	170	99 700	1 141	1 250
7 (Prolonged cough → CXR → Xpert)	79	12	171	99 738	671	1 266
8 (Any symptom → Xpert)	177	319	73	99 431	3 112	565
9 (Any symptom → CXR → Xpert)	174	80	76	99 670	1 750	575
10 (CXR → Xpert)	225	249	25	99 501	4 125	445

CXR: chest X-ray; SSM: sputum-smear microscopy; Xpert: Xpert MTB/RIF assay (Cepheid, Sunnyvale, CA, United States).

^a The algorithms are described in detail in Chapter 2.

^b This is the number needed to screen to detect one true case.

Table A3. Yield and costs per true case of TB detected by triage algorithms in a hypothetical population of 100 000 with a TB prevalence of 500 cases/100 000^a

Algorithm	True positive	False positive	False negative	True negative	Cost per case	Number needed to screen ^b
1 (Prolonged cough → SSM)	107	100	393	99 400	143	935
2 (Prolonged cough → CXR → SSM)	105	25	395	99 475	320	953
3 (Any symptom → SSM)	235	637	265	98 863	296	426
4 (Any symptom → CXR → SSM)	230	159	270	99 341	795	435
5 (CXR → SSM)	299	498	201	99 002	1 842	335
6 (Prolonged cough → Xpert)	161	50	339	99 450	575	622
7 (Prolonged cough → CXR → Xpert)	158	12	342	99 488	347	633
8 (Any symptom → Xpert)	354	318	146	99 182	1 562	283
9 (Any symptom → CXR → Xpert)	347	80	153	99 420	887	289
10 (CXR → Xpert)	451	249	49	99 251	2 065	222

CXR: chest X-ray; SSM: sputum-smear microscopy; Xpert: Xpert MTB/RIF assay (Cepheid, Sunnyvale, CA, United States).

^a The algorithms are described in detail in Chapter 2.

^b This is the number needed to screen to detect one true case.

Table A4. Yield and costs per true case of TB detected by triage algorithms in a hypothetical population of 100 000 with a TB prevalence of 1 000 cases/100 000^a

Algorithm	True positive	False positive	False negative	True negative	Cost per case	Number needed to screen ^b
1 (Prolonged cough → SSM)	214	99	786	98 901	73	468
2 (Prolonged cough → CXR → SSM)	209	25	791	98 975	166	479
3 (Any symptom → SSM)	470	634	530	98 366	149	213
4 (Any symptom → CXR → SSM)	460	158	540	98 842	401	218
5 (CXR → SSM)	598	495	402	98 505	922	168
6 (Prolonged cough → Xpert)	322	50	678	98 950	295	311
7 (Prolonged cough → CXR → Xpert)	316	12	684	98 988	185	317
8 (Any symptom → Xpert)	708	317	292	98 683	786	142
9 (Any symptom → CXR → Xpert)	694	79	306	98 921	453	145
10 (CXR → Xpert)	902	248	98	98 752	1 039	111

CXR: chest X-ray; SSM: sputum-smear microscopy; Xpert: Xpert MTB/RIF assay (Cepheid, Sunnyvale, CA, United States).

^a The algorithms are described in detail in Chapter 2.

^b This is the number needed to screen to detect one true case.

Annex 2.**Proportion of TB cases detectable through screening with chest X-ray or by screening for chronic cough**

The results of several recent prevalence surveys are presented below (Table B1), including the proportion of TB cases identified that reported various TB symptoms and that had abnormal CXR results. In particular, the column “Percentage positive only by CXR” demonstrates the potential contribution of CXR to TB detection over screening or triage based on symptoms alone.

Table B1. Proportion of TB cases detectable through screening with chest X-ray or by screening for chronic cough among persons diagnosed with bacteriologically confirmed TB in recent TB prevalence surveys^a

Country	No. of bacteriologically confirmed cases	No. symptom positive, CXR positive	No. symptom positive, CXR negative	No. symptom negative, CXR positive	Percentage positive by both CXR and symptoms	Percentage positive only by CXR	Percentage positive only by symptoms
Cambodia	314	90	3	216	29	69	1
Ethiopia	110	45	12	53	41	48	11
Gambia	77	32	12	33	42	43	16
Ghana	202	67	15	85	33	42	7
Indonesia	426	220	25	181	52	42	6
Laos	237	111	7	119	47	50	3
Malawi	132	25	67	40	19	30	51
Mongolia	248	44	7	190	18	77	3
Myanmar	311	65	1	231	21	74	0
Nigeria	144	76	16	52	53	36	11
Pakistan	341	157	40	142	46	42	12
Rwanda	40	15	4	21	38	53	10
Sudan	112	44	8	43	39	38	7
Thailand	142	42	6	94	30	66	4
Uganda	160	63	16	81	39	51	10
Zambia	265	115	46	104	43	39	17
Zimbabwe	107	29	10	64	27	60	9

CXR: chest X-ray.

^a Data from country prevalence surveys or personal communication. Information about sources available upon request.

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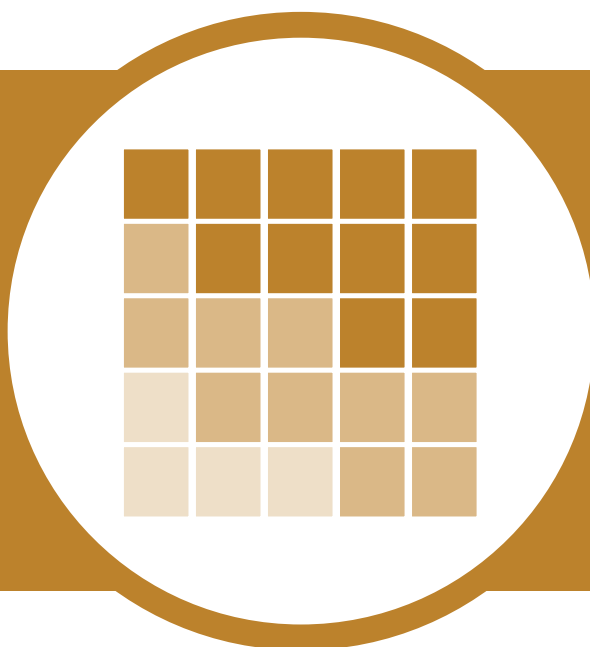
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management in primary health care**



Risk-based CVD management



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Abbreviations

BMI	body mass index
CAD	coronary artery disease
CVD	cardiovascular disease
DM	diabetes mellitus
ESC	European Society of Cardiology
GBD	Global Burden of Disease
HTN	hypertension
HDL-C	high-density lipoprotein cholesterol
IHME	Institute for Health Metrics and Evaluation
IHRMS	INTERHEART Modifiable Risk Score
LMIC	low- and middle-income countries
MI	myocardial infarction
NCD-RisC	NCD Risk Factor Collaboration
PEN	Package of Essential Noncommunicable Disease Interventions
SBP	systolic blood pressure
TC	total cholesterol
WHO/ISH	World Health Organization / International Society of Hypertension



HEARTS Technical Package

More people die each year from cardiovascular diseases (CVDs) than from any other cause. Over three quarters of heart diseases and stroke-related deaths occur in low- and middle-income countries (1). The HEARTS technical package provides a strategic approach to improving cardiovascular health. It comprises six modules and an implementation guide. This package supports Ministries of Health to strengthen CVD management in primary care and aligns with World Health Organization's Package of Essential Non-communicable Disease Interventions (WHO PEN).

The WHO HEARTS modules are intended for use by policy-makers and programme managers at different levels within Ministries of Health who can influence CVD primary care delivery. Different sections of each module are aimed at different levels of the health system and different cadres of providers. All modules will require adaptation at country level.

The people who will find the modules most useful are:

- **National level** – Ministry of Health noncommunicable disease (NCD) policy-makers responsible for:
 - Developing strategies, policies and plans related to service delivery for CVD
 - Setting national targets on CVD, and monitoring and reporting progress
- **Subnational level** – Health/NCD programme managers responsible for:
 - Planning, training, implementing and monitoring service delivery
- **Primary care level** – Facility managers and primary health care providers and trainers responsible for:
 - Assigning tasks, organizing training and ensuring the facility is running smoothly
 - Collecting facility-level data on indicators of progress towards CVD targets.

Target users may vary, based on context, existing health systems and national priorities.

MODULES OF THE HEARTS TECHNICAL PACKAGE				
Module	What does it include?	Who are the target users?		
		National	Subnational	Primary care
H ealthy-lifestyle counselling	Information on the four behavioural risk factors for CVD is provided. Brief interventions are described as an approach to providing counselling on risk factors and encouraging people to have healthy lifestyles.		✓	✓
E vidence-based protocols	A collection of protocols to standardize a clinical approach to the management of hypertension and diabetes.	✓	✓	✓
A ccess to essential medicines and technology	Information on CVD medicine and technology procurement, quantification, distribution, management and handling of supplies at facility level.	✓	✓	✓
R isk-based CVD management	Information on a total risk approach to the assessment and management of CVD, including country-specific risk charts.	✓	✓	✓
T eam-based care	Guidance and examples on team-based care and task shifting related to the care of CVD. Some training materials are also provided.		✓	✓
S ystems for monitoring	Information on how to monitor and report on the prevention and management of CVD. Contains standardized indicators and data-collection tools.	✓	✓	✓



Introduction

Development of cardiovascular disease (CVD) is influenced by risk factors such as: tobacco use, an unhealthy diet, physical inactivity, obesity (which can result from a combination of unhealthy diet, physical inactivity, and other factors), elevated blood pressure (hypertension), abnormal blood lipids (dyslipidaemia) and elevated blood glucose (diabetes mellitus). Continuing exposure to these risk factors leads to further progression of atherosclerosis, resulting in clinical manifestations of these diseases, including angina pectoris, myocardial infarction, heart failure and stroke. Total CVD risk depends on the individual's overall risk-factor profile.

In 2007, the WHO published Guidelines for the assessment and management of cardiovascular risk (2) that provide guidance for reducing disability and premature deaths from CVD in people at high risk who have not yet experienced a cardiovascular event. These 2007 guidelines were used as a framework for the development of national guidance on CVD prevention, and provide the WHO and International Society of Hypertension (WHO/ISH) cardiovascular (CVD) risk prediction charts for regional CVD risk prediction. WHO updated the CVD risk charts in 2019 (3). This Risk-based CVD management module (HEARTS-R) will present the updated CVD risk charts for the assessment and management of CVD risk. Relevant information from the 2007 guidelines is reproduced.

Timely, affordable and sustained healthy-lifestyle interventions and, when needed, drug treatment will reduce the risk of heart attack and stroke in people with a high total risk of CVD, and hence will reduce premature morbidity, mortality and disability. Those who have their elevated blood pressure or cholesterol treated can reduce their risk by more than one-quarter to one-third (4) and have their risk reduced by half if both are treated (5). For many years we have had inexpensive medications for hypertension (HTN). More recently, generic statins have been added to the WHO Essential Medicines list and have become available in many more countries. Despite the increased availability of drugs, less than 10% of the global population has its blood pressure controlled (6).

Poor control of risk factors is in part a result of a lack of awareness among individuals of their risk status. The identification of risk level by health care providers is therefore a useful means of detecting those with high CVD risk and identifying who could benefit from treatment of high blood pressure, abnormal blood lipids and elevated blood glucose (7). Recommendations for assessment and management of CVD risk factors have evolved over time in CVD guidelines.

This evolution in CVD recommendations and guidelines is a response to:

- an improved ability to quantify and identify those at highest risk
- an increase in the population affected by these risk factors
- changes in generic therapies available
- a recognition that despite these changes we are failing globally to significantly prevent CVD mortality, particularly in low- and middle-income countries (LMICs).

One change in guidelines has been a greater use of overall CVD risk levels to determine intensity of treatments. The additional focus on overall risk has occurred for three main reasons. First, there was recognition that an individual who has multiple risk factors, even if each factor is only moderately elevated, may, when all risk factors are taken into account, be at an overall higher risk than someone with an elevated level for a single risk factor. For example, a middle-aged female with moderate levels of several risk factors could have a 5- to 10-fold increased risk of having CVD in the next 10 years, when compared to a young male with an isolated elevated risk factor.

Second, targeting of those individuals at highest risk was recognized to lead to greater efficiency in benefit in terms of number of events avoided, because the relative risk reduction would be applied to a higher baseline risk (8). Therefore, targeting patients with a high risk is the highest priority in a risk stratification approach (9).

Third, improved computing and statistical methods have allowed researchers to pool large global data sets to create more accurate risk prediction tools for various populations.

As the cost of medicines is a significant component of total preventive health care costs, it is particularly important to base drug treatment decisions principally on an individual's risk level rather than on criteria such as ability to pay, or on blanket preventive strategies. In addition, guidelines based on total CVD risk, which use risk scoring methods, have been shown to be both less expensive and more effective than guidelines based on single risk-factor levels to treat the same number of patients (10).

Drug therapy (including glycaemic control for diabetes mellitus, control of HTN and cholesterol, using a total risk approach) and counselling to individuals who have had a heart attack or stroke and to persons with high risk ($\geq 20\%$) of a fatal and non-fatal cardiovascular event in the next 10 years is one of the "best buys" for tackling NCDs (11). For those with a pre-existing condition such as prior CVD, risk identification is simple. For others, the use of clinical algorithms or "risk charts" can be used to further stratify patients.

A risk stratification approach is particularly suitable to settings with limited resources, where saving the greatest number of lives at lowest cost becomes imperative. However, in most guidelines, the use of risk tools has not replaced the knowledge needed to treat individual risk factors but has rather been used to augment the decision of whom to treat, so that those at the highest level of risk are easily identified for treatment. This module helps to identify those who would benefit from lifestyle changes and basic medical treatment to lower blood pressure, cholesterol, and manage diabetes mellitus in an integrated manner. Guidance on HTN and diabetes management can be found in other HEARTS Modules, as well.

Application of an approach that considers the total CVD risk will be better informed through country-level implementation research. Many areas, including the level of the facility at which CVD risk can be assessed, thresholds for treatment, adaptation of protocols, follow-up intervals and other logistics are best defined in the local context.

1 Updated WHO cardiovascular risk charts

2007 WHO/ISH cardiovascular risk prediction charts

Many risk-prediction models have been developed over the years. However, most equations are derived, recalibrated and validated in limited settings. The majority are derived from or recalibrated on populations of European descent living in high-income countries. As a result, these models might not be directly translatable or valid in low-resource and non-European settings. The WHO/ISH cardiovascular risk prediction charts were developed in 2007 to estimate 10-year risk of a fatal or non-fatal major cardiovascular event (myocardial infarction or stroke), according to age, sex, smoking status, blood pressure, total blood cholesterol and presence or absence of diabetes mellitus for 14 WHO epidemiological sub-regions (12). There are two sets of charts. One set could be used in settings where blood cholesterol can be measured, and the other in settings where blood cholesterol cannot be measured. Both sets require the status of diabetes in the individual to be known (13).

2019 Updated WHO cardiovascular disease risk charts

The WHO updated the 2007 WHO/ISH cardiovascular risk prediction charts through the formation of a cross-sectorial collaboration of academics, policy-makers and end users of CVD risk scores. The model revision took place in three steps.

First, risk prediction algorithms were developed using individual-participant data from 85 prospective cohort studies with long-term follow-up in the Emerging Risk Factors Collaboration. Endpoint definitions of fatal and non-fatal CVD outcomes used for the model are described in Annex 4.

Second, to adjust the algorithms to the contemporary circumstances of different global regions, they were recalibrated using age-specific and gender-specific incidence rates and risk-factor values obtained from the Global Burden of Disease (GBD) Study (14) and the NCD Risk Factor Collaboration (NCD-RisC) (15).

Third, performance of the algorithms was assessed by external validation, using individual-participant data from a further 19 prospective cohort studies (i.e. studies distinct from those used in the algorithm derivation) (3).

The updated WHO CVD risk prediction charts were developed and presented for 21 global regions, defined by GBD to maximize between-region variability and minimize heterogeneity in mortality and major drivers of health outcomes within each region (3) (Annex 1). The charts are intended to allow the introduction of a total risk-stratification approach for management of CVD. They are presented as laboratory-based and non-laboratory-based charts. Laboratory-based algorithms include information on age, sex, smoking status, systolic blood pressure, history or evidence of diabetes mellitus, and the total cholesterol value. In the non-laboratory-based algorithms, body mass index (BMI) is included; information on diabetes mellitus and cholesterol is not necessary for these charts.

Countries can find the charts relevant to them according to the appropriate GBD regional groups (Annex 2 and Annex 3).

WHO CVD risk (laboratory-based) charts

These are CVD risk charts that include measurements of total cholesterol and information on diabetes mellitus (Annex 2). The laboratory-based CVD risk charts should be used for treatment decisions. This is the indicated risk chart in a setting where laboratory facilities, and human and financial resources are accessible. These charts will facilitate health providers to initiate an intervention and treatment regimen, and to implement an appropriate follow-up plan based on the patient's total risk status. As an example, Figure 1 presents the WHO CVD risk charts for North Africa and Middle East Region for use when information on total cholesterol and diabetes is available.

WHO CVD risk (non-laboratory-based) charts

Many low-resource settings have limited testing facilities or limited financial and physical capacity for biochemical measurements (e.g. blood sugar determination and cholesterol assays). WHO CVD risk (non-laboratory-based) charts can be used to predict total CVD risk without information on total cholesterol and diabetes (Annex 3). Only age, sex, smoking status, systolic blood pressure and body mass index (BMI) will be needed to predict cardiovascular risk. These non-laboratory-based WHO CVD risk charts are aimed at stratification in low-resource communities and office settings and can be used for decisions regarding referral. In population samples, there was moderate agreement between WHO CVD risk predictions using laboratory and non-laboratory algorithms. Of those at >20% risk using the laboratory-based algorithm, >97% of men and women were also identified at >10% by using the non-laboratory-based algorithm. However, when using a 20% threshold with non-laboratory-based algorithm, only approximately 65% of men and 35% of women were identified as at the same level of risk when compared to the laboratory-based algorithm. This discrepancy is largely due to the fact that the non-laboratory-based algorithm does not allow for the extra CVD risk associated with diabetes mellitus and substantially underestimates CVD risk in individuals with diabetes mellitus. For example, among individuals with diabetes classified as being at greater than 20% risk with the laboratory-based models, about 45% of men and 25% of women were classified as being at greater than 20% risk with the non-laboratory-based models (whereas in individuals without diabetes, about 85% of men and 95% of women showed such agreement) (3). As an example, Figure 2 is the WHO CVD risk (non-laboratory-based) charts for North Africa and Middle East.

Figure 1: Example of WHO CVD risk (laboratory-based) chart

North Africa and Middle East

Afghanistan, Algeria, Bahrain, Egypt, Iran (Islamic Republic of), Iraq, Jordan, Kuwait, Lebanon, Libya, Morocco, occupied Palestinian territory, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, Tunisia, Turkey, United Arab Emirates, Yemen

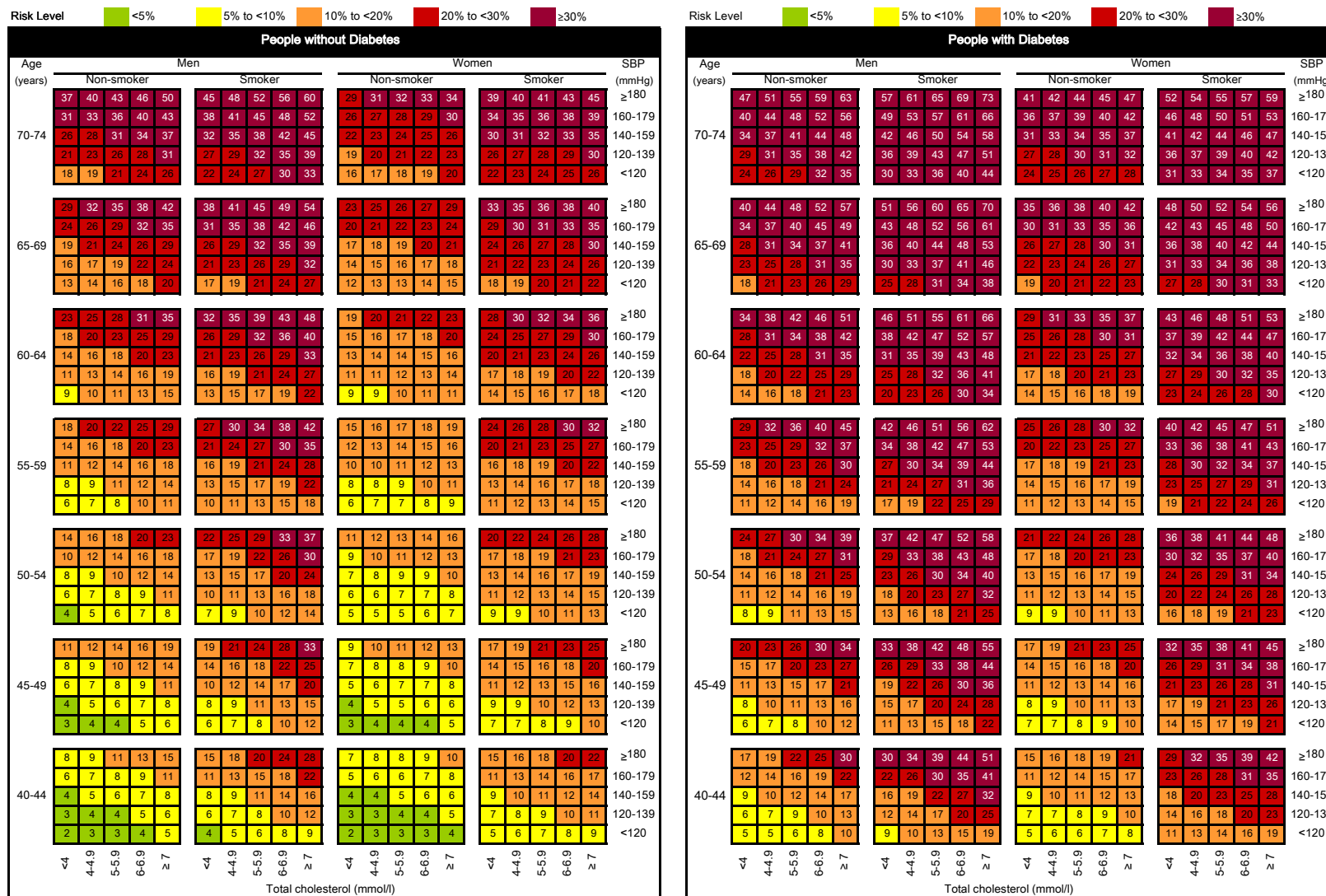
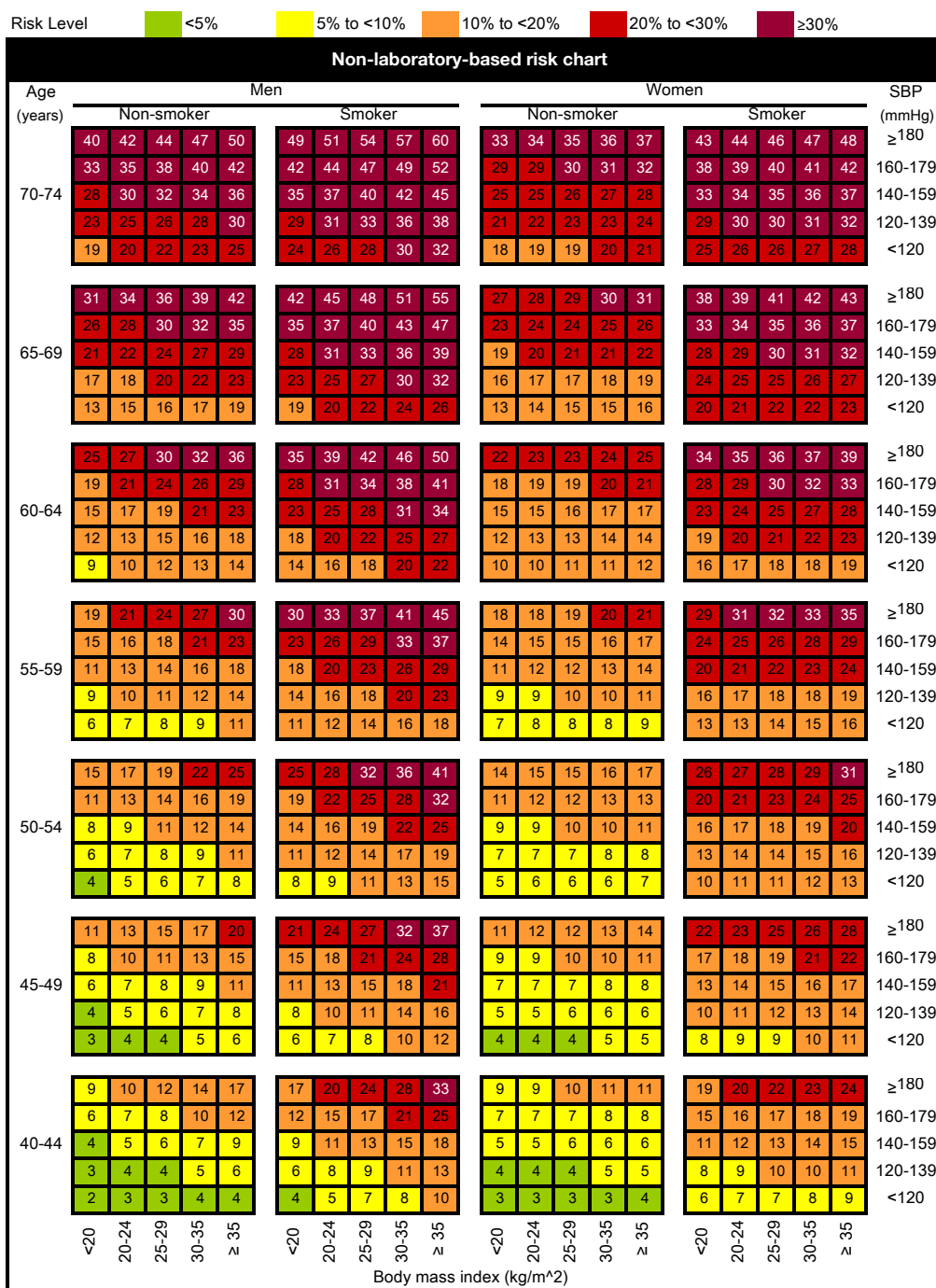


Figure 2: Example of WHO CVD risk (non-laboratory-based) chart

North Africa and Middle East

Afghanistan, Algeria, Bahrain, Egypt, Iran (Islamic Republic of), Iraq, Jordan, Kuwait, Lebanon, Libya, Morocco, occupied Palestinian territory, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, Tunisia, Turkey, United Arab Emirates, Yemen



North Africa and Middle East

Comparison between 2007 WHO/ISH risk prediction charts and 2019 WHO CVD risk charts

The updated CVD risk charts are presented in an analogous manner to the previous WHO/ISH risk charts to facilitate their continuity of use with some adaptation. However, there are several advantages of the updated WHO CVD risk charts over the previous WHO/ISH risk prediction charts and other global, regional and national CVD risk prediction models available (Table 1).

First, compared with the 2007 charts, the new risk charts use data with more scientific rigour, from diverse populations, which is much more globally relevant.

Second, the charts use more contemporary estimates of CVD incidence and risk factors. This, together with simple and generalizable methods, allows for recalibration of the models for many different populations. This means that the CVD risk estimate for an individual in a given country is likely to be much more accurate than in the older charts.

Third, the recalibration approach used allows for rapid revision of CVD models, enabling a flexible updating of models as new relevant epidemiological data for geographical areas arise.

Fourth, the new charts produce estimates for 21 Institute for Health Metrics and Evaluation (IHME) Global Burden of Disease (GBD) regions compared to the 14 WHO regions in the old charts.

The risk stratification for the updated WHO charts differs from those used in the 2007 WHO/ISH CVD risk prediction charts to recognize changes in reported GBD CVD incidence. Stratification is now as follows: <5% (green), 5% to <10% (yellow), 10% to <20% (orange), 20% to <30% (red), and $\geq 30\%$ (dark red). The category >40% is not retained in the updated charts as fewer people in the new risk charts were in that group. This risk stratification is aligned to the WHO recommendations for management of CVD risk.

Annexes 5a, 5b and 6 present the comparison of different risk charts and an approach to identifying suitable risk charts.

Table 1: Comparison between 2007 WHO/ISH risk prediction charts with 2019 WHO CVD risk charts

Parameter	2007 WHO/ISH risk prediction charts			2019 WHO CVD risk charts		
Presentation	For 14 WHO epidemiological subregions			21 IHME GBD regions with more homogenous grouping of countries		
Types of charts	Two sets: One set can be used in settings where blood cholesterol can be measured. The other set is for settings in which blood cholesterol cannot be measured.			Two sets: Laboratory-based charts Non-laboratory-based charts		
Variables	1. With individual cholesterol value: • age • sex • smoking • systolic blood pressure • presence or absence of diabetes • total cholesterol 2. Without individual cholesterol value: • age • sex • smoking • systolic blood pressure • presence or absence of diabetes • national average cholesterol value			1. Laboratory-based: • age • sex • smoking • systolic blood pressure • presence or absence of diabetes • total cholesterol 2. Non-laboratory-based: • age • sex • smoking • systolic blood pressure • BMI		
Risk levels and colour code		2007			2019	
		Green	<10%		Green	<5%
		Yellow	10% to <20%		Yellow	5% to <10%
		Orange	20% to <30%		Orange	10% to <20%
		Red	30% to <40%		Red	20% to <30%
		Deep red	>40%		Deep red	≥30%
Difference in interpretation of risk levels	Green was <10%			Green is <5% and corresponding changes in other risk levels.		

2 Assessment and management of total CVD risk

This section is based on Protocol 1 in the WHO Package of Essential NCD interventions (16). This protocol remains current except for the change in the risk level in the updated WHO cardiovascular (CVD) risk charts. Assessment of total CVD risk can be used for routine management of hypertension (HTN) and diabetes mellitus (DM), and for targeting the following categories of people:

- age >40 years
- smokers
- obesity
- known to have HTN
- known to have DM
- history of premature CVD in first-degree relative
- history of DM or kidney disease in first-degree relative.

Instructions for using the WHO CVD risk (laboratory- based) charts

Table 2 and Figure 3 present a step-by-step guide to applying the WHO CVD (laboratory-based) risk charts. These charts are to be used only for individuals whose status regarding diabetes and total cholesterol is available. Tests for diabetes and cholesterol can be carried out at the time of assessment. If the information on diabetes and total cholesterol is not available, then refer to the instructions on use of non-laboratory-based risk charts.

Table 2: Using the WHO CVD risk (laboratory-based) charts

Action		
Select the regional chart covering your country: • REGION NAME is printed at the top of the charts. • Countries included in each region can be found in Annex 1.		
Have the following information ready: • age • sex • smoker* or non-smoker • presence or absence of diabetes† • systolic blood pressure • total blood cholesterol‡		
Using the charts		
STEP 1: Select the section of the chart as relevant for people with or without diabetes.		
STEP 2: Select the table for men or women, as appropriate.		
STEP 3: Select smoker or non-smoker column.		
STEP 4: Select age-group.		
STEP 5: Within the selected box find the cell where the person's systolic blood pressure and total blood cholesterol intersect.		
STEP 6: The colour of the cell indicates the 10-year risk of a fatal or non-fatal CVD event. The value within the cell is the risk percentage. Colour coding is based on the grouping.		Green
		Yellow
		Orange
		Red
		Deep red
		<5%
		5% to <10%
		10% to <20%
		20% to <30%
		≥30%
STEP 7: Record CVD risk percentage in person's chart.		
STEP 8: Counsel, treat and refer according to risk level		

* Current smoker

† Fasting plasma glucose ≥ 7.0 mmol/L (126 mg/dL), or 2-h plasma glucose ≥ 11.1 mmol/L (200 mg/dL), or HbA1c $\geq 6.5\%$ or known diabetes

‡ Cholesterol values are to be entered in the chart as mmol/L (To convert mg/dL to mmol/L, multiply by 0.02586. e.g 200 mg/dL x 0.02586 = 5.172 mmol/L.)

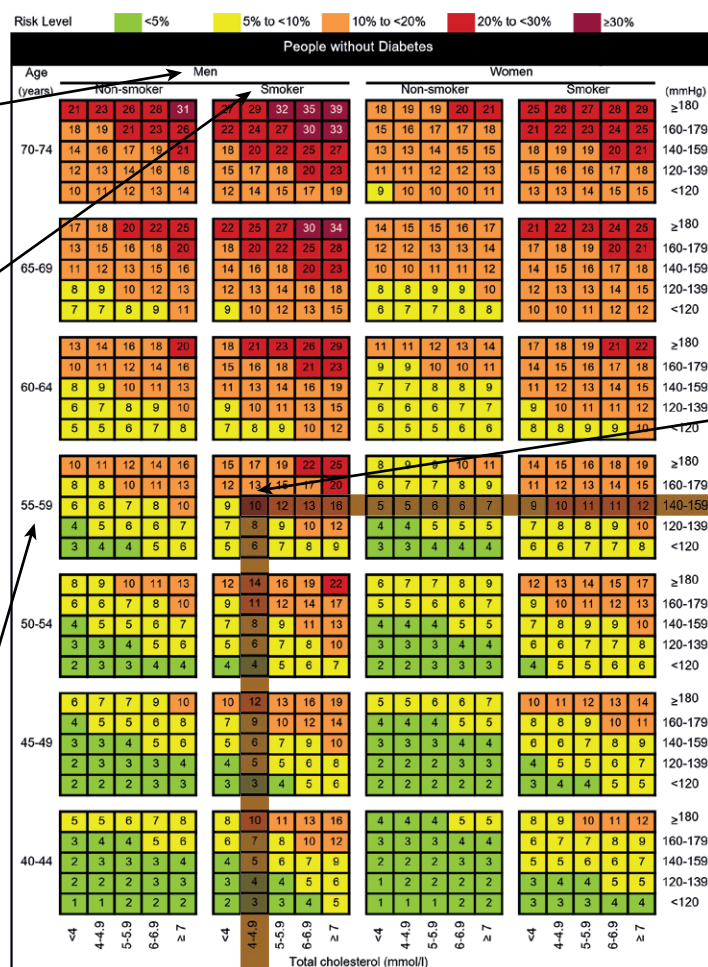
Figure 3: Illustration of how to use the WHO CVD risk (laboratory-based) chart

STEP 1: Select the section of the chart for people with or without diabetes.

STEP 2: Select the table for men or women, as appropriate.

STEP 3: Select the column for non-smoker or smoker.

STEP 4: Select the relevant age group.



STEP 5: Within the selected box, find the cell where the person's systolic blood pressure and serum cholesterol intersect.

Integrated management of total CVD risk

Management of total CVD risk in people with different risk thresholds is presented in Table 3. This is based on the WHO PEN Protocol (16) except for the change in risk levels, which are adjusted to the updated CVD risk charts. The purpose of management is to motivate and assist individuals with high risk levels to lower their total CVD risk. All individuals with blood pressure at or above 160/100 mmHg, or blood pressure raised to a lesser degree but with target organ damage should have drug treatment and specific lifestyle advice to lower their blood pressure and risk of CVD.

All patients should receive counselling on diet (which includes lipid-lowering diet and low sodium), physical activity, tobacco cessation (smoking and smokeless) and avoiding harmful use of alcohol (1). For further details on counselling please refer to the HEARTS Module "H" on Healthy-lifestyle counselling (17).

Table 3: Management guidance for total CVD risk

Management of total CVD risk (adapted from WHO PEN Protocol 1)	
Risk <10%	Counsel on diet, physical activity, smoking cessation and avoiding harmful use of alcohol. If risk <5%, follow up in 12 months. If risk 5% to <10%, follow up every 3 months until targets are met, then 6–9 months thereafter.
Risk 10% to <20%	Counsel on diet, physical activity, smoking cessation and avoiding harmful use of alcohol. Persistent BP $\geq$ 140/90 mmHg consider drugs (see below). Follow up every 3–6 months.
Risk $\geq$20%	Counsel on diet, physical activity, smoking cessation and avoiding harmful use of alcohol. Persistent BP $\geq$ 130/80, consider drugs (see below). Give a statin. Follow up every 3 months. If there is no reduction in cardiovascular risk after six months of follow-up refer to next level.
Important practical points	Management of hypertension and diabetes: • For management of hypertension refer to HEARTS technical package E module (18): https://www.who.int/cardiovascular_diseases/hearts/en/ • For management of diabetes mellitus type 2, refer to the HEARTS module Diagnosis and management of diabetes (19): https://www.who.int/publications-detail/who-ucn-ncd-20.1
	Consider drug treatment for following categories: • All patients with established DM and CVD (coronary heart disease, myocardial infarction, transient ischaemic attacks, cerebrovascular disease or peripheral vascular disease), renal disease. If stable, should continue the treatment already prescribed and be considered as having risk $\geq$20%. • People with albuminuria, retinopathy, left ventricular hypertrophy. • All individuals with persistent raised BP $\geq$160/100 mmHg. • All individuals with total cholesterol at or above 8 mmol/L (320 mg/dL).
	Follow-up visits: • Ask about: new symptoms, adherence to advice on tobacco and alcohol use, physical activity, healthy diet, medications etc. • Assess (physical exam). • Estimate cardiovascular risk. • Refer if necessary. • Counsel all and treat as shown in protocol.

Countries may set different thresholds for initiating treatment, based on the distribution of CVD risk in the population (Annex 7). If, after a reasonable amount of time, the non-drug treatment alone is not able to meet the hypertension (HTN) control goals, consider drug treatment.

Follow-up

Follow-up frequency will depend on the capacity of the health system and will have to be decided based on the local context. National protocols are important for the management as they will help to standardize the treatment and bring about efficiencies. A schedule of follow-up based on WHO PEN Protocol 1 is provided. If risk is <5%, follow up in 12 months. If risk is 5%–10%, follow up every 3 months, then 6–9 months thereafter (16). For patients who are not at high enough risk for medical therapy, CVD risk should be reassessed every 12 months, or earlier if clinical symptoms develop.

Instructions for using WHO CVD risk (non-laboratory-based) charts

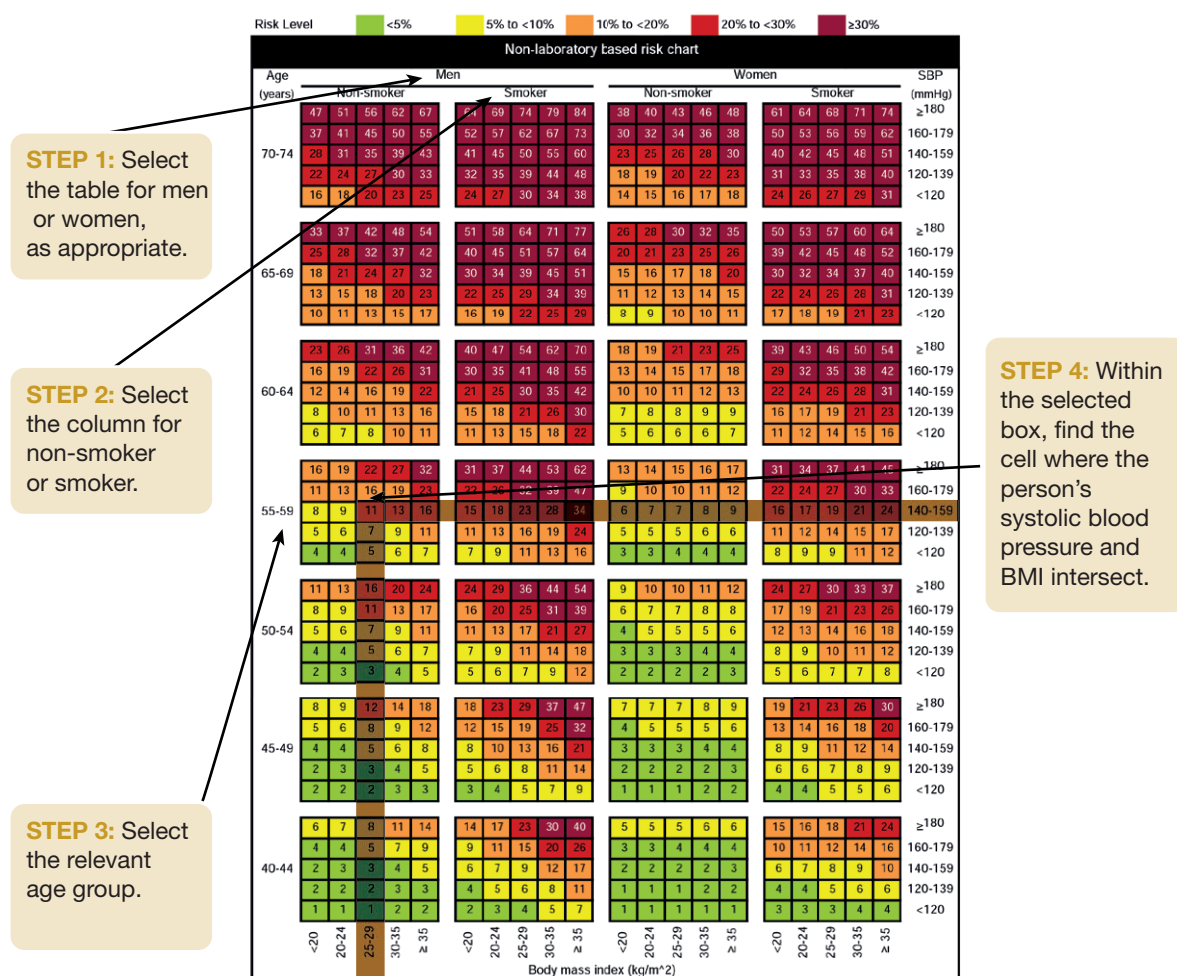
These charts are for the use of settings where diabetes and cholesterol cannot be measured. They can also be used to identify people at high risk who can be taken up for further investigations. Table 4 and Figure 4 present the steps to apply the non-laboratory WHO CVD risk charts.

Table 4: Using the WHO CVD risk (non-laboratory-based) charts

Action		
Select the regional chart covering your country: - REGION NAME is printed at the top of the charts. - Countries included in each region can be found in Annex 1.		
Have the following information ready: - age - sex - smoker* or non-smoker - systolic blood pressure - BMI (body mass index) = weight (kg) ÷ height (m)²		
Using the charts		
STEP 1: Select the table for men or women, as appropriate.		
STEP 2: Select smoker or non-smoker column.		
STEP 3: Select age-group.		
STEP 4: Within the selected box find the cell where the person's systolic blood pressure and body mass index (BMI) intersect.		
STEP 5: The colour of the cell indicates the 10-year risk of a fatal or non-fatal CVD event. The value within the cell is the risk percentage. Colour coding is based on the grouping.		Green <5%
		Yellow 5% to <10%
		Orange 10% to <20%
		Red 20% to <30%
		Deep red ≥30%
STEP 6: Record CVD risk percentage in person's chart.		
STEP 7: Counsel, treat and refer according to risk level		

* Current smoker

Figure 4: Illustration of how to use the WHO CVD risk (non-laboratory-based) chart



Use of WHO CVD risk (non-laboratory-based) charts

Currently, the WHO CVD risk (non-laboratory-based) charts can be considered for identifying a subset of the population who might benefit from laboratory-based risk assessment. Where laboratory testing may be available but extremely limited due to costs or distance, use of the non-laboratory charts could allow for a two-stage process that reduces the number of people at lower levels of risk who are subjected to unwarranted testing. Additionally, the non-laboratory risk charts can be used for education and advocacy regarding total CVD risk in areas where lab testing remains currently unavailable. In these areas, health care providers and policy-makers can use the non-laboratory risk charts to assess the general risk of the population, and to advocate for more resources if risk appears high. Lastly, some studies have shown that use of the non-laboratory-based tool for determining future risk performs as well as laboratory-based tools and may even be cost-effective for decisions regarding medical therapy (4, 5). As more data become available to confirm these findings, the WHO CVD risk (non-laboratory-based) charts may be recommended for use in treatment decisions as well. Individuals with a total CVD risk level of 10% and above should receive an assessment using laboratory-based charts after measurement for diabetes and cholesterol. Advice on lifestyle modification should be given as needed.

Annex 1: GBD regions

The first step when using the WHO CVD risk charts is to identify which Global Burden of Disease (GBD) region a country is in (Table 5).

Table 5: GBD regional groups

GBD region	Countries	Page number of charts	
		Lab-based	Non-lab-based
High-income North America	Canada, Greenland, United States of America	25	46
Caribbean	Antigua and Barbuda, Bahamas, Barbados, Belize, Bermuda, Cuba, Dominica, Dominican Republic, Grenada, Guyana, Haiti, Jamaica, Puerto Rico, Saint Lucia, Saint Vincent and the Grenadines, Suriname, Trinidad and Tobago	26	47
Central Latin America	Colombia, Costa Rica, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, Panama, Venezuela (Bolivarian Republic of)	27	48
Andean Latin America	Bolivia, Ecuador, Peru	28	49
Tropical Latin America	Brazil, Paraguay	29	50
Southern Latin America	Argentina, Chile, Uruguay	30	51
Western Europe	Andorra, Austria, Belgium, Cyprus, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Israel, Italy, Luxembourg, Malta, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom	31	52
Central Europe	Albania, Bosnia and Herzegovina, Bulgaria, Croatia, Czech Republic, Hungary, Montenegro, North Macedonia, Poland, Romania, Serbia, Slovakia, Slovenia	32	53
Eastern Europe	Belarus, Estonia, Latvia, Lithuania, Republic of Moldova, Russian Federation, Ukraine	33	54
North Africa and Middle East	Afghanistan, Algeria, Bahrain, Egypt, Iran (Islamic Republic of), Iraq, Jordan, Kuwait, Lebanon, Libya, Morocco, occupied Palestinian territory, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, Tunisia, Turkey, United Arab Emirates, Yemen	34	55

GBD region	Countries	Page number of charts	
		Lab-based	Non-lab-based
Western Sub-Saharan Africa	Benin, Burkina Faso, Cabo Verde, Cameroon, Chad, Cote d'Ivoire, Gambia, Ghana, Guinea, Guinea-Bissau, Liberia, Mali, Mauritania, Niger, Nigeria, Sao Tome and Principe, Senegal, Sierra Leone, Togo	35	56
Central Sub-Saharan Africa	Angola, Central African Republic, Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon	36	57
Eastern Sub-Saharan Africa	Burundi, Comoros, Djibouti, Eritrea, Ethiopia, Kenya, Madagascar, Malawi, Mozambique, Rwanda, Somalia, Uganda, United Republic of Tanzania, Zambia	37	58
Southern Sub-Saharan Africa	Botswana, Eswatini, Lesotho, Namibia, South Africa, Zimbabwe	38	59
Central Asia	Armenia, Azerbaijan, Georgia, Kazakhstan, Kyrgyzstan, Mongolia, Tajikistan, Turkmenistan, Uzbekistan	39	60
East Asia	China, Democratic People's Republic of Korea	40	61
South Asia	Bangladesh, Bhutan, India, Nepal, Pakistan	41	62
South-East Asia	Cambodia, Indonesia, Lao People's Democratic Republic, Malaysia, Maldives, Mauritius, Myanmar, Philippines, Seychelles, Sri Lanka, Thailand, Timor-Leste, Viet Nam	42	63
High-income Asia Pacific	Brunei Darussalam, Japan, Republic of Korea, Singapore	43	64
Australasia	Australia, New Zealand	44	65
Oceania	Fiji, Kiribati, Marshall Islands, Micronesia (Federated States of), Papua New Guinea, Samoa, Solomon Islands, Tonga, Vanuatu	45	66

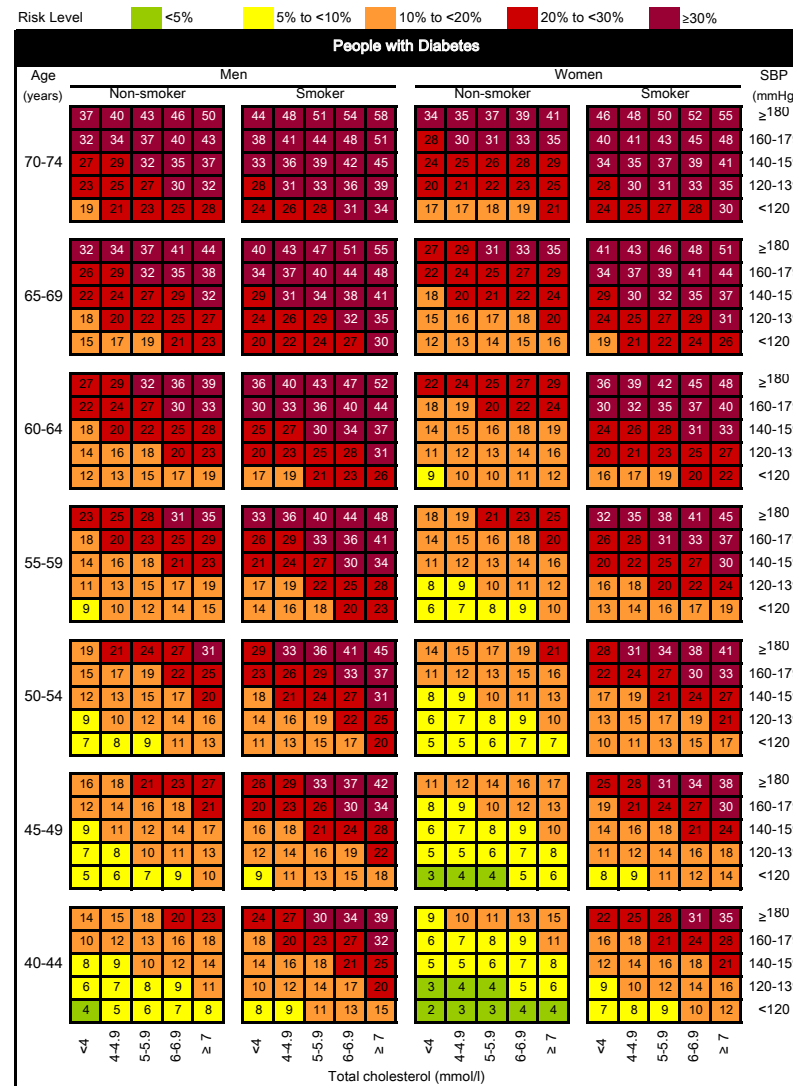
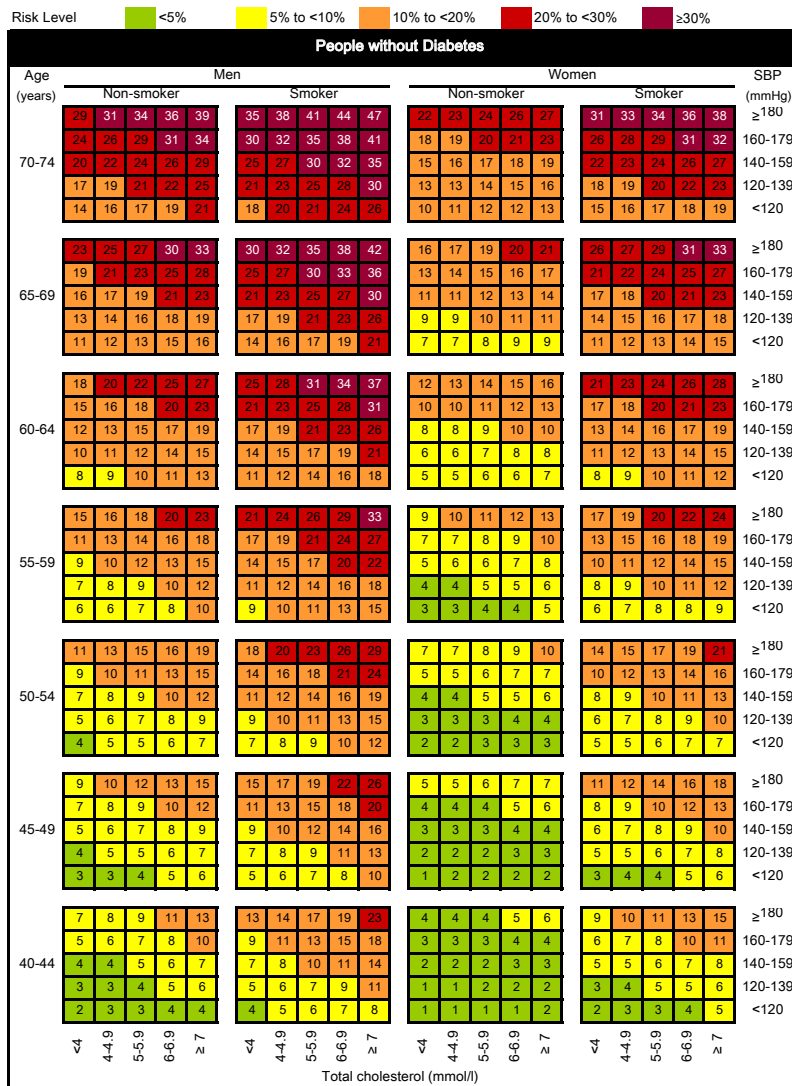
WHO CVD risk charts for a specific region can be downloaded from:
<https://www.who.int/news-room/detail/02-09-2019-who-updates-cardiovascular-risk-charts>

Annex 2: WHO CVD risk (laboratory-based) charts

WHO cardiovascular disease risk laboratory-based charts

High-income North America

Canada, Greenland, United States of America

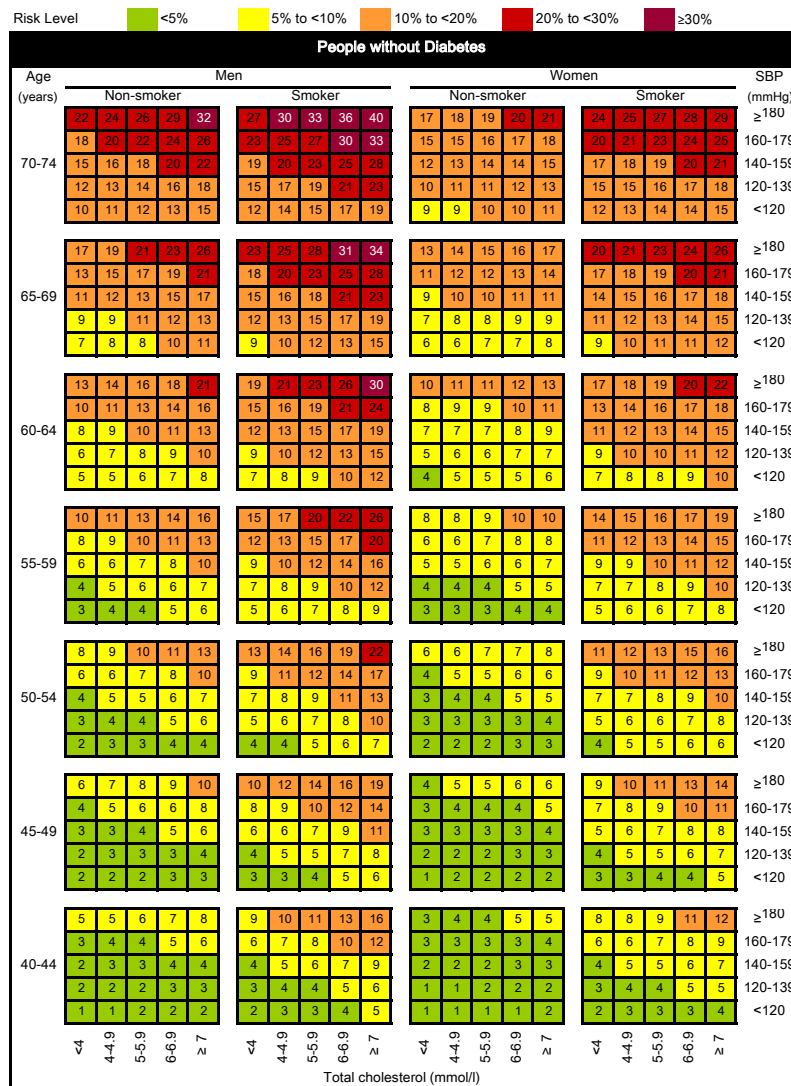


High-income North America

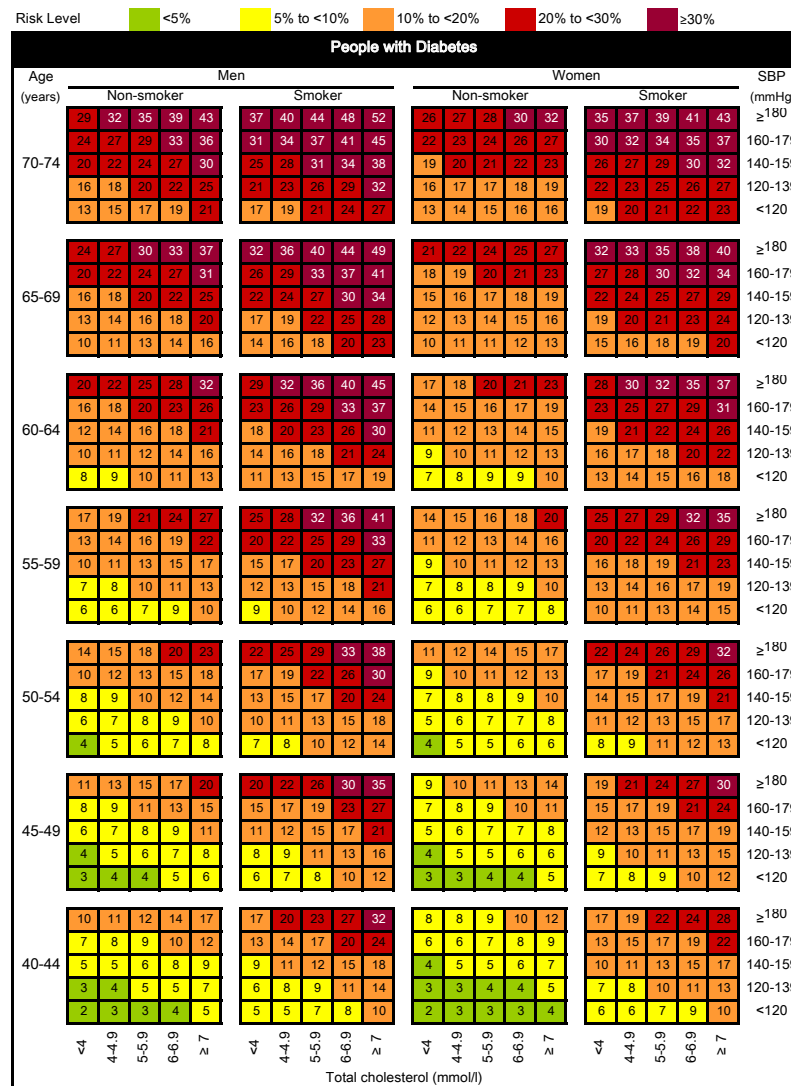
WHO cardiovascular disease risk laboratory-based charts

Caribbean

Antigua and Barbuda, Bahamas, Barbados, Belize, Bermuda, Cuba, Dominica, Dominican Republic, Grenada, Guyana, Haiti, Jamaica, Puerto Rico, Saint Lucia, Saint Vincent and the Grenadines, Suriname, Trinidad and Tobago



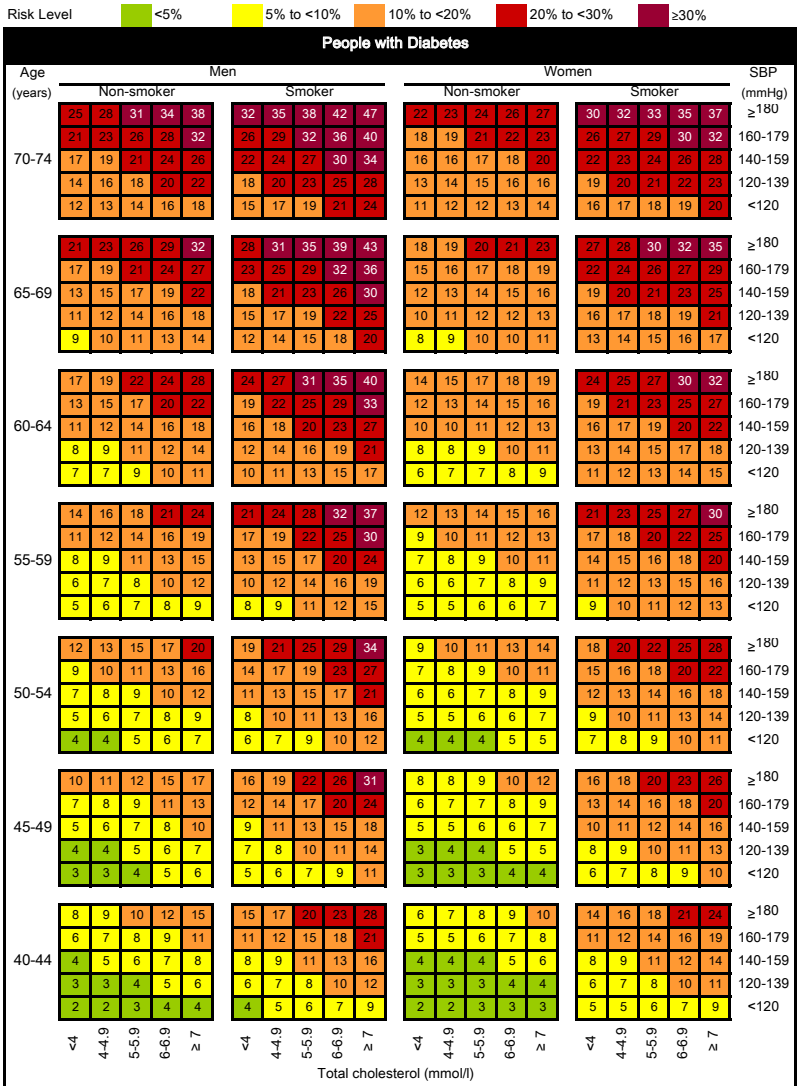
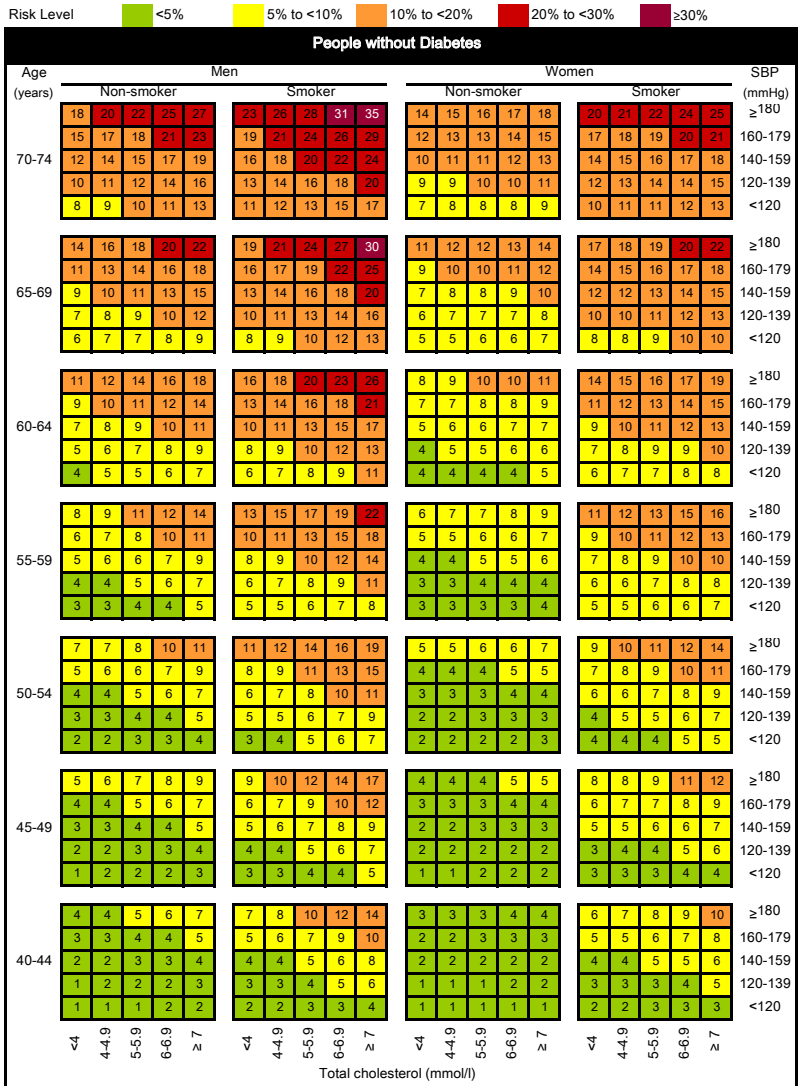
Caribbean



WHO cardiovascular disease risk laboratory-based charts

Central Latin America

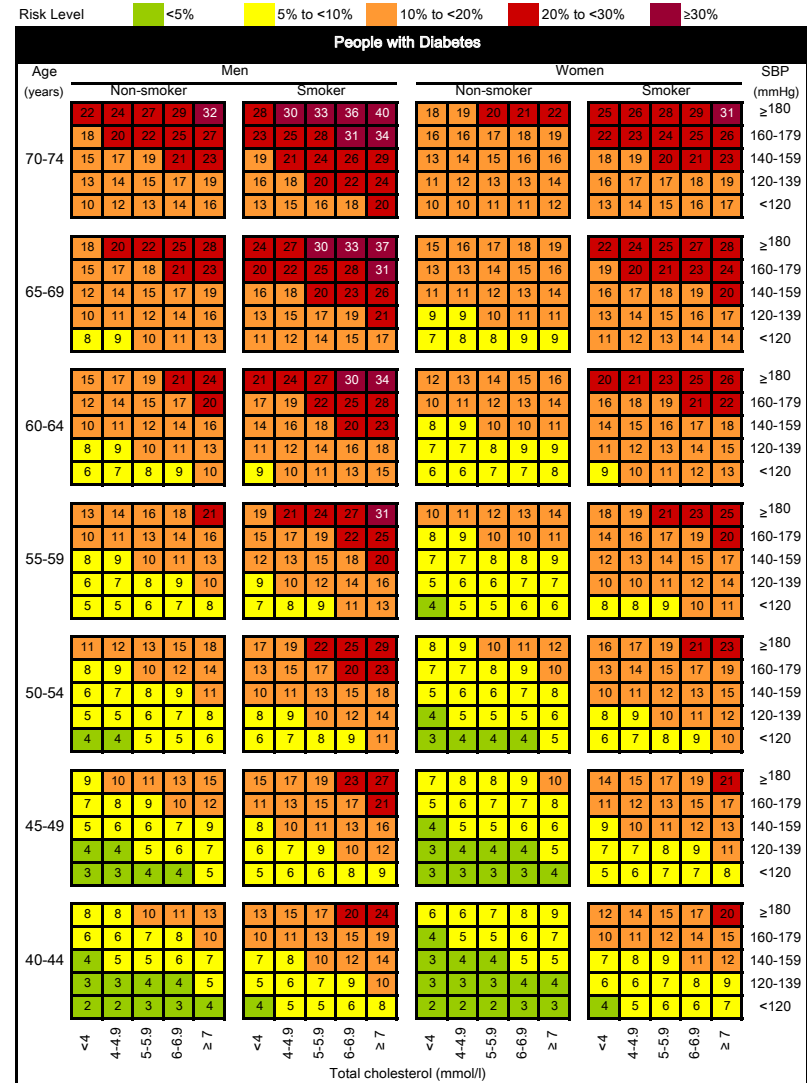
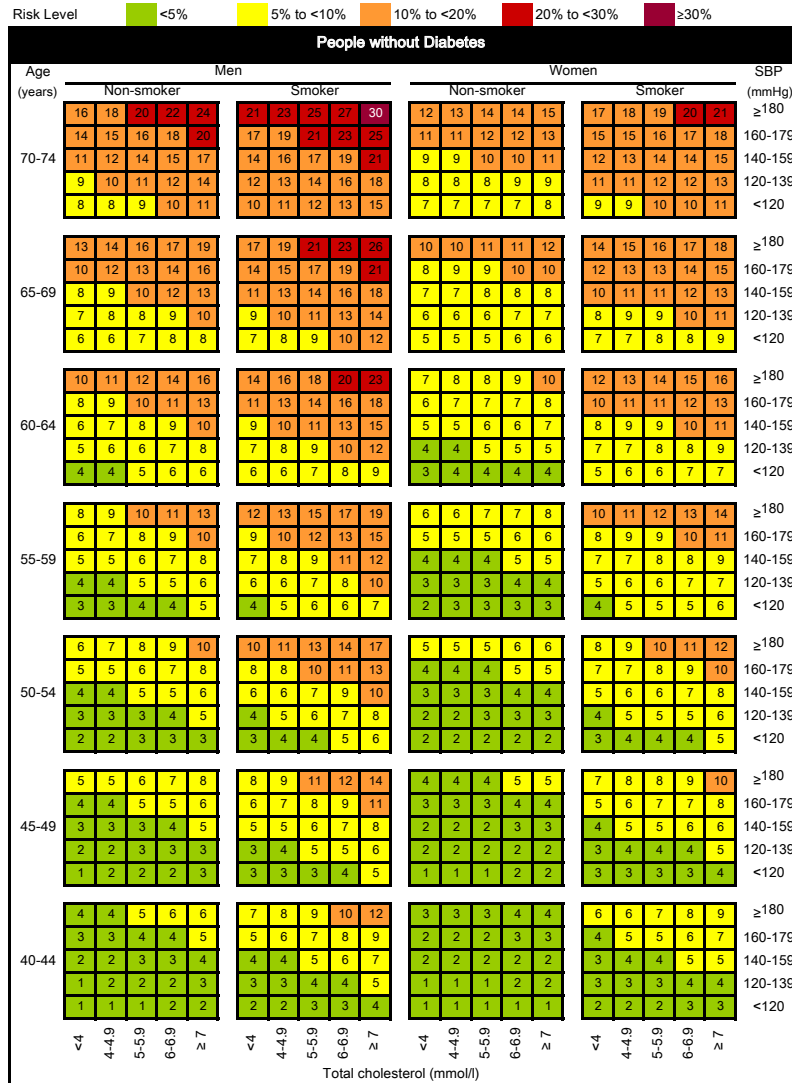
Colombia, Costa Rica, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, Panama, Venezuela (Bolivarian Republic of)



WHO cardiovascular disease risk laboratory-based charts

Andean Latin America

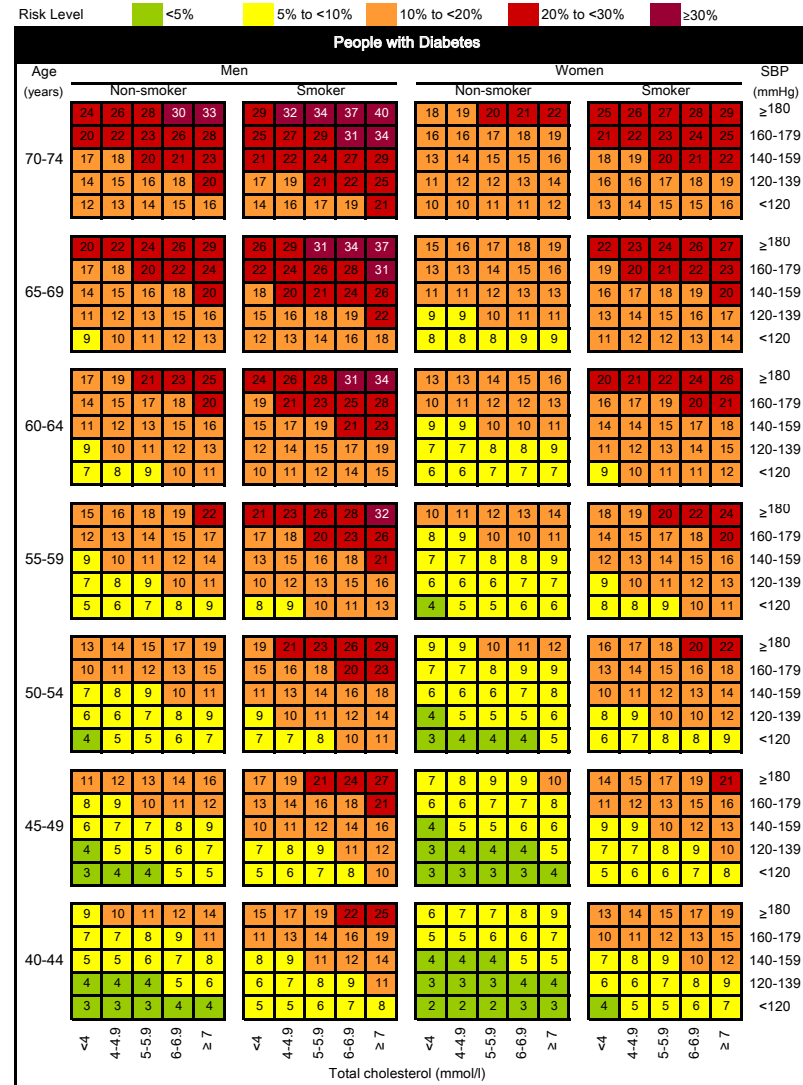
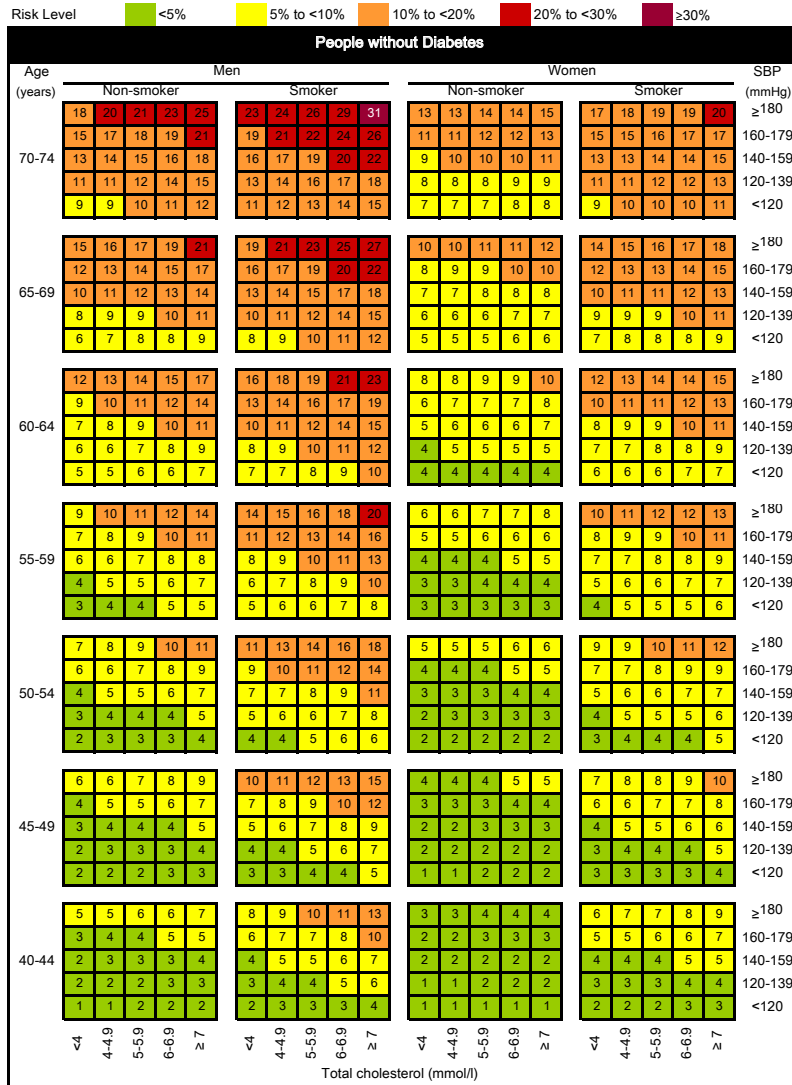
Bolivia, Ecuador, Peru



Andean Latin America

WHO cardiovascular disease risk laboratory-based charts

Tropical Latin America
Brazil, Paraguay

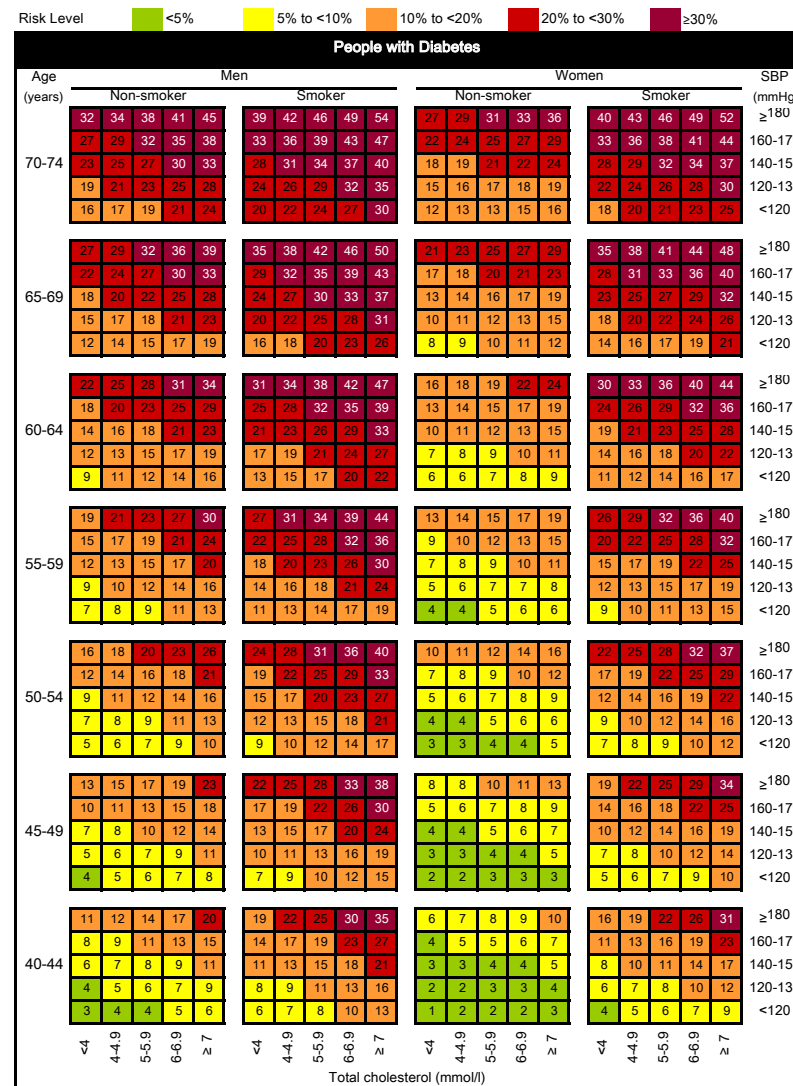
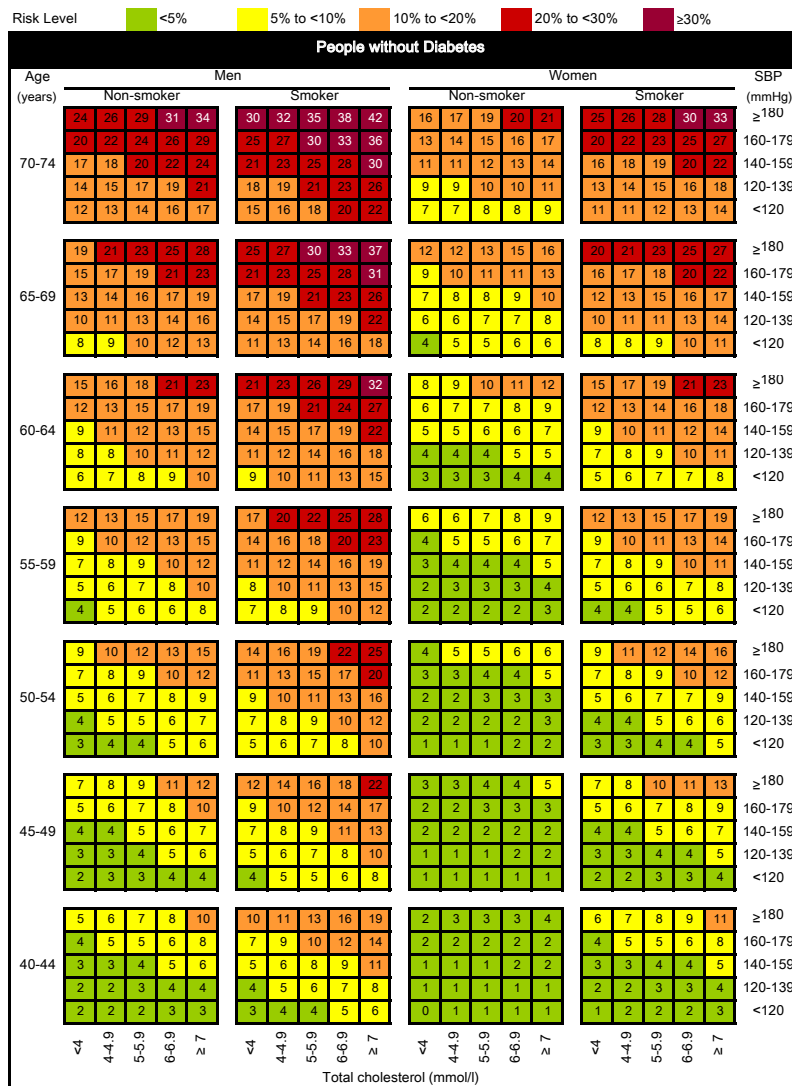


Tropical Latin America

WHO cardiovascular disease risk laboratory-based charts

Southern Latin America

Argentina, Chile, Uruguay

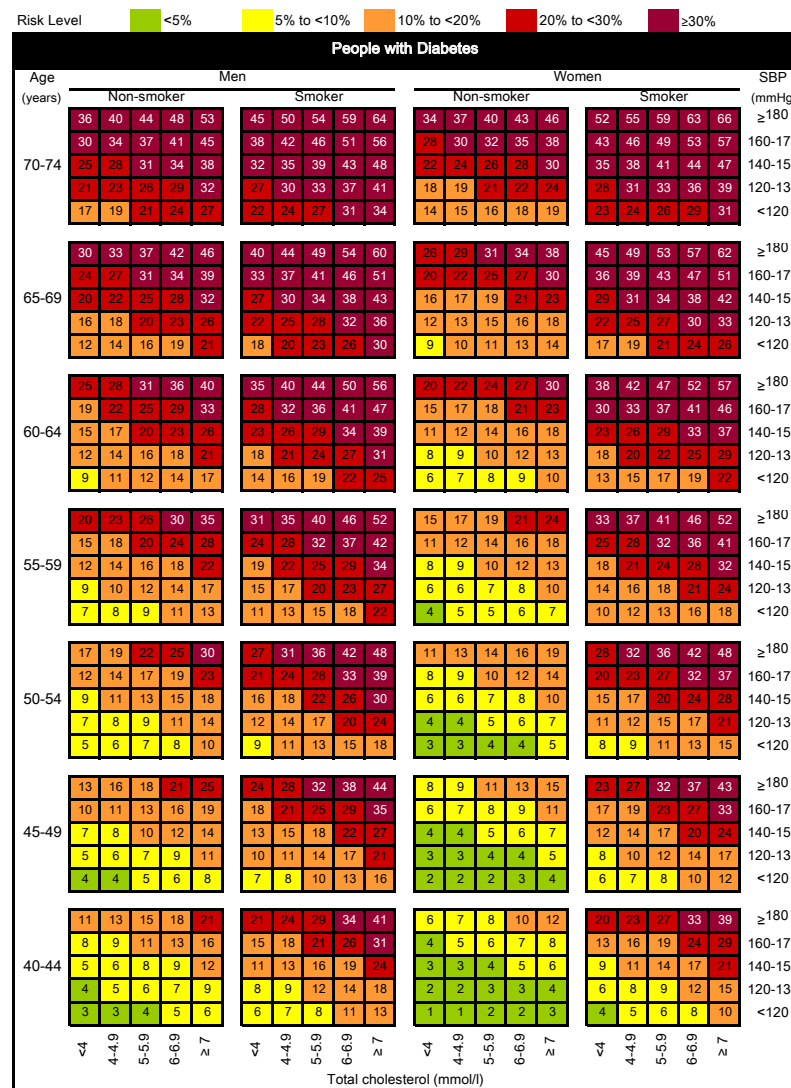
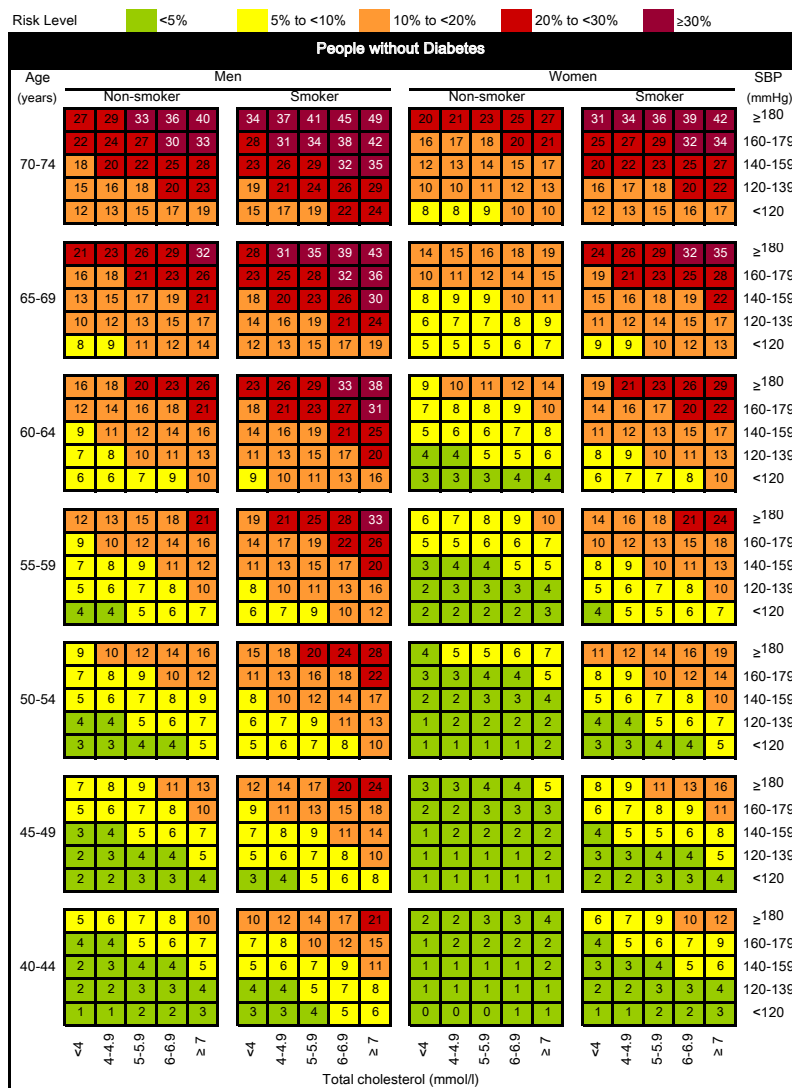


Southern Latin America

WHO cardiovascular disease risk laboratory-based charts

Western Europe

Andorra, Austria, Belgium, Cyprus, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Israel, Italy, Luxembourg, Malta, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom

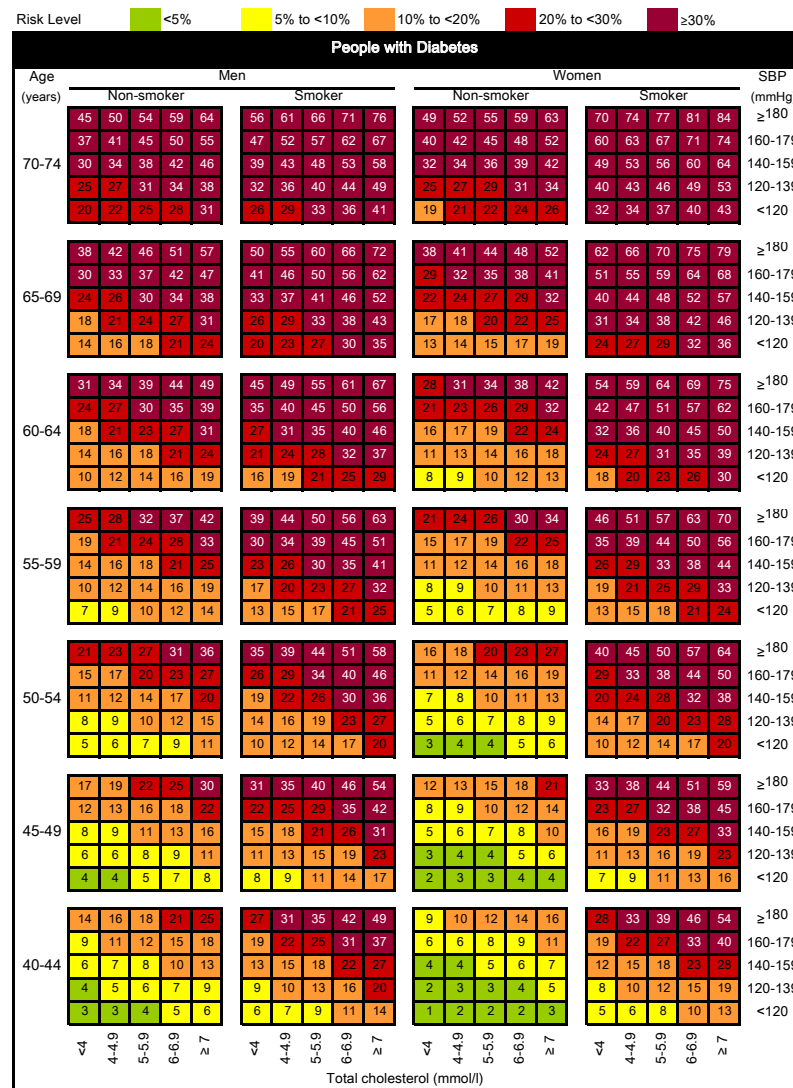
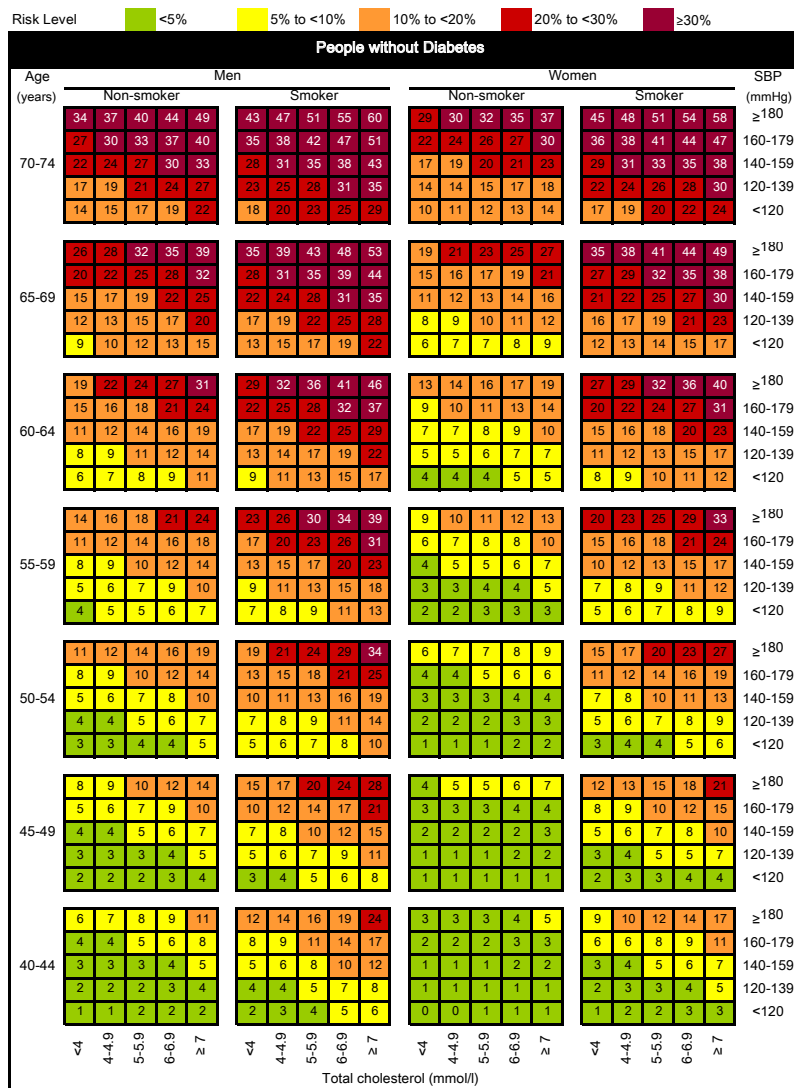


Western Europe

WHO cardiovascular disease risk laboratory-based charts

Central Europe

Albania, Bosnia and Herzegovina, Bulgaria, Croatia, Czech Republic, Hungary, Montenegro, North Macedonia, Poland, Romania, Serbia, Slovakia, Slovenia

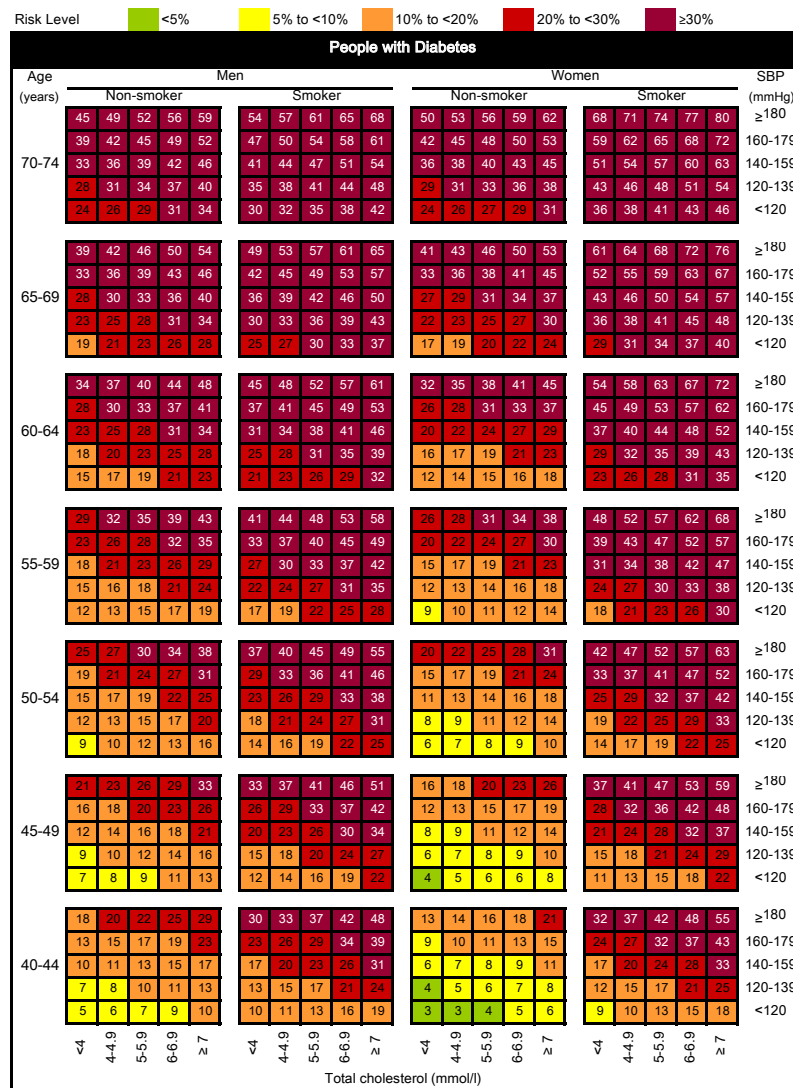
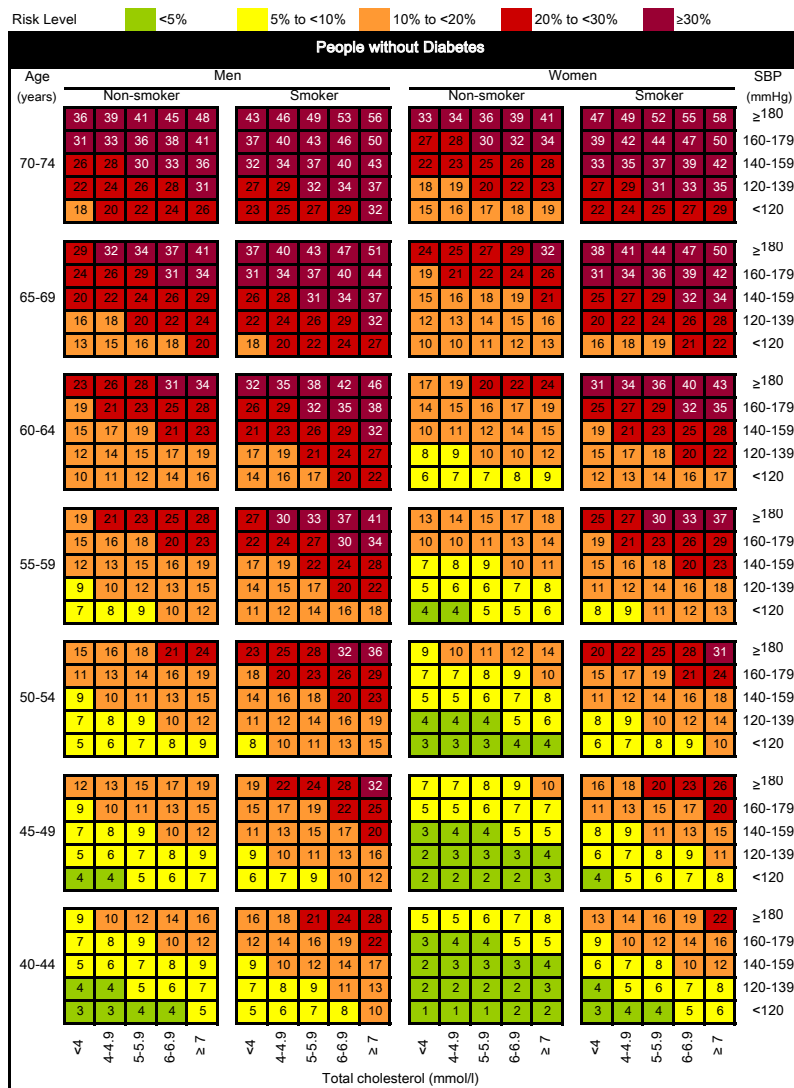


Central Europe

WHO cardiovascular disease risk laboratory-based charts

Eastern Europe

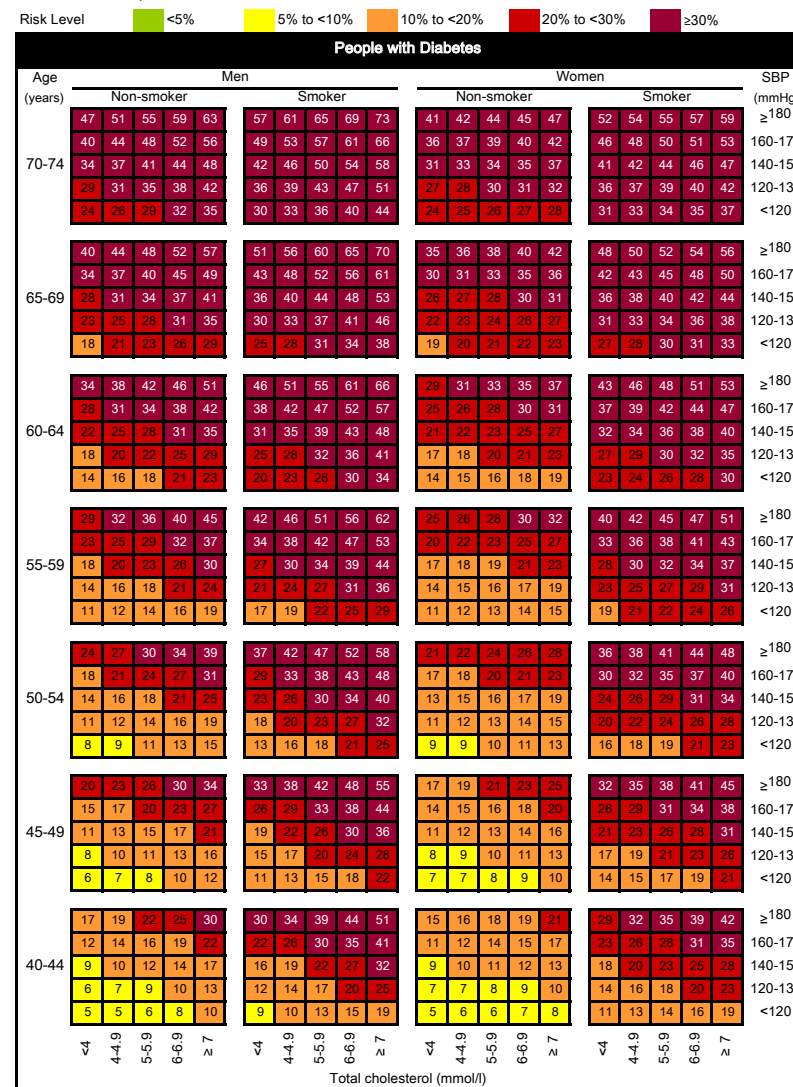
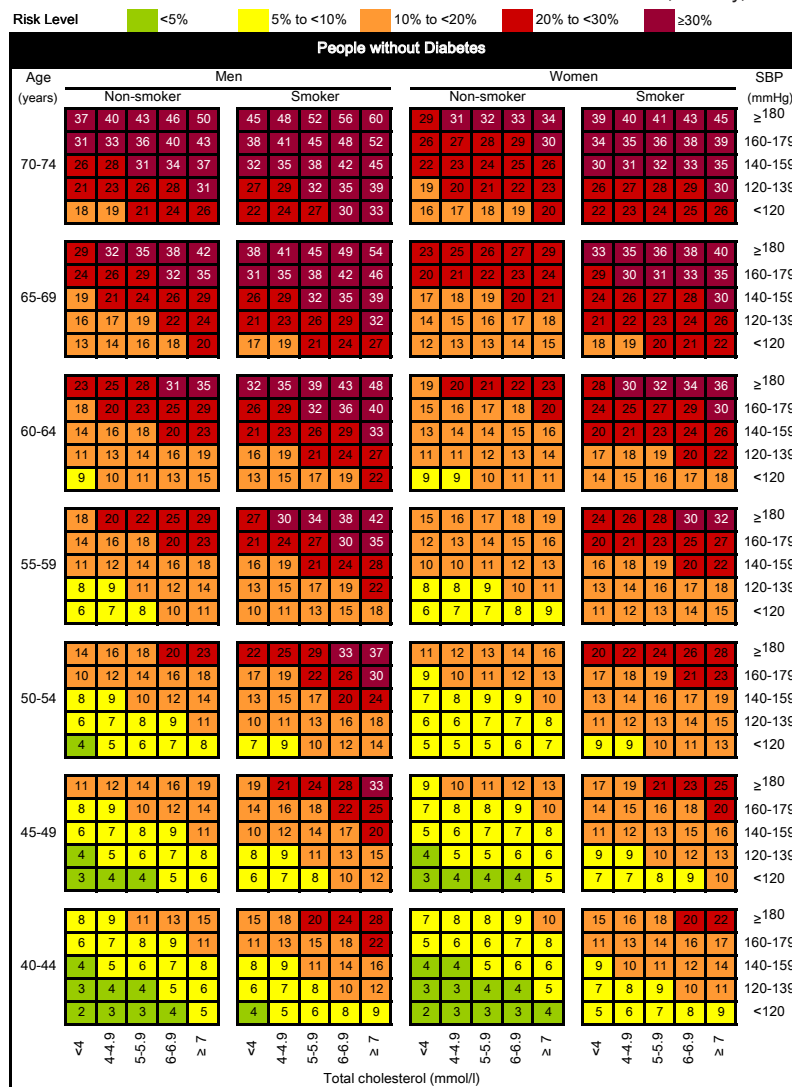
Belarus, Estonia, Latvia, Lithuania, Republic of Moldova, Russian Federation, Ukraine



WHO cardiovascular disease risk laboratory-based charts

North Africa and Middle East

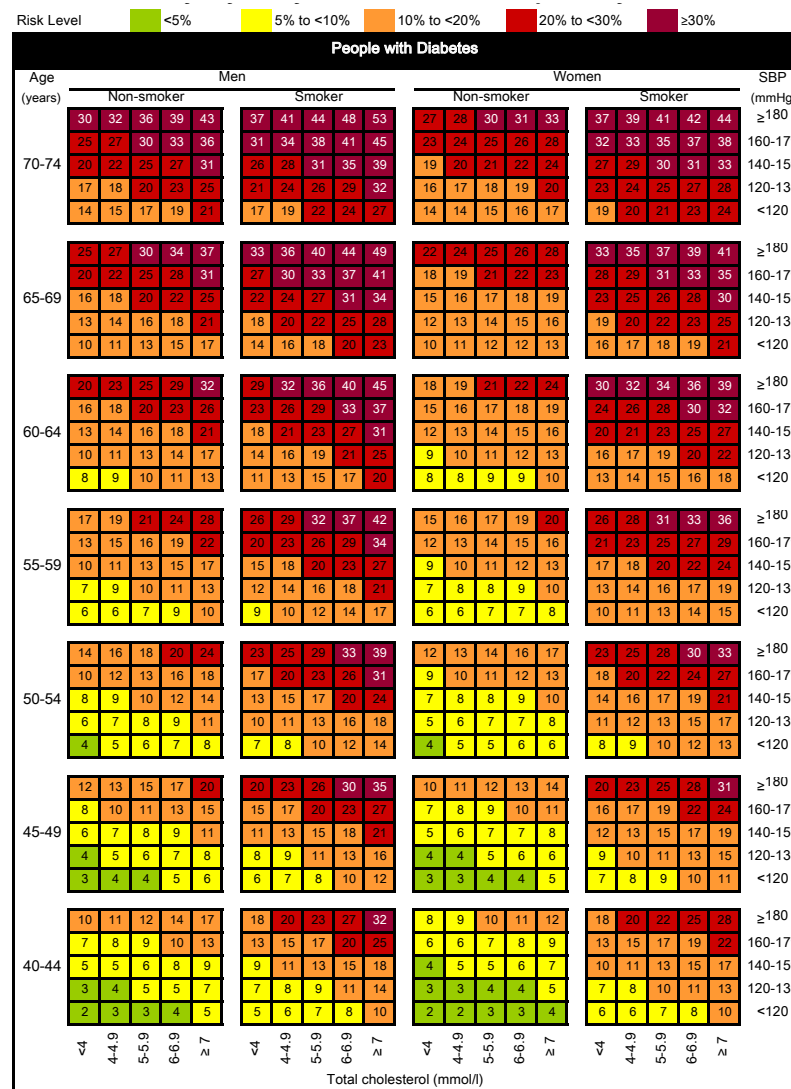
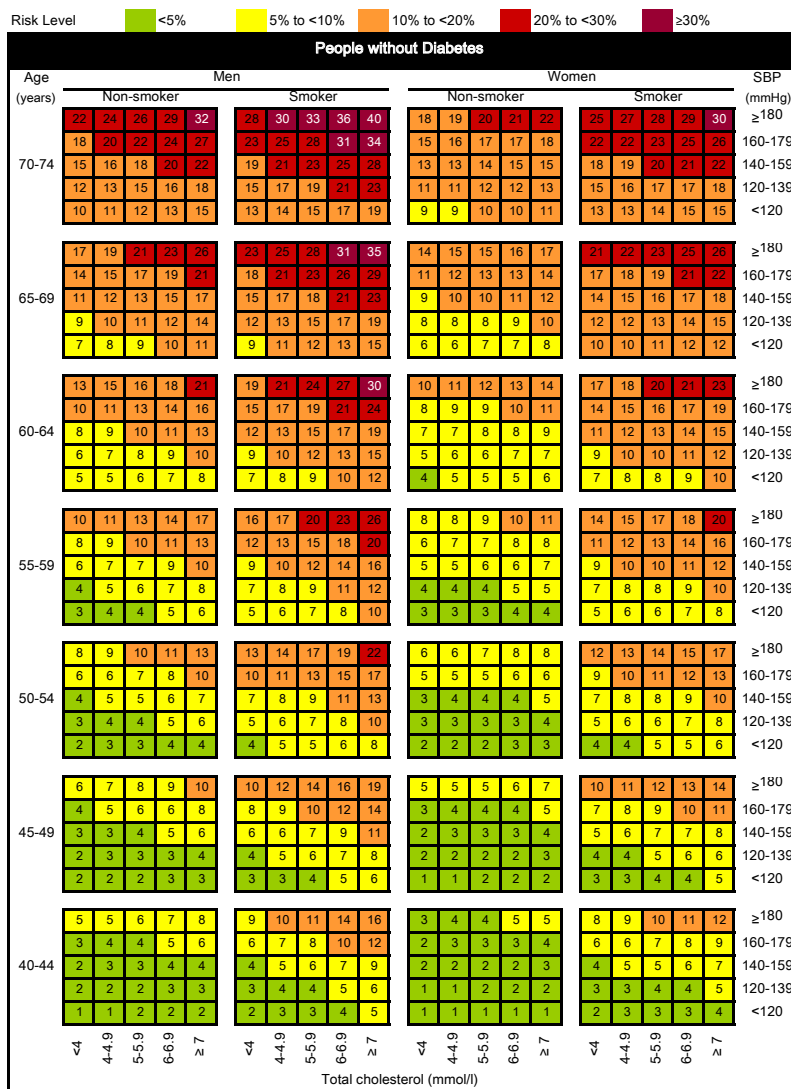
Afghanistan, Algeria, Bahrain, Egypt, Iran (Islamic Republic of), Iraq, Jordan, Kuwait, Lebanon, Libya, Morocco, occupied Palestinian territory, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, Tunisia, Turkey, United Arab Emirates, Yemen



WHO cardiovascular disease risk laboratory-based charts

Western Sub-Saharan Africa

Benin, Burkina Faso, Cabo Verde, Cameroon, Chad, Cote d'Ivoire, Gambia, Ghana, Guinea, Guinea-Bissau, Liberia, Mali, Mauritania, Niger, Nigeria, Sao Tome and Principe, Senegal, Sierra Leone, Togo

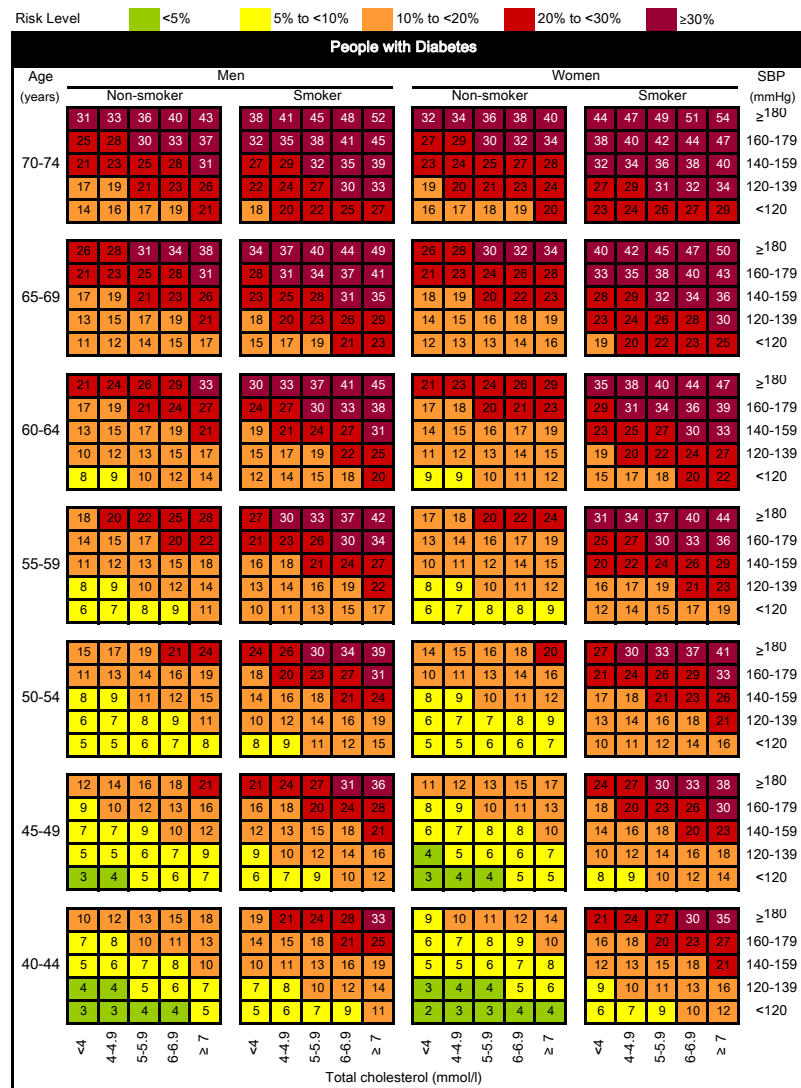
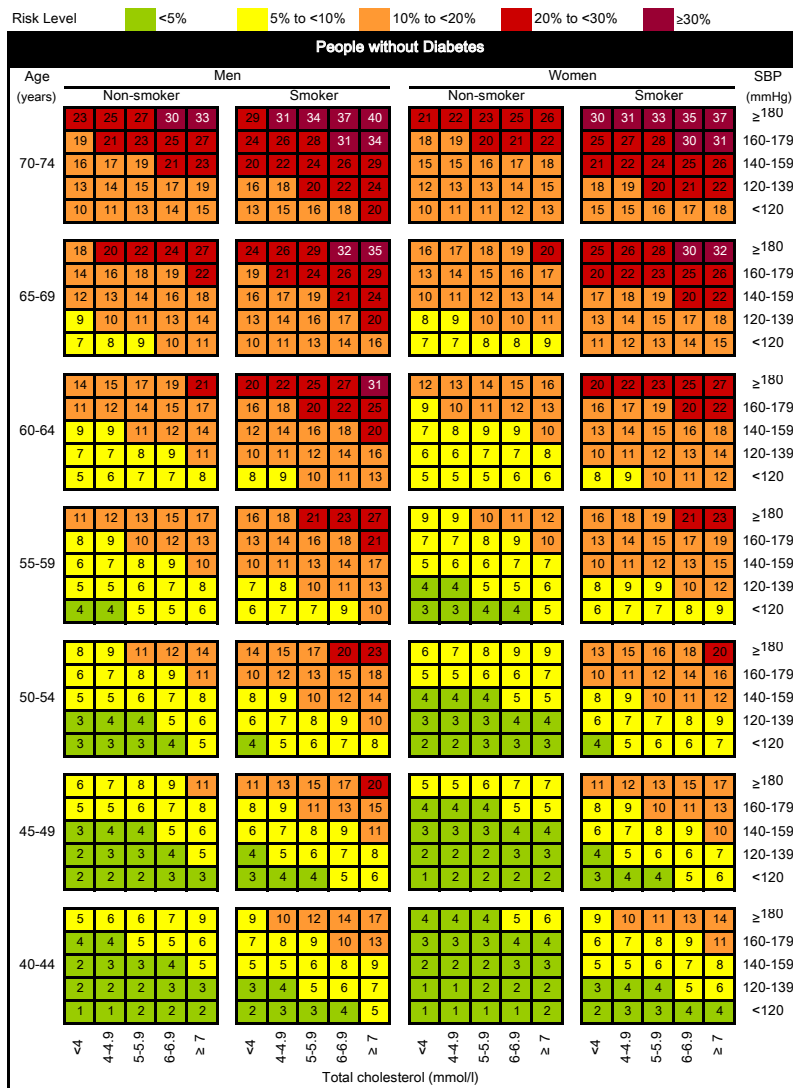


Western Sub-Saharan Africa

WHO cardiovascular disease risk laboratory-based charts

Central Sub-Saharan Africa

Angola, Central African Republic, Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon

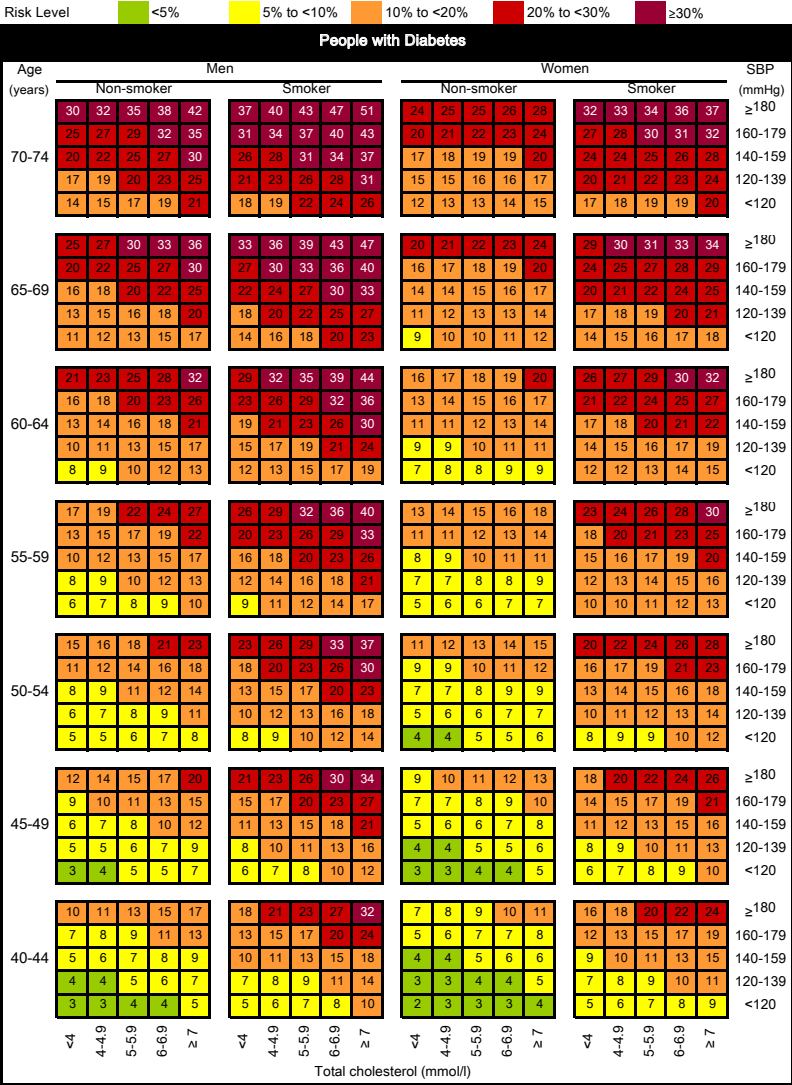
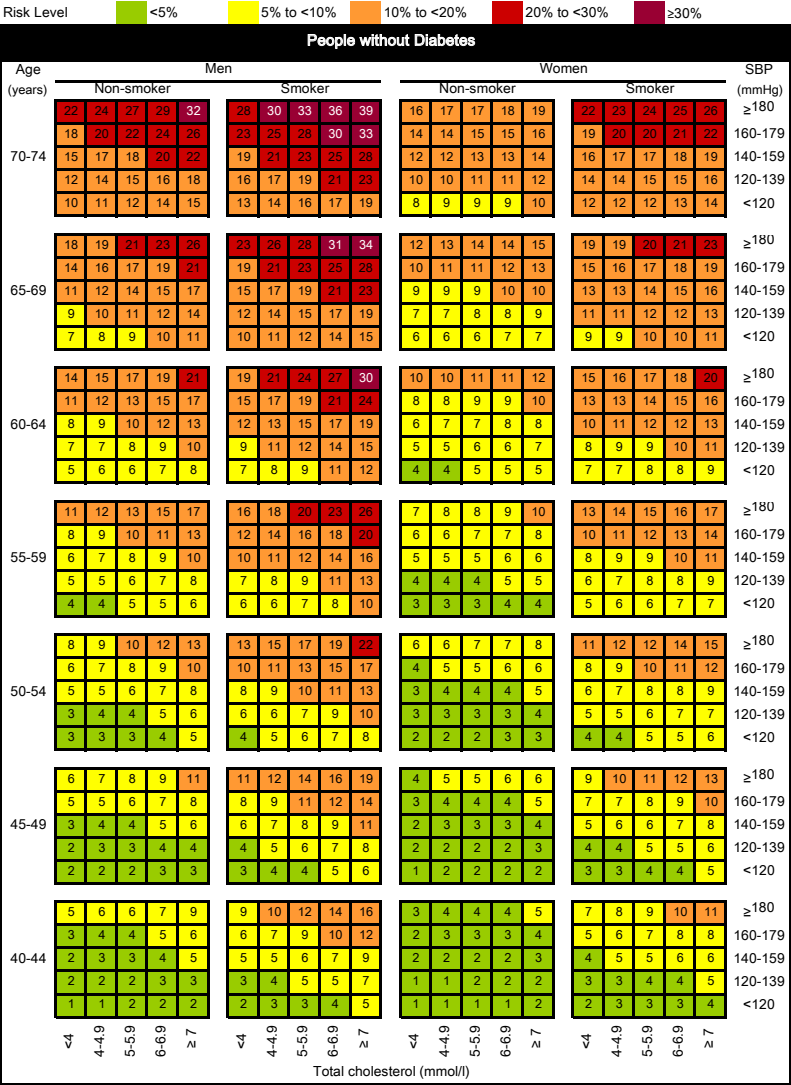


Central Sub-Saharan Africa

WHO cardiovascular disease risk laboratory-based charts

Eastern Sub-Saharan Africa

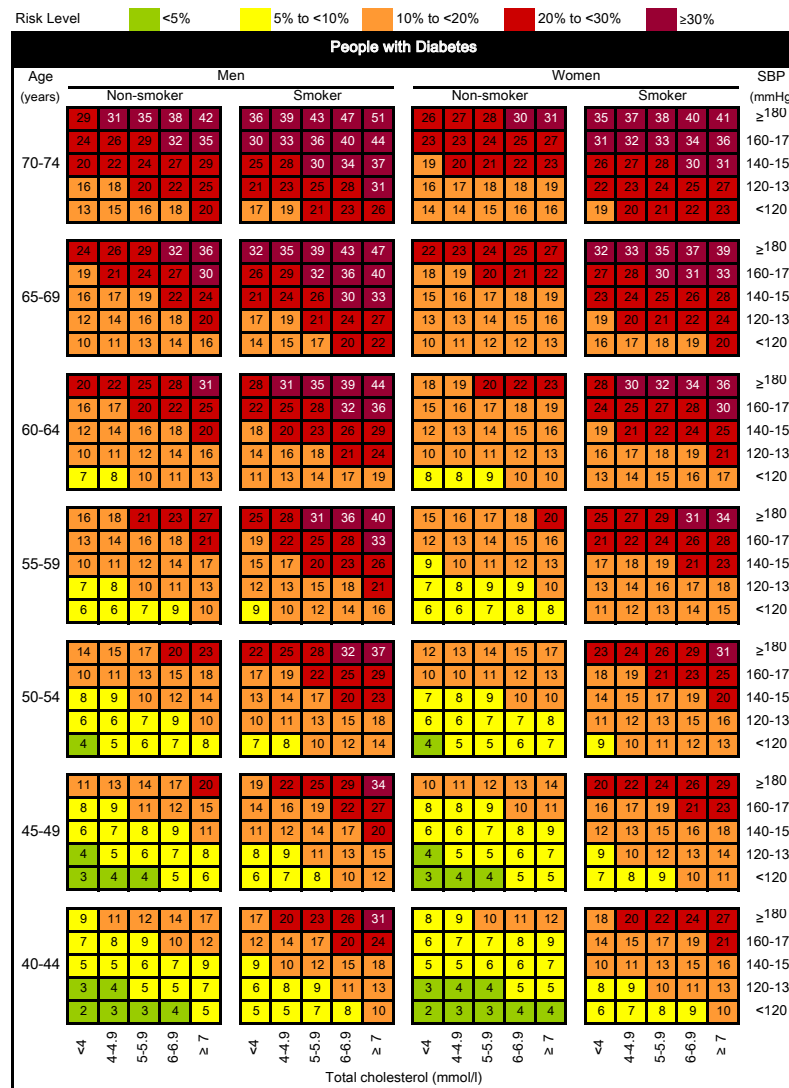
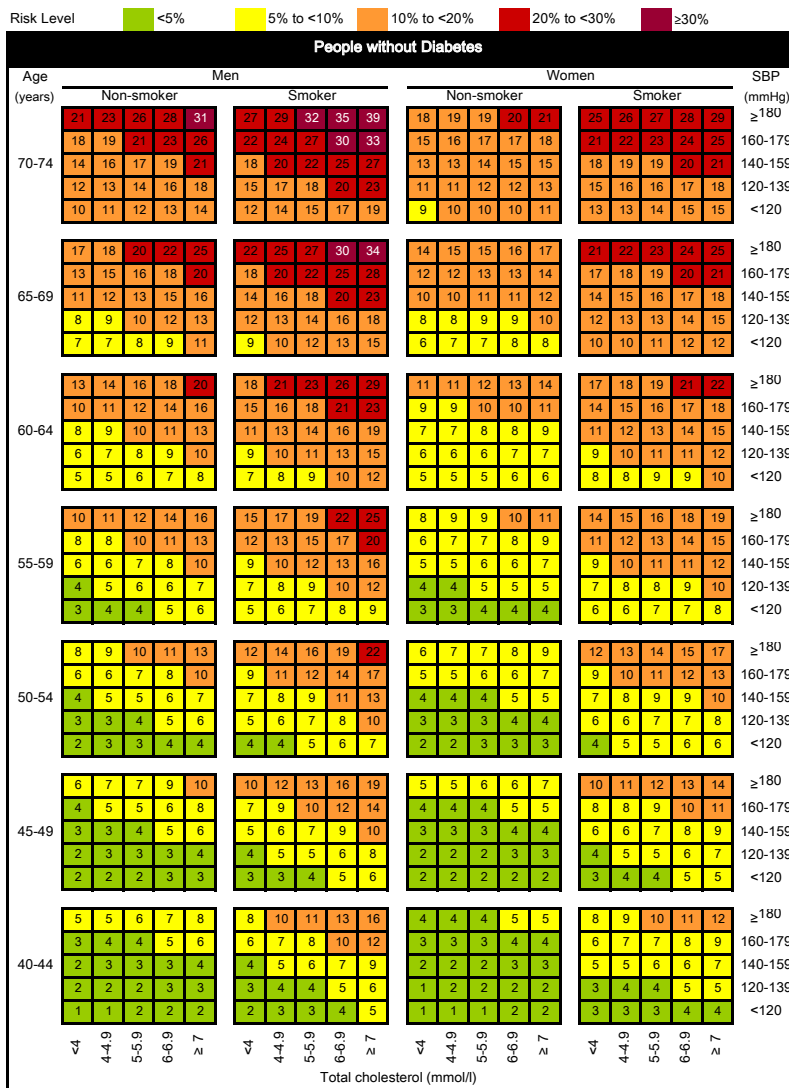
Burundi, Comoros, Djibouti, Eritrea, Ethiopia, Kenya, Madagascar, Malawi, Mozambique, Rwanda, Somalia, Uganda, United Republic of Tanzania, Zambia



WHO cardiovascular disease risk laboratory-based charts

Southern Sub-Saharan Africa

Botswana, Eswatini, Lesotho, Namibia, South Africa, Zimbabwe

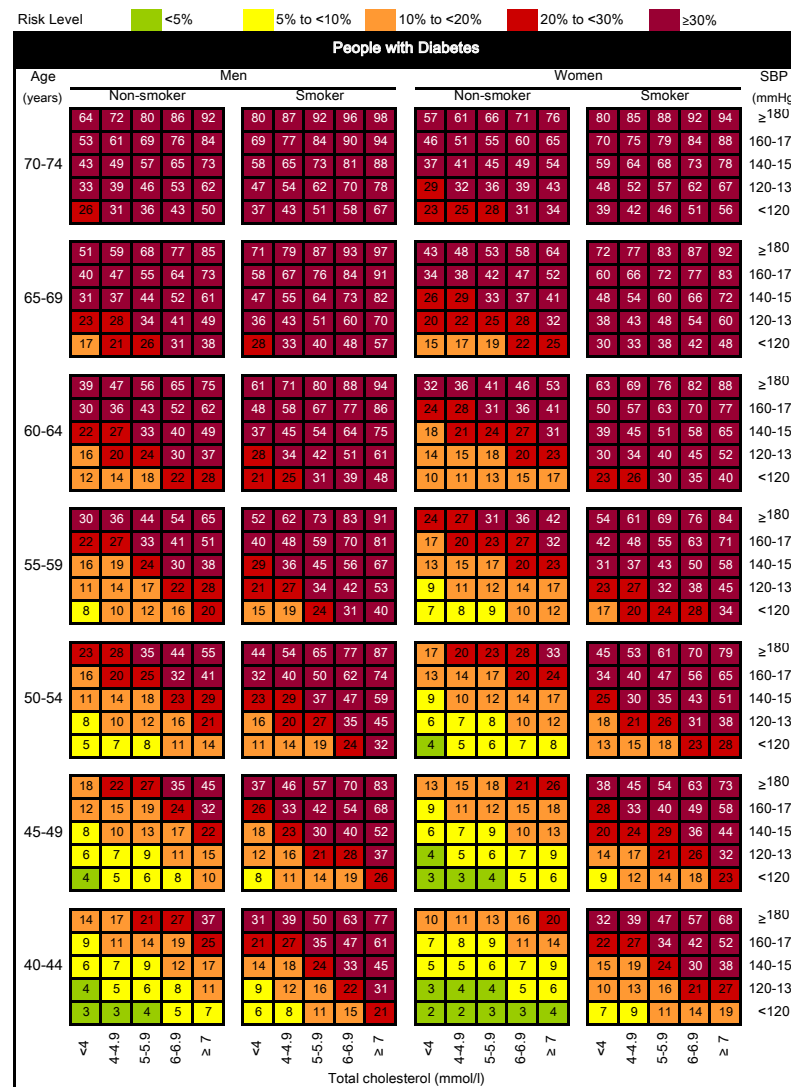
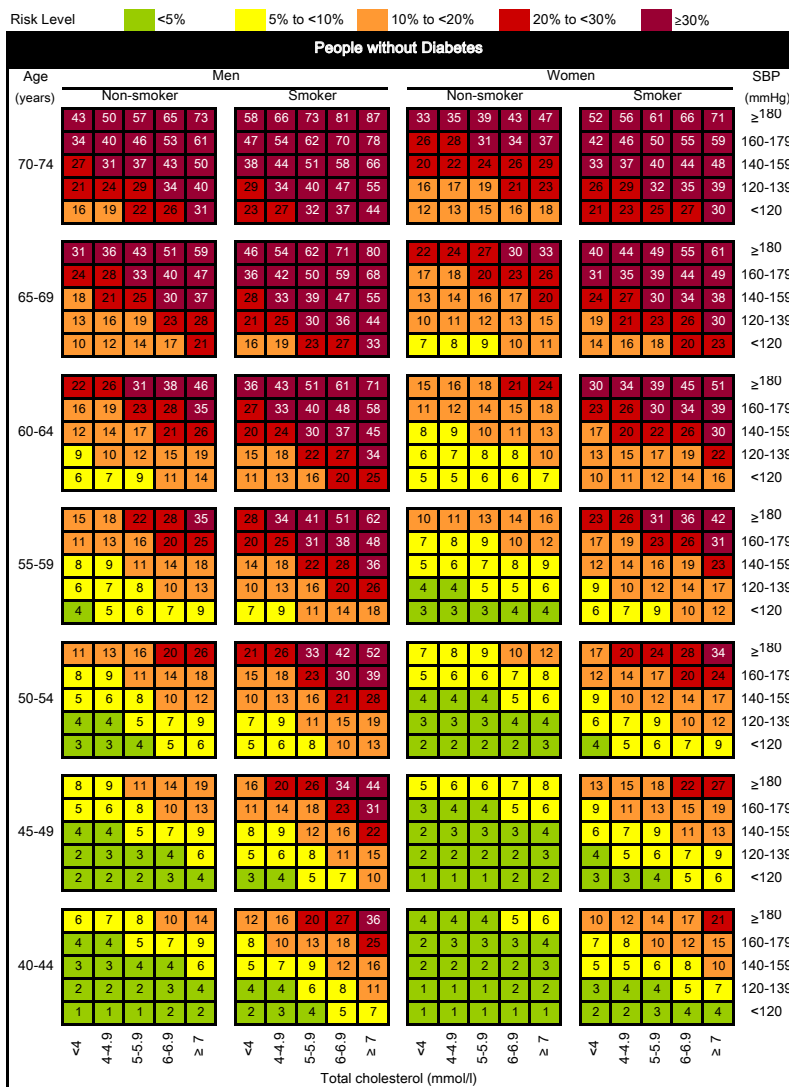


Southern Sub-Saharan Africa

WHO cardiovascular disease risk laboratory-based charts

Central Asia

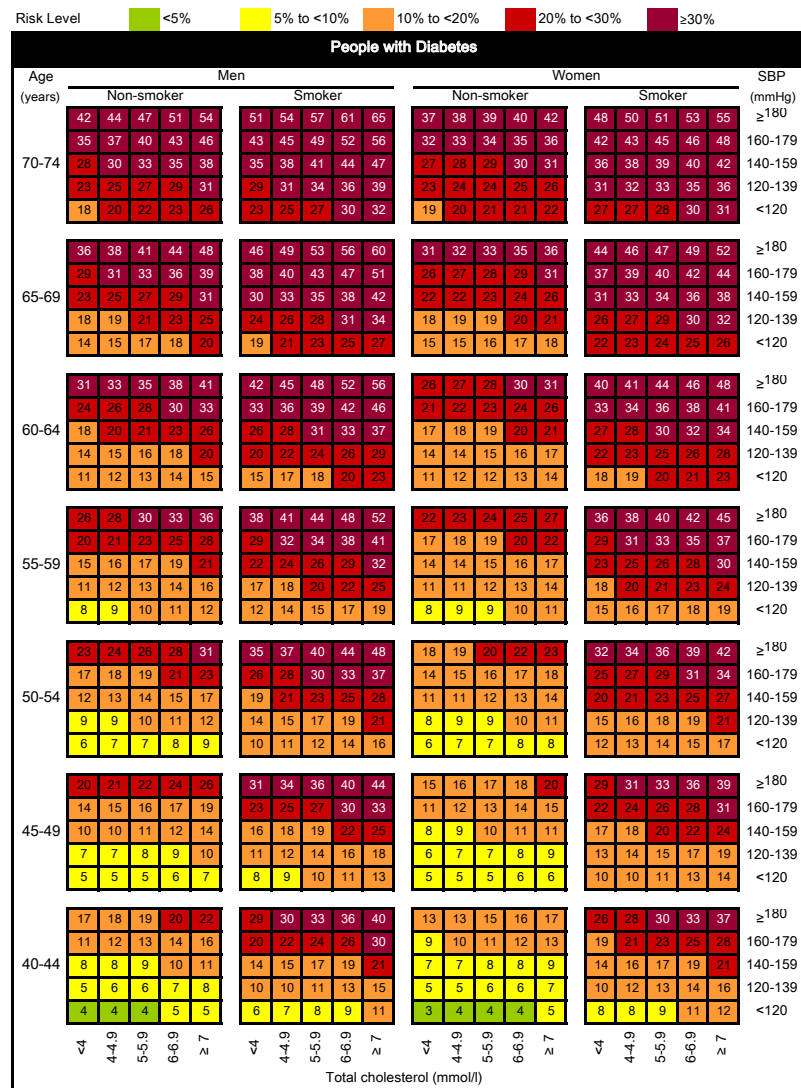
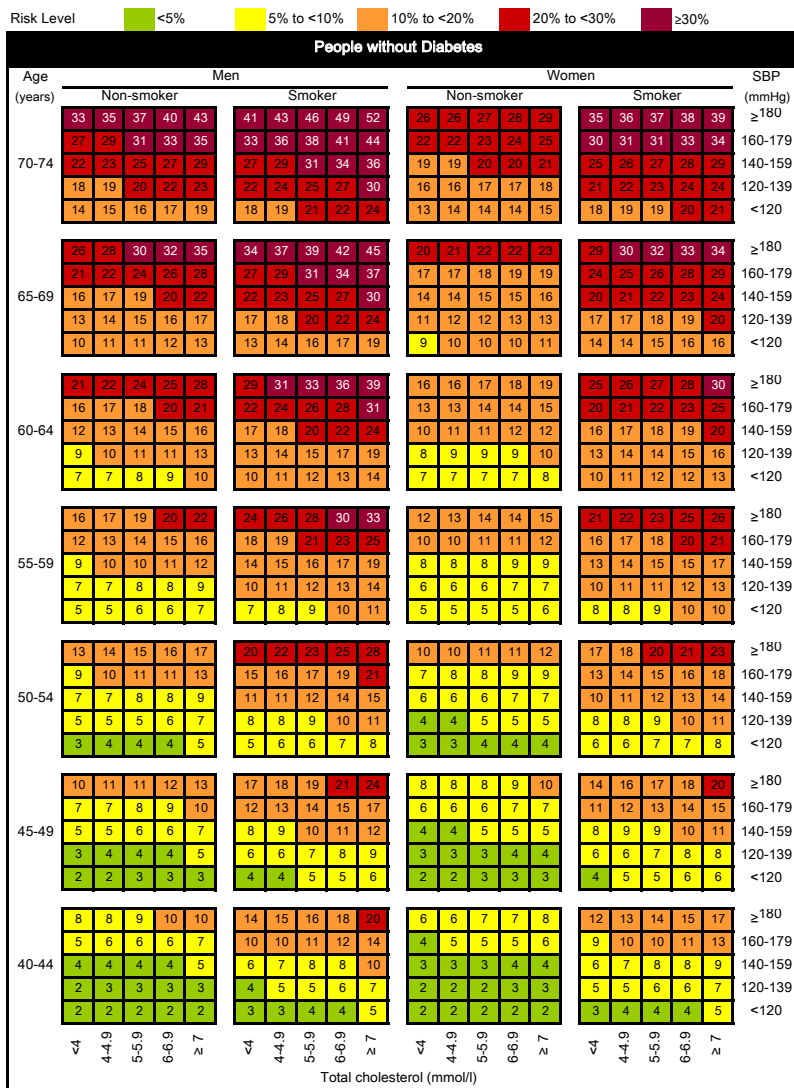
Armenia, Azerbaijan, Georgia, Kazakhstan, Kyrgyzstan, Mongolia, Tajikistan, Turkmenistan, Uzbekistan



WHO cardiovascular disease risk laboratory-based charts

East Asia

China, Democratic People's Republic of Korea

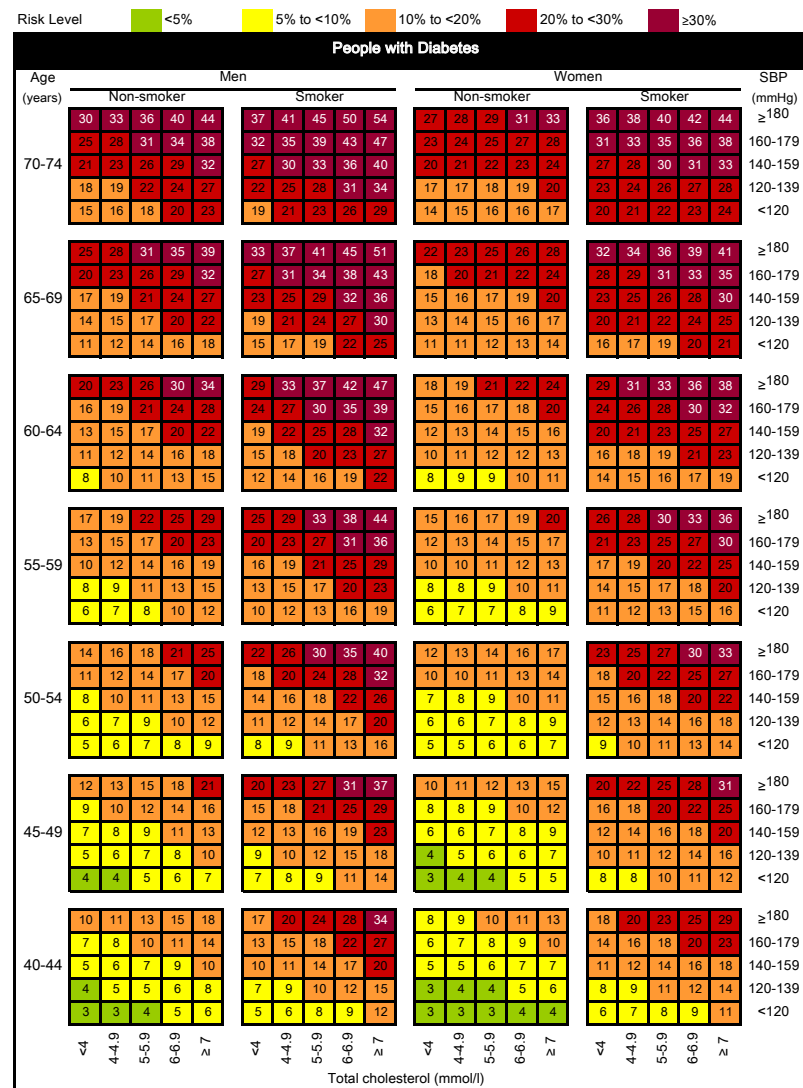
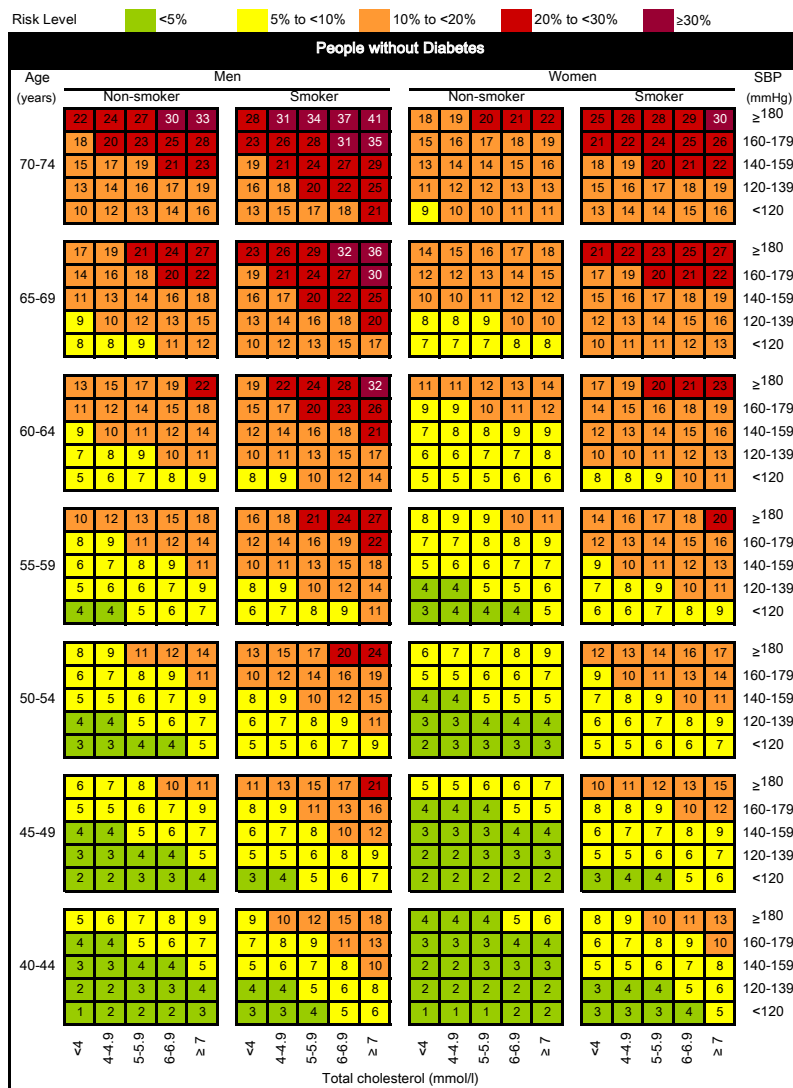


East Asia

WHO cardiovascular disease risk laboratory-based charts

South Asia

Bangladesh, Bhutan, India, Nepal, Pakistan

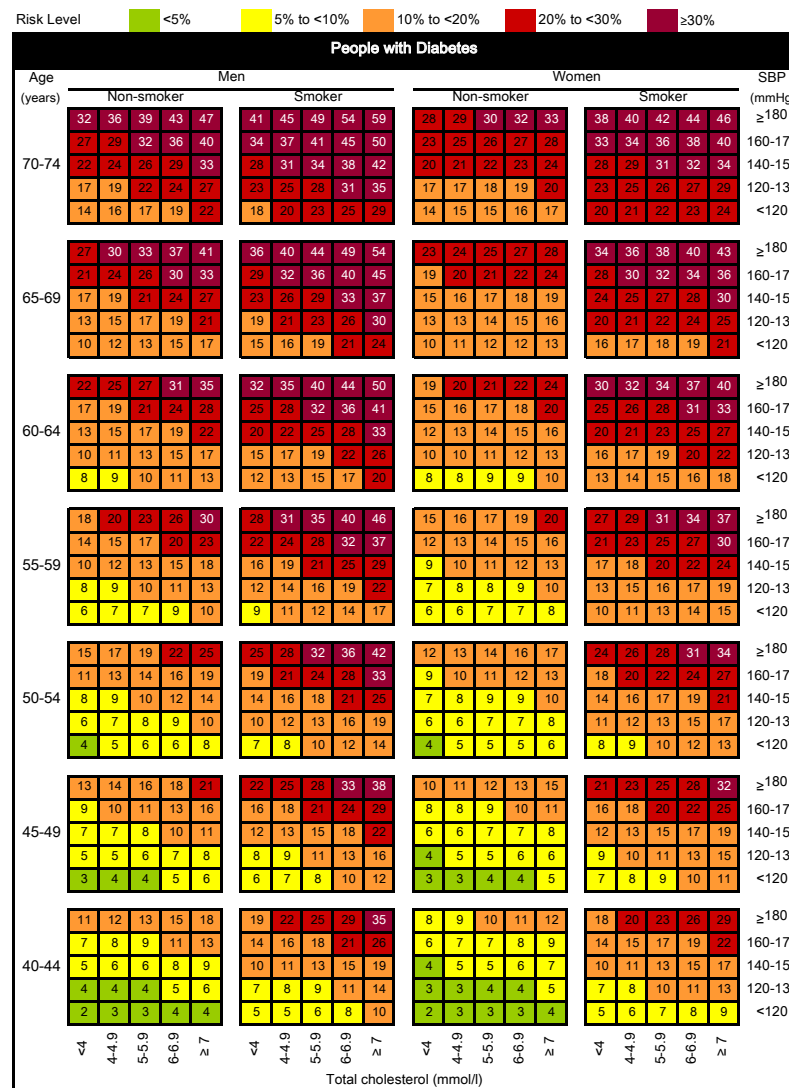
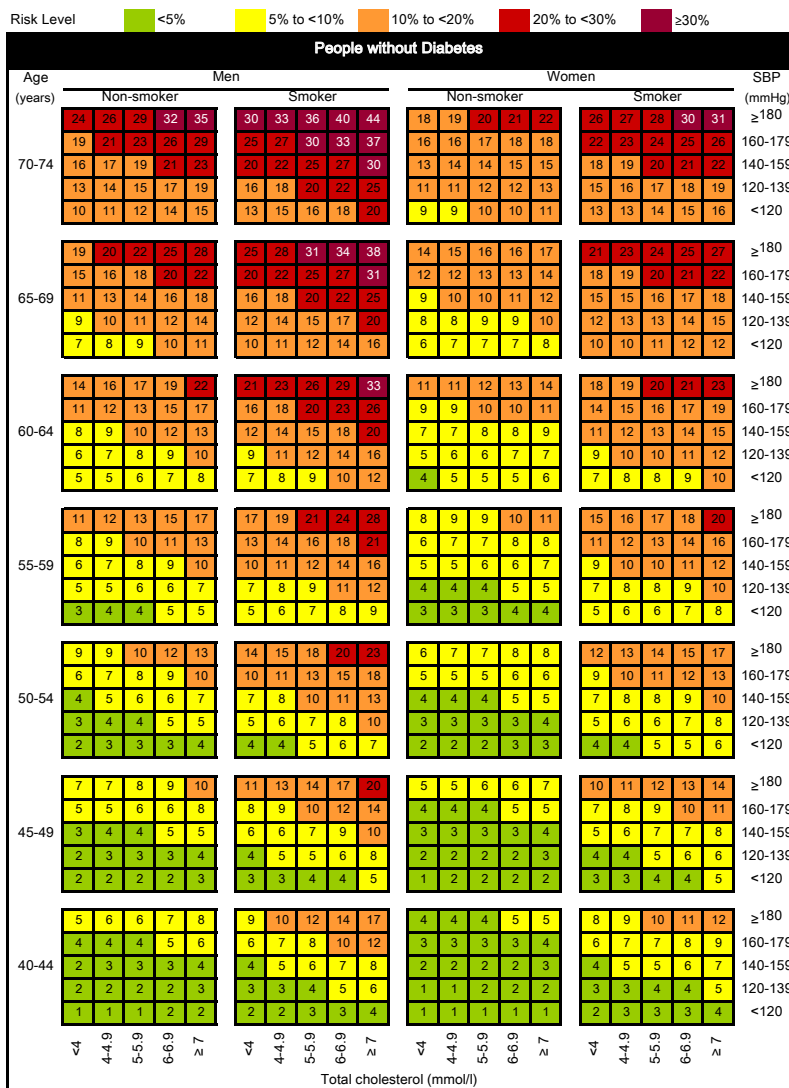


South Asia

WHO cardiovascular disease risk laboratory-based charts

South-East Asia

Cambodia, Indonesia, Lao People's Democratic Republic, Malaysia, Maldives, Mauritius, Myanmar, Philippines, Seychelles, Sri Lanka, Thailand, Timor-Leste, Viet Nam

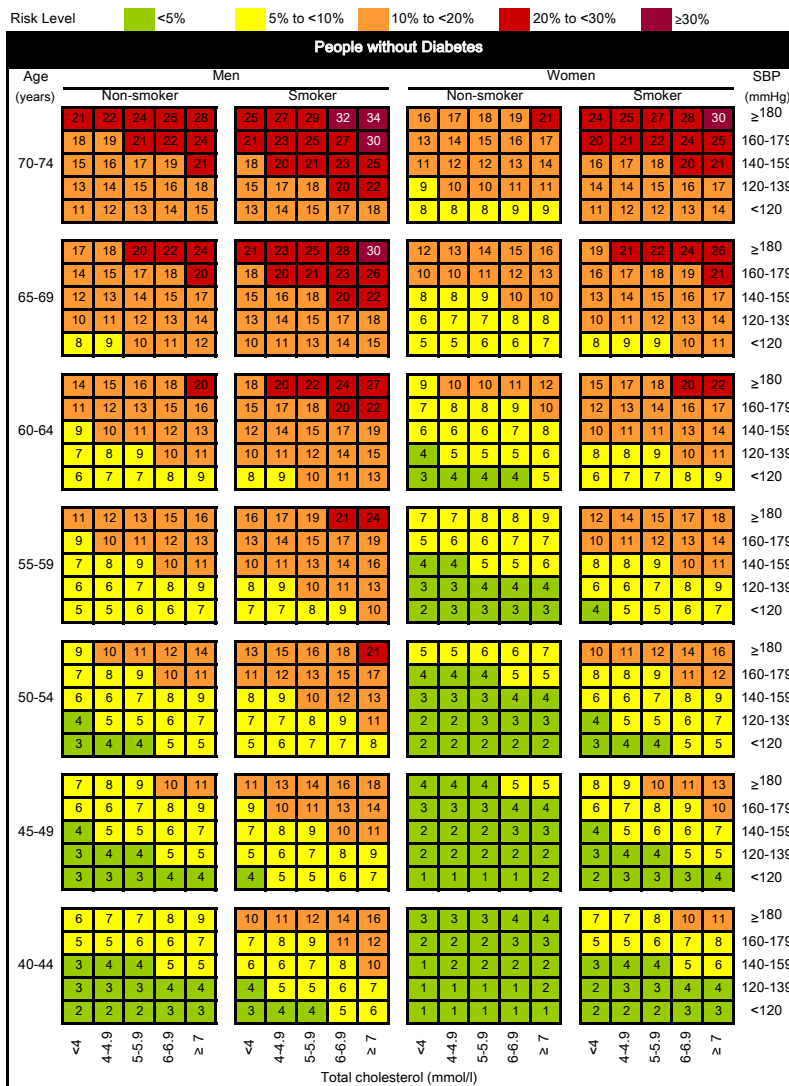


South-East Asia

WHO cardiovascular disease risk laboratory-based charts

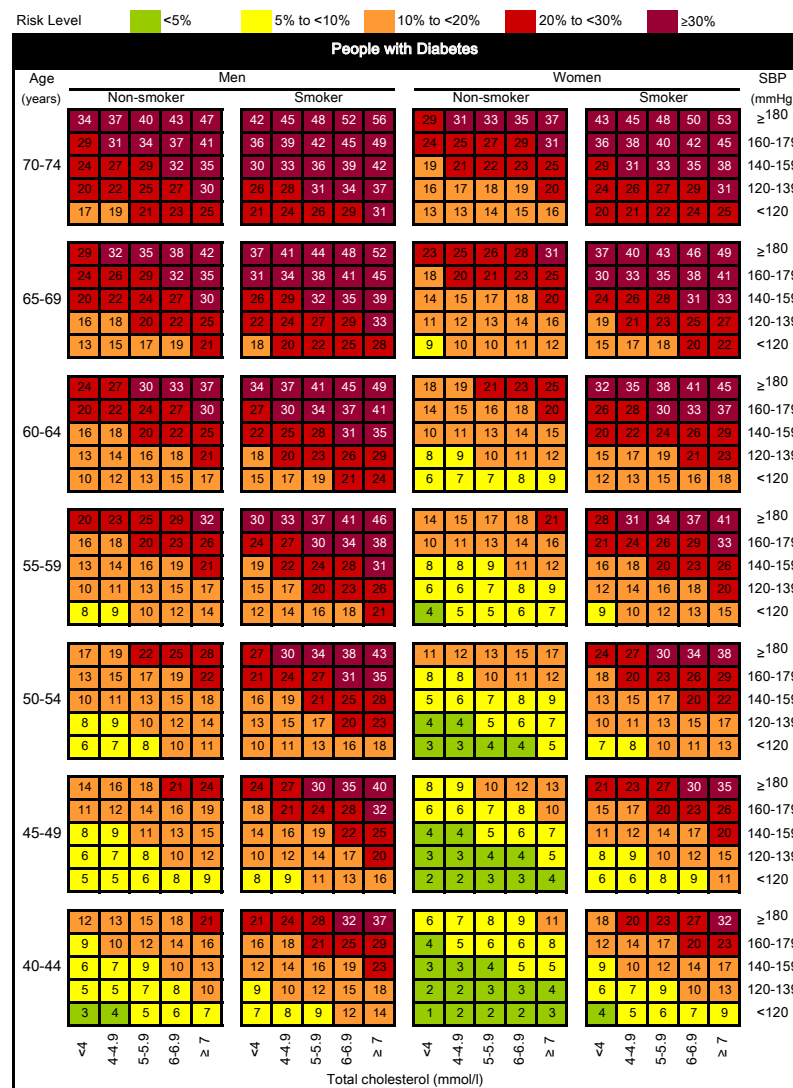
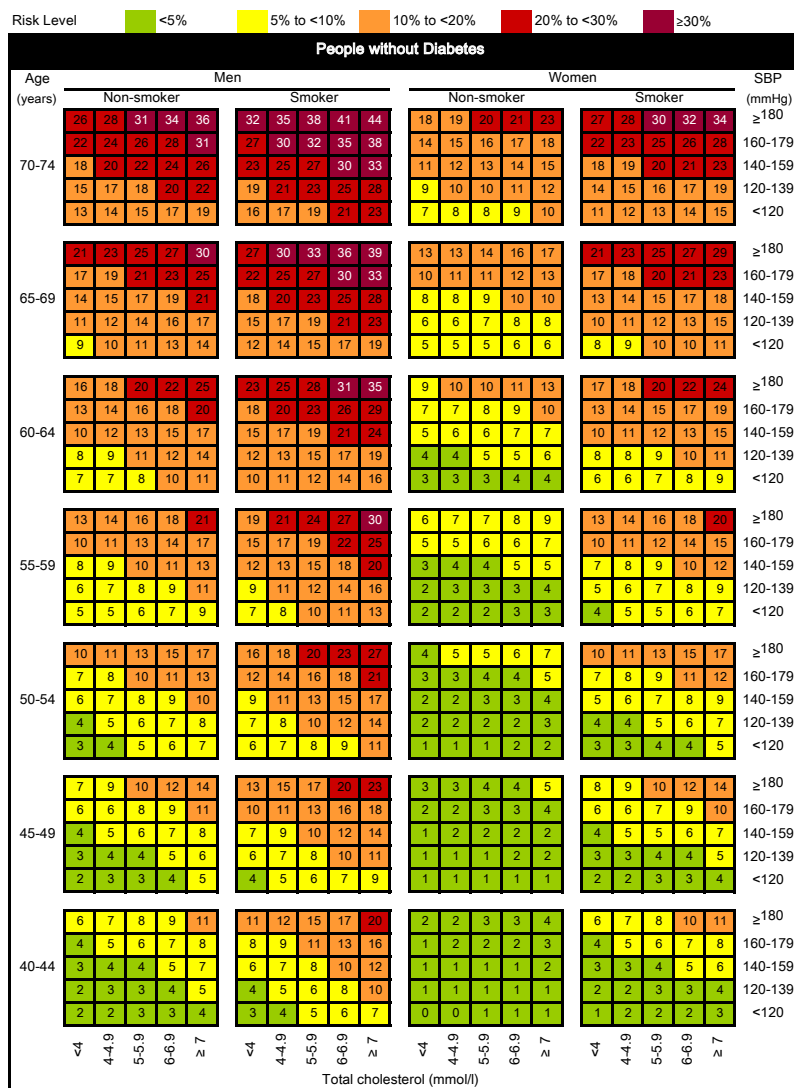
High-income Asia Pacific

Brunei Darussalam, Japan, Republic of Korea, Singapore



WHO cardiovascular disease risk laboratory-based charts

Australasia
Australia, New Zealand

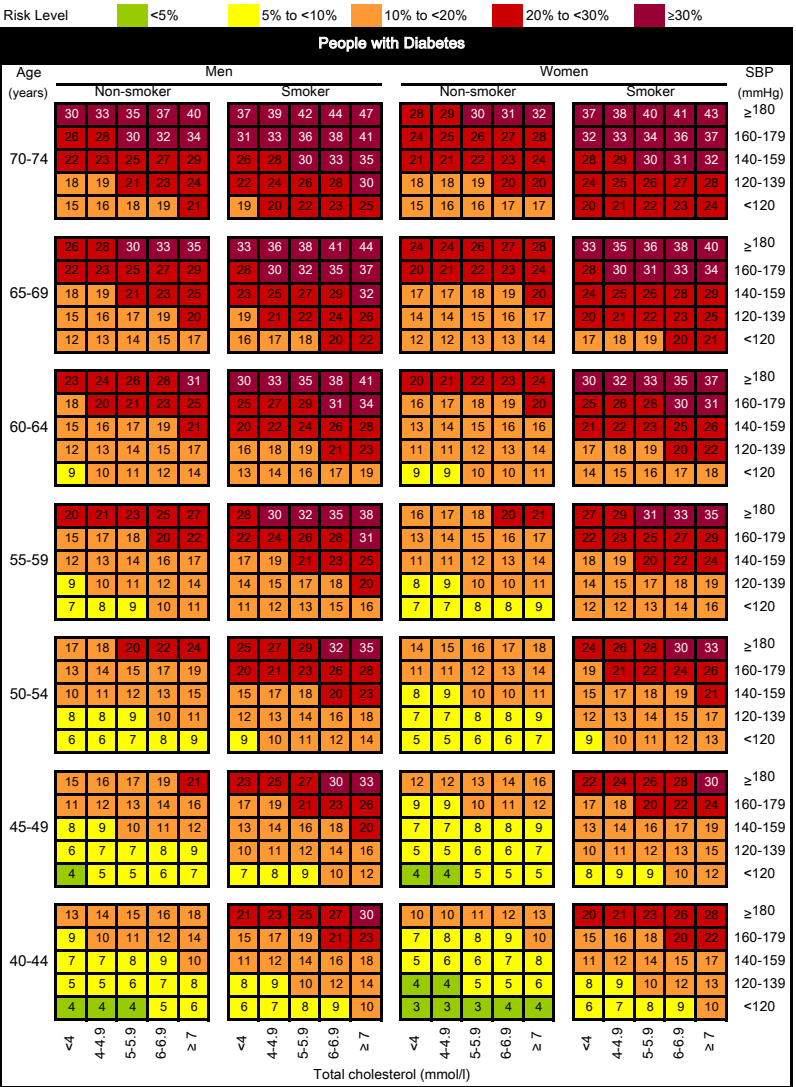
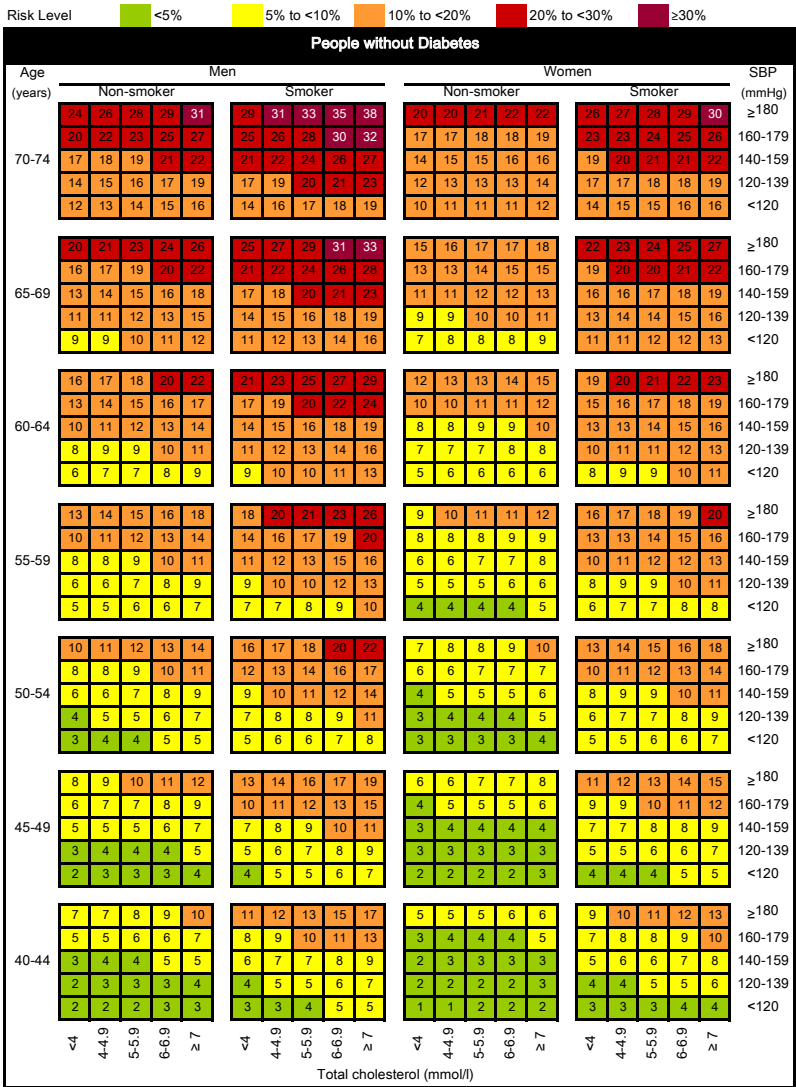


Australasia

WHO cardiovascular disease risk laboratory-based charts

Oceania

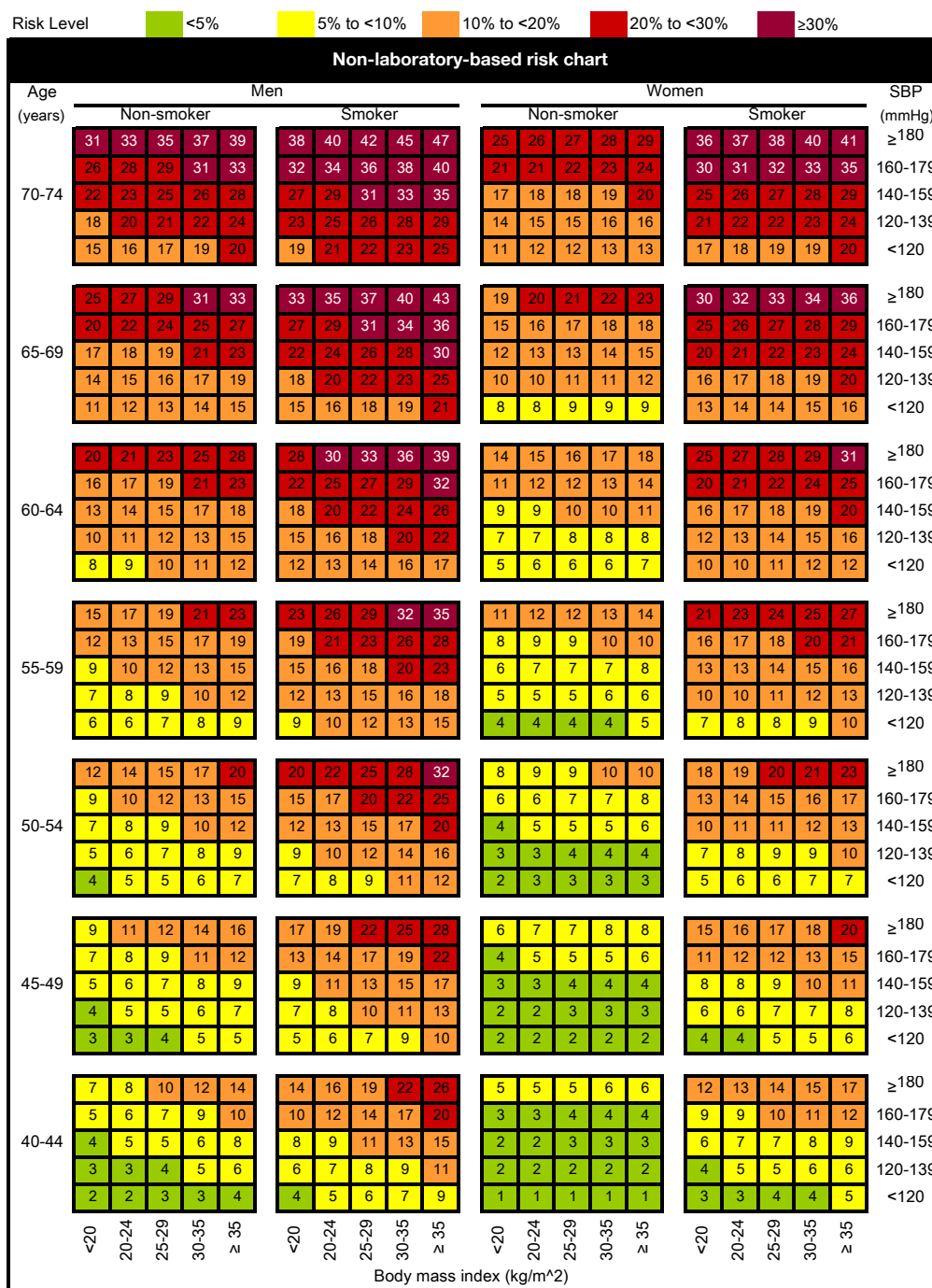
Fiji, Kiribati, Marshall Islands, Micronesia (Federated States of), Papua New Guinea, Samoa, Solomon Islands, Tonga, Vanuatu



Annex 3: WHO CVD risk (non-laboratory-based) charts

WHO cardiovascular disease risk non-laboratory-based charts

High-income North America
Canada, Greenland, United States of America

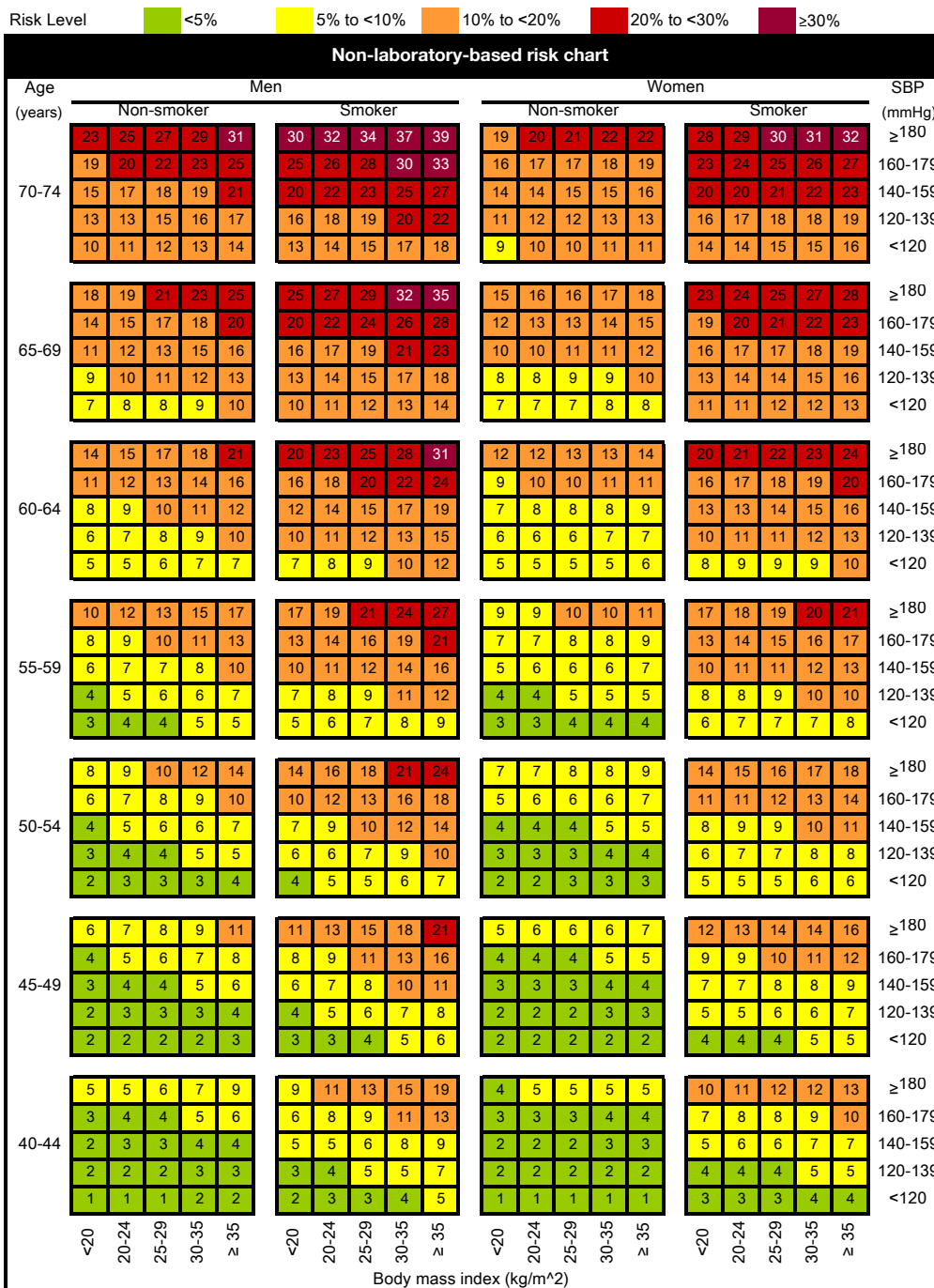


High-income North America

WHO cardiovascular disease risk non-laboratory-based charts

Caribbean

Antigua and Barbuda, Bahamas, Barbados, Belize, Bermuda, Cuba, Dominica, Dominican Republic, Grenada, Guyana, Haiti, Jamaica, Puerto Rico, Saint Lucia, Saint Vincent and the Grenadines, Suriname, Trinidad and Tobago

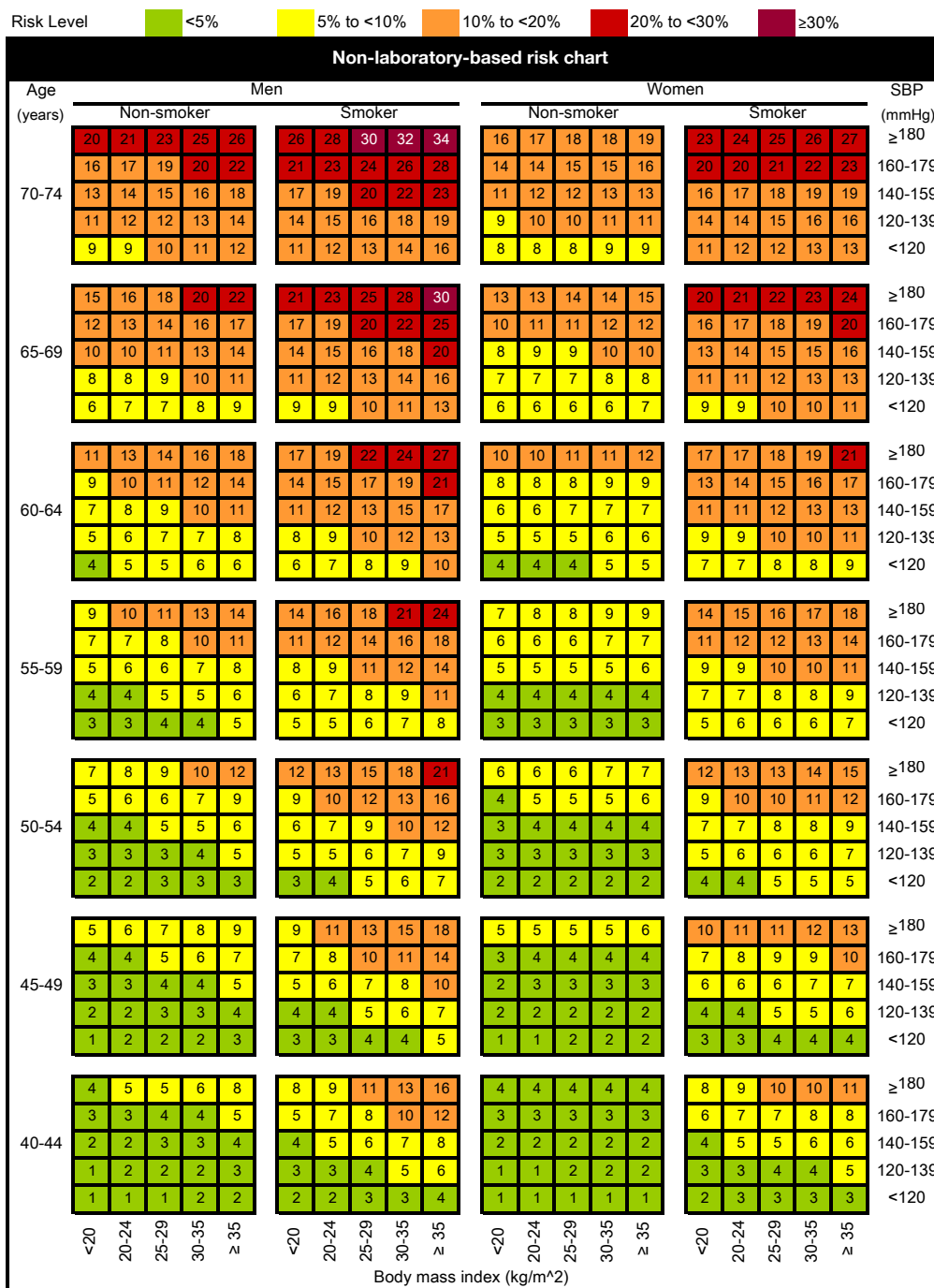


Caribbean

WHO cardiovascular disease risk non-laboratory-based charts

Central Latin America

Colombia, Costa Rica, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, Panama, Venezuela (Bolivarian Republic of)

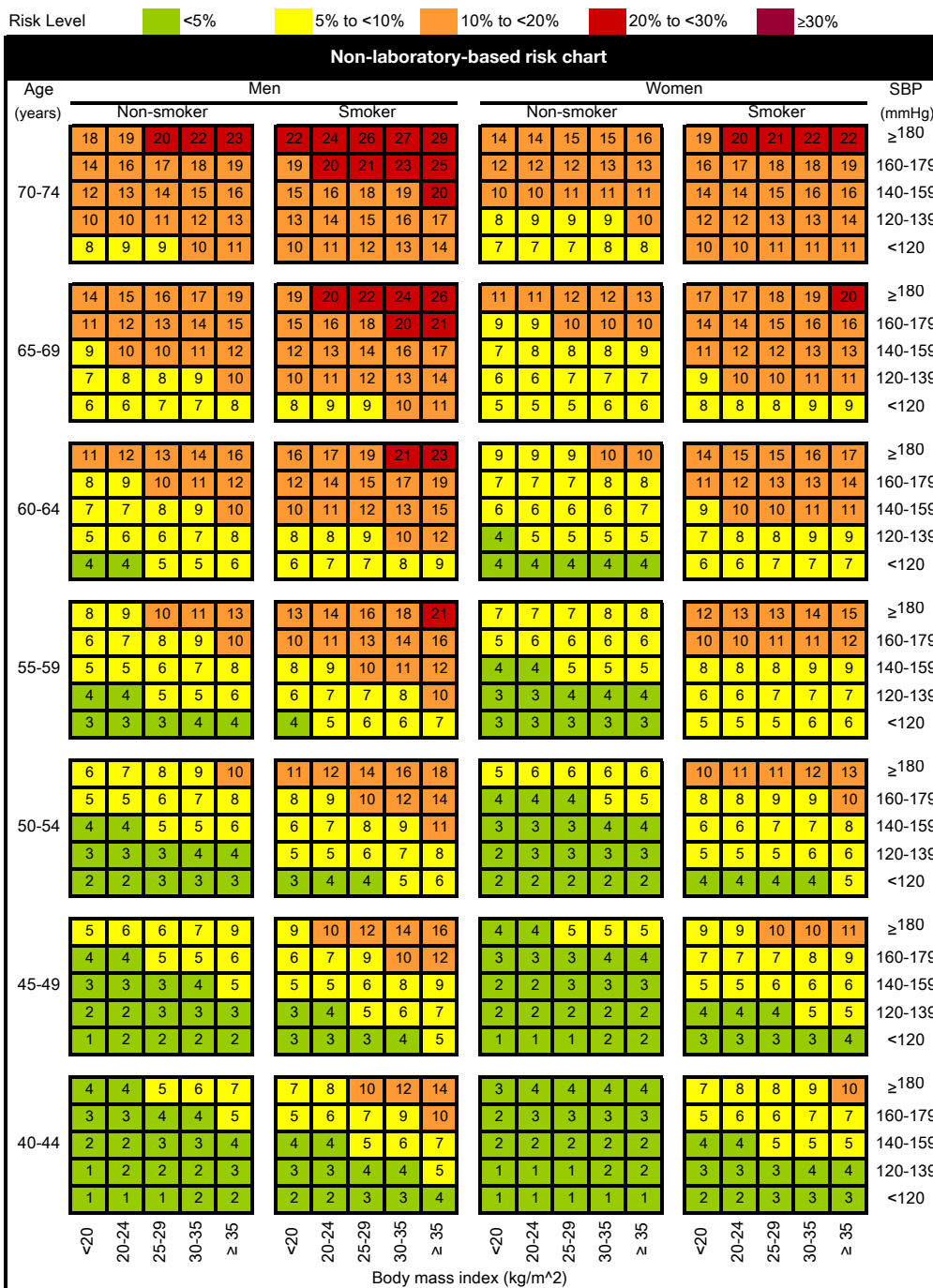


Central Latin America

WHO cardiovascular disease risk non-laboratory-based charts

Andean Latin America

Bolivia, Ecuador, Peru

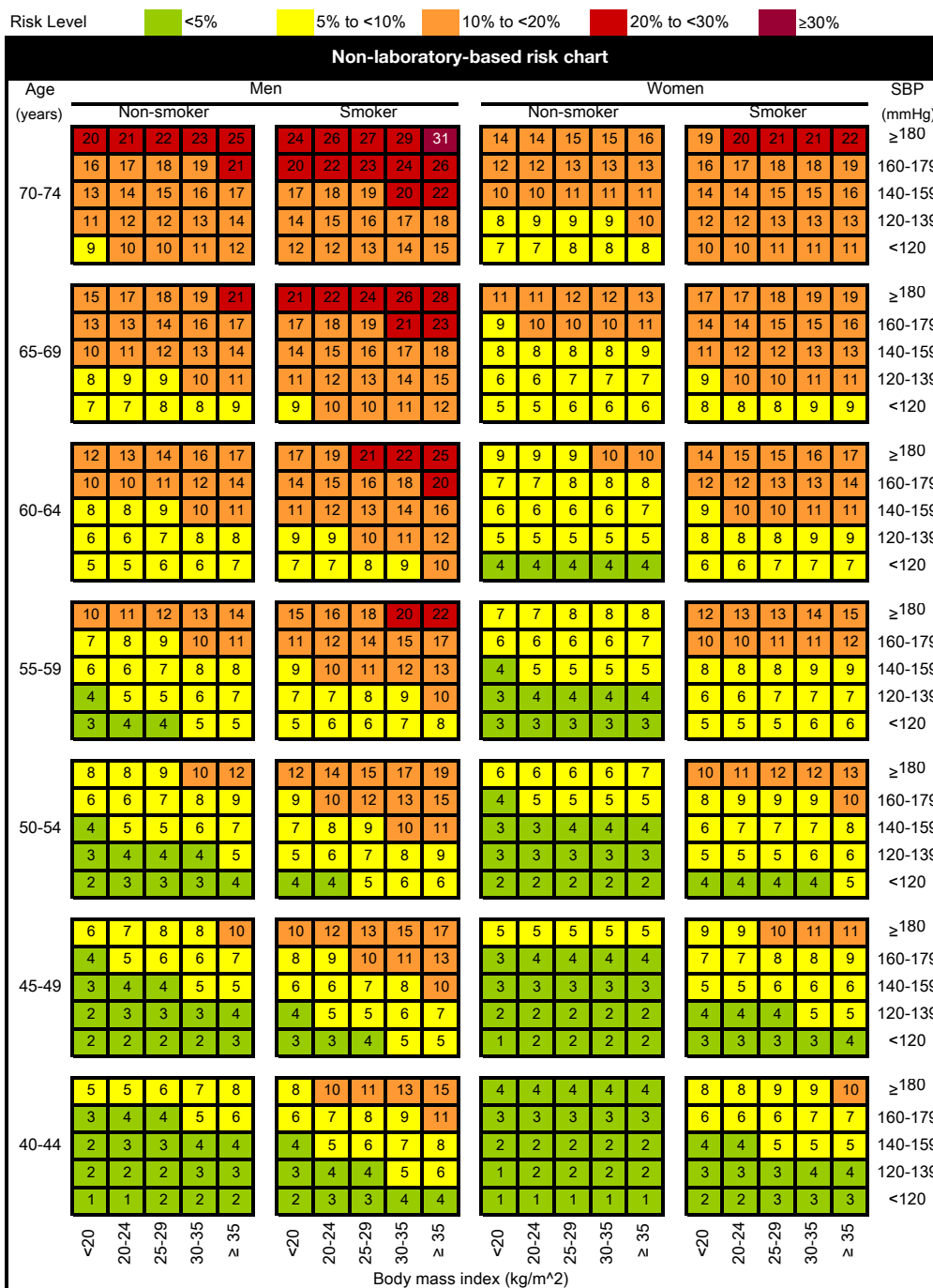


Andean Latin America

WHO cardiovascular disease risk non-laboratory-based charts

Tropical Latin America

Brazil, Paraguay

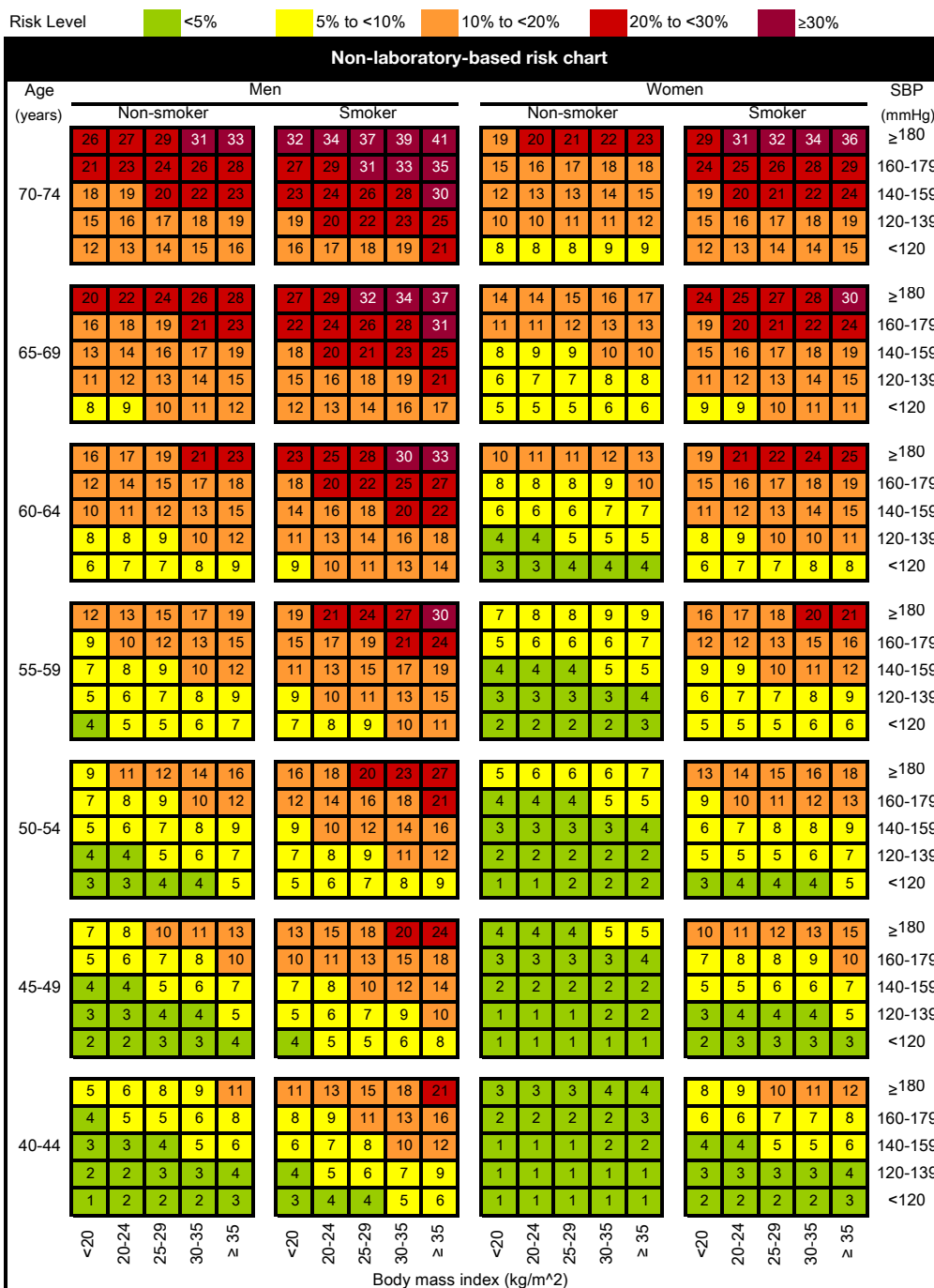


Tropical Latin America

WHO cardiovascular disease risk non-laboratory-based charts

Southern Latin America

Argentina, Chile, Uruguay

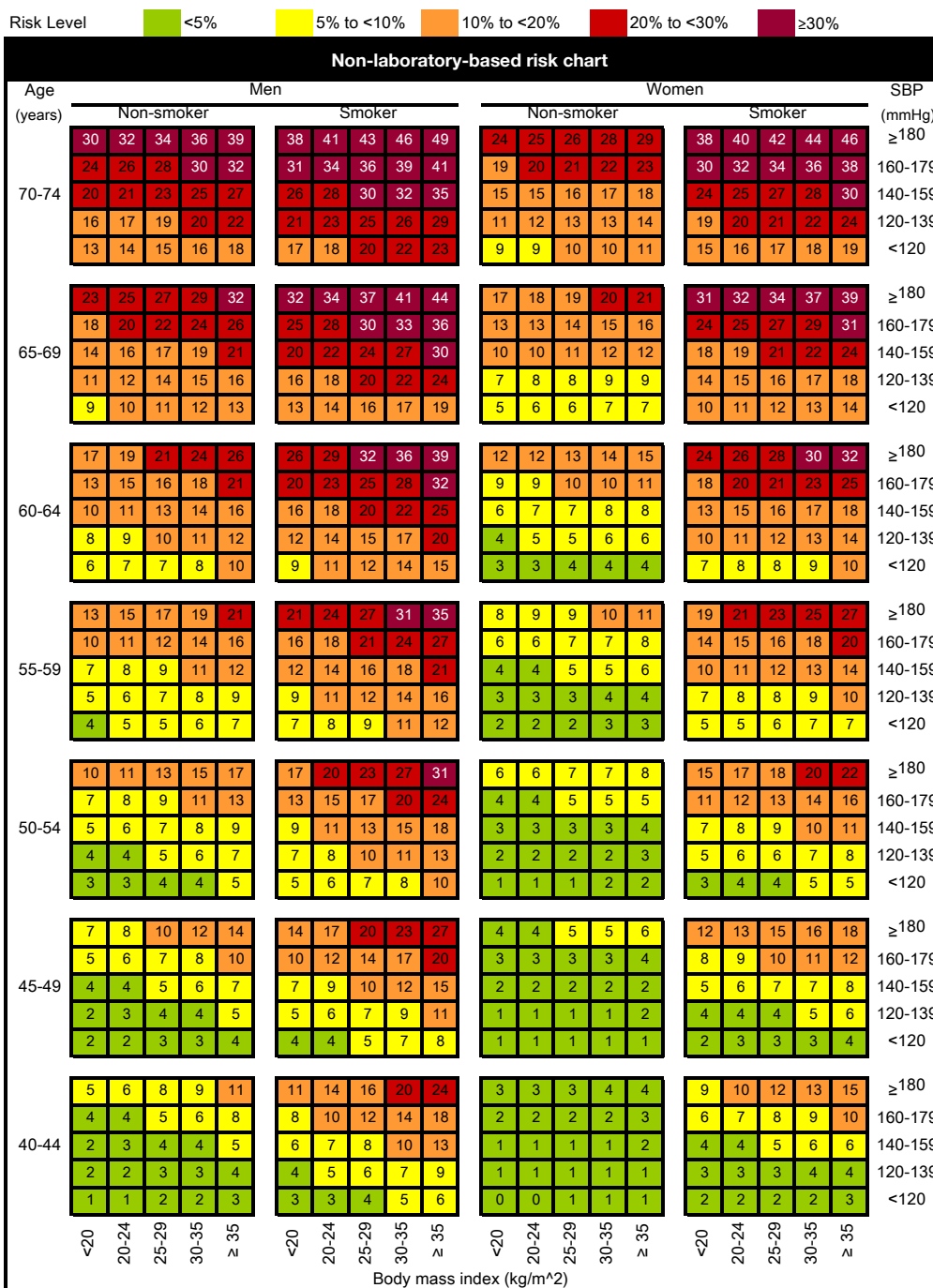


Southern Latin America

WHO cardiovascular disease risk non-laboratory-based charts

Western Europe

Andorra, Austria, Belgium, Cyprus, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Israel, Italy, Luxembourg, Malta, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom

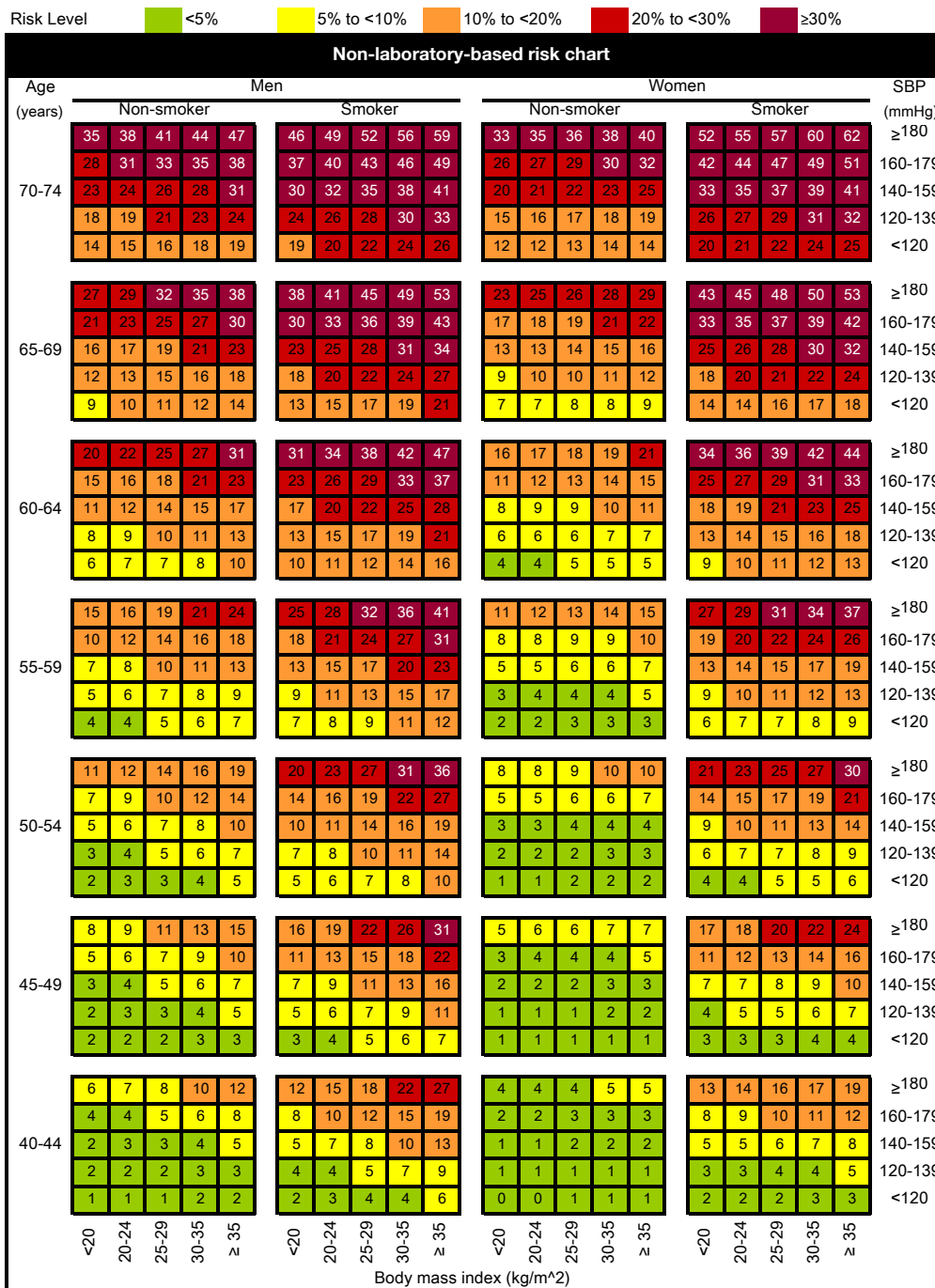


Western Europe

WHO cardiovascular disease risk non-laboratory-based charts

Central Europe

Albania, Bosnia and Herzegovina, Bulgaria, Croatia, Czech Republic, Hungary, Montenegro, North Macedonia, Poland, Romania, Serbia, Slovakia, Slovenia

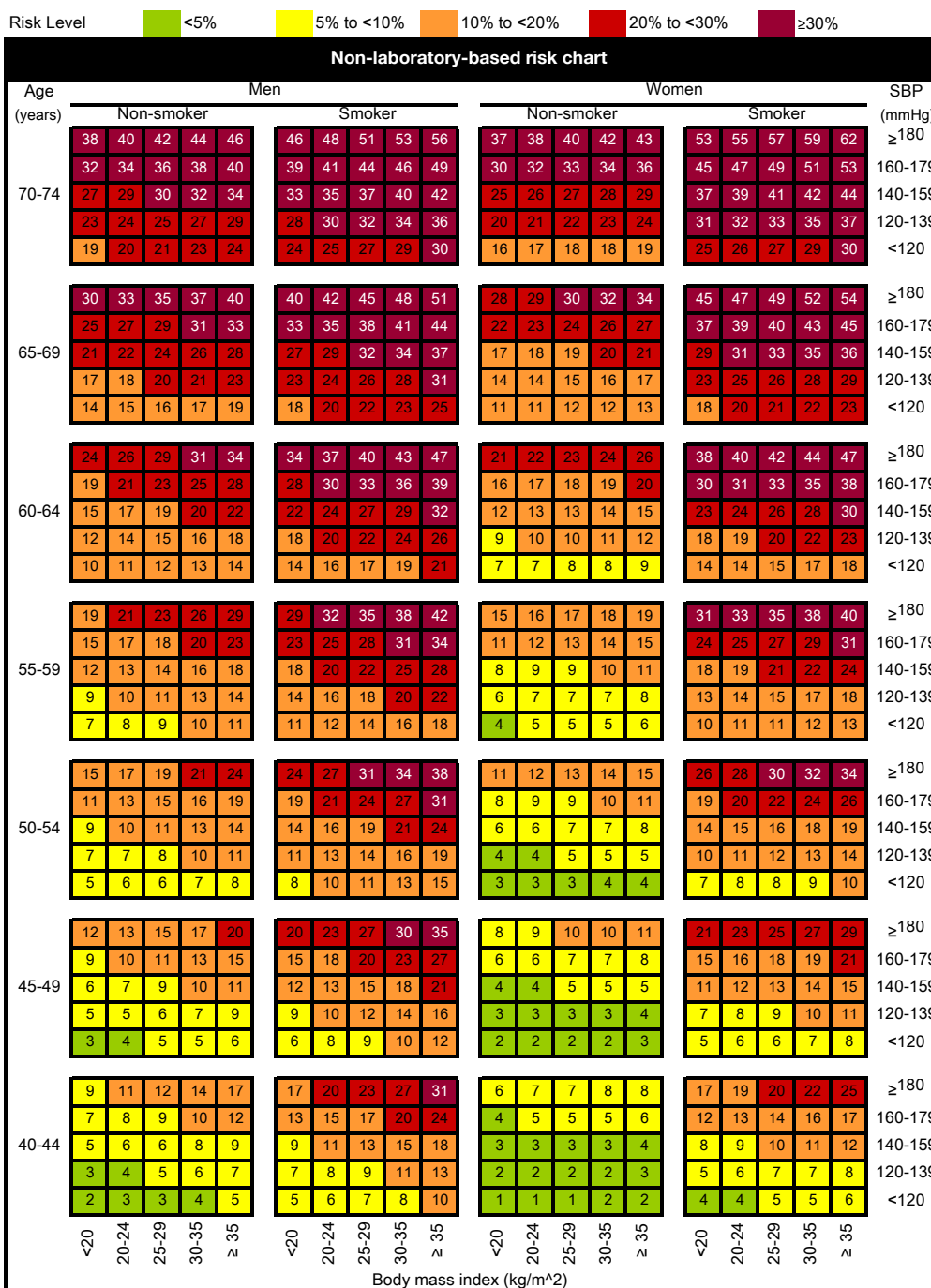


Central Europe

WHO cardiovascular disease risk non-laboratory-based charts

Eastern Europe

Belarus, Estonia, Latvia, Lithuania, Republic of Moldova, Russian Federation, Ukraine

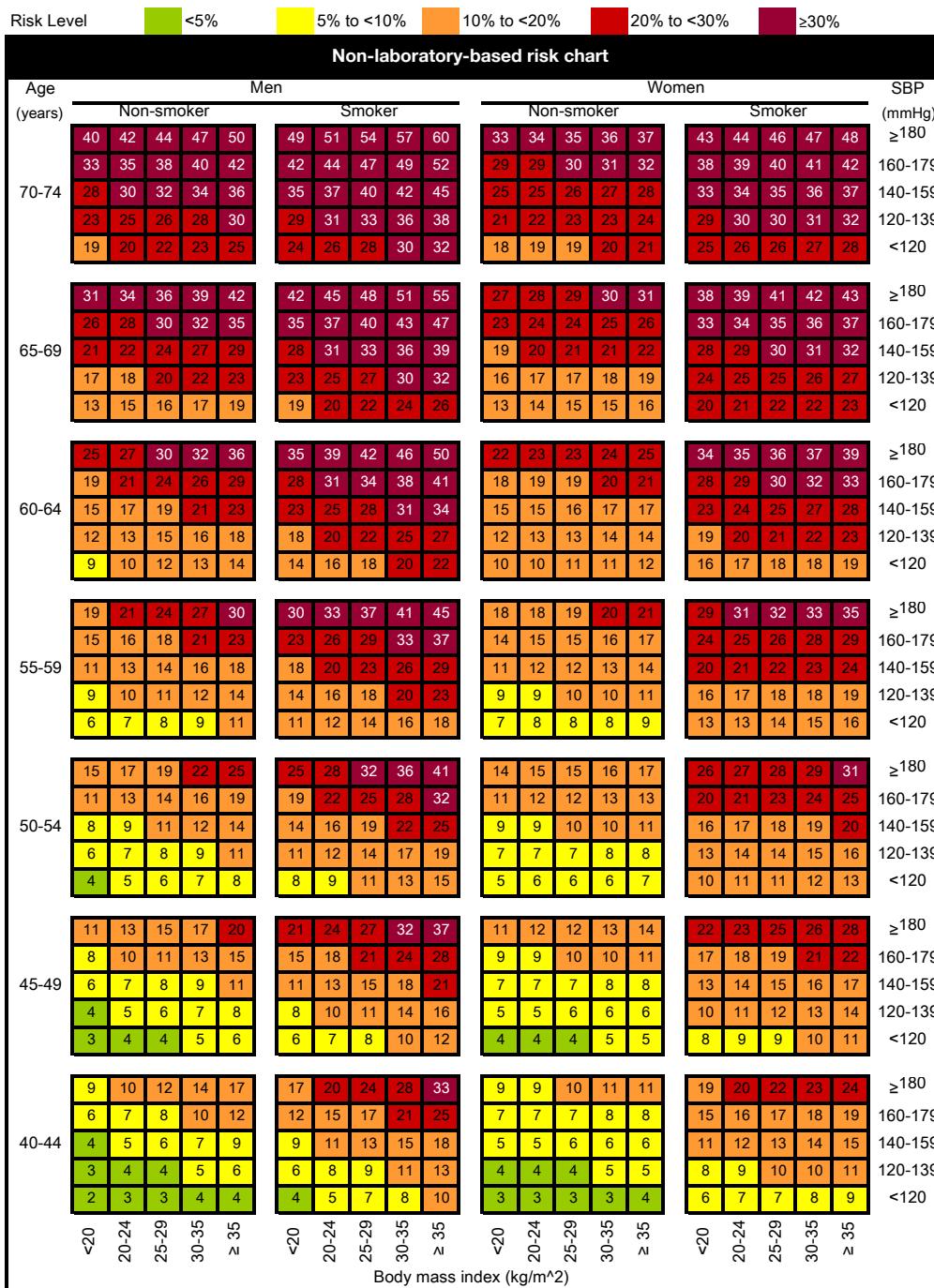


Eastern Europe

WHO cardiovascular disease risk non-laboratory-based charts

North Africa and Middle East

Afghanistan, Algeria, Bahrain, Egypt, Iran (Islamic Republic of), Iraq, Jordan, Kuwait, Lebanon, Libya, Morocco, occupied Palestinian territory, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, Tunisia, Turkey, United Arab Emirates, Yemen

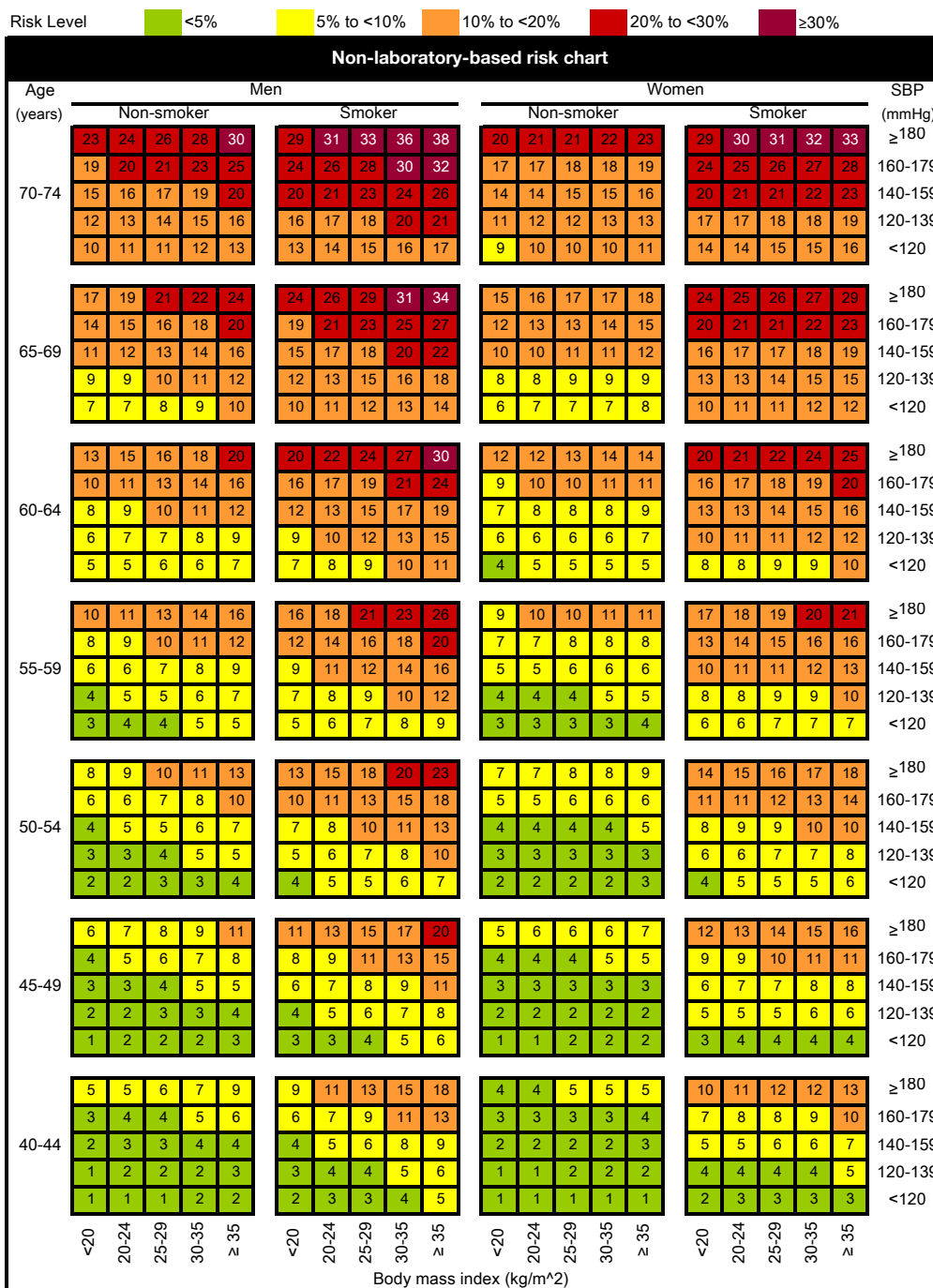


North Africa and Middle East

WHO cardiovascular disease risk non-laboratory-based charts

Western Sub-Saharan Africa

Benin, Burkina Faso, Cabo Verde, Cameroon, Chad, Cote d'Ivoire, Gambia, Ghana, Guinea, Guinea-Bissau, Liberia, Mali, Mauritania, Niger, Nigeria, Sao Tome and Principe, Senegal, Sierra Leone, Togo

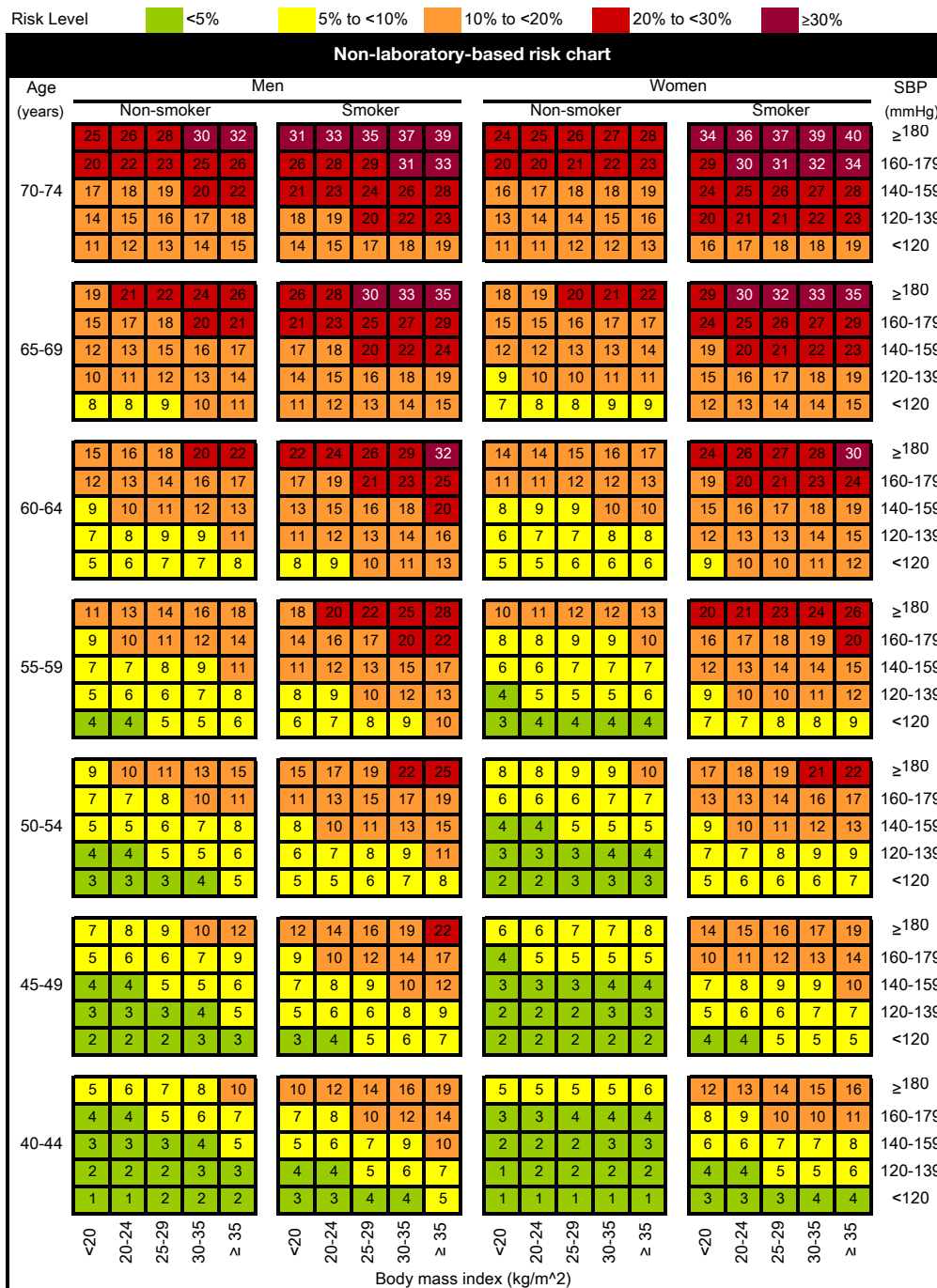


Western Sub-Saharan Africa

WHO cardiovascular disease risk non-laboratory-based charts

Central Sub-Saharan Africa

Angola, Central African Republic, Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon

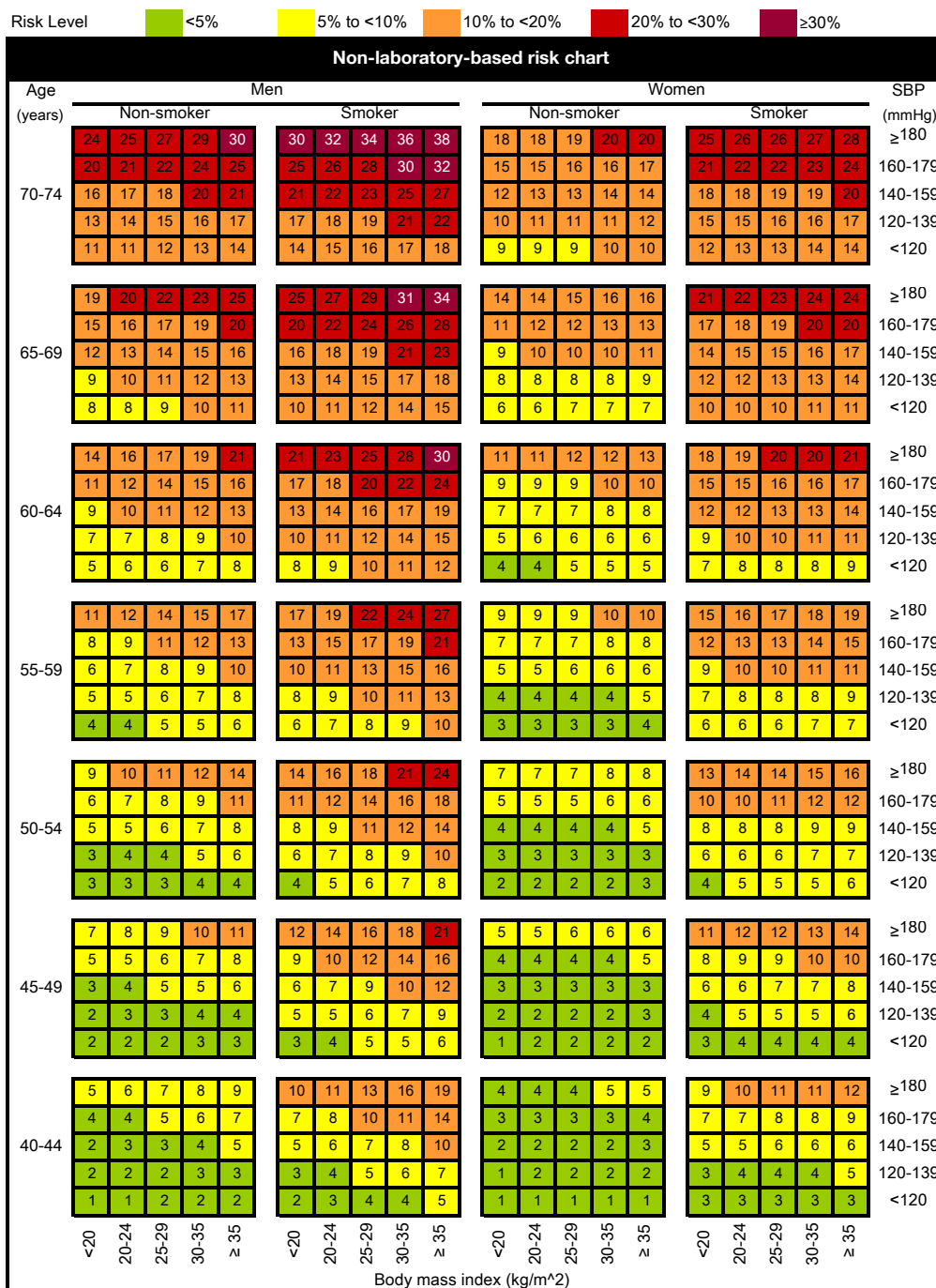


Central Sub-Saharan Africa

WHO cardiovascular disease risk non-laboratory-based charts

Eastern Sub-Saharan Africa

Burundi, Comoros, Djibouti, Eritrea, Ethiopia, Kenya, Madagascar, Malawi, Mozambique, Rwanda, Somalia, Uganda, United Republic of Tanzania, Zambia

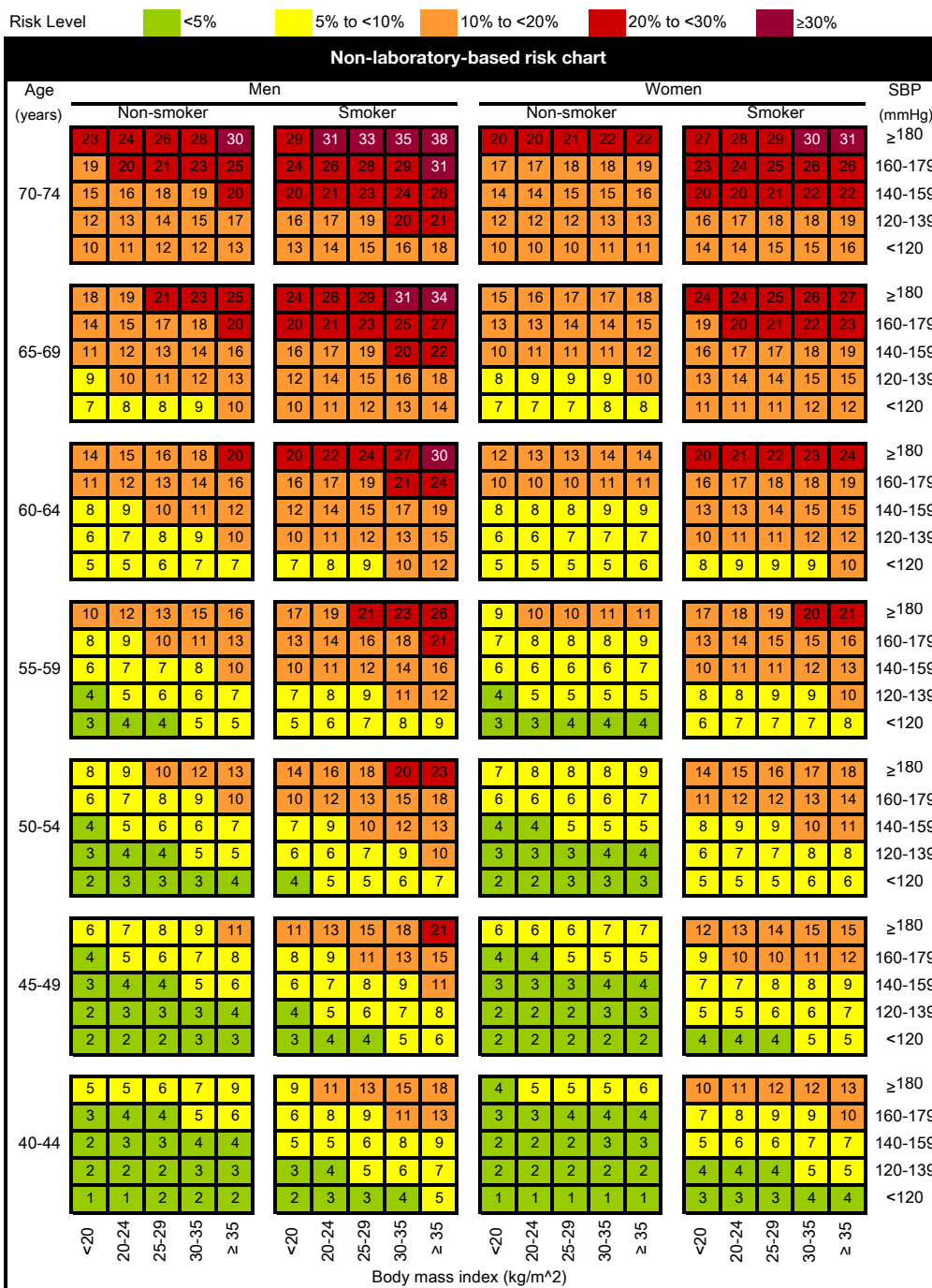


Eastern Sub-Saharan Africa

WHO cardiovascular disease risk non-laboratory-based charts

Southern Sub-Saharan Africa

Botswana, Eswatini, Lesotho, Namibia, South Africa, Zimbabwe

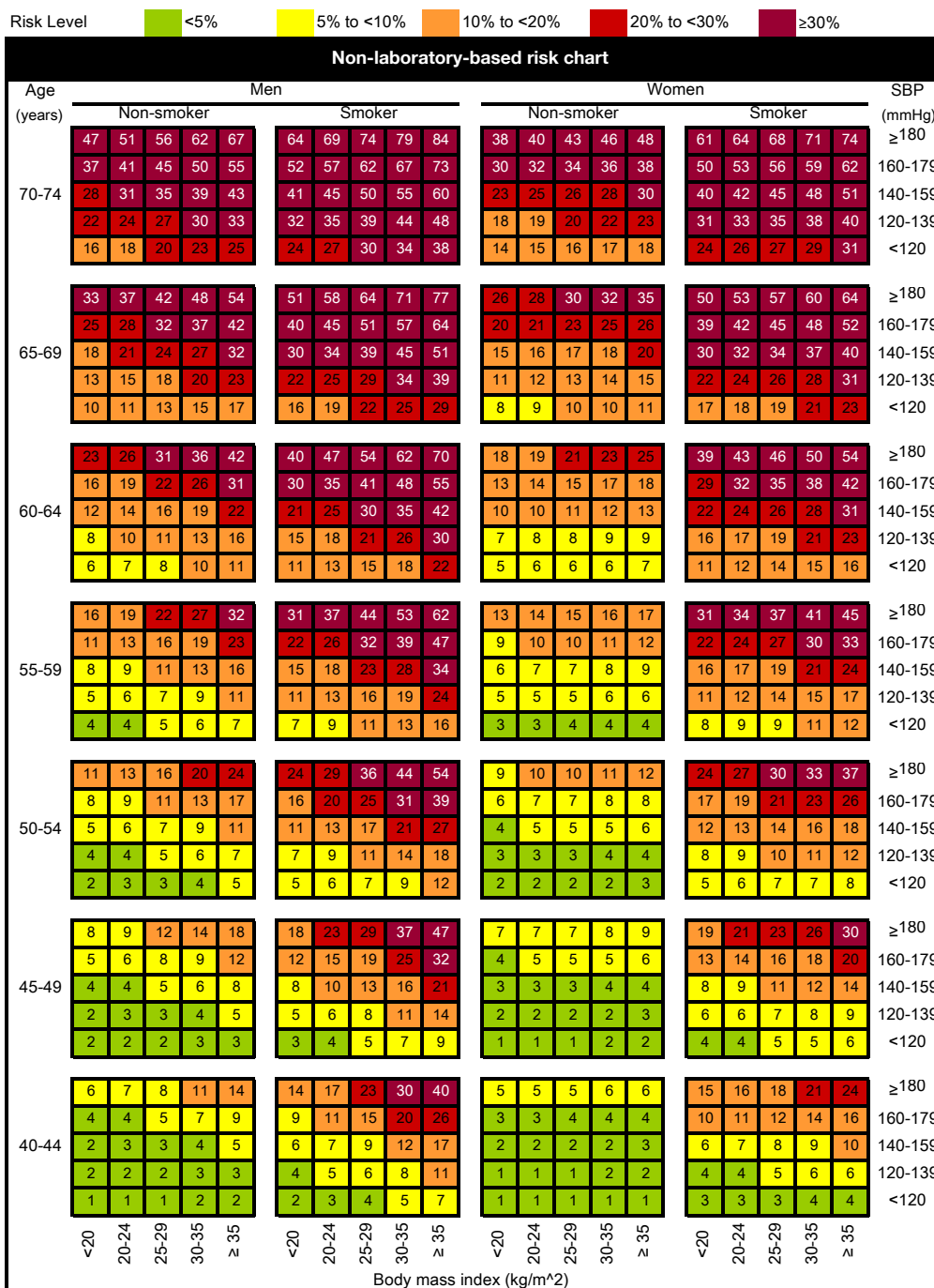


Southern Sub-Saharan Africa

WHO cardiovascular disease risk non-laboratory-based charts

Central Asia

Armenia, Azerbaijan, Georgia, Kazakhstan, Kyrgyzstan, Mongolia, Tajikistan, Turkmenistan, Uzbekistan

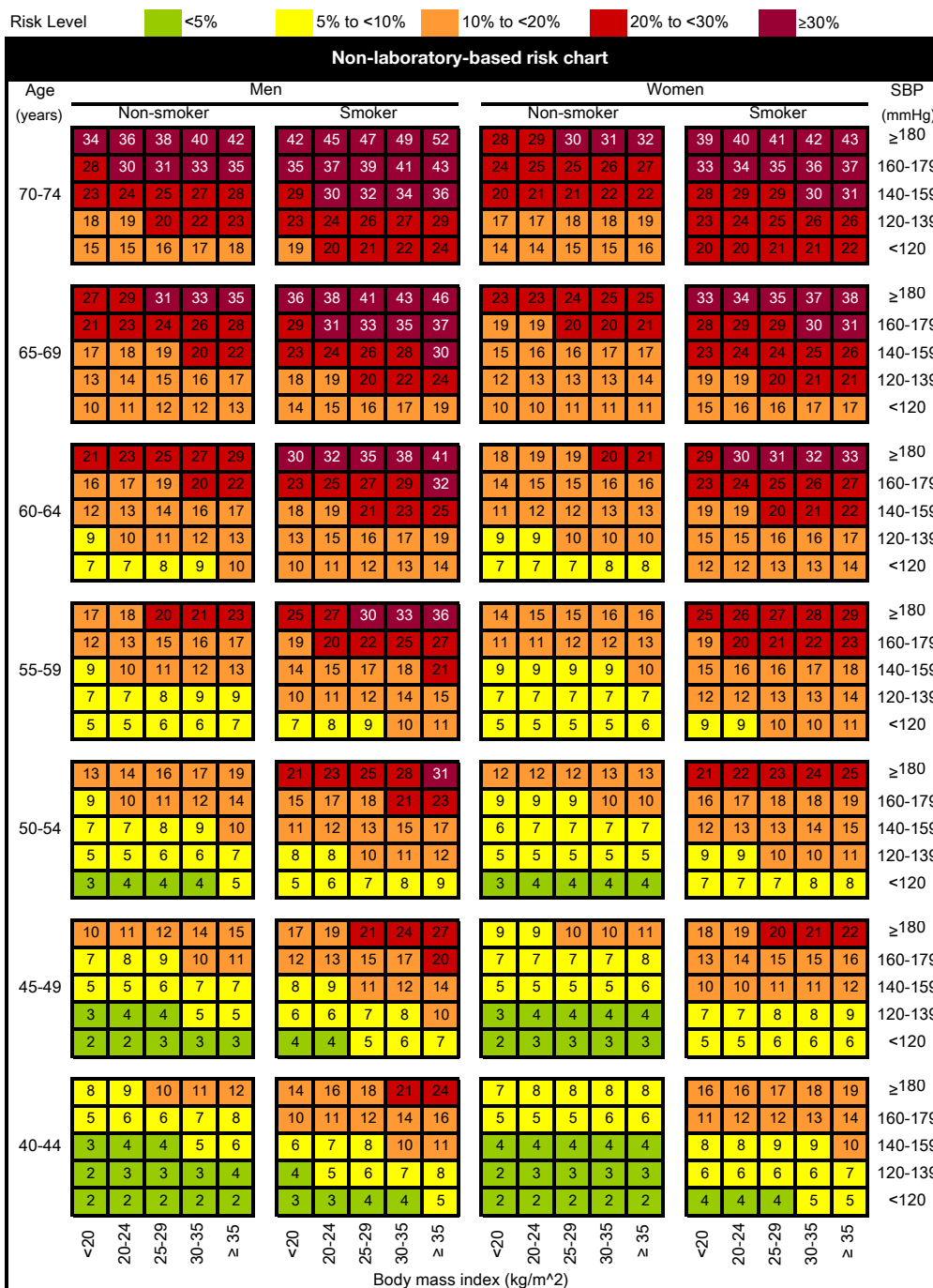


Central Asia

WHO cardiovascular disease risk non-laboratory-based charts

East Asia

China, Democratic People's Republic of Korea

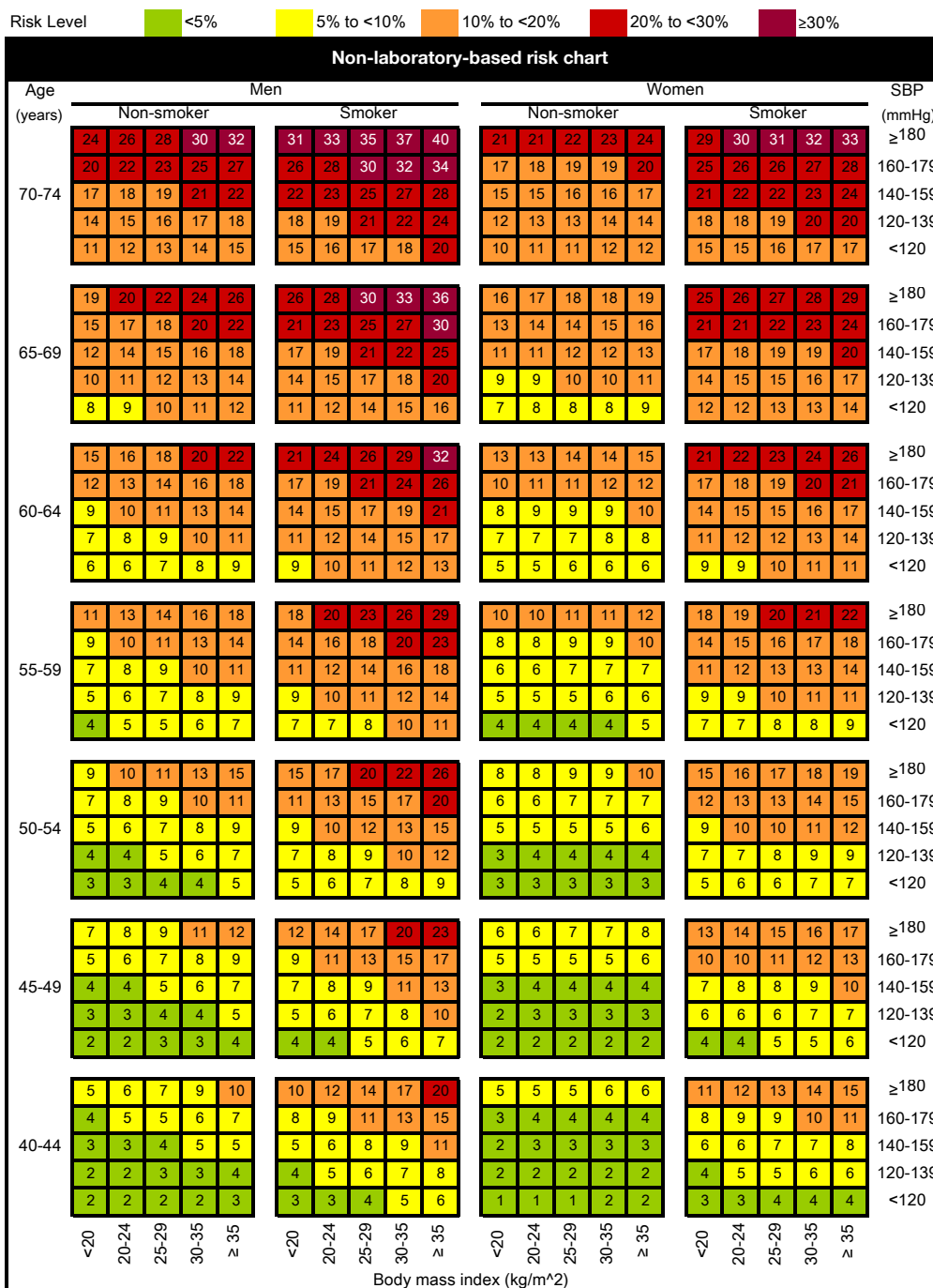


East Asia

WHO cardiovascular disease risk non-laboratory-based charts

South Asia

Bangladesh, Bhutan, India, Nepal, Pakistan

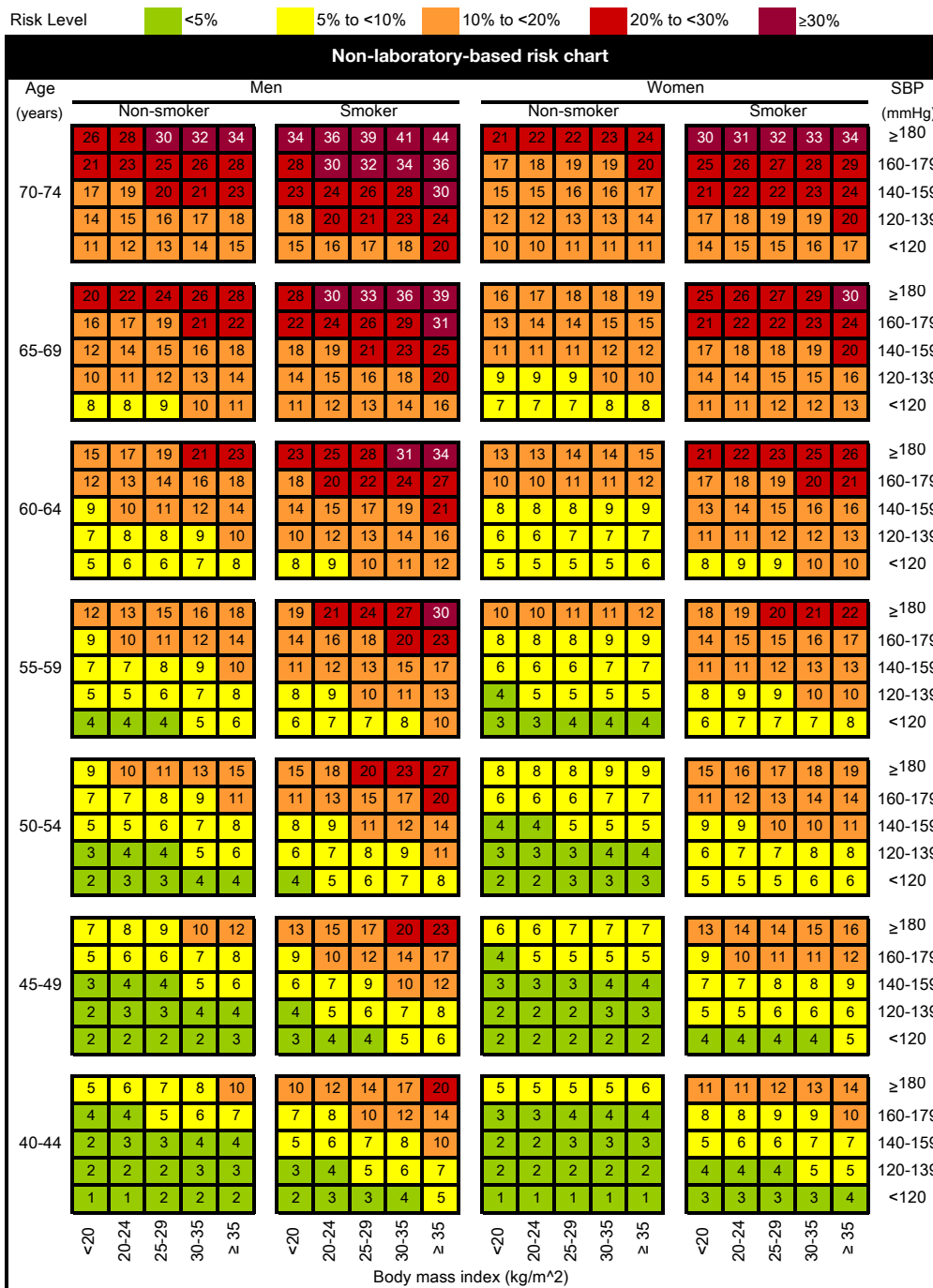


South Asia

WHO cardiovascular disease risk non-laboratory-based charts

South-East Asia

Cambodia, Indonesia, Lao People's Democratic Republic, Malaysia, Maldives, Mauritius, Myanmar, Philippines, Seychelles, Sri Lanka, Thailand, Timor-Leste, Viet Nam

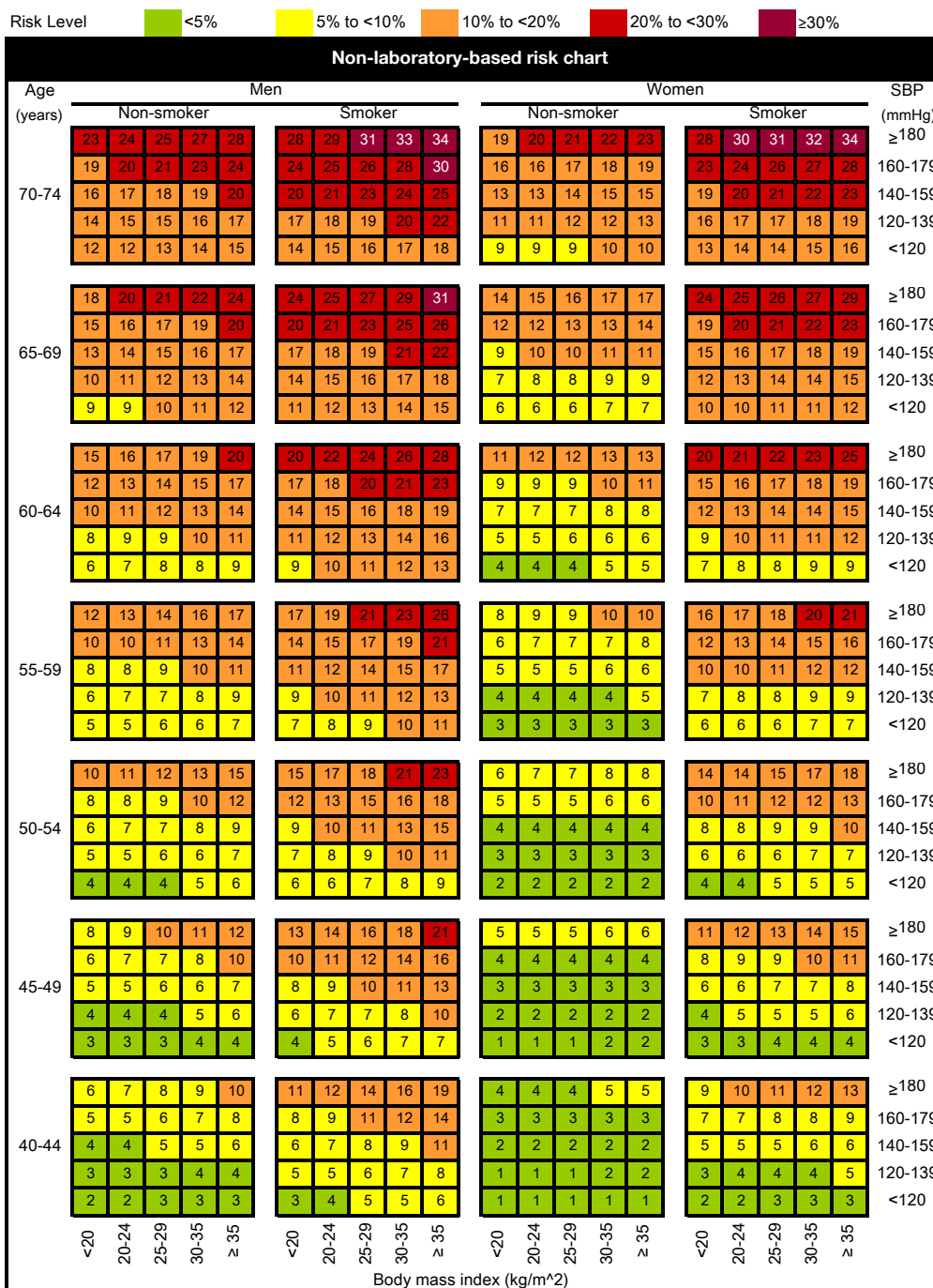


South-East Asia

WHO cardiovascular disease risk non-laboratory-based charts

High-income Asia Pacific

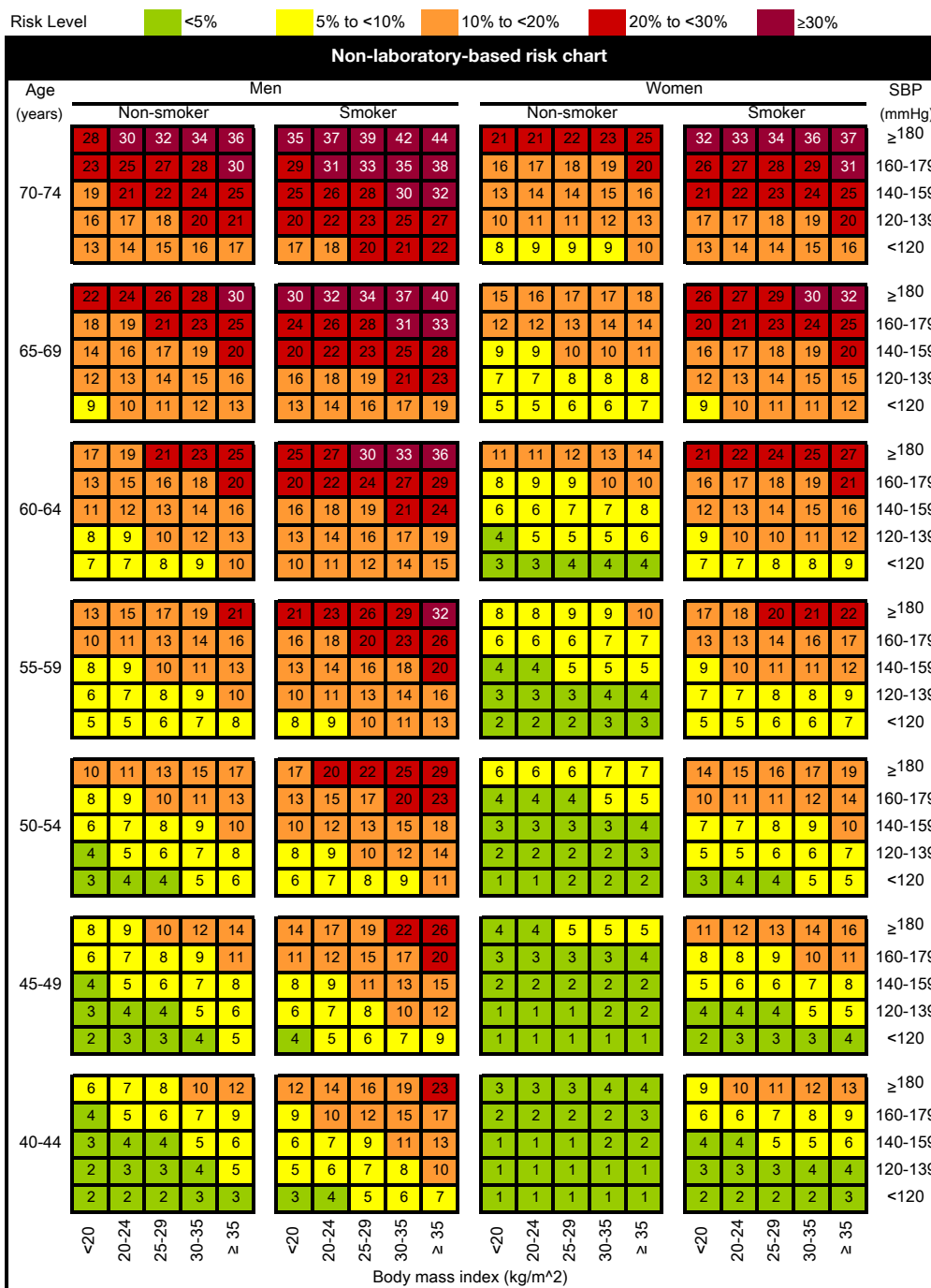
Brunei Darussalam, Japan, Republic of Korea, Singapore



High-income Asia Pacific

WHO cardiovascular disease risk non-laboratory-based charts

Australasia
Australia, New Zealand

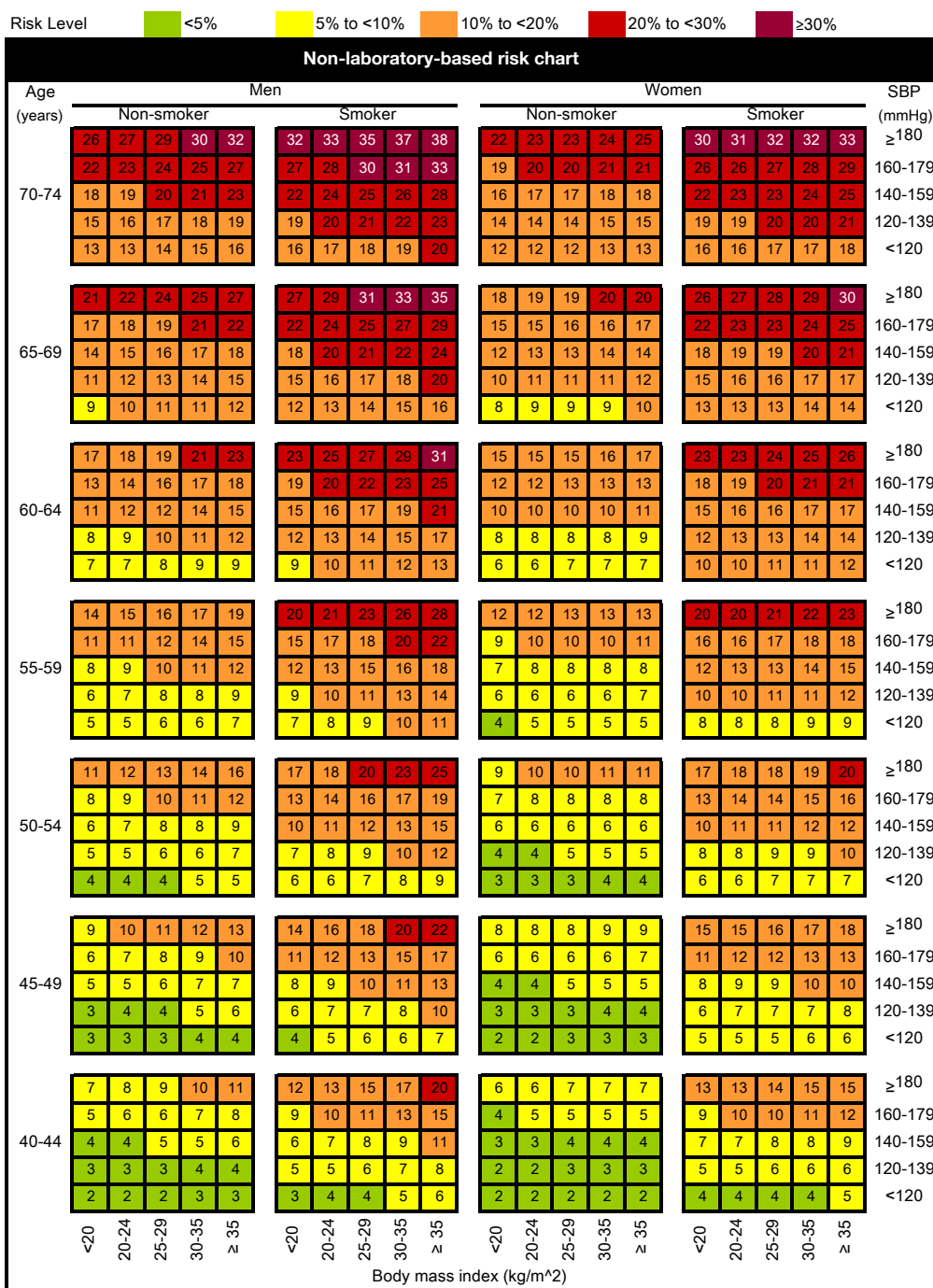


Australasia

WHO cardiovascular disease risk non-laboratory-based charts

Oceania

Fiji, Kiribati, Marshall Islands, Micronesia (Federated States of), Papua New Guinea, Samoa, Solomon Islands, Tonga, Vanuatu



Annex 4: Endpoint definitions used for incidence rates

Myocardial infarction

Case definition

Acute myocardial infarction (MI) – definite and possible MI according to the third universal definition of myocardial infarction:

- When there is clinical evidence of myocardial necrosis in a clinical setting consistent with myocardial ischaemia OR
- Detection of a rise and/or fall of cardiac biomarker values and with at least one of the following: i) symptoms of ischaemia, ii) new or presumed new ST-segment-T wave changes or new left bundle branch block, iii) development of pathological Q waves in the ECG, iv) imaging evidence of new loss of viable myocardium or new regional wall motion abnormality, or v) identification of an intracoronary thrombus by angiography or autopsy.
- Sudden (abrupt) unexplained cardiac death, involving cardiac arrest or no evidence of a noncoronary cause of death
- Prevalent MI is considered to last from the onset of the event to 28 days after the event and is divided into an acute phase (0–2 days) and subacute (3–28 days).

Stroke

Case definition

Stroke was defined according to WHO criteria – rapidly developing clinical signs of focal (at times global) disturbance of cerebral function lasting more than 24 hours or leading to death with no apparent cause other than that of vascular origin (20). Data on transient ischaemic attack (TIA) were not included.

- *Acute stroke*: stroke cases are considered acute from the day of incidence of a first-ever stroke through day 28 following the event.
- *Ischaemic stroke*: an episode of neurological dysfunction caused by focal cerebral, spinal, or retinal infarction.
- *Intracerebral haemorrhage*: a focal collection of blood within the brain parenchyma or ventricular system that is not caused by trauma.

Definitions are taken from GBD 2017 Disease and Injury Incidence and Prevalence Collaborators, Supplement 1 (20).

Annex 5a: Guide for comparison of CVD risk tools

Checklist: Cardiovascular disease risk prediction models in clinical practice		
From model derivation to implementation		Considerations
Derivation	Population	- Does the dataset used for model derivation contain information of a sufficient number of individuals (i.e. is the sample size large enough)? - Are the characteristics of individuals in the derivation sample sufficiently aligned with those of the target screening population to allow transferability of estimated risk ratios (e.g. age range, prior disease status)?
	Risk factors	- Are the relevant risk factors included? Note: main CVD risk factors (e.g. smoking, blood pressure) are key for risk prediction. - Are they accurately measured in sufficient numbers in the derivation sample? For example, use of sphygmomanometer for hypertension is more accurate than self-reported. - Can the risk factors be accurately measured in the target population?
	End points	- Does the model predict the relevant endpoint (e.g. overall CVD, stroke, CHD)? - Is endpoint collection in the derivation sample systematic and well-validated?
	Follow up	Is the follow-up time in the derivation sample sufficient to allow risk estimation over the timeframe of interest (usually 10 years)?
	Statistical Model	- Are appropriate statistical models used for the type of risk estimation? - Are relevant assumptions tested (e.g. proportional hazards assumption)?
	Internal validation	- Discrimination – is the model able to predict order of CVD event among individuals? - Calibration – is there good agreement between predicted and observed incidence (absolute risk)? - Has the model been checked for overfitting (relevant for smaller sample sizes and can be checked by cross-validation)? - Reclassification – is there appropriate movement of individuals through relevant risk categories when comparing alternative risk models?

Checklist: Cardiovascular disease risk prediction models in clinical practice		
From model derivation to implementation		Considerations
Recalibration	Recalibrated	Has the model been recalibrated for use in different populations?
	Recalibration data	Are the data used for recalibration appropriate? For example, do the recalibration data share the same characteristics as the target population?
	Recalibration methods	Is there a methodological framework proposed / provided for future recalibration in response to changing trends with time as well as divergent CVD rates across regions / populations? For example, is a guide or statistical code provided for recalibration?
		- What is the ease of recalibration? - Are additional data needed for recalibration and is this available?
External validation		Is the model proven transferable to a new relevant setting / population other than used in the model derivation?
Usability	Format	Is the format appropriate to be used in the population to which the model is applied (e.g. online risk calculator, colour-coded charts)?
	Risk-factor measurement	Can all risk factors be feasibly measured and are alternative formats available for resource constraint settings (e.g. models using risk factors that don't require laboratory measurements)?
	Country settings	Is the model available for different country settings?
Implementation	Guidelines	Has the model been recommended for use by relevant guidelines?
	Health gains*	- Has the model been evaluated for health gains when used to assess total CVD risk and guide interventions (such as using statins) in high-risk populations? - Has use of the risk-prediction model resulted in significant health gains when used?
	Cost-effectiveness*	Has use of the risk-prediction model been shown to be cost-effective?

* Cost-effectiveness and health gains of (different) CVD risk models in clinical practice are highly dependent on, for example, the target population and the clinical interventions for different thresholds (e.g. statin allocation).

Table based on Cooney et al. (21), Rosello et al. (22), 2016 European Guidelines on CVD (23) and the TRIPOD guidelines (24). If of interest, the TRIPOD guidelines can be used for a more detailed checklist for statistical risk-prediction modelling (24).

Annex 5b: Comparison of CVD risk charts

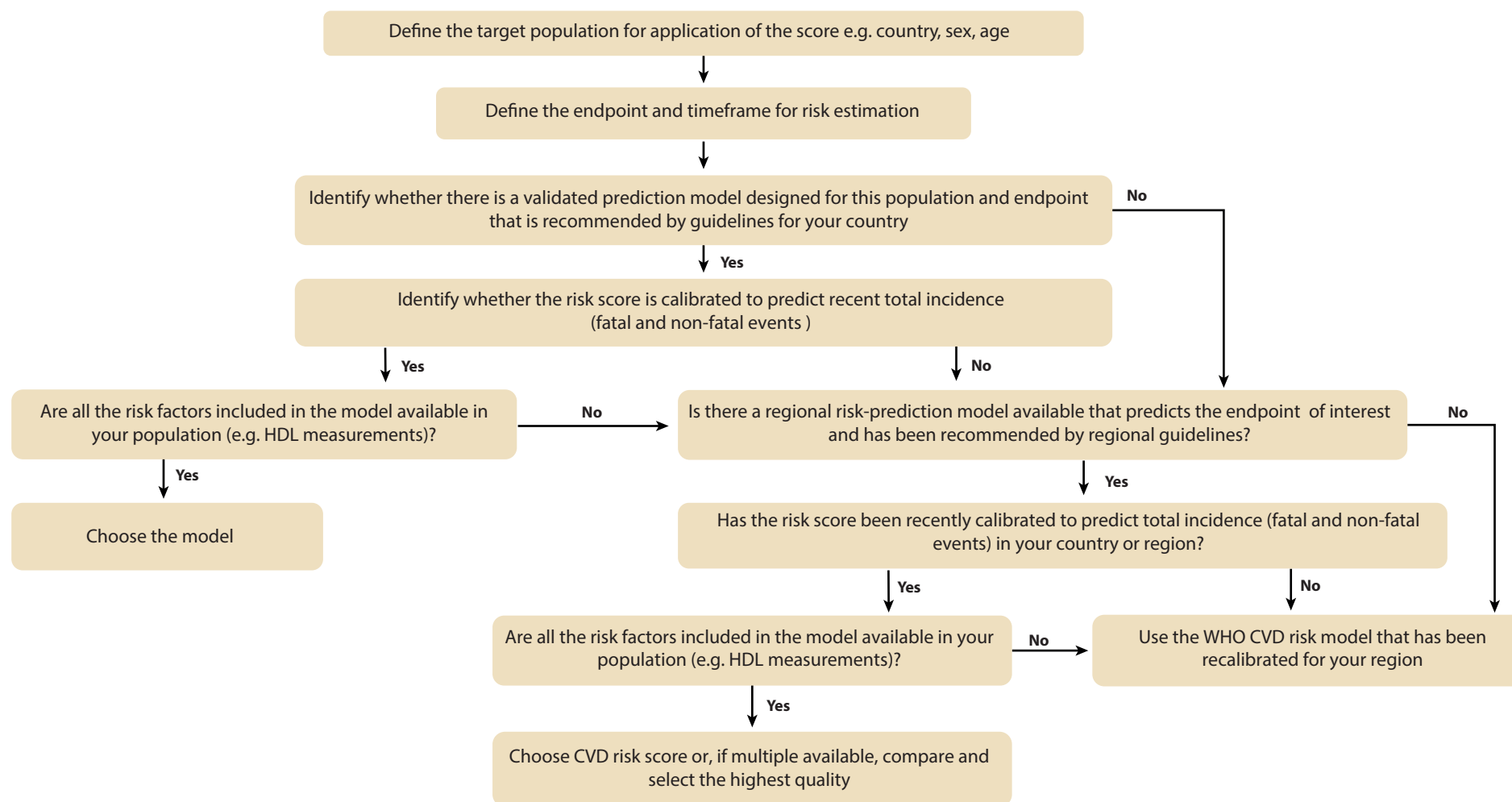
	WHO CVD risk 2019		WHO/ISH 2007	Globorisk 2015	IHMRS 2011	SCORE 2016
Derivation	Population	<i>Study design:</i> Prospective cohorts (85) with 376 177 individuals, 19 333 events <i>Age range:</i> 40–80 yrs <i>Date of baseline survey:</i> 1960–2013 <i>Location:</i> Europe, North America, Japan, Australia	<i>Study design:</i> No single derivation cohort. Risk-factor distribution, relative risks, CVD incidence from various sources combined. <i>Date of baseline survey:</i> Not applicable <i>Location:</i> Incidence, risk factors from 14 WHO regions	<i>Study design:</i> Prospective cohorts (8) with 50 129 individuals, 6042 events <i>Age range:</i> 40–84 yrs <i>Date of baseline survey:</i> 1948–1993 <i>Location:</i> North America	<i>Study design:</i> Case control study, 5349 cases 7423 controls <i>Median age:</i> 58 (49–67) yrs <i>Date of baseline survey:</i> 1999–2003 <i>Location:</i> 52 countries	<i>Study design:</i> Prospective cohorts (12) with 205 178 individuals, 7934 fatal events <i>Age range:</i> 40–65 yrs <i>Date of baseline survey:</i> 1967–1991 <i>Location:</i> Europe
	Risk factors (lab)	Age, sex, smoking, SBP, TC, DM	Age, sex, smoking, SBP, TC, DM	Age, sex, smoking, SBP, TC, DM	Age, sex, smoking, DM, HTN, Apolipoprotein B/A1 or TC/HDL-C ratios	Age, sex, smoking, SBP, TC or TC/HDL-C
	Risk factors (non-lab)	Age, sex, SBP, smoking, BMI	Sex, age, SBP, smoking, DM	Age, sex, smoking status, SBP, BMI	Age, sex, smoking, DM, HTN, family, diet, history, lifestyle, psychosocial factors	Not available
	Time horizon and outcomes	10-yr risk of fatal and nonfatal CVD (CHD or stroke)	10-yr risk of fatal and nonfatal CVD (CHD or stroke)	10-yr risk of fatal and nonfatal CVD (CHD or stroke)	Risk of fatal and nonfatal MI	10-yr risk of fatal CVD (CHD or stroke)
	Follow up	>10-yr follow-up in most cohorts	No actual follow-up, hypothetical 10-yr	>10-yr follow-up in 7 of 8 cohorts	No follow up, case-control design	>10-yr follow-up in all cohorts
	Statistical model	Cox survival models	Combination of several relative and absolute risks in 'Cox model type' structure	Cox survival model	Unconditional logistic regression	Weibull survival models
	Internal validation	Well-validated internally	Not applicable	Well-validated internally	Well-validated internally	Well-validated internally

	WHO CVD risk 2019		WHO/ISH 2007	Globorisk 2015	IHMRS 2011	SCORE 2016
Recalibration	Recalibrated	Integral part of model development; calibrated to 21 global regions	Integral part of construction; calibrated for 14 global regions	Integral part model development; calibrated for 187 countries	Not recalibrated for different settings	High/low risk charts provided; recalibrated post hoc in several countries
	Recalibration data	Region-specific incidences from GBD and country-specific risk factors from NCD-RisC	Risk-factor distributions from WHO Comparative Risk Assessment study; Region-specific incidence from GBD	Used country-specific mortality rates and case-fatality to estimate incidence rates based on age trends in Swedish data; rates then modelled over past years and projected into future 10 years	Not applicable	Country-specific cohorts
	Recalibration methods	Simple framework and statistical code provided. Can be applied using routinely available data	No specific method provided	Intuitive framework provided. Can be applied with routinely available data. Requires modelling if future projections of rates desired as used in original score	Not applicable since model do not predict absolute risk	No standardized approach recommended
External validation	External validation	Well validated in several external cohorts	No adequate external validation	Well-validated in several external cohorts	Well-validated in several external cohorts	Well-validated in several external cohorts
Implementation	Format	Colour-coded charts and software code	Colour-coded charts	Colour-coded charts, online calculator	Calculation charts	Colour-coded charts and online calculator
	Low resource setting	Yes	Yes	Yes	Yes	No
	Website	WHO CVD risk	WHO/ISH	Globorisk		SCORE
	Country-specific versions available?	Different charts for 21 worldwide regions.	Different charts for 14 worldwide regions	Different charts available per country.	-	Low-risk charts and high-risk charts, for grouped European countries
	Guidelines	WHO on CVD prevention 2019	WHO guidelines for CVD prevention 2007	-	-	ESC Guidelines 2019

	WHO CVD risk 2019		WHO/ISH 2007	Globorisk 2015	IHMRS 2011	SCORE 2016
Advantages	Key advantages	Risk-prediction charts provided for different ethnic-geographic regions Simplicity of recalibration approach with code provided to allow efficient updating A non-laboratory variant is available Externally validated in numerous studies Risk distribution assessed in datasets representing 79 countries	Risk-prediction charts provided for different ethnic-geographic regions A non-laboratory variant is available	Risk-prediction charts provided for different ethnic-geographic regions Intuitive systematic recalibration approach A non-laboratory variant is available Externally validated in numerous studies	Included large number of women, youth and people from LMICs in derivation A non-laboratory variant is available Externally validated in numerous studies	Existence of low-risk and high-risk charts for European countries Externally validated in numerous studies
Limitations	Key limitations	Data used in model derivation were mostly from HICs	Absence of individual-level population data; models based on summary inputs No internal or external validation of the model in epidemiological cohorts	Rates used in recalibration rely on many modelling steps, assumptions and projections Only USA-based data in model derivation	Case-control rather than cohort design used; may induce bias in RR estimates, prevents estimation of absolute risk Not available for different settings	Only estimates risk of fatal CVD No non-laboratory variant available Model derived only in European cohorts
ESC: European Society of Cardiology; IHMRS; INTERHEART Modifiable Risk Score, WHO/ISH; World Health Organization / International Society of Hypertension; MI: myocardial infarction; TC: total cholesterol; SBP: systolic blood pressure; HTN: hypertension; DM: diabetes mellitus; BMI: body mass index; GBD study: Global Burden of Disease Study; CVD: cardiovascular disease; CAD: coronary artery disease; NCD-RisC: NCD Risk Factor Collaboration LMIC: Low and middle income countries						

Based on: WHO, 2007 (2), Cooney et al, 2009 (21), Rossello et al, 2019 (22), 2016 European Guidelines on CVD (23), the TRIPOD guidelines (24), Mendis et al, 2007 (25), Hajifathalian et al, 2015 (26), McGorrian et al, 2011 (27), European Society of Cardiology, SCORE risk charts (28).

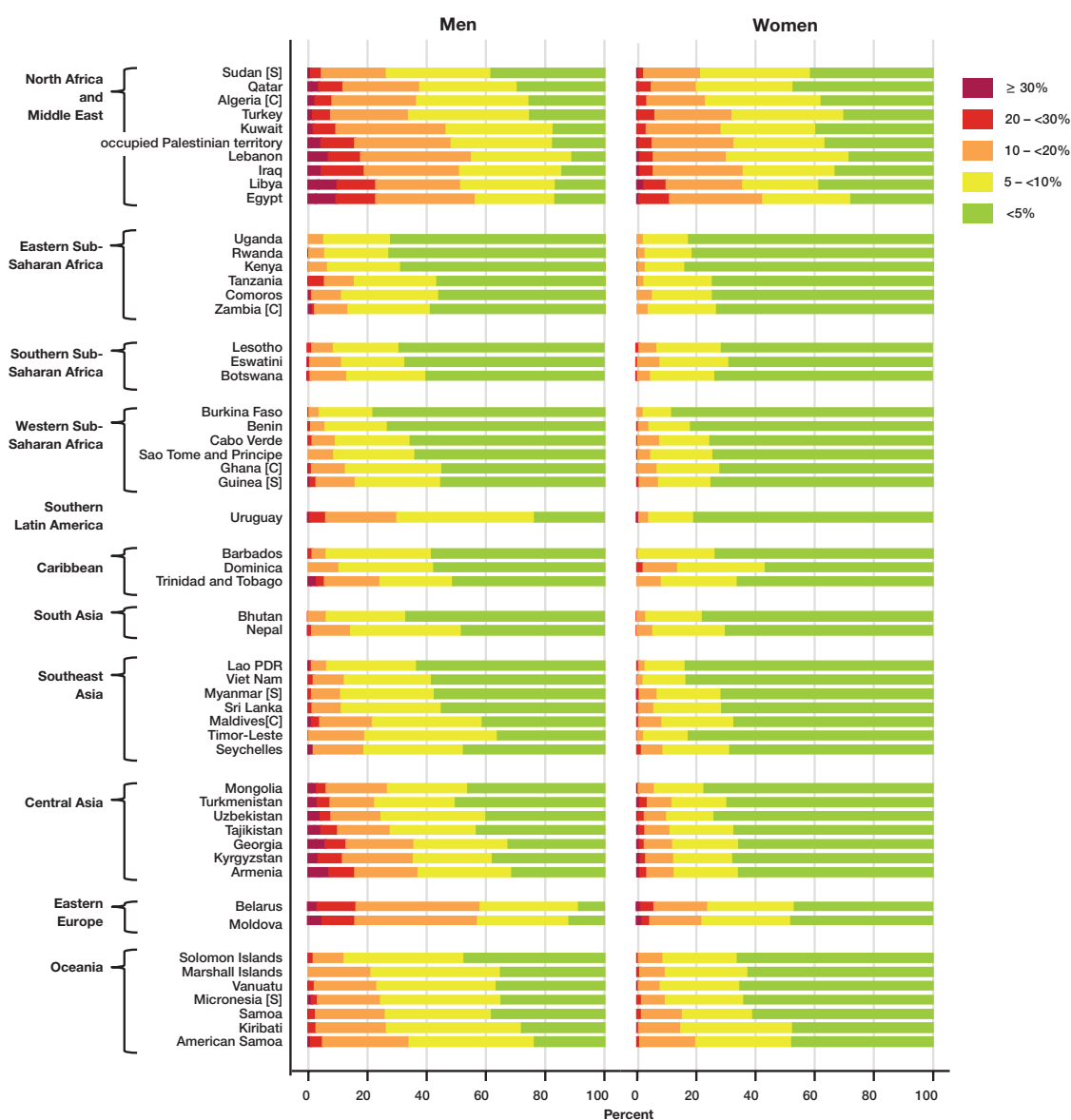
Annex 6: Identifying a CVD risk prediction model



Annex 7: Considerations for initiation of therapy

The appropriate threshold of an individual's total risk at which intensive lifestyle interventions and drug treatment are initiated depends on the availability of resources as well as the risk level. The percentage of patients from representative countries in each region who would be eligible for treatment based on each cut-off level is presented in Figure 5 (3). As an example, <5% of the population of Uganda is at 10% or greater CVD risk, whereas approximately 50% of Egypt's population is above 10% CVD risk.

Figure 5: Distribution of 10-year CVD risk according to recalibrated laboratory-based WHO risk prediction models for individuals aged 40–64 years from example countries



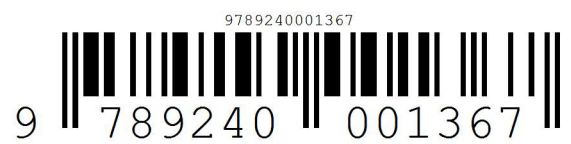
Data from all countries are from adults aged 40–64 years with total cholesterol concentrations of 2.6–10.3 mmol/L and from samples representative of the national population, unless otherwise specified as subnational (S) or community based (C).

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RECOGNIZING NEGLECTED TROPICAL DISEASES THROUGH CHANGES ON THE SKIN



BURULI ULCER CUTANEOUS LEISHMANIASIS POST KALA-AZAR DERMAL LEISHMANIASIS LEPROSY LYMPHATIC
FILARIASIS MYCETOMA ONCHOCERCIASIS PODOCONIOSIS SCABIES YAWS (ENDEMIC TREPONEMATOSIS) BURULI ULCER
CUTANEOUS LEISHMANIASIS POST KALA-AZAR DERMAL LEISHMANIASIS LEPROSY LYMPHATIC FILARIASIS
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A TRAINING GUIDE FOR FRONT-LINE HEALTH WORKERS



**World Health
Organization**

Recognizing neglected tropical diseases through changes on the skin: a training guide for front-line health workers.

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1. INTRODUCTION

The skin of a patient is the first and most visible structure of the body that any health-care worker encounters during the course of an examination. To the patient, it is also highly visible, and any disease that affects it is noticeable and will have an impact on personal and social well-being. The skin is therefore an important entry point for both diagnosis and management. Many diseases of humans are associated with changes to the skin, ranging from symptoms such as itching to changes in colour, feel and appearance. The major neglected tropical diseases (NTDs) frequently produce such changes in the skin, re-enforcing the feelings of isolation and stigmatization experienced by patients affected by these diseases. Indeed, these are often the first signs that patients will notice, even before changes in internal organs or other systems occur. The following NTDs all have prominent skin changes at some stage in their evolution.

Purpose of this guide

This training guide explains how to identify the signs and symptoms of neglected tropical diseases of the skin through their visible characteristics. It also contains information on how to diagnose and manage common skin problems that front-line health workers may encounter.

Target audience

The guide is intended for use by front-line health workers without specialist knowledge of skin diseases. Pathways for more detailed investigation and/or management are described but, often, this will involve referral to a specialist service.

2. NEGLECTED TROPICAL DISEASES OF THE SKIN

This section summarizes the characteristics of the major neglected tropical diseases of the skin, or the so-called skin NTDs.

2.1 Buruli ulcer

Buruli ulcer is an infection mainly affecting the skin caused by *Mycobacterium ulcerans*. The mode of transmission is not known, although in many cases there is associated exposure to rivers, streams or wetlands. The first changes are often raised lumps on the skin that subsequently ulcerate. These ulcers are usually single and large, with a yellowish surface appearance and an underlying red moist base; sometimes 2–3 ulcers may appear. The edges of the ulcer are often undermined, making it is easy to insert a blunt probe for a short distance underneath it. During the early stages of infection, extensive swelling may extend around the emerging ulcerated area; this is characteristic of Buruli ulcers. Most patients living in areas where the disease is endemic are children aged under 15 years. **Fig.1.**

2.2 Cutaneous leishmaniasis

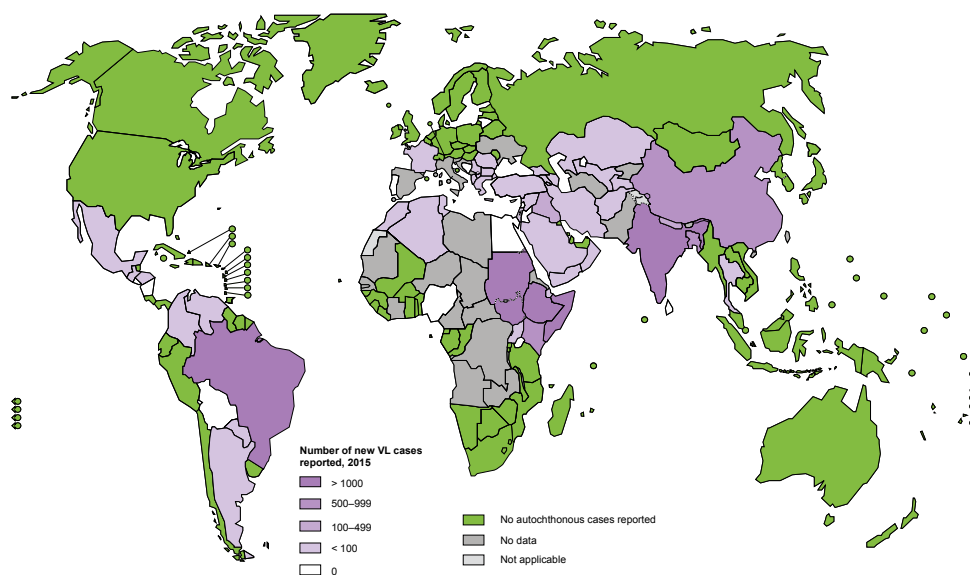
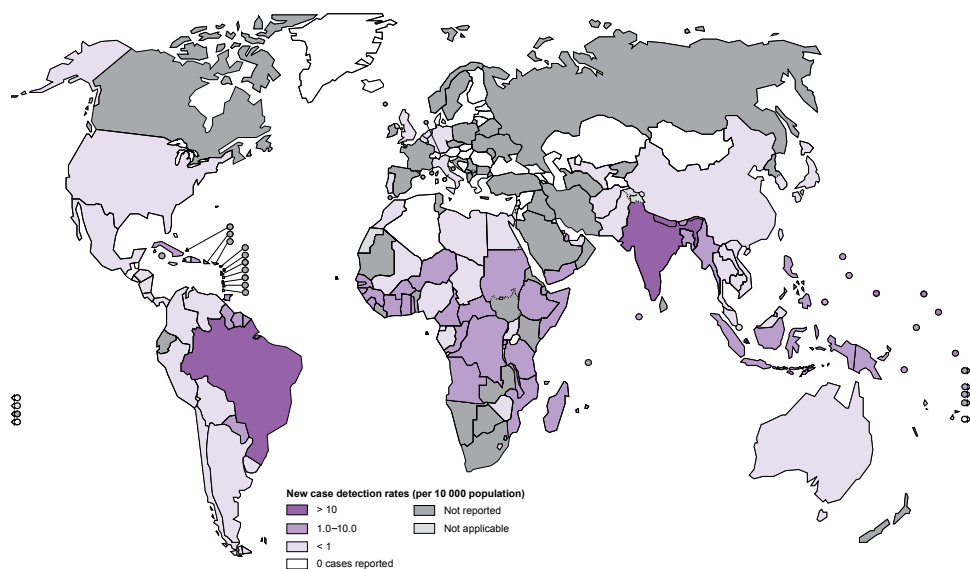
Cutaneous leishmaniasis is an infection of the skin or oral mucosa caused by *Leishmania* protozoa. The infection is spread through biting insects, sandflies of the genera *Phlebotomus*, *Lutzomyia* and *Psychodopygus*. Reservoir hosts are wild animals, domestic animals and humans. It also presents with lumps on the skin that can be rounded or flat and which may ulcerate in the centre – the base of the ulcer is usually red and bleeds easily. Ulcer edges or borders are often raised. Lesions may be single or multiple, usually occurring on exposed sites such as the face or limbs and containing variable numbers of *Leishmania* parasites. Cutaneous leishmaniasis may affect any age group; while infection normally occurs sporadically, large numbers of people may be affected in areas where displaced populations are housed. The visible changes may vary in different parts of the endemic range; for instance, in areas of South America lesions may also occur in the mouth or nasal passages. **Fig.2.**

2.3 Post kala-azar dermal leishmaniasis

Post kala-azar dermal leishmaniasis is a complication of systemic or visceral infection with *Leishmania* that is restricted to a few countries. Some 5–20% of patients who have visceral leishmaniasis develop skin lesions as a late complication of the internal infection. Although it is not completely clear whether these skin lesions are caused by the presence of viable organisms, they do contain a variable number of *Leishmania* parasites and are treated with anti-*Leishmania* medications. These lesions are usually either pale patches, or small or large lumps. This form of leishmaniasis may occur at any age but develops in around 10–30% of patients some 6–12 months after apparent resolution of visceral (internal) leishmaniasis. **Fig.3.**

2.4 Leprosy

Leprosy is a systemic infection caused by the bacterium *Mycobacterium leprae*, which targets the skin and nerves. Although much of the damage caused by leprosy results from destruction of nerves, the skin is affected early in the course of the infection. This disease is transmitted directly through close contact over long periods. Leprosy is transmitted via droplets, from the nose and mouth, during close and frequent contacts with untreated cases. Infection can occur at any age. The first signs are usually skin patches of variable sizes that are usually dry and whose colour may be slightly paler than the rest of the skin. Another presentation of leprosy takes the form of multiple lumps, also in variable sizes. Leprosy destroys nerves, and it is easy to damage the skin because patients cannot feel hot objects or rubbing shoes, and ulcers form at these sites. If untreated, these ulcers (usually on the hands or on the feet), which are known as neuropathic ulcers, can proceed to destroy other structures in the area including bone. **Fig.4.**

Fig. 3. Distribution of visceral leishmaniasis, worldwide, 2016**Fig. 4. Leprosy new case detection rates, 2016**

2.5 Lymphatic filariasis

Lymphatic filariasis is a systemic infection caused by the filarial nematode worms *Wuchereria bancrofti* and, in limited geographical areas, *Brugia malayi* or *Brugia timori*. It is transmitted by mosquitoes such as *Culex* and *Anopheles*. The filarial worms that cause lymphatic filariasis live in, dilate and obstruct the lymphatic channels, usually in the lower parts of the body such as the groin and genital area and thighs. Adult worms produce immature microfilariae that can be detected in blood at times specific to the periodicity of the parasite. Infections impair lymph flow, leading to progressive swelling or lymphoedema of the lower limbs and, in males, swelling of the scrotum (hydrocele). Other parts of the body can be involved such as the breasts or vulva in females. Although lymphatic filariasis often starts in childhood, it becomes prominent in adults where the results of progressive lymphatic obstruction are apparent. A clinical diagnosis is suggested if one leg becomes swollen and/or if the leg on one side is more swollen than the other. Hygiene and skin care are important to prevent secondary bacterial infections which cause “acute attacks” of acute inflammation of the skin, lymph vessels and lymph glands accompanied by debilitating pain, fever and swelling. Acute attacks, also known as adenolymphangitis or acute dermatolymphangioadenitis, progress the severity of disease. The affected area is warm, reddish and painful. With time the skin may become thickened, covered in small lumps giving a cobbled appearance and susceptible to recurrent infections, **Fig.5**.

2.6 Mycetoma

Mycetoma is an infection of the subcutaneous tissue and bone caused by a variety of different environmental bacteria and fungi. The disease follows implantation of fungi or filamentous bacteria from plant debris or soil through a deep skin injury such as a thorn prick. In mycetoma, there is an infection under the skin that causes the development of a large lump which is usually painless. As this enlarges it becomes covered in smaller lumps, some of which are filled with pus; these discharge yellow or blood-stained fluid that may contain the microcolonies of the infecting organisms seen as tiny black or pale pieces of grit. In advanced mycetoma, the bone in the area is invaded with local destruction and the part of the limb or body affected becomes swollen and deformed. Adults are mainly affected, although rare childhood cases occur. **Fig.6**. Other deep mycoses seen affecting the skin include chromoblastomycosis where there are thick raised patches often with a rough surface like very large warts.

Fig. 5. Countries where lymphatic filariasis is endemic and status of preventive chemotherapy in those countries, 2016

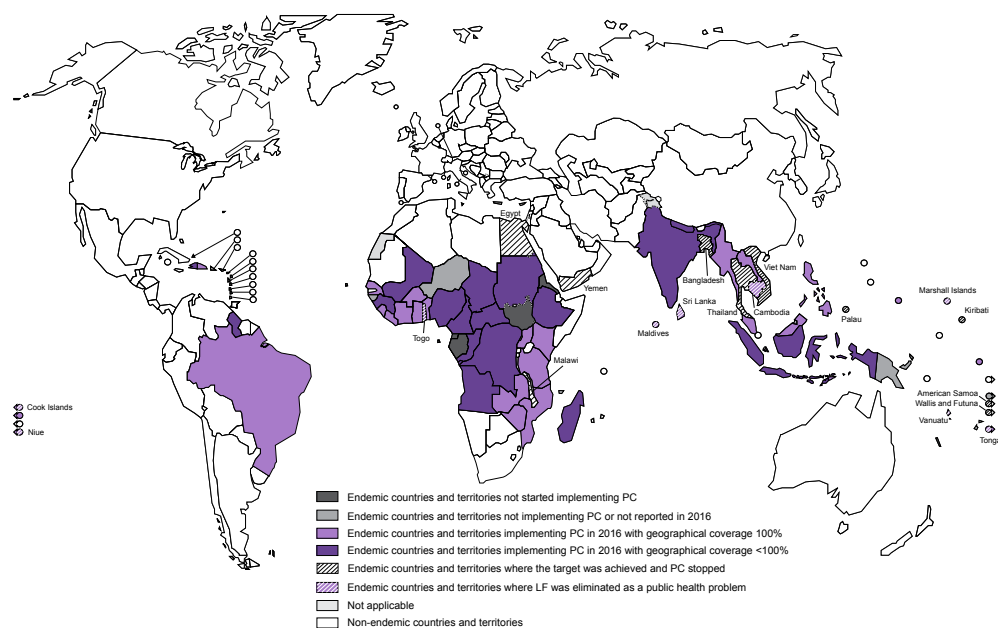


Fig. 6. Prevalence of mycetoma per 100 000 population, latest year available



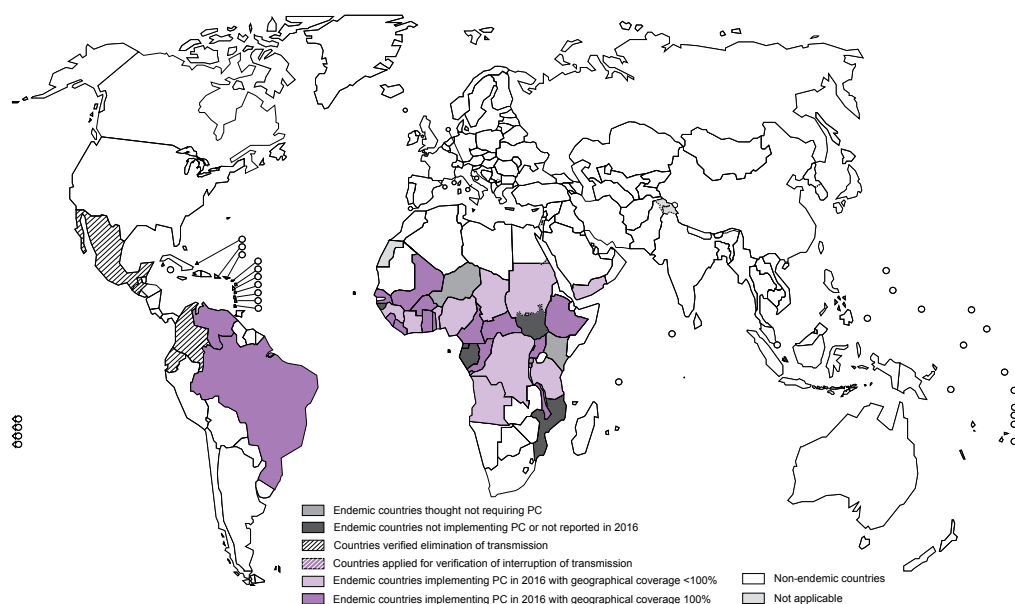
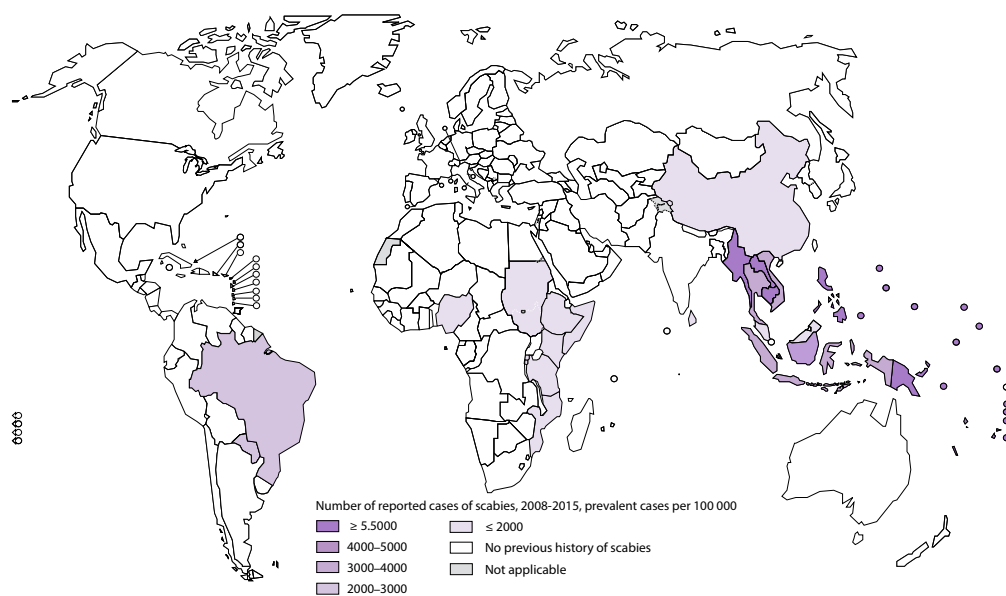
2.7 Onchocerciasis

Onchocerciasis is a systemic filarial infection caused by *Onchocerca volvulus* that affects the skin and the eye. Infection may lead to blindness or debilitating skin disease. The filarial worm *Onchocerca volvulus* is transmitted by black flies of the genus *Simulium* that breed near fast running streams and rivers. The adult female worms are surrounded by fibrous tissue to form firm subcutaneous or deeper large lumps or nodules, which do not typically cause symptoms. People are most likely to complain of troublesome itch (severe pruritus). Symptomatic skin signs are produced by the inflammatory response to dying microfilaria (an early larval form of the parasite). Similarly, eye diseases such as iritis (inflammation near the pupil of the eye), corneal scarring and retinal damage (retinopathy) result from the inflammatory response to microfilariae. Many patients with onchocerciasis do not have marked skin symptoms, although as the disease progresses the skin becomes wrinkly and skin folds can enlarge and sag, for example in the groin (hanging groin), and visible depigmentation may occur (leopard skin). However, in many endemic areas onchocerciasis is accompanied by small itchy lumps on the skin, often around the shoulders or waist and thighs; constant scratching is common. Sometimes, small itchy lumps appear more densely on one limb than the other and the skin of the affected limb becomes rough, chronically itchy and with thickened patches that are darker in colour than the rest of the skin; these patches are intensely itchy. Onchocerciasis can occur at all ages, although most cases are recognized in older children or adults. **Fig.7.**

2.8 Scabies and other parasitic infections of the skin

Scabies is a very itchy infection caused by the human itch mite *Sarcoptes scabiei*. It is common and endemic in many communities in resource-poor settings and is associated with household overcrowding. The infection is transmitted by human to human contact and although it can affect all age groups, it is particularly common in infants and children. The skin in certain areas is covered with small lumps, some containing pus as well as small squiggly lines where the mites burrow into the skin. Hands, feet, external genitalia, buttocks and shoulders are all commonly affected and usually other members of the household are infected and itch. Secondary bacterial (streptococcal) infection of scabies may cause kidney inflammation or nephritis, an immunological reaction to the bacteria, in children. **Fig.8.**

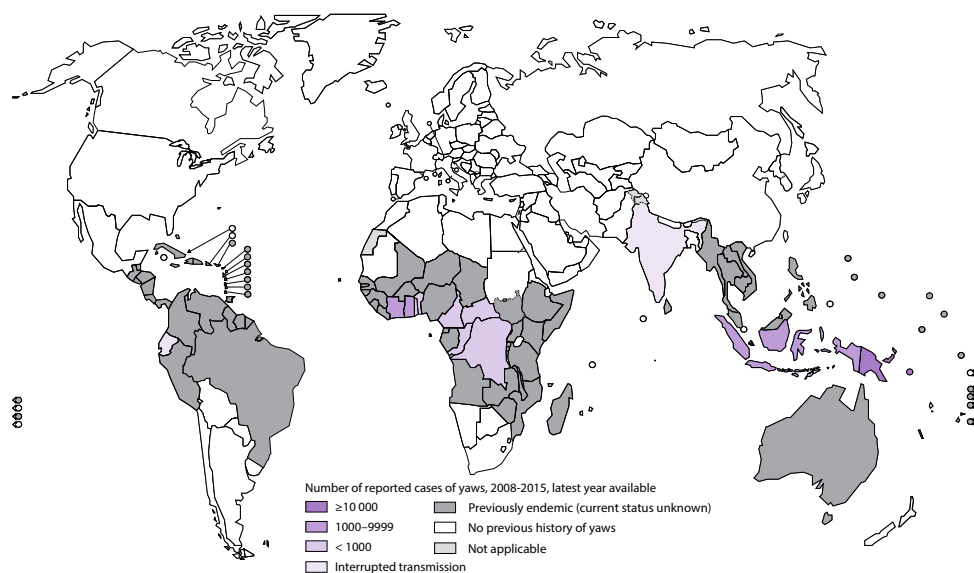
Tungiasis, an itchy infection caused by a tropical sand flea, *Tunga penetrans*, causes itchy lumps, usually with a central dark or pus-filled plug, most commonly on the feet after contact with contaminated sandy soil, for example in homes or around villages. However, it can occur at almost any site, particularly in areas where there are body folds, and develop at any age.

Fig. 7. Status of preventive chemotherapy for onchocerciasis, worldwide, 2016**Fig. 8. Distribution of scabies, worldwide, 2016**

2.9 Yaws (endemic treponematosi)

Yaws, a systemic infection affecting the skin and, more rarely, bone, is caused by the bacterium *Treponema pertenue*, a subspecies of *T. pallidum* which causes syphilis. However, yaws is not sexually transmitted. The disease is often first recognized in children but may affect all ages; it is spread by direct contact between individuals. Yaws presents with one or multiple skin lesions, which appear as small lumps or nodules. These may have a red or yellow surface colour and often feel soft, pebbled or bumpy. These lumps may break down to leave an area of ulceration in the centre of the nodule or appear in the shape of a small volcano. **Fig.9.**

Fig. 9. Distribution of yaws, worldwide, 2008–2016



2.10 Other common non-NTD skin diseases

Skin disease is one of the most common of all human afflictions, affecting about 900 million people in the world at any time. By using changes to the skin as a means of detecting NTDs at their earliest stage, other diseases causing skin changes will be seen. Yet, in most communities, five common conditions account for over 80% of skin disease. It is possible to identify the most common of these other skin diseases and, where feasible, to start treatment as their management is simple and they often respond to medications that are widely available.

The commonest of these diseases are bacterial skin infections or pyodermas such as boils or impetigo (an infection of the skin surface) and fungal skin infections. Other common skin diseases such as eczema are non-infective. These diseases are included in the diagnostic charts and a brief description of their treatments is provided (**Section 5**).

Bacterial skin infections caused by *Staphylococci* (such as impetigo or boils) or *Streptococci* (impetigo) can be transmitted from human to human in homes or schools. They are commonest in children but can affect adults.

Superficial fungal skin infection or tinea is caused by fungi that infect the outer layers of the skin to cause scaly, itchy patches on the body, legs or scalp; in the head there is also loss of hair. They are often transmitted from human to human in homes or schools and can affect all ages.

Eczema, often called dermatitis, is a common non-infectious skin problem causing scattered itchy dry patches with scaling on the skin of the arms (such as the inner surface of the elbows), legs (behind the knees), face and body. Where it is scratched and constantly rubbed the skin becomes darker and thicker. Eczema is common in children. Other common causes of itching are insect bites, which are scattered itchy lumps on exposed sites; these may become infected (see above).

Ulceration or loss of the surface skin is also a common sign of skin or internal disease and important in the differential diagnosis; it is discussed in the next section.

Note

Not every patient affected by a neglected tropical disease will have skin changes, but they are sufficiently common to be a very useful aid to diagnosis and a starting point for using more specialist diagnostic methods such as antigen detection card tests or serology.

3. COMMON CHANGES IN THE SKIN

In order to make a diagnosis by looking at the skin, it is important to recognize that there are a limited number of possible changes in the appearances of the skin due to any disease. A more detailed summary of these changes is provided in the **Annex**, but for the purposes of this guide the descriptions are restricted to the four major changes in the skin, namely: (i) lumps, (ii) ulcers, (iii) swollen limbs and (iv) patches. These changes may occur with or without itch, change in colour or loss of sensation. For the skin NTDs, these are described below.

3.1 Lumps

A lump is a **small** or **large** raised bump that can occur at any site on the skin. Sometimes lumps are hard but they may be soft to touch, which generally indicates that they contain fluid or pus. They can also have a smooth surface or appear rough and wart-like. Lumps may also be painful or itch.

Fig. 3.1.1. Bacterial skin infection – Bullous impetigo with soft fluid-filled lumps



Fig. 3.1.2. Buruli ulcer – Lump in early nodule



Fig. 3.1.3. Cutaneous leishmaniasis – Smooth glazed lump



Fig. 3.1.4. Post kala-azar dermal leishmaniasis – Multiple smooth lumps



Fig. 3.1.5. Lepromatous Leprosy – Small hard lumps around ear lobe



Fig. 3.1.6. Lymphatic filariasis – Swollen leg to show cobbled change seen with chronic lymph obstruction



Fig. 3.1.7. Mycetoma – lumps, openings of draining sinuses discharging pus and blood



Fig. 3.1.8. Acute papular onchodermatitis – Multiple small itchy lumps



Fig. 3.1.9. Chronic papular onchodermatitis – Flat-topped lumps

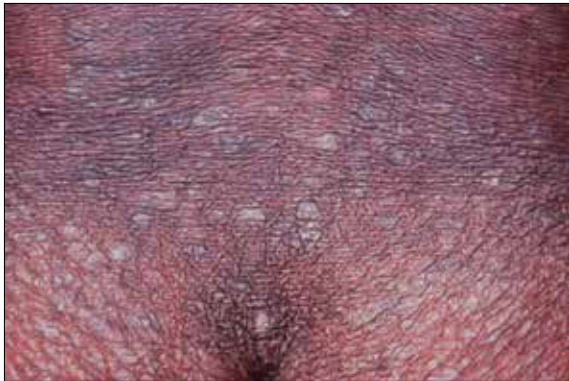


Fig. 3.1.10. Onchocercal nodule – Large hard lump



Fig. 3.1.11. Podoconiosis – Warty lumps on swollen legs



Fig. 3.1.12. Scabies – Small itchy lumps and burrows



Fig. 3.1.13. Tungiasis – Itchy lumps with black centres around toes



Fig. 3.1.14. Yaws – Large lump with raspberry-like surface (papilloma)



Fig. 3.1.15. Yaws – Large lump with raspberry-like surface (papilloma)



Fig. 3.1.16. Buruli ulcer – Lump, an early nodule



Fig. 3.1.17. Cutaneous leishmaniasis – Multiple crusted lumps



Fig. 3.1.18. Tungiasis – Multiple lumps along margin of foot



3.2 Ulcers

An ulcer occurs where there is a break in the surface of the skin. Ulcers are usually round or oval in shape, but they may become irregular and are usually more than 0.5 cm deep. On inspection, the base is often red and raw-looking or it may be covered by dead tissue or a yellow surface crust. Ulceration is an important feature of many skin NTDs.

Other common causes of ulceration include venous or varicose (stasis) ulcers caused by poor drainage through the lower leg veins and arterial ulcers, which develop because of a faulty arterial blood supply. With venous ulceration prominent dilated veins on the leg (varicose veins) are seen and, with arterial ulceration, arterial pulses on the legs cannot be felt. Common infective ulcers caused by bacteria such as *Fusobacterium* and *Haemophilus* occur in areas where yaws is endemic. Clues to the diagnosis of these conditions is included in the diagnostic charts (see **Section 5**).

Fig. 3.2.1. Buruli ulcer – Showing undermined edge



Fig. 3.2.2. Cutaneous leishmaniasis – Ulcer forming on a lump



Fig. 3.2.3. Tuberculoid leprosy – Ulcer in an area with sensory loss



Fig. 3.2.4. Yaws – Ulcerated lesion



Fig. 3.2.5. Ulcers (tropical) – Tropical ulcer caused by *Fusobacterium ulcerans*, this appears over a few days



Fig. 3.2.6. Ulcers (tropical) – Tropical ulcer caused by *Haemophilus ducreyi*, this develops slowly



3.3 Swollen limbs, face or body

Swelling of the skin often occurs when it contains too much fluid.

This change is easiest to see on the limbs as it is possible to compare the size of one limb such as a leg with the other, but large swellings on the face or head as well as the body can also follow inflammation and accumulation of fluid in these areas.

Any swelling that appears as an increase in the circumference of an arm or leg is often best seen when comparing one side with the other. Swollen limbs may be hard to touch or soft, leading to a dent on the skin surface where pressed gently but firmly with a finger, and they can be painful (tender) and warm to the touch – so it is important to ask the patient about pain before touching the limb. Patients will often say that the limb is more swollen by the end of the day and it may reduce when they lie down. It may also be difficult to put on clothes and shoes as these feel tight. In lymphatic filariasis male patients may also present with a swollen scrotum; in females the breast is sometimes affected.

In some areas endemic for NTDs such as the highland areas of Ethiopia and Rwanda, podoconiosis is an important cause of foot or leg swelling (lymphoedema) in children and adults. Podoconiosis is a genetically determined inflammatory reaction to certain soils and often affects both legs.

Leg swelling can occur in other conditions such as heart failure where the presence of other symptoms such as breathlessness, weakness and palpitations may be found.

Fig. 3.3.1. Buruli ulcer – Extensive swelling of the left arm



Fig. 3.3.2. Lymphatic filariasis – Limb oedema



Fig. 3.3.3. Mycetoma – Swelling of the foot and ankle



Fig. 3.3.4. Podoconiosis – Bilateral foot swelling



Fig. 3.3.5. Lymphatic filariasis – Swollen arm for more than 5 years



Fig. 3.3.6. Lymphatic filariasis – Scrotal swellings



3.4 Patches

A patch is a distinctive area of skin that is either raised above the skin surface or is different in colour or to feel (texture) compared with the surrounding skin; for example, it may feel rough to the touch. Patches may also itch. Raised patches vary in height above the skin surface from a few millimetres to one centimetre or more.

The ability to recognize changes in skin colour associated with skin lesions largely depends on whether the skin is naturally dark. Darkening or lightening of skin colour is more noticeable in darker skin types. Increases in skin pigmentation associated with skin inflammation may persist for months. Increased redness or erythema may be more difficult to distinguish in darker skin.

Fig. 3.4.1. Buruli ulcer – Patch



Fig. 3.4.2. Cutaneous leishmaniasis – Patch (pebbly) around ear



Fig. 3.4.3. Eczema – Itchy patch (lichenified)



Fig. 3.4.4. Borderline tuberculoid leprosy – Multiple pale patches with loss of sensation



Fig. 3.4.5. Onchocerciasis – Hypopigmented onchodermatitis (leopard skin)



Fig. 3.4.6. Lichenified onchodermatitis – Thickened very itchy patches; worse on one side than the other



Fig. 3.4.7. Post kala-azar dermal leishmaniasis – Pigmented flat patch



Fig. 3.4.8. Fungal infection – Ring shaped patch of *Tinea corporis*; a superficial fungal infection



Fig. 3.4.9. Scalp ringworm or fungal infection – Patch of hair loss and scaling in a child with *Tinea capitis*



Fig. 3.4.10. Yaws – Pale patch (rough surface)



Fig. 3.4.11. Buruli ulcer – Patch over area where ulcer will form (hard to the touch)



Fig. 3.4.12. Fungal infection – Pale scaly patches of pityriasis versicolor



3.5 Other important signs and symptoms of skin involvement

3.5.1 Itchy skin

Itch is the commonest symptom associated with any skin disease and is most often seen where there is active inflammation. Itching may vary in its severity but, in many patients, it is not trivial and may disrupt their daily lives and sleep. Although there are no specific signs of itchy skin, there may be scratch marks or patients may be observed scratching while waiting to be seen.

3.5.2 Loss of sensation

Loss of sensation in an area of skin is a key feature of leprosy and forms a very important part of the examination. Sensory loss of the skin, particularly the feet, can also occur in diabetes.

3.5.3 Changes in skin pigmentation

One of the features of any disease that causes inflammation of the skin is either an increase in skin colour (hyperpigmentation) or a loss of skin colour (hypo- or de-pigmentation). As these are general signs and seldom specific to one disease they will be used as additional changes seen in skin affected by NTDs.

Note

Any of the above signs and symptoms can occur in combination. For instance, lumps on the skin may ulcerate. This is also true of symptoms and signs that accompany inflammation, such as itching and pigment change, both of which may be experienced by patients with ulcers, patches, lumps or swollen limbs.

4. HISTORY TAKING ASKING PATIENTS ABOUT THEIR CONDITION AND EXAMINING THE SKIN

This section describes how to take a medical history from and examine a patient who has problems of the skin.

4.1 Talking to patients with skin problems

Talking to people about their skin condition can reveal much about the disease, how it started and how it has developed further. But what may be obvious to you may not be to the patient. The presence of symptoms such as itching or pain often draws the patient's attention to an area, but the back and backs of the legs or arms are not visible without a mirror and changes, particularly if they do not itch and are not raised, may not be noticed.

Greet the patient and introduce yourself.

Try to use simple terms that are used locally to describe the changes that you (and they) see. The following are points to concentrate on:

- Where is the location of the skin problem/lesion(s)?
- When did you first notice this?
- How did the condition change over time?
- Does it itch or is it painful or warm?
- Where were you living when this first occurred?

- Do you swim or wash in a lake or river?
- Has this been treated with any medicines or home remedies? (Both can be easy to obtain without seeing a health worker in many countries and their use or misuse can alter the appearances of skin lesions.)
- Does anyone else in the family or among your friends have a similar problem?

Remember to ask about reduced sensation in the limbs or in the vicinity of the lesion.

4.2 Examining the skin

Remember these points when conducting an examination of the skin, taking care to avoid any discomfort to the patient:

- Examine the skin, where possible, in good light. Using a poorly lit house or room does not provide the best environment to see the changes.
- Observe the general condition of the patient. Is the patient comfortable or appear weak, in pain, or having difficulty breathing or walking?
- Look at as much of the skin surface as possible and not just the area pointed out by the patient. Step back and examine from a distance. Are there colour changes (e.g. dark brown, pale)? Are most of the skin lesions located in one area of the body or do they affect many parts of the skin?
- Look closely at the following characteristics of skin lesions:
 - What types of skin lesions do you see? Lumps, patches, ulcers, or swelling?
 - What is their colour? Flesh-coloured, red, white, brown, black, yellow?
 - What is the size of most of the lesions? Are they tiny or large? If possible, measure the size of ulcers.
 - Are there any scratch marks?
- Feel the texture of the lesions. Remember to ask if the lesions are painful before touching them. Are they hard or soft? Is the surface rough? Wear gloves to examine broken skin such as ulcers.
- It may be necessary to ask the patient to remove clothes. Bear in mind that examining the skin should also be subject to the normal standards of consent and behaviour. Most patients are modest and shy about exposing themselves, even to health workers, and so providing them with suitable privacy to carry out the examination is important.
- For female patients, examination is often best carried out by female staff. Equally, local traditions of social behaviour have to be respected as it may be difficult to view intimate areas such as genitalia and breasts even where the presence of lesions may help with the diagnosis.

As you will read later, it is often important to add specific additional steps to your questions or examination.

For instance, if you suspect leprosy it will be important to test the lesion or the hands and feet for sensory loss as well as feeling for (palpating) enlarged peripheral nerves. If you suspect that scabies may be important, look for tiny tracks (short scaly lines) or burrows in the skin – these are best seen between the fingers, on the wrists and around the ankles.

In interpreting your findings, remember the natural geography and distribution of NTDs. Many only occur or are more common in certain countries. Statistics on disease prevalence may not be accurate, and even in areas where individual diseases have been reduced to a low level, it is important to be vigilant for the odd case that has evaded detection, as recognizing these patients will prevent re-emergence of the disease.

Table 1. Distribution of targeted skin NTDs by region

Diseases	Region					
	Africa	Asia	Americas	Europe	Eastern Mediterranean	Pacific
Buruli ulcer	++	+	+	–	–	+
Cutaneous leishmaniasis	+	+	++	+	++	–
Post kala-azar dermal leishmaniasis	+	+	–	–	+	–
Leprosy	+	+	+	–	+	+
Lymphatic filariasis	++	++	+	–	+	++
Mycetoma	+	+	+	–	+	–
Onchocerciasis ¹	+	–	+	–	+	–
Scabies	++	++	++	+	+	++
Yaws	+	+	+	–	–	+

++ = Common; + = Present but not common; – = Absent

¹ = Common in endemic areas

5. USING KEY SKIN SIGNS TO MAKE THE DIAGNOSIS

This section is subdivided for each of the main clinical signs in the skin: lumps, ulcers, swollen limbs, patches and itching. Diagnostic charts are included and should be used in conjunction with the NTD distribution maps (Figs 1–9).

Table 2. Summary of skin NTDs and the four main clinical signs and one symptom in the skin; itch may accompany any of these signs

Diseases	Lumps	Ulcers	Swollen limbs	Patches	Itchy skin
Buruli ulcer	+	+	+	+	–
Cutaneous leishmaniasis	+	+	–	+	–
Post kala-azar dermal leishmaniasis	+	–	–	+	–
Leprosy	+	+	–	+	–
Lymphatic filariasis	+	+	++	+	– ¹
Mycetoma	+	+	+	–	–
Onchocerciasis ²	+	–	–	+	+
Scabies	+	–	–	–	+
Yaws	+	+	–	+	–

++ = Common; + = Present but not common; – = Absent

¹ = Occasionally, itch in lymphatic filariasis if inflamed.

² = In some areas, patients with onchocerciasis have few skin changes.

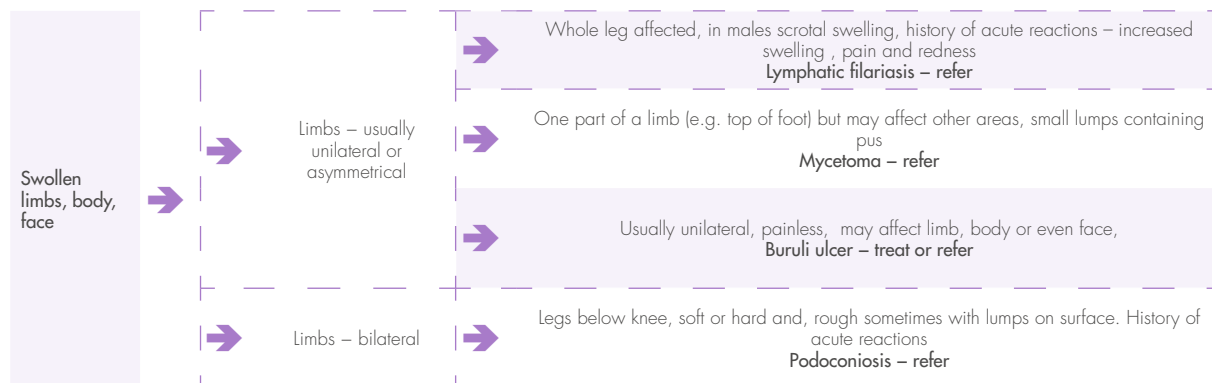
Lumps on or under the skin

Lumps on the skin	→	1–3	→	Solid (large)	→	Single or few solid lumps under skin Onchocerciasis – refer
					→	Usually limbs, firm, not painful and ulcerate later Buruli ulcer (early stage) – treat or refer
					→	Anywhere (exposed sites), ulcerates later Cutaneous leishmaniasis – treat or refer
					→	Single site liable to injury (e.g. foot), single hard lump Mycetoma – treat or refer
					→	Lumps on multiple sites, bumpy surface or yellow crust Yaws – treat
					→	Soft lumps, any area, contains pus, warm surface Abscess – treat or refer
	→	> 3	→	Soft, contains pus or fluid	→	Multiple, small lumps, smooth surface Post kala-azar dermal leishmaniasis – refer
					→	Multiple solid lumps – anywhere – enlarged nerves Leprosy – refer
					→	Multiple small itchy lumps, trunk, buttocks and/or multiple large lumps under skin Onchocerciasis – refer
					→	Multiple small, very itchy lumps on body, hands, feet, external genitalia, family affected Scabies – treat
					→	Multiple small soft pus-filled lumps over swollen limb Mycetoma – refer
					→	Small soft pus-filled lumps or dried crusts – common on face, itch Impetigo (pyoderma) – treat
	→					One to many small lumps; dark or pus-filled centre, feet, itchy Tungiasis – treat

Skin ulcers

Skin ulcers Ulcers can develop on swollen limbs e.g. lymphatic filariasis and podoconiosis	→	→	Ulcer raised above skin surface /with red, pebbly base	→	Cutaneous leishmaniasis – refer
		→	Red moist ulcer, clear or crusted base	→	Yaws – treat
			Legs or lower arms, ulcer of variable onset but regular oval to round	→	Other tropical ulcers such as <i>Fusobacterium</i> or <i>Haemophilus</i> infections – treat or refer
		→	Painless ulcer reaching large size – undermined edge	→	Buruli ulcer – treat or refer
			→	→	Other skin patches or lumps, enlarged peripheral nerves Leprosy ulcer – refer
		→	Reduced sensation around ulcer	→	Known diabetic Diabetic ulcer – refer
			→	→	Arterial ulcer – refer
		→	Arterial pulses absent	→	→
			→	→	Stasis or varicose ulcers – refer
		→	Irregular ulcers on lower legs associated with enlarged veins	→	→

Swollen limbs, bodies and faces



Patches on the skin

Patches	→	Non-itchy	→	Scaly dry surface, reduced sensation, reduced pigmentation Leprosy – refer
			→	Raised, scaly, increased pigmentation Leishmaniasis – refer
			→	Flat on skin surface, small oval patches joined together Post kala-azar dermal leishmaniasis – refer
			→	Front of shins or legs, speckled appearance with loss of pigment Onchocerciasis – refer
			→	Anywhere, with paler pigmentation but no loss of sensation; lumps or ulcers may be present Yaws – treat
	→	Itchy	→	Raised, dark and scaly, mainly in one area (e.g. one leg) Onchocerciasis – refer
			→	1–3 patches, scaly, raised edge, scalp in children with hair loss Fungal skin Infection – treat
			→	> 1 or more patches, scaly and raised, body folds Eczema – treat or refer

5.1 Lumps

There is a logical approach to the diagnosis using this feature. In examining the patient ask yourself these questions:

1. If the patient has a lump on the skin surface, is it small or large? (Small lesions measure under 0.5 cm in diameter, e.g. the diameter of a pencil.)
2. How many lumps are there? Group them into (i) 1 and up to 3 or (ii) more than 3.
3. Where are the lumps? On the arms or legs or are they more widely scattered over the body?
4. Do they feel soft or firm/hard?
5. If soft, do they contain fluid or pus?
6. Are they painful or warm when they are touched? .
7. Are they itchy?
8. Is the surface smooth or bumpy?
9. Are there multiple lumps on swollen limbs?

Most NTDs are associated with lumps on the skin. Their descriptions are as follows:

- A. **Yaws.** Usually single or up to three; scattered; yellow crusts may cover a soft, red bumpy (like a raspberry) surface. But there may be more lesions in some cases
- B. **Onchocerciasis.** Onchocercal nodules – single/few; overlying bony prominences, for example iliac crest of pelvic girdle. Nodules may also be seen on other sites including the shoulders, side of the chest or even head; firm, smooth surface. Onchocerciasis may also present with multiple small very itchy lumps on the trunk and limbs. Often the tops of these small lumps have been scratched
- C. **Buruli ulcer (early stage).** Single/few; any part of the body; usually, firm, smooth surface, slightly mobile, painless
- D. **Leishmaniasis.** Single/few, if more than 1, in groups anywhere on the skin surface; hard, bumpy or smooth surface. The surface may be covered by a crust or may ulcerate
- E. **Leprosy.** (less common as cause of lumps) – multiple; anywhere on the skin, hard, smooth. Ear lobes, eye brows may be affected. Check for enlarged nerves and peripheral sensory loss. Test for sensory loss using a pointed object such as a pen. Explain to the patient what you are going to do
- F. **Mycetoma.** Single and large; anywhere on the body but feet are most commonly affected; often warm to touch; surface covered with small bumps (pus-filled or oozing)
- G. **Post kala-azar dermal leishmaniasis.** Single/few. Smooth surface; any site but may also present with multiple smooth small lumps

- H. **Scabies.** Multiple small itchy lumps; anywhere on the body apart from the head except in babies – common sites are wrists, fingers, ankles, shoulders, buttocks, external genitalia (males); soft, often containing pus, small curved or squiggly lines (tunnels) may be seen that correspond to the burrows where the mites reside in the skin; very itchy – other household members often affected
- I. **Tungiasis.** Single or multiple small lumps; commonest on the feet – soles or on or between the toes – but also may occur in other sites such as body folds e.g. groin. These are itchy and may be filled with pus or have a central dark spot

Common skin diseases that also present with lumps:

- J. **Abscesses.** Single or few; large, anywhere; soft and containing pus; painful, red and warm to touch
- K. **Multiple small pebbly lumps or warty growths** may occur on long standing lymphoedema of lower limbs caused by other conditions including lymphatic filariasis. These are firm with a rough warty surface resembling the skin of elephants. It is called elephantiasis).

5.2 Ulcers

Ulcers can occur anywhere on the skin although they are commoner in areas of the body that are subject to trauma. A number of different NTDs cause skin ulceration such as Buruli ulcer, cutaneous leishmaniasis, dracunculiasis (guinea-worm disease) and yaws. In some cases, ulceration starts from a pre-existing lump on the skin. Additionally, a number of common infections also cause skin ulcers and can occur in areas where NTDs are endemic; these include other tropical ulcers caused by *Haemophilus* (slow onset) or *Fusobacterium* (rapid onset), *Corynebacteria* (diphtheria) and tuberculosis of the skin. Also if the sensation of the skin is damaged because of nerve damage (neuropathy) ulceration can occur, usually following unnoticed trauma. Neuropathic ulceration can occur in leprosy, but also follows nerve damage due to other diseases such as diabetes.

In examining patients with skin ulcers, it is important to ask for information about the presence or absence of pain and the speed of development – has the ulcer appeared suddenly, for example within 1–2 days? The overall appearance is also important. For instance, is the shape of the ulcer edge regular or irregular? Look closely at the edges. Are these undermined? In other words, is it possible to insert a sterile probe under the edge of the ulcer?

Questions to ask in reaching a diagnosis

1. Are the ulcers single or multiple? Group them into (i) 1 and up to 3 or (ii) more than 3
2. Are they localized to one body region such as the lower leg or arm or are they scattered across different body sites?
3. Was there any preceding trauma?
4. Have they appeared slowly or rapidly, i.e. within a few days?

5. Are the edges undermined, i.e. can you insert a probe under the edge of the ulcer?
6. Is there any swelling around the ulcer?
7. Is the base red and clean or covered with a coloured crust, e.g. yellow?
8. Was there a preceding lump at the same site?
9. Are they painful?

Specific NTDs

- A. **Buruli ulcer.** Usually 1 but up to 3; mainly limbs; slow development from lump; deep, undermined edges, red base often covered by dead-looking tissue (in the beginning); painless; sometimes associated with swelling
- B. **Cutaneous leishmaniasis.** 1–3; scattered – exposed parts of the body; slow development from a lump; edges are raised above the skin surface; clean or crusted base; not usually painful
- C. **Yaws.** Usually single (sometimes more); development from a lump (ulcers occasionally co-exist with existing lumps); edges are raised above the skin surface; clean or yellow crusted base
- D. **Leprosy.** 1–3 lesions; often on feet and hands; irregular edge and crusted base; painless; loss of sensation in area; often associated with injury (trauma, burns, ill-fitting shoes, etc) but the patient does not notice. Look for other signs of leprosy: lumps, patches and enlarged nerves.
- E. **Lymphatic filariasis ulcers.** This follows trauma such as rubbing. Usually one lesion on swollen limb. Ulcers may also develop with other NTDs that present with limb swelling.

Other causes (differential diagnosis) of ulceration include other tropical infective ulcers e.g. *Haemophilus*, *Fusobacterium*, diabetes, arterial and stasis/venous ulcers as follows:

- F. **Other tropical infective ulcers, e.g. those caused by *Fusobacterium* or *Haemophilus*.** 1–3 lesions; legs or hands; variable onset; sometimes painful; clean red base
- G. **Diabetic ulcers.** Usually 1–3 on feet at sites of rubbing from shoes; painless; irregular with crusted base; there is diminished sensation indicating nerve damage in the area around the ulcer. Feel for arterial pulses. Ask about history of diabetes.
- H. **Arterial ulcers.** Similar in appearance to diabetic ulcers although they may also be painful. A key feature is the reduction or absence of arterial pulses on the foot. It is important to feel for these pulses in all ulcer patients as tight bandaging is contraindicated where there is arterial damage.
- I. **Stasis ulcers.** 1–2; may be bilateral at ankles; irregular shape, ulcer base covered with crust; may be painful. Presence of varicose (enlarged and visible) veins higher up on leg.

5.3 Swollen limbs, face or body

Swelling of the limbs or appendages such as the scrotum or elsewhere on the body is usually a sign of the accumulation of fluid in the tissue. Where there is a lot of fluid the swelling is soft and when pressed firmly with a finger the imprint of the finger remains on the skin surface. Even with fluid accumulation, with time the swelling becomes hard due to fibrosis or scarring and the skin surface may become irregular and covered with small lumps, a condition often known as “mossy foot”. This is simply a sign of chronic fluid accumulation in the skin, hyperkeratosis or skin thickening that has become secondarily infected with bacteria. The surface may appear bumpy and fluid may ooze from between the bumps. Inflammation in the skin can also cause swelling, although usually the area is warm to the touch and often tender when pressed – this usually indicates infection. Chronically swollen limbs may become secondarily infected as an “acute event” on top of the existing swelling which becomes red and sore and may be associated with systemic illness such as fever.

Useful discriminating questions include noting whether the swelling affects both sides such as both legs and, in males, whether there is scrotal swelling.

Questions to ask in reaching a diagnosis

1. Does the swelling affect one side e.g. one leg or arm but not, or less, on the other?
2. Is the swelling hard or soft?
3. How long has it been present?
4. Does it change (reduce) after lying down overnight?
5. Is the swelling irregular with one or more lumps and to patients is the limb heavy and difficult to lift, e.g. while walking?
6. Is the swelling uniform, with a smooth surface and soft to the touch?
7. Is it tender (painful) to touch?
8. Are there small lumps which may contain pus in the swollen area?
9. Is there any discharge from the surface of the swollen area?

Specific NTDs

- A. **Lymphatic filariasis.** Unilateral or bilateral swelling on legs – in males scrotal swelling or hydrocele and in women vulval swelling may occur; non-tender unless seen during an acute attack; swelling soft and reducible in early cases, but hard (firm to touch) and irreversible in older lesions. There may be a discharge which is often foul smelling
- B. **Mycetoma.** Unilateral and localized swelling to a single site; feet commonly affected but may occur elsewhere e.g. chest wall or head; not painful; surface of swelling may be studded with small soft pus-containing lumps
- C. **Buruli ulcer.** Before ulceration occurs on a limb, the surface of the skin is usually swollen. Also, once an ulcer has formed there is usually a lot of swelling around the ulcer.

The differential diagnosis of limb swelling includes podoconiosis.

D. Podoconiosis. Usually bilateral on legs – usually limited to below the knee. It is generally non-tender unless seen during an acute episode. The swelling is soft and reducible in some cases, but hard, stiff and not compressible in other lesions. There is usually no discharge apart from leaking lymphatic fluid. Later, hard nodules may develop on the surface.

NB. Swelling of both legs may occur in heart failure but the patient usually has additional symptoms such as breathlessness, tiredness and palpitations.

5.4 Patches on the skin

Patches on the skin are flat areas where the appearance and texture of the skin, and sometimes the colour, has altered. Patches may be slightly raised above the rest of the skin surface or distinctive in other ways from surrounding skin, because the surface is rough or the colour different. The surface is sometimes smooth but it may be rough or scaly. The pigmentation is also important as in some patients the patch is lighter in colour than the surrounding skin. It is always important to test the sensation of patches (see above Lumps – leprosy). If they indicate that the sensation has altered, more detailed testing of sensation is indicated along with a search for enlarged peripheral nerves. Another important discriminating sign is the presence of itching, dryness and hair loss.

NTDs associated with patches include leprosy, onchocerciasis, yaws, and cutaneous leishmaniasis and post kala-azar dermal leishmaniasis.

Use as discriminators: Itch, sensory loss, colour change.

Questions to ask in reaching a diagnosis

1. Are the patches single or multiple?
2. Is the surface raised?
3. Does the patch itch?
4. Is there any change in sensation on the lesions?
5. Is there any loss of pigmentation?
6. Is there any increase in pigmentation?
7. Is the surface scaly?

Specific NTDs

A. Leprosy. Single or multiple patches. These may be raised or flat on the skin surface, surface dry with reduced pigmentation (paler), non-itchy. Sometimes the patches may be difficult to see so also look at the patient from several steps away. Test for loss of sensation and look for enlarged peripheral nerves.

- Touch the skin lightly with the pen and ask the person to point to where they felt the pen. Then ask them to close their eyes so that they cannot see what you are doing. Lightly touch the centre of the most prominent skin patch and ask them to point to where they felt the pen. Repeat the procedure on normal skin and on the same patch again. If the person feels nothing on the skin patch, it suggests leprosy.
- B. **Onchocerciasis.** Raised very itchy patches involving large parts of a single limb, darker than surrounding skin. Surface rough and dry. Examine for large lumps in the area (onchocercal nodules – see Lumps – Onchocerciasis).
 - C. **Leishmaniasis.** The development is similar to that seen with lumps (see above); edges are raised above the skin surface; often ulcerate; not usually painful
 - D. **Post kala-azar dermal leishmaniasis.** Multiple small flat patches which are pale in colour and may join together are also a presenting feature.
 - E. **Yaws.** Multiple flat patches which may be lighter in colour than surrounding skin. This usually occurs in the presence of typical forms of the disease such as lumps or ulcers.

Some common skin disease also present with patches on the skin. These include fungal infections and eczema. It is also very common for patients to present with discolouration – increased pigmentation of the skin following any form of skin inflammation.

- F. **Fungal skin infection.** Usually 1–3 patches. Dry scaly and itchy. The edge of the patch is more prominent, ie it sticks out and is more scaly, than the central areas like a ring. In children there may be single or multiple patches with loss of hair in the scalp – these patches may also be scaly.
- G. **Eczema.** Single or multiple very itchy patches. Scaling across the entire surface; may involve the hand or feet or the flexures of the skin – the creases behind the knees, elbows or around the neck – common in children.

5.5 Itchy skin

Itching of the skin is an important feature which is helpful in making a diagnosis. It may affect all the body surface (generalized) and the patients' first description of the condition is often that it is very itchy. In other cases, the itching is confined to a specific lesion such as a patch or a lump. Patients who itch usually scratch the affected area and leave visible scratch marks where the broken skin has been bleeding. In very persistent itchy lesions, the skin becomes thickened and may form a raised patch with darkened skin colour. This is the direct result of persistent scratching and rubbing and is known as lichenification (this process is listed under "patches" in this guide). This form of thickened skin is seen on one limb in some forms of onchocerciasis.

Another important point is to examine patients with itchy skin for any visible skin changes. If there are none, also ask if there was a lumpy rash (“wheals”) that comes and goes within a short time; this is a change known as urticaria and it may occur with certain parasitic infections such as in the early stages of schistosomiasis. A further key question is to ask if any other members of the household are itching, as this indicates that the condition can be passed from one person to another. This is typical of scabies.

Useful questions to include:

- Is the itching confined to one area or does it affect many parts of the body?
- Are other members of the household itching?
- Are there patches or lumps or wheals?

A. Onchocerciasis. Itching is associated with multiple small lumps on the trunk and limbs or patches affecting one limb only. Examine for large lumps under the skin – onchocercal nodules.

Common skin diseases associated with itching include **scabies**, **fungal infections** and **eczema**. Their features are described above. Remember that itching in the other household members is associated with scabies.

5.6 When to consider underlying HIV infection

HIV/AIDS is prevalent in some regions of the world. Skin presentations in HIV-positive patients are common although most of the NTDs discussed above are not often associated with HIV. However, HIV can be associated with changes in the severity of infections such as cutaneous leishmaniasis and in their presentation, which is usually more widespread. Some simple clues that might alert the health worker to investigate for the possible presence of HIV are:

- Sore mouth – white patches on the tongue and sides and roof of the mouth – oral candidiasis
- Persistent genital ulcerations – these may occur with sexually-transmitted infections including syphilis and herpes simplex
- Widespread and sudden appearance of red patches with scaly surfaces – psoriasis (often itchy) or secondary syphilis (non-itchy)
- Scaly patches on the face – such as around the nose, ears and in the eyebrows – seborrhoeic dermatitis. But may spread to affect large areas of the body.
- Red/purple patches or lumps on the trunk or limbs that appear like bruises and do not have a scaling surface, although some are wart-like. Examine the inside of the mouth and see if there are similar bruises on the roof of the mouth – Kaposi sarcoma.

- Multiple non-symptomatic small lumps on the face or body. Many of these have a small depression on the surface which reveals a soft centre. These may be caused by the virus infection – molluscum contagiosum – but they may also be the presenting sign of a systemic fungal infection such as cryptococcosis.
- Itchy skin – small lumps with extensive itchiness, can be localized or generalized – pruritic papular eruption or eosinophilic folliculitis.

More information is available from:

http://www.who.int/maternal_child_adolescent/documents/skin-mucosal-and-hiv/en/

<http://hivinsite.ucsf.edu/InSite?page=kb-04-01-0>

6. NEXT STEPS

The diagnostic charts shown in **Section 5** are intended to allow health-care workers to reach the next phase of diagnosis. It is helpful to use these in combination with the distribution maps. The common skin diseases such as scabies, fungal infections, pyoderma and eczema are seen in all regions. In the case of NTDs, further investigation or management often involves referral to a specialist centre or trained team. Generally, this will apply to all patients who present with either ulcers or swollen limbs or bodies. The diagnostic charts indicate which of those patients presenting with lumps or patches should be referred to a specialist service which, depending on local organization, may be the nearest health centre/hospital or a tertiary referral centre. In most of these patients, arriving at the diagnosis depends on the use of specialist tests (blood tests, direct microscopic examination, histopathology, culture). Tests currently in use are listed in **Table 4**.

For the common skin infections, it is usually not necessary to carry out any diagnostic tests, although they are available in some centres. These diseases are indicated in the diagnostic flow charts (Treat). The conditions respond well to treatments (**Table 5**). If the disease is not responding to treatment, seek advice on the next steps.

Table 3. Common epidemiological and clinical diagnostic clues

Diseases	Age	Gender	Pain	Fever	Lesion location	Transmission
Buruli ulcer	Child/Adult	Both	No	No	Mainly limbs	Unknown ? water
Cutaneous leishmaniasis	Child/Adult	Both	No	No	Exposed sites	Sandfly bite
Post kala-azar dermal leishmaniasis	Mainly Adult	Both	No	No	Anywhere	Sandfly bite
Leprosy	Child/Adult	Both	Loss of sensation	No – apart from reactions	Anywhere – ulcers on hands or feet	Close contact over time
Lymphatic filariasis	Mainly Adult	Both	Only in acute attacks	Only during acute attacks	Swelling in lower limbs, scrotum	Mosquito Patients often no longer infected
Mycetoma	Child/Adult	Both	No	No	Limbs or site of trauma	Trauma e.g. thorn injury
Onchocerciasis	Child/Adult	Both	No but may be itchy	No	Anywhere – but limbs are commonly affected	Blackfly bite
Scabies	Child/Adult	Both	No but itchy	No	Anywhere – hands, external genitalia, feet, buttocks	Other household members who also itch
Yaws	Child/Adult	Both	No	No	Anywhere, but often on the limbs	Contact

Table 4. Summary diagnosis and treatment

Diseases	Investigation	Laboratory tests	Medical treatment	Supportive care
Buruli ulcer	Swab	fTLC and PCR of skin swab/FNA samples	Oral rifampicin + oral clarithromycin for 8 weeks	Wound dressing, physiotherapy as needed depending on site and size
Cutaneous leishmaniasis	Swab/biopsy	Microscopy and PCR of skin swab	Stibogluconate, paromomycin, miltefosine, amphotericin B	No
Post kala-azar dermal leishmaniasis	Slit skin smear or biopsy	Microscopy, PCR	Miltefosine, amphotericin B	
Leprosy	Slit skin smear or skin biopsy	Direct microscopy or histology	Multidrug antibiotic therapy for 6 or 12 months. e.g. rifampicin, dapsone, clofazimine	Care of insensitive areas (loss of feeling) plus regular wound inspection appropriate footwear, contact survey
Lymphatic filariasis ¹	Blood test	Rapid diagnostic tests ² , thick blood film (collected during hours of peak mf circulation)	Oral diethylcarbamazine for 12 days ± doxycycline for 4–6 weeks ³	Limb cleaning, exercise, wound care, anti-bacterial/anti-fungal creams when needed. Oral antibiotics and antipyretics during acute attacks
Mycetoma	Biopsy	Microscopy histology and culture of grains/biopsy	Depends on species. Long term antibiotic or antifungal ± surgery	Walking support
Onchocerciasis	Skin snip biopsy	Microscopy, PCR	Oral ivermectin, doxycycline in areas with Loiasis unless known to be low risk for adverse events	Emollients
Scabies	Clinical diagnosis, dermoscopy, videomicroscopy	Presence of mites scraping	Permethrin cream, sulphur (4%), ointment 15%, benzyl benzoate, oral ivermectin	Treat family and contacts
Yaws	Swabs	DPP test and PCR of skin swab	Single oral dose of azithromycin (2nd line: injectable benzathine penicillin)	Wound dressing, Treat family and contacts

BRT, *Brugia* rapid test; DPP, dual path platform; FNA, fine needle aspiration; fTLC, fluorescent thin layer chromatography; FTS, filarial test strip; PCR, polymerase chain reaction.

¹ For lymphoedema and hydrocele patients, diagnosis of infection is not required for case management.

² Alere Filariasis Test Strip for *W. bancrofti* (Abbott Scarborough (MN), USA); *Brugia* Rapid Test for *Brugia* spp. (Reszon Diagnostics International, Malaysia).

³ This regimen is altered in those regions where there is onchocerciasis or *Loa loa*. Single dose with regimen used in mass drug administration also acceptable.

Many skin lesions, from eczema to ulcers, can be secondarily infected by bacteria and this may be the reason for failure to respond to therapy. Bacterial entry points (BEP) should be treated in all lymphoedema patients. An important complication of lymphoedema is an acute episode of fever and redness of the leg (cellulitis). This is caused by the entry of bacteria through breaks in the skin such as athletes foot (cracking between the toes), wounds or cuts. Patients should be asked to check these entry points regularly to minimize the risk of infection. Eliminating cracking of the skin in the toe web spaces is the most difficult part of entry point care. Where there is athletes foot (a fungal infection between the toe) an antifungal cream can be used. Many patients have associated eczema of the legs (stasis eczema). (see below). Skin wounds, including ulcers, may also be entry points for more serious infections such as tetanus, and it is important therefore to ask about the patient's current immunization status.

Although treating skin disease in often simple be careful not too use powerful corticosteroid creams without being certain that the disease is not caused by an infection. Also patients can often obtain strong corticosteroid creams without prescription and their use will alter the appearance of infections. So it is important to ask patients if they have already treated their skin with any medication.

Table 5. Investigations and treatment of common non-NTD skin diseases and scabies

Diseases	Investigation*	Medical treatment	Supportive care
Bacterial skin infection – abscess	Swab for culture	Drain abscess. Oral cloxacillin, trimethoprim	Wound care
Bacterial skin infection – impetigo	Swab for culture	Topical creams (mupirocin) or antiseptics (povidone)	Treat contacts
Eczema		Emollients and medium strength steroid cream for 2–4 weeks	Avoid irritants. May relapse
Fungal infection	Scraping for mycology microscopy culture	Topical azole (clotrimazole, miconazole) creams. In severe cases, oral terbinafine or griseofulvin itraconazole	If in scalp do not share caps or hats
Scabies	Skin smear/dermoscopy	Permethrin cream, sulphur (4%) ointment. 1.5% benzyl benzoate. Oral ivermectin	Treat family and contacts
Tropical ulcers caused by <i>Fusobacteria</i> or <i>Haemophilus</i>	Smear for PCR	Depends on cause. Penicillin for 5 days or azithromycin	Wound dressing/grafting

*Often only available in specialist centres.

PCR, polymerase chain reaction

If, after going through the various tables and algorithms, there is still doubt about the diagnosis of the skin lesion, refer to a specialist service.

ANNEX. DESCRIPTIVE TERMS FOR SKIN LESIONS

The descriptive terms below adapted from the 2016 International League of Dermatological Societies' revised glossary for the description of cutaneous lesions <http://dx.doi.org/10.1111/bjd.14419>.

Included under lumps

Nodule. An elevated, solid, palpable lesion > 1 cm usually located primarily in the dermis and/or subcutis. The greatest portion of the lesion may be exophytic or beneath the skin surface

Papule. An elevated, solid, palpable lesion that is ≤ 1 cm in diameter

Included under patches

Plaque. A circumscribed, palpable lesion > 1 cm in diameter; most plaques are elevated.. But plaques may result from a coalescence of papules

Macule. A flat, circumscribed, nonpalpable lesion that differs in colour from the surrounding skin. It can be any colour or shape

Included under ulcers

Ulcer. Full-thickness loss of the epidermis plus at least a portion of the dermis; it may extend into the subcutaneous tissue

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Podoconiosis

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Additional links and resources

Neglected Tropical Diseases

http://www.who.int/neglected_diseases/en

International Foundation for Dermatology

<http://www.ifd.org/>

International League of Dermatology Societies

<http://web.ilds.org/>

Skin App

<http://thespindle.org/project/skin-app/>

Dermatología comunitaria (Mexican web site)

<http://www.dermatologiacomunitaria.org.mx/>

Africa Teledermatology project

<http://africa.telederm.org/>

Regional Dermatology Training Centre (Tanzania)

<http://rdtc.go.tz/>

The skin of a patient is the first and most visible structure of the body that any health-care worker encounters during the course of an examination. To the patient, it is also highly visible, and any disease that affects it is noticeable and will have an impact on personal and social well-being. The skin is therefore an important entry point for both diagnosis and management. Many diseases of humans are associated with changes to the skin, ranging from symptoms such as itching to changes in colour, feel and appearance. The major neglected tropical diseases (NTDs) frequently produce such changes in the skin, re-enforcing the feelings of isolation and stigmatization experienced by patients affected by these diseases. Indeed, these are often the first signs that patients will notice, even before changes in internal organs or other systems occur. The following NTDs all have prominent skin changes at some stage in their evolution.

BURULI ULCER CUTANEOUS LEISHMANIASIS POST KALA-AZAR DERMAL LEISHMANIASIS LEPROSY LYMPHATIC FILARIASIS MYCETOMA ONCHOCERCIASIS PODOCONIOSIS SCABIES YAWS (ENDEMIC TREPONEMATOSIS) BURULI ULCER CUTANEOUS LEISHMANIASIS POST KALA-AZAR DERMAL LEISHMANIASIS LEPROSY LYMPHATIC FILARIASIS MYCETOMA ONCHOCERCIASIS PODOCONIOSIS SCABIES YAWS (ENDEMIC TREPONEMATOSIS) BURULI ULCER CUTANEOUS LEISHMANIASIS POST KALA-AZAR DERMAL LEISHMANIASIS LEPROSY LYMPHATIC FILARIASIS MYCETOMA ONCHOCERCIASIS PODOCONIOSIS SCABIES YAWS (ENDEMIC TREPONEMATOSIS) BURULI ULCER CUTANEOUS LEISHMANIASIS POST KALA-AZAR DERMAL LEISHMANIASIS LEPROSY LYMPHATIC FILARIASIS MYCETOMA ONCHOCERCIASIS PODOCONIOSIS SCABIES YAWS (ENDEMIC TREPONEMATOSIS)

This training guide explains how to identify the signs and symptoms of neglected tropical diseases of the skin through their visible characteristics. It also contains information on how to diagnose and manage common skin problems that frontline health workers may encounter.

The guide is intended for use by frontline health workers without specialist knowledge of skin diseases. Pathways for more detailed investigation and/or management are described but, often, this will involve referral to a specialist service.

DEPARTMENT OF
CONTROL OF NEGLECTED TROPICAL DISEASES

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Revised WHO classification and treatment of childhood pneumonia at health facilities

• EVIDENCE SUMMARIES •



**World Health
Organization**

Revised WHO classification and treatment of childhood pneumonia at health facilities

EVIDENCE SUMMARIES



**World Health
Organization**

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Executive Summary

In the early 1980s, the global burden of childhood mortality due to pneumonia led the World Health Organization (WHO) to develop a pneumonia control strategy suitable for countries with limited resources and constrained health systems. Management of pneumonia cases formed the cornerstone of this strategy. Simple signs were identified to classify varying severities of pneumonia in settings with little or no access to diagnostic technology; the classifications determined the appropriate case management actions. Children with fast breathing were classified as having “pneumonia” and were given an oral antibiotic (at that time oral cotrimoxazole) to take at home for five days. Children who had chest indrawing with or without fast breathing were classified as having “severe pneumonia” and were referred to the closest higher-level health facility for treatment with injectable penicillin. Children who had any general danger signs were classified as having “severe pneumonia or very severe disease”. These children received a first dose of oral antibiotic and were then urgently referred to a higher-level health facility for further evaluation and treatment with parenteral antibiotics.

These pneumonia classification and management guidelines had been developed based on evidence generated in the 1970s and early 1980s, and were incorporated into the original version of Integrated Management of Childhood Illness (IMCI). In the intervening time, new evidence has emerged which prompted the development of revised guidelines.

Research results provided solid scientific evidence to guide and support the revision of the pneumonia guidelines. During two related consultations, a panel of experts assessed the new evidence according to the GRADE methodology (“Grading of Recommendations, Assessment, Development and Evaluation”). The consultations aimed to summarize the new WHO recommendations for policy and practice, to review GRADE evidence profiles, and to discuss the factors that determined the strength of the recommendations. The first consultation resulted in updated recommendations for preventing and managing pneumonia in HIV-infected and -exposed infants and children; these were published in 2010.¹ The second resulted in updated recommendations for managing pneumonia in non-HIV affected infants and children, published in 2012.²

The revisions include changing the recommendation for the first-line antibiotic and re-defining the classification of pneumonia severity. The data show that oral amoxicillin is preferable to oral cotrimoxazole for the treatment of “fast breathing pneumonia” and is equivalent to injectable penicillin/ampicillin in cases of “chest indrawing pneumonia”. Hence, in a programmatic context, the distinction between previously defined “pneumonia” (fast breathing) and “severe pneumonia”

¹ Integrated Management of Childhood Illness (IMCI). WHO recommendations on the management of diarrhoea and pneumonia in HIV-infected infants and children. Geneva: World Health Organization; 2010 (http://www.who.int/maternal_child_adolescent/documents/9789241548083/en).

² Recommendations for management of common childhood conditions, Evidence for technical update of pocket book recommendations. Geneva: World Health Organization; 2012 (http://www.who.int/maternal_child_adolescent/documents/management_childhood_conditions/en).

(chest indrawing) loses its significance. The new classification is therefore simplified to include only two categories of pneumonia; “pneumonia” with fast breathing and/or chest indrawing, which requires home therapy with oral amoxicillin, and “severe pneumonia”, pneumonia with any general danger sign, which requires referral and injectable therapy.

Dosages for pneumonia treatment at health facilities have been revised to reflect three age bands: 2 months up to 12 months (4–<10 kg); 12 months up to 3 years (10–<14 kg); 3 years up to 5 years (14–19 kg). Dosages and age bands for treatment of fast breathing pneumonia by community health workers (CHWs) have not changed.

National child health programmes will benefit from the revised recommendations and are encouraged to incorporate them into their existing guidelines for care at health facilities. The recommendations concerning the use of amoxicillin should also be included in guidelines for integrated community case management (iCCM). Programmes should recognize the importance of these revisions, which will result in a substantially lower need for referral, and in better treatment outcomes. Local adaptations may be required, particularly the arrangements to include amoxicillin as the first-line therapy; facility-level health workers will also need to be re-trained in the new system of classification and treatment.

The purpose of this document is to provide a summary of WHO-approved recommendations,^{1,2} and the evidence supporting them, and to assist national child health programmes in revising their guidelines to conform to the new recommendations.

The revised recommendations are:

Recommendation 1

Children with fast breathing pneumonia with no chest indrawing or general danger sign should be treated with oral amoxicillin: at least 40mg/kg/dose twice daily (80mg/kg/day) for five days. In areas with low HIV prevalence, give amoxicillin for three days.

Children with fast-breathing pneumonia who fail on first-line treatment with amoxicillin should have the option of referral to a facility where there is appropriate second-line treatment.

Recommendation 2

Children age 2–59 months with chest indrawing pneumonia should be treated with oral amoxicillin: at least 40mg/kg/dose twice daily for five days.

Recommendation 3

Children aged 2–59 months with severe pneumonia should be treated with parenteral ampicillin (or penicillin) and gentamicin as a first-line treatment.

- Ampicillin: 50 mg/kg, or benzyl penicillin: 50 000 units per kg IM/IV every 6 hours for at least five days
- Gentamicin: 7.5 mg/kg IM/IV once a day for at least five days

Ceftriaxone should be used as a second-line treatment in children with severe pneumonia having failed on the first-line treatment.

Recommendation 4

Ampicillin (or penicillin when ampicillin is not available) plus gentamicin or ceftriaxone are recommended as a first-line antibiotic regimen for HIV-infected and -exposed infants and for children under 5 years of age with chest indrawing pneumonia or severe pneumonia.

For HIV-infected and -exposed infants and for children with chest indrawing pneumonia or severe pneumonia, who do not respond to treatment with ampicillin or penicillin plus gentamicin, ceftriaxone alone is recommended for use as second-line treatment.

Recommendation 5

Empiric cotrimoxazole treatment for suspected *Pneumocystis jirovecii* (previously *Pneumocystis carinii*) pneumonia (PCP) is recommended as an additional treatment for HIV-infected and -exposed infants aged from 2 months up to 1 year with chest indrawing or severe pneumonia.

Empirical cotrimoxazole treatment for *Pneumocystis jirovecii* pneumonia (PCP) is not recommended for HIV-infected and -exposed children over 1 year of age with chest indrawing or severe pneumonia.

Introduction

Pneumonia continues to be the biggest killer worldwide of children under five years of age. Although the implementation of safe, effective and affordable interventions has reduced pneumonia mortality from 4 million in 1981 (1) to just over one million in 2013 (2,3), pneumonia still accounts for nearly one-fifth of childhood deaths worldwide.

Case management is a cornerstone of pneumonia control strategies (4). It consists of classifying the severity of illness using simple clinical signs such as fast breathing, chest indrawing and general danger signs,³ and then applying the appropriate treatment. Treatment includes home care advice, antibiotics for home therapy, or referral to a higher-level health facility.

The recommendations for the management of pneumonia in health facilities have recently been modified (5–7). The revisions are a result of new evidence generated from research carried out in the last decade including in those low- and middle-income countries with the largest burden of pneumonia mortality. Integrated Management of Childhood Illness (IMCI) at health facility level has been updated accordingly. Integrated Community Case Management (iCCM) by lay community health workers (CHWs) has not been changed.

The original guidelines classified the respiratory symptoms of children 2 to 59 months of age into four categories (8,9). Children with cough and cold who did not have signs of pneumonia were classified as “no pneumonia”, and their caregivers were advised on appropriate home care. Children with fast breathing were classified as “pneumonia” and were given an oral antibiotic (at that time oral cotrimoxazole) to take at home for five days. Children who had chest indrawing with or without fast breathing were classified as “severe pneumonia” and were referred to the closest health facility for treatment with injectable penicillin. Children who had any general danger signs were classified as “severe pneumonia or very severe disease”. These children received a first dose of oral antibiotic and were then urgently referred to a health facility for further evaluation and treatment with parenteral antibiotics.

Data shows that the majority of childhood pneumonia deaths are due to severe pneumonia/severe disease (10); management of these cases requires early identification, prompt referral and the availability of good-quality higher-level care. However, in many low-resource settings, referral is difficult and often does not take place (11–15). On the basis of this information, WHO undertook a review of the evidence, with the aim of developing a simplified approach that could increase the number of children receiving correct treatment for pneumonia.

³ Not able to drink, persistent vomiting, convulsions, lethargic or unconscious, stridor in a calm child or severe malnutrition.

In two consultations, experts updated recommendations for 1) preventing and managing pneumonia in HIV-infected and -exposed infants and children (9) and 2) for managing pneumonia in non-HIV-affected infants and children (5). The recommendations were published in 2010 and 2012 respectively.

This document presents a summary of recommendations and of the research that forms the technical basis of the revised guidelines; it also suggests implications for implementation in countries. Its purpose is to assist national child health programmes in revising their guidelines to conform to the new recommendations. These revisions, implemented in countries with a high pneumonia burden, will result in a higher proportion of children receiving care at the outpatient or community levels, and a reduced number of pneumonia-related deaths.

1. Scientific basis of WHO recommendations for treatment of pneumonia

1.1 Recommendation 1

Children with fast breathing pneumonia⁴ with no chest indrawing or general danger sign should be treated with oral amoxicillin: at least 40 mg/kg/dose twice daily (80mg/kg/day) for five days. In areas with low HIV prevalence, give amoxicillin for three days.

Children with fast-breathing pneumonia who fail on first-line treatment with amoxicillin should have the option of referral to a facility where there is appropriate second-line treatment.

1.1.1 The effectiveness of various antibiotics in community-acquired childhood pneumonia: a systematic review

Kabra et al (16) systematically reviewed literature comparing various antibiotics, in order to identify effective drug therapy for community-acquired pneumonia in children. All randomized controlled trials (RCTs) that compared at least two antibiotics for community-acquired pneumonia in children in hospital or ambulatory settings were selected.

The review showed that for treatment of pneumonia, cotrimoxazole was inferior in efficacy to both amoxicillin (clinical failure rates odds ratio (OR) 1.33; 95% CI 1.05 to 1.67) and procaine penicillin (clinical cure rates OR 2.64; 95% CI 1.57 to 4.45). Injectable penicillin in conjunction with injectable gentamicin was more effective than injectable chloramphenicol alone (re-hospitalization rates OR 1.61; 95% CI 1.02 to 2.55). Co-amoxycyclavulanic acid was more effective than amoxicillin alone (cure rates OR 10.44; 95% CI 2.85 to 38.21). There were no differences between injectable penicillin and oral amoxicillin (failure rates OR 1.03; 95% CI 0.81 to 1.31); between azithromycin and erythromycin; between cefodoxime and amoxicillin; or between azithromycin and co-amoxycyclavulanic acid.

The authors concluded that for treatment of ambulatory patients with community-acquired pneumonia, amoxicillin was more effective than cotrimoxazole. Moreover, in hospitalized patients, penicillin was found to be more effective than cotrimoxazole; and the combination of penicillin and gentamicin was superior to chloramphenicol. Injectable penicillin and oral amoxicillin had similar clinical failure rates.

Another systematic review (17) was conducted by nine clinicians and researchers with extensive experience in the field of childhood pneumonia. The review aimed to identify appropriate first- and second-line antimicrobial agents for the empirical treatment of fast breathing pneumonia in children by first-level health care providers, in addition to defining treatment failure and criteria for referral. For the first-line treatment oral amoxicillin was found to be most effective in the treatment of fast breathing pneumonia. The review suggested that treatment failure for fast breathing pneumonia should be redefined as “clinical deterioration” only, instead of changing therapy when the child’s condition is the “same” on follow-up after two or three days of treatment. It was sug-

⁴ Previously classified as pneumonia (IMCI Chart Booklet 2008).

gested that once classified as “treatment failure” each child should be assessed independently for referral as not all cases will mandate immediate referral. The review recommended the development of an algorithm to determine the best course of action when treatment failure occurs.

1.1.2 A three-day course of antibiotics is as effective as a five-day course in treating children with fast breathing pneumonia

A systematic review by Haider et al (18) reviewed three randomized controlled trials evaluating the efficacy of short-course versus long-course treatment of fast breathing pneumonia in children age 2–59 months. Two of the studies (19,20) examined treatment with oral amoxicillin, while the third (21) examined treatment with oral cotrimoxazole. All studies compared the duration of two courses of treatment, one lasting three days and the other five days, while holding the antibiotic constant. The three studies were double blind, randomized, placebo controlled trials with a total of 6210 participants. Data was available for 5763 cases. The primary outcome was clinical cure at the end of the treatment. The studies found no significant differences in clinical cure rates between the two groups (RR 0.99; 95% CI 0.97 to 1.01). Nor were there significant differences in either the treatment failure or relapse rates between groups. The review recommended a shorter course of antibiotic therapy, keeping in mind the benefits to the individual as well as to the health system, especially in settings with limited resources.

1.2 Recommendation 2

Children age 2–59 months with chest indrawing pneumonia⁴ should be treated with oral amoxicillin: at least 40mg/kg/dose twice daily (80mg/kg/day) for five days.

Previous guidelines by WHO for the management of chest indrawing pneumonia in children (22) recommended parenteral antibiotics for at least three days. A study by Straus et al (23) published in 1998 found that treatment failure rate with oral amoxicillin was significantly lower than with oral cotrimoxazole (18% and 33% respectively) in children with chest indrawing pneumonia ($p=0.009$). It was concluded that although oral cotrimoxazole was effective in fast breathing pneumonia, it was less effective in treating chest indrawing pneumonia. Multi-country studies, using a randomized controlled trial design and an adequate sample size, were then undertaken to assess the effectiveness of oral amoxicillin in chest indrawing pneumonia. A summary of the evidence that has led to the revised recommendation is presented below.

1.2.1 Oral amoxicillin is as effective as injectable penicillin in the treatment of chest indrawing pneumonia in children in low-resource settings

A multicentre, randomized, open-label trial by Addo-Yobo et al (24) (APPIS Study) compared the efficacy of oral amoxicillin and injectable penicillin. The study was conducted at nine tertiary health facilities in eight developing countries and enrolled 1702 children age from 3 to 59 months with chest indrawing. Children were randomly assigned to receive either a five-day course of oral amoxicillin ($n=857$) or parenteral penicillin ($n=845$). Evaluations were carried out at 48 hours, five days and 14 days. The primary outcome of the study was clinical treatment failure at 48 hours. The study found treatment failure rates of 19% in each treatment group ($n=161$ penicillin; $n=167$ amoxicillin; risk difference -0.4%; 95% CI -4.2 to 3.3). The study described a number of advantages of oral treatment over parenteral treatment, including reduced risk of injection-related morbidity, and fewer needs for medical supplies such as needles, and suggested that oral treatment be considered as an equally effective alternative to parenteral treatment. Although this study proved the

effectiveness of oral amoxicillin, it did not fully address the issue of safety, as all enrolled children were hospitalized for 48 hours and kept under close supervision. More research was needed to test the hypothesis under a wider range of circumstances.

1.2.2 Oral amoxicillin is equally effective for pneumonia of various severities in a high-resource setting

A study in England in 2007 (PIVOT Trial) (25) compared oral amoxicillin and intravenous (IV) benzyl penicillin in the management of severe pneumonia. The randomized, controlled, non-blinded equivalence trial was conducted in eight paediatric centres (district general and tertiary hospitals) and enrolled children with all but the most severe cases of pneumonia. Additional exclusion criteria were: wheeze, oxygen saturation less than 85%, shock, immunodeficiency, pleural effusion at presentation requiring drainage, chronic lung condition (excluding asthma), penicillin allergy, and age less than 6 months. Children were randomly assigned to a 7-day treatment of either oral amoxicillin or IV benzyl penicillin. The primary outcome was the time required for temperature to be below 38 °C for 24 continuous hours. The study found the two treatments to be equivalent, each having a median time of 1.3 days to achieve the primary outcome. While equivalence between oral and parenteral antibiotics itself was not a new development, this study was unique in the scope of varying severities of pneumonia treated. It was also the first study of its kind for children in a high-resource setting with radiologically confirmed pneumonia. The study recommended that children be treated with oral amoxicillin instead of IV benzyl penicillin, as oral treatment was both painless and non-invasive.

1.2.3 It is safe to treat chest indrawing pneumonia at home with oral amoxicillin

Hazir et al (26) (NO-SHOTS Study) carried out a randomized, open-label equivalency trial at seven sites in Pakistan, comparing hospitalization with parenteral ampicillin to home treatment with oral amoxicillin. Children with chest indrawing pneumonia were randomized to treatment in hospital (n=1012) with two days of injectable ampicillin followed by three days of oral amoxicillin (80–90 mg/kg/day), or were sent home with a five-day twice daily course of oral amoxicillin (n=1025). The primary outcome was clinical treatment failure. Follow-up assessments were conducted at 1, 3, 6, and 14 days after enrolment. At day 6 there were 87 (8.6%) treatment failures in the hospitalized group and 77 (7.5%) treatment failures in the home-treatment group (risk difference 1.1%, 95% CI – 1.3–3.5). In addition to finding equivalence between the two treatments, the authors suggested that in cases of chest indrawing pneumonia without underlying complications, home treatment with a short course of high-dose oral amoxicillin was preferable to parenteral treatment because of the associated reduction in referral, admission, and treatment costs as well as the reduced invasiveness of oral treatment.

1.2.4 Home therapy with oral amoxicillin is effective in a range of settings

To test whether the interpretation that chest indrawing pneumonia can be treated safely and effectively at home was generalizable across communities and geographic regions, a multicentre observational study was conducted in Bangladesh, Egypt, Ghana and Viet Nam (27). Outcome was ascertained on a total of 823 children age 3 to 59 months old, with chest indrawing pneumonia, who were given oral amoxicillin (80–90 mg/kg/day) twice daily for five days. Follow-up at home was carried out on days 1, 2, 3 and 6 to assess treatment failure, and at the health facility on day 14 to assess relapse. Overall treatment failure was 9.2%, varying from 6.4% in Ghana to 13.2% in Vietnam. The common reasons for failure were persistence of chest indrawing at day 6 (3.8%),

abnormally sleepy or difficult to wake (1.3%) and central cyanosis (1.3%). The authors concluded that among children with chest indrawing pneumonia treated at home with oral amoxicillin, clinical treatment failure and adverse event rates did not differ across geographic areas. This argues in favour of the option of home-based therapy of chest indrawing pneumonia in a wide variety of settings.

In these studies, the children with chest indrawing pneumonia were identified and treated on an outpatient basis by qualified doctors, and were followed up either in an outpatient facility or at home. This made it difficult to generalize to situations where pneumonia is treated within the community, by CHWs who may have little or no formal education. Data were needed to assess the effectiveness of community case management of chest indrawing pneumonia.

1.2.5 Efficacy of higher dose (80–90 mg/kg/day) vs. lower dose (45 mg/kg/day) of amoxicillin

1.2.5.1 Amoxicillin is more effective when given in higher doses

The major parameter determining the *in vivo* efficacy of many antibiotics is the duration of time that serum level exceeds the mean inhibiting concentration (MIC), and studies have demonstrated the effect of increasing concentrations of antimicrobials on their bactericidal activity. A clinical review and other reports showed that for *Streptococcus pneumoniae* and *Haemophilus influenzae* in patients with otitis media (28–34), serum levels must be above the MIC for more than 40% of the time the child is in treatment in order to achieve a bacteriological cure rate of 85% to 100%. The review also showed that amoxicillin and cefuroxime provided adequate duration above the MIC for penicillin-intermediate resistant strains. In contrast, for penicillin-resistant strains, only amoxicillin provided levels above the MIC for more than 40% of the dosing interval. The review also showed that a four-fold higher dose given every eight hours results in a much higher peak/MIC ratio than a dose administered every two hours.

The evidence-based practice guidelines of the American Academy of Pediatrics and the American Academy of Family Physicians for the treatment of community-acquired pneumonia recommend an amoxicillin dose of 75–100 mg/kg/day (35). This recommendation was based on extrapolation from microbiology studies on acute otitis media (AOM). Amoxicillin can be given in a twice-daily regimen.

The main causal organisms of AOM and childhood pneumonia are *S. pneumoniae* and *H. influenzae*. Keeping in mind the penicillin-intermediate and penicillin resistant strains of these organisms, the revised guidelines recommend 80mg/kg/day of amoxicillin in two divided doses for the treatment of chest indrawing pneumonia.

1.2.5.2 Amoxicillin can be given twice instead of thrice daily for children with fast breathing and chest indrawing pneumonia

Fonseca W et al (36) compared levels of oral amoxicillin in a 15 mg/kg/body weight/dose given thrice daily with a regimen of 25 mg/kg/dose twice daily in 66 children age 3 to 59 months. Amoxicillin concentrations were determined by high performance liquid chromatography after the first daily dose on days 1 and 3. For amoxicillin, the mean area under the concentration time curve after the 25 mg/kg/dose was 54.7 µg/ml x h, whereas after the 15 mg/kg/dose it was 24.9 µg/ml x h. The study concluded that oral amoxicillin twice daily is a feasible alternative to thrice daily dosing. It was suggested that in order to lengthen the time above the MIC at higher concentration levels, a 30 to 40 mg/kg/dose twice daily should be considered instead of the 25 mg/kg/dose used in this study.

Valtonen M et al (37) compared the clinical efficacy and side effects of amoxicillin (40 mg/kg/day) in two groups of children randomly assigned to a regimen of either two or three doses per day. In the group with two daily doses, 82% of the patients with otitis media were cured; in the group with three daily doses this figure was 86%. Side effects were equal in both groups. The trial concluded that “the same total daily dose of amoxicillin given either three times daily or two times daily is comparably effective and tolerated in children with acute respiratory infections”.

Daschner FD et al (38) compared the effects of giving amoxicillin (50 mg/kg/day) twice or four times per day in 34 children with respiratory tract infections. Peak and trough antibiotic concentrations were determined. Eradication of bacteria, clinical improvement and side effects were compared in both groups. The authors concluded that the same total daily dosage of amoxicillin given either twice or four times daily was equally effective and safe.

The Catchup study group (39) compared the clinical effectiveness of twice-daily oral amoxicillin with twice-daily oral cotrimoxazole in 1459 children with fast breathing pneumonia in a randomized, double blind trial in seven outpatient departments and one community health service in Pakistan. 730 children were randomly assigned to receive a 25 mg/kg/dose of amoxicillin, and 741 to receive 4 mg/kg trimethoprim plus 20 mg/kg sulphamethoxazole (cotrimoxazole). Treatment failure in the amoxicillin group was 16.1% as compared to 18.9% in the cotrimoxazole group (OR 0.83, 95%CI 0.63–1.08, $p=0.160$). The authors concluded that both amoxicillin and cotrimoxazole provided equally effective therapy for fast breathing pneumonia in twice-daily dosing regimens.

Hazir et al (26) compared high dose oral amoxicillin (80–90 mg/kg/day) with injectable penicillin in children with chest indrawing pneumonia, using a twice-daily regimen. This study showed that amoxicillin given two times a day was effective in treating children with chest indrawing pneumonia.

These studies demonstrate that amoxicillin given in a twice-daily dosage regimen is as effective as regimens of three- or four-times daily, provided that the total daily dosage of amoxicillin is the same. A twice-daily schedule has advantages for caregivers and programmes as it may result in improved adherence.

1.3 Recommendation 3

Children aged 2–59 months with severe pneumonia¹ should be treated with parenteral ampicillin (or penicillin) and gentamicin as a first-line treatment.

- **Ampicillin: 50 mg/kg, or benzyl penicillin: 50 000 units per kg IM/IV every six hours for at least five days**
- **Gentamicin: 7.5 mg/kg IM/IV once a day for at least five days**

Ceftriaxone should be used as a second-line treatment in children with severe pneumonia having failed on the first-line treatment.

One systematic review (40) that included two large randomised controlled trials (RCT) comparing beta-lactam and gentamicin versus chloramphenicol for very severe pneumonia showed high-quality evidence that ampicillin/penicillin and gentamicin reduce clinical failure rates compared to

¹ Previously classified as very severe pneumonia (Pocket book of hospital care for children: Guidelines for the management of common illnesses with limited resources. First edition. Geneva: World Health Organization; 2005).

chloramphenicol. There was moderate-quality evidence of a trend towards reduced death rates for treatment with ampicillin/penicillin and gentamicin compared to chloramphenicol.

1.3.1 Penicillin/gentamicin vs chloramphenicol at high altitude

The first of these RCTs (41), conducted in the highlands of Papua New Guinea (1600–1800 m above sea level), included 1116 children aged 1–59 months with WHO-defined very severe pneumonia (modified to include heart failure as a danger sign). Enrolled children had a median oxygen saturation of 71%. Five hundred and fifty-nine (559) children were treated with 100 mg/kg/day chloramphenicol; 557 children were treated with penicillin (200 mg/kg/day) plus gentamicin (7.5 mg/kg/day). Duration of treatment was 14 days. Measured outcomes were: death, treatment failure by day 5, and readmission. More children in the penicillin/gentamicin group required a change of antibiotic (60 versus 49), while 147 (26%) children treated with chloramphenicol and 123 (22%) treated with penicillin and gentamicin had adverse outcomes ($p = 0.11$, not significant). Thirty-six children treated with chloramphenicol and 29 treated with penicillin and gentamicin died (difference not significant). More children treated with chloramphenicol presented again with severe pneumonia within one month of hospital discharge ($p = 0.03$), as compared to those treated with penicillin and gentamicin.

1.3.2 Penicillin/gentamicin vs chloramphenicol at low altitude

The second of the RCTs (42) was a multi-country study, with 80% of children residing at sea level. Children aged 2–59 months were enrolled: 479 were randomized to receive chloramphenicol (75 mg/kg/day), and 479 to receive ampicillin (200 mg/kg/day) and gentamicin (7.5 mg/kg/day). Median oxygen saturation on admission was higher than for the Papua New Guinea study (88%) (41). Duration of treatment was 10 days. More children in the chloramphenicol group required a change in antibiotic (45 versus 26). Measured outcomes were death or treatment failure by days 5, 10, and 21. More children failed treatment with chloramphenicol at day 5 (16% versus 11%; relative risk 1.43, 95% CI 1.03 to 1.97) as well as at days 10 and 21. There was a trend towards reduced death rates in children treated with ampicillin plus gentamicin, but this was not significant.

1.3.3 Ceftriaxone as second-line treatment

Although there were no data on the use of ceftriaxone in the treatment of pneumonia with general danger signs, the WHO Guidelines Development Group recognized the need to include ceftriaxone as a second-line treatment for children with severe pneumonia with general danger signs, especially for hospital care.

1.4 Recommendation 4

Ampicillin (or penicillin when ampicillin is not available) plus gentamicin or ceftriaxone are recommended as a first-line antibiotic regimen for HIV-infected and -exposed infants and for children under 5 years of age with chest indrawing pneumonia or severe pneumonia.

For HIV-infected and -exposed infants and for children with chest indrawing pneumonia or severe pneumonia, who do not respond to treatment with ampicillin or penicillin plus gentamicin, ceftriaxone alone is recommended for use as second-line treatment.

While there has been no randomized controlled trial designed on the basis of an a-priori hypothesis to examine the efficiency of antibiotic regimens or case management, a subgroup analysis was conducted in one randomized controlled trial in which oral amoxicillin was compared with paren-

teral penicillin for chest indrawing pneumonia in children (43). The response rates with the two regimens were comparable, but the treatment failure rate was significantly higher for HIV-infected infants at day 14 (40.7% versus 24.3%; OR, 2.8; 95% CI 1.35; 3.5).

Therefore, it was recommended that HIV-infected or -exposed children with chest indrawing pneumonia be hospitalized and treated as patients with severe pneumonia (pneumonia with general danger signs).

1.5 Recommendation 5

Empiric cotrimoxazole treatment for suspected *Pneumocystis jirovecii* (previously *Pneumocystis carinii*) pneumonia (PCP) is recommended as an additional treatment for HIV-infected and -exposed infants aged from 2 months up to 1 year with severe or very severe pneumonia.

Empirical cotrimoxazole treatment for *Pneumocystis jirovecii* pneumonia (PCP) is not recommended for HIV-infected and -exposed children over 1 year of age with chest indrawing or severe pneumonia.

Two reviews led to this decision, that was based on those guidelines previously approved by the WHO Guidelines Review Committee (9). One review examined the evidence for the management of pneumonia in HIV-infected children (44). The second review, which evaluated etiological agents for pneumonia (45), was comprised of nine descriptive studies, two post-mortem studies, and 15 analytical studies, of which five were post-mortem.

In a cohort study of HIV-infected children in Malawi (46), the incidence of severe pneumonia was high, mostly in children older than two years. No cases of PCP were diagnosed in this age group. In autopsy studies in South Africa and Zimbabwe (47,48) the mean age of all 46 cases of PCP was reported to be 3–4 months. The findings of these and subsequent autopsy and clinical studies in the region provided consistent evidence that most cases of PCP occur in young infants. Of 130 autopsy cases of PCP reported, only 5% were in children over 12 months of age.

In the 1990s, data reported from Europe, Thailand and the United States of America also showed that the peak prevalence of PCP in children was in infants under 12 months of age, with the highest prevalence under 6 months (49,50). A report from Ireland and the United Kingdom showed that, before 1998, 27% of HIV-infected infants had PCP or cytomegalovirus as a first indicator; 91% of these were young infants (51).

Based on the systematic review, the WHO Guidelines Development Group recognized the importance of covering clinically suspected PCP in infants younger than 1 year of age and thus recommended empirical treatment with cotrimoxazole of all HIV-infected and -exposed children under one year of age.

1.6 Additional information on implementing the management of pneumonia at community level

1.6.1 Management of pneumonia at community level

According to the WHO/UNICEF joint statement on management of pneumonia in community settings (52), an important strategy to increase access to quality care for pneumonia is to train and deploy CHWs to assess and treat children with pneumonia. There is strong scientific and programmatic evidence to support the effectiveness of this approach.

Two large-scale studies, in Bangladesh and Nepal, assessed the ability of volunteer CHWs with intensive basic training and close supervision to properly diagnose and treat pneumonia. Hadi et al (53) showed that the sensitivity, specificity, and overall agreement rates in pneumonia diagnosis and treatment were significantly higher among those health volunteers in Bangladesh who had intensive basic training and routine supervision than among those who had not. Ghimire et al (54) showed a significant trend towards a decrease in the proportion of pneumonia and severe pneumonia cases from 2004 to 2006 in districts of Nepal where the volunteers were given special training to manage pneumonia.

In addition, numerous countries implement community case management of childhood pneumonia. For example, the Gambia has a nationwide programme addressing pneumonia in the community (52). In Honduras, pneumonia treatment has been incorporated into the national integrated community child care programme where community volunteers – in addition to their other responsibilities – provide treatment for pneumonia and diarrhoea in more than 1800 communities (52). In 2008, the government of Malawi initiated activities to deliver community-level treatment of common childhood illnesses including suspected pneumonia. By September 2011, 3296 health surveillance assistants (HSAs) had been trained, and 2709 village health clinics were functional. Evaluation has shown that HSAs are able to treat sick children with a quality similar to that provided by professional health personnel in fixed facilities. Monitoring data also showed that communities were utilising the HSA services (55). Large-scale, national-level programmes supporting CHWs to treat illness including childhood pneumonia are also implemented in Ethiopia and Pakistan.

The current WHO/UNICEF tools for CHWs recommend oral amoxicillin in two daily doses. Because the CHW is not expected to treat chest indrawing pneumonia, these guidelines will retain two age bands and will not be revised at the present time. Some data from Asia (56,57) shows that CHWs can manage chest indrawing pneumonia with oral amoxicillin. However, current CHW guidelines will not be changed until more evidence becomes available from additional regions and countries.

1.6.2 Community management of chest indrawing pneumonia

Two studies have shown that CHWs, when properly trained and supported, can effectively and safely treat chest indrawing pneumonia at home with oral amoxicillin.

Bari et al (56) aimed to compare community management of children with chest indrawing pneumonia by lady health workers (LHWs), using oral amoxicillin, to the standard of care: a first dose of cotrimoxazole followed by referral. Twenty-eight clusters in the Haripur district of Pakistan were randomly assigned equally in a 1:1 ratio to either intervention or control. Children age 2 to 59 months with chest indrawing pneumonia were included. In the intervention clusters, LHWs provided mothers with oral amoxicillin (80–90 mg/kg/day in two divided doses) for five days, and gave specific guidance on its use. In the control clusters, LHWs gave the first dose of oral cotrimoxazole and referred the child for standard care to the nearest health facility. The primary outcome was treatment failure by day 6. A total of 1995 children participated in the 14 intervention clusters and 1477 in the 14 control clusters. Treatment failure rates were significantly reduced in the intervention clusters (165 [9%] vs 241 [18%], risk difference –8.9%). Most of the risk reduction was in the occurrence of fever and chest indrawing on day 3 (–6.7%, –10.0 to –3.3). Two deaths were reported in the control clusters and one in the intervention cluster. It was concluded that community case management by LHWs could result in appropriate treatment for children with chest indrawing pneumonia, reduce delays in initiating treatment, and reduce costs for families and health systems.

Similar evidence on the management of severe pneumonia by LHWs was generated in a cluster-randomized controlled trial in Matiari district of rural Sindh, Pakistan (57). Public-sector LHWs undertook community management of chest indrawing pneumonia. In the intervention clusters children with suspected pneumonia were screened by LHWs, and identified cases of chest indrawing pneumonia were prescribed oral amoxicillin (90 mg/kg per day in two doses) for five days at home. Children in control clusters were given one dose of oral cotrimoxazole and were referred to their nearest health facility for further management. In both groups, follow-up visits at home were conducted at days 2, 3, 6, and 14 by the LHWs. The primary outcome was treatment failure by day 6 after enrolment. 2341 children in the intervention clusters and 2069 children in the control clusters participated in the study. Treatment failure by day 6 was 8% in the intervention group and 13% in the control group. After adjusting for clustering, the risk difference for treatment failure was -5.2% (95% CI -13.7% to 3.3%). Two deaths by day 6 and one between days 7 and 14 were recorded; no serious adverse events were witnessed. The authors concluded that properly trained LHWs were able to satisfactorily diagnose and treat chest indrawing pneumonia at home in rural Pakistan. This strategy could effectively increase access to care for pneumonia in settings where referral is difficult, and could become a key component of community detection and management strategies for childhood pneumonia.

2. Costs of treating pneumonia using the revised recommendations

One potential barrier to access to case management in low-resource settings is the cost of treatment. Understanding the relevant costs can help public sector programmes and ministries of health reduce them by rationalizing the use of available resources and focusing on those interventions with the greatest public health benefit.

A number of studies carried out in countries with a high burden of childhood pneumonia compared the cost of pneumonia therapy at household, outpatient and hospital levels. The general conclusion is that costs of hospitalization are far greater than the costs of home therapy.

2.1 Household treatment costs for pneumonia

Sadrudin et al (58) compared household costs for chest indrawing pneumonia cases referred to a health facility with those managed directly by LHWs in Pakistan. Data on direct and indirect costs were collected through interviews and record reviews. The average household cost of each LHW-managed case was US\$ 1.46, while the average cost for a referred case was US\$ 7.60. After excluding the cost of the antibiotic provided, the cost per case came to US\$ 0.25 for each LHW-managed case and US\$ 7.51 per referred case; this is a 30-fold difference. The authors concluded that expanding the treatment of chest indrawing pneumonia to community level could significantly reduce household costs, improve access to treatment, and ultimately prevent many deaths.

2.2 Outpatient treatment costs for pneumonia

Chola et al (59) collected data on annual economic and financial costs from urban health centres in Zambia in 2005–06. The average cost of providing outpatient services was US\$ 3 per visit, while the cost per outpatient visit for childhood pneumonia was US\$48. In a study in the Northern Areas of Pakistan, Hussain H et al (60,61) calculated that the average cost of treating childhood pneumonia was US\$ 13.44.

2.3 Inpatient treatment costs for pneumonia

Chola et al (59) calculated the cost of inpatient treatment in Zambia to be US\$ 18 per bed day; for childhood pneumonia this increased to US\$ 215 per bed day. Hussain et al (60,61) calculated that in northern Pakistan each in-patient episode of pneumonia cost the health system US\$ 71; each in-patient case of severe pneumonia cost US\$ 235. Outpatient costs averaged US\$ 13 and US\$ 86 respectively for pneumonia and severe pneumonia. The same authors showed that, when the provider costs (excluding inpatient care) and household costs were added together, the total societal average cost per episode was US\$ 22.62 for fast breathing pneumonia, and US\$ 142.90 for chest indrawing pneumonia (60,61). It should be clarified here that the costs quoted in the latter example are per episode of illness; thus they are slightly higher than the cost per visit cost shown in the earlier paper.

A cost minimization analysis from eight paediatric centres in England (62) published in 2010 concluded that treatment of community-acquired pneumonia with oral amoxicillin could result in savings of between £ 473 and £ 518 per case.

2.4 Comparison of inpatient costs for pneumonia at different tiers of health facilities

Madsen et al (63) looked at the health-care provider cost and household cost of the treatment of severe pneumonia in infants and young children admitted to secondary and tertiary level health-care facilities in Vellore, India. The total cost to the health system for one episode of hospitalized childhood pneumonia treated at secondary level was US\$ 84.0 (INR 3524); an episode treated at tertiary level cost US\$ 147.0 (INR 6158). At both levels the greatest single cost was the hospital stay itself, comprising 74% and 56% of the total costs respectively.

Ayieko et al (64) calculated the costs of treatment for childhood pneumonia at different tier hospitals in Kenya. A sensitivity analysis was conducted using WHO-CHOICE values for cost per bed day. From the provider perspective the mean cost per admission for childhood pneumonia at the national hospital was US\$ 177.0, at district hospitals was between US\$ 54.1 to US\$ 99.3 and at mission hospitals ranged from US\$ 43.4 to US\$ 142.2. In public sector hospitals, households provide partial subsidies to the provider costs through payment of user fees; within the private sector the total amount of provider costs may be passed on to households. The authors suggested that these findings could be used in cost effectiveness analysis of health interventions.

3. Flexible Solid Oral Dosage: Dispersible formulations of amoxicillin

Amoxicillin has been recognized as a “priority essential” medicine by WHO and UNICEF (65). The dosage of amoxicillin is based on the child’s weight. There is a potential risk of microbial resistance with under-dosing, and of toxicity with over-dosing. It is therefore crucial that the paediatric formulation have flexibility for dose adjustment, while at the same time remaining within the effective therapeutic range. One of the current challenges is the suitability of existing formulations of amoxicillin for children in resource-constrained settings. The use of solid preparations often involves breaking an adult tablet into smaller pieces, then crushing and adding it to food or liquid; this can lead to inaccuracies in dosing. There is a widespread expectation from both end-users and clinicians that medicines for children should be liquid, as it is generally believed that children prefer this formulation. In addition, liquid formulations make weight-based dosing much easier. However, the accurate administration of a liquid medicine is not assured even if the dose is correctly calculated. There is evidence that measuring spoons and other devices supplied with liquid medicines are not always accurate and that significant under- or over-dosing can occur (66–68).

In order to reach consensus on the most suitable formulations of medicines for children, with particular attention to conditions in developing countries, in 2006 WHO hosted a meeting of pediatricians, pharmacists, clinical pharmacologists, and representatives of the European Medicines Agency, the International Federation of Pharmaceutical Manufacturers and Associations, the Medicines for Malaria Venture, the National Institutes of Health, UNICEF, and the Bill & Melinda Gates Foundation. The group recommended that in general, the dosage forms of medicines that are likely to prove most ‘suitable’ particularly for developing countries are flexible solid dosage forms, such as tablets that are oro-dispersible and/or that can be used for preparation of oral liquids (for e.g. suspension or solution). Provided the product can be dispersed in breast milk from the mother, it could potentially be used for very young children (0–6 months). This type of product is feasible to manufacture in facilities that produce conventional tablets, but requires excipients that ensure stability and palatability. Examples of existing dispersible tablet product suggest that they can be more affordable than standard liquid dosage forms (69).

In September 2010, the UNICEF Supply Division and WHO brought together experts in paediatric medicine from academia, the pharmaceutical industry (both innovator and generic), regulators, programme managers and implementers. The meeting aimed to reach consensus on priority essential medicines for children, and to identify the steps required to ensure that these medicines are available and affordable. The group concluded, “Liquid formulations can be difficult to use and store, which can make their use impractical in some areas. Solid formulations and flexible solid oral dosage forms, such as dispersible tablets, that can be given whole to older children or dispersed in water for ease of administration to younger children are preferred” (65).

Based on these consultations, WHO recommended ‘Flexible Solid Oral Dosage’ forms as the optimum formulation for children’s medicines administered orally, as they have better stability and shelf life than liquids, in addition to being less bulky to ship and store (70).

4. Implications for implementation

The revised recommendations present a number of advantages.

- Oral amoxicillin can be used to treat both fast breathing pneumonia and chest indrawing pneumonia (see Table).
- Pneumonia classification and management are simplified to two categories instead of three (see Figure).
- Access to antibiotic treatment closer to home is increased.
- The need for referrals to higher level facilities is decreased.
- The probability of hospitalization and thus the risk of nosocomial and injection borne diseases is reduced.
- The probability of antimicrobial resistance is diminished, due to better adherence to the simplified treatment.
- Training of health workers is simplified.

Adoption of the revised guidelines will have a number of implications for policy as well as for implementation at different levels of the health system.

4.1 Implications for policy

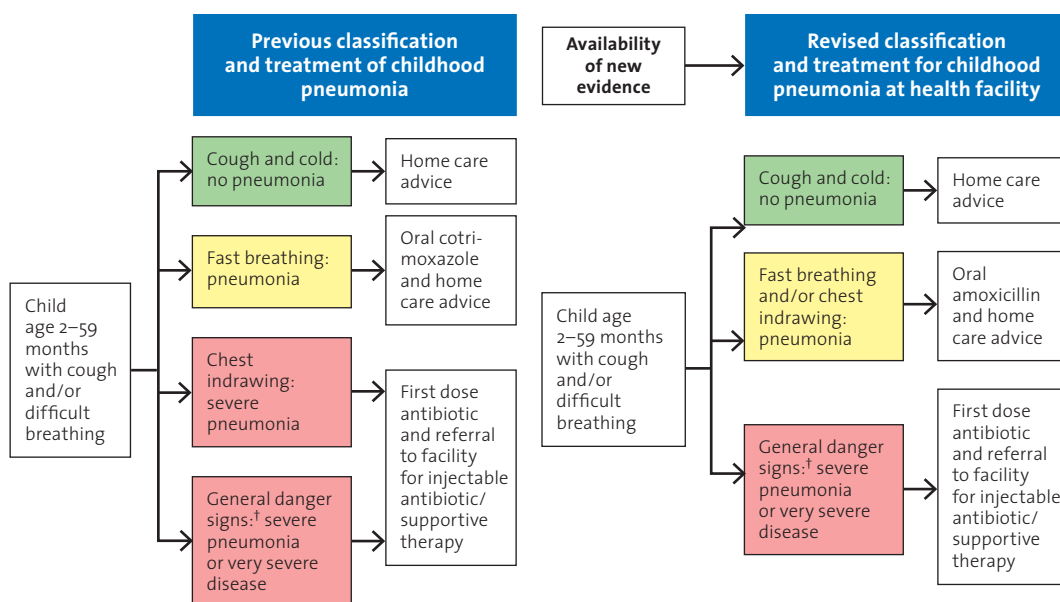
- National programmes will need to switch from oral cotrimoxazole to oral amoxicillin. Because of this:
 - ✓ *retraining of healthcare providers will be required (with training materials and job aids appropriately revised);*
 - ✓ *reduction in overall costs of pneumonia treatment by promoting outpatient/home therapy will help offset the additional costs of switching to oral amoxicillin from oral cotrimoxazole.*
- Combining the two previous categories of “pneumonia” (fast breathing) and “severe pneumonia” (chest indrawing) under a single classification of “pneumonia” simplifies classification. Because of this:
 - ✓ *training will be simplified;*
 - ✓ *the process of local adaptation of training materials and job aids will be easier.*
- It is crucial that the policy makers, administrators and officials of relevant programmes understand the significance of these modifications. Because of this:
 - ✓ *widespread dissemination of technical updates is of utmost importance to ensure adoption and adherence.*
- In order to pre-empt the possible over-use of antibiotics, programmes will need to invest in careful monitoring procedures. Because of this:
 - ✓ *close monitoring will be essential in areas implementing the new guidelines at large scale.*

Table

Doses of amoxicillin for children 2–59 months of age with pneumonia

TOOLS	CATEGORY OF PNEUMONIA	AGE/WEIGHT OF CHILD	DOSAGE OF AMOXICILLIN DISPERSIBLE TABLETS (250 mg)
iCCM tool for community health workers: no change	Fast breathing pneumonia	2 months up to 12 months (4–<10 kg)	1 tab twice a day x 5 days (10 tabs)
		12 months up to 5 years (10–19 kg)	2 tabs twice a day x 5 days (20 tabs)
IMCI tool for professional health workers at health facilities: revised	Fast breathing and chest indrawing pneumonia	2 months up to 12 months (4–<10 kg)	1 tab twice a day x 5 days (10 tabs)
		12 months up to 3 years (10–<14 kg)	2 tabs twice a day x 5 days (20 tabs)
		3 years up to 5 years (14–19 kg)	3 tabs twice a day x 5 days (30 tabs)

Figure

Comparison of previous and revised classification and treatment of childhood pneumonia at health facility

† Not able to drink, persistent vomiting, convulsions, lethargic or unconscious, stridor in a calm child or severe malnutrition.

4.2 Implications for implementation at community level (integrated Community Case Management)

The WHO/UNICEF joint statement *Management of pneumonia in community settings* (52) recommends the training and deployment of CHWs as a central strategy for increasing access to quality care for pneumonia. Research showed that educated community members could be trained to detect and manage pneumonia in their communities; large-scale studies confirmed that the sensitivity, specificity, and overall agreement rates in pneumonia diagnosis and treatment were high among health volunteers who had intensive basic training and routine supervision. This led to a substantial decrease in the proportion of severe pneumonia cases and deaths.

It has also been shown that properly trained and supervised CHWs can appropriately manage chest indrawing pneumonia with oral amoxicillin at community level, resulting in increased access to treatment and reduced costs for families. As evidence is limited, the iCCM guidelines are not currently being updated to include the management of chest indrawing pneumonia. In situations where referral is not possible, however, CHWs may treat chest indrawing pneumonia with oral amoxicillin, if local health policy allows them to do so.

Thus, at the **community level** the revised recommendations imply that:

- all children with fast breathing are classified as having “pneumonia” and treated with oral amoxicillin;
- children with “chest indrawing” pneumonia should be referred to a higher level. However, in situations where referral is not possible and if local health policy allows, CHWs may treat chest indrawing pneumonia with oral amoxicillin;
- dispersible amoxicillin is the preferred treatment for children.

4.3 Implications for implementation at health facility Level

The revised recommendations can be encapsulated into the following statements, each with a short description of the supporting evidence:

It is safe to treat chest indrawing pneumonia at home with oral amoxicillin. A comparison of the use of parenteral ampicillin in hospital to oral amoxicillin at home found that the two treatments were equivalent. In cases of chest indrawing pneumonia without underlying complications, home treatment with a short course of high-dose oral amoxicillin is preferable to parenteral treatment because of the associated reduction in referral, admission, and treatment costs as well as the reduced invasiveness of oral treatment.

A three-day course of antibiotics is as effective as a five-day course in treating children with fast breathing pneumonia in areas of low HIV prevalence. A systematic review of trials evaluating the efficacy of short-course versus long-course treatment of fast breathing pneumonia in children age 2 to 59 months recommended a shorter course of antibiotic therapy, keeping in mind the benefits to the individual as well as to the health system, especially in settings with limited resources.

Amoxicillin is more effective when given in higher doses. Increasing concentrations of antimicrobials improves their bactericidal activity. Clinical reviews show that, for community-acquired pneumonia, oral amoxicillin should be given in a regimen of at least 80 mg/kg/day in two divided doses.

Amoxicillin can be given twice instead of thrice daily for children. Amoxicillin given in a twice-daily dosage regimen is as effective as regimens of three- or four-times daily, provided that the total daily dosage of amoxicillin is the same. A twice-daily schedule has advantages for caregivers and programmes as it may result in improved adherence to treatment.

Dispersible amoxicillin is the most suitable formulation for children. Because the dosage of amoxicillin is based on the child's weight, and because of the risks associated with under- or over-dosing, it is crucial that the paediatric formulation have flexibility for dose adjustment. The use of solid preparations often involves breaking an adult tablet into smaller pieces, then crushing and adding it to food or liquid; this can lead to inaccuracies in dosing. Liquid formulations make weight-based dosing much easier, however there is evidence that measuring devices supplied with liquid medicines are not accurate and that significant under- or over-dosing can occur. A multidisciplinary consensus supports the recommendation of 'Flexible Solid Oral Dosage' forms for children's oral medicines, as they are less costly than tablets, they have better stability and shelf life than liquids, and they are less bulky to ship and store.

Thus, at the **health facility level**, changes in the management of ARI implied by these new recommendations can be summarized as follows:

- all children with fast breathing and/or chest indrawing are classified as having "pneumonia" and treated with oral amoxicillin; the recommended dosage is 80 mg/kg for five days (40 mg/kg twice a day); in settings of low HIV prevalence the duration of treatment for 'fast breathing pneumonia' can be reduced to three days;
- only those children who have either general danger signs or who are HIV positive and have chest indrawing need to be referred to higher level facility for inpatient treatment with injectable antibiotics;
- dispersible amoxicillin is the preferred treatment for children.

4.4 Implications for implementation at hospital level

Children with chest indrawing pneumonia in settings of low HIV prevalence will be treated at the outpatient level and will no longer need to be referred for hospitalization.

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WHO consolidated guidelines on tuberculosis

Module 2: Screening

**Systematic screening for
tuberculosis disease**



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Organization

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Web Annex A. Methods and Expert Panels

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The following persons contributed to the reviews and summarized evidence for the guidelines using the Population, Intervention, Comparator and Outcomes (PICO) framework (see **Section 1.3** for more information about the PICO questions).

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Abbreviations and acronyms

ART	antiretroviral treatment
CAD	computer-aided detection of TB-related abnormalities on chest radiography
CI	confidence interval
CRP	C-reactive protein
CXR	chest radiograph (chest X-ray)
ERG	External Review Group
GDG	Guideline Development Group
GRADE	Grading of Recommendations Assessment, Development and Evaluation
IPD	individual patient data
LF-LAM	lateral flow urine lipoarabinomannan assay
mWRD	molecular WHO-recommended rapid diagnostic test
NNS	number needed to screen (to detect one person with tuberculosis disease)
PICO	population, intervention, comparator and outcomes (framework)
RR	relative risk
TB	tuberculosis
TPT	TB preventive treatment
W4SS	WHO-recommended four-symptom screen
WHO	World Health Organization

Definitions

The definitions below apply to terms as used in these guidelines, and they may have different meanings elsewhere.

Active (tuberculosis) case-finding (ACF): Provider-initiated screening and testing in communities by mobile teams, often using mobile X-ray and rapid molecular tests. The term is sometimes used synonymously with “systematic screening”.

Computer-aided detection (CAD): The use of specialized software to interpret abnormalities on chest radiographs that are suggestive of TB. The results are expressed as abnormality scores. CAD may be used for screening or triage.

Enhanced (tuberculosis) case-finding: Health information or education, or awareness campaigns to provide information about what type of health-seeking behaviour is appropriate when people experience symptoms of TB; this type of case-finding may be combined with improving access to diagnostic services. Enhanced case-finding may or may not be combined with screening.

Initial screening: The first screening test, examination or other procedure applied in the population eligible for screening.

Number needed to screen (NNS): The number of persons that need to undergo screening in order to diagnose one person with TB disease.

Passive case-finding: A patient-initiated pathway to TB diagnosis involving: (1) a person with TB disease who experiences symptoms that he or she recognizes as serious; (2) the person having access to and seeking care, and presenting spontaneously at an appropriate health facility; (3) a health worker correctly assessing that the person fulfils the criteria for presumptive TB; and (4) successful use of a diagnostic algorithm with sufficient sensitivity and specificity to diagnose TB.

Repeat screening: This refers to rescreening the same population at a given interval.

Risk groups: Any group of people in which the prevalence or incidence of TB is significantly higher than in the general population.

Screening test, examination or procedure for tuberculosis disease: A test, examination, or other procedure used to distinguish people with a high likelihood of having TB disease from people who are highly unlikely to have TB. A screening test is not intended to be diagnostic. People with positive results on a screening test should undergo further evaluation, depending on the screening algorithm used.

Second screening: A second screening test, examination or other procedure undergone by persons whose results were positive during the initial screen.

Systematic screening for TB disease: The systematic identification of people at risk for TB disease, in a predetermined target group, by assessing symptoms and using tests, examinations or other procedures that can be applied rapidly. For those who screen positive, the diagnosis needs to be established by one or several diagnostic tests and additional clinical assessments. This term is sometimes used interchangeably with “active tuberculosis case-finding”. It should be distinguished from testing for TB infection (with a TB skin test or interferon- γ release assay).

Triage: The process of deciding the diagnostic and care pathways for people based on their symptoms, signs, risk markers and test results. Triaging involves assessing the likelihood of various differential diagnoses as a basis for making clinical decisions. It can follow more- or less-standardized protocols and algorithms, and it may be done in multiple steps.

Triage test for TB: A test that can be rapidly conducted among people presenting to a health facility to differentiate those who should have further diagnostic evaluation for TB from those who should undergo further investigation for non-TB diagnoses.

Tuberculosis disease: The disease state caused by *Mycobacterium tuberculosis*. It is usually characterized by clinical manifestations, which distinguish it from TB infection without signs or symptoms (previously referred to as latent TB infection). Also referred to as active tuberculosis.

Executive summary

Tuberculosis (TB) is a leading cause of death from a single infectious agent, despite being largely curable and preventable. In 2019 an estimated 2.9 million of the 10 million people who fell ill with TB were not diagnosed or reported to the World Health Organization (WHO). The Political Declaration adopted by the United Nations General Assembly in September 2018 at the High-Level Meeting on the Fight Against Tuberculosis commits to, among other goals, diagnosing and treating 40 million people with TB by 2022. In order to achieve this ambitious target, there is an urgent need to deploy strategies to improve the diagnosis and initiation of care for people with TB. One of these strategies is systematic screening for TB disease, which is included in the End TB Strategy as a central component of its first pillar aimed at ensuring early diagnosis for all with TB.

To facilitate the implementation of TB screening at the country level, WHO published guidelines on *Systematic screening for active tuberculosis: principles and recommendations* in 2013. Since then, there have been important new studies evaluating the impact of screening interventions on both individual-level and community-level outcomes related to TB, as well as new research evaluating innovative tools for screening for TB – including the use of computer-aided detection of TB on digital radiographs, C-reactive protein and molecular WHO-recommended rapid diagnostic tests for TB – among important populations at high risk for TB disease.

In view of these new developments and due to requests by countries for more guidance, WHO convened a Guideline Development Group (GDG) in 2020 to examine the evidence and update the 2013 guidelines. The GDG met in virtual sessions between June and October 2020 and proposed several new and updated recommendations related to TB screening. WHO gratefully acknowledges the work of the GDG members, the evidence reviewers, representatives of national TB and HIV programmes, WHO colleagues, technical and funding partners, civil society representatives, patients and all others who contributed to the data used to inform this guideline update.

The evidence reviewed to address the guideline questions was derived from several trials and other studies, programmatic data, surveys and modelling work. The certainty of the evidence and the strength of the recommendations were assessed using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) method. Decisions about the strength of a recommendation and the evidence depend on the level of confidence in the estimates, as well as on other critical considerations, such as acceptability, feasibility, resource use and impact on health equity.

The use of **chest radiography** (chest X-ray, or CXR) as a screening tool for detecting TB disease in several populations was reviewed, including in the general public, people living with HIV, people younger than 15 years who are contacts of TB patients and other high-risk groups. Across all populations considered, CXR was found to be a sensitive screening tool that, while lacking sufficient specificity to confirm a TB diagnosis, has an important role in the early detection of TB in children and adults who are at higher risk of TB, as well as the potential to reduce the population burden of TB disease when combined with early treatment.

In recent years **computer-aided detection** (CAD) software packages have been developed and introduced to automate the interpretation of digital CXR images and produce a numerical score indicating the likelihood of TB. Three independent evaluations of CAD were reviewed to develop recommendations for both screening and triage for TB. The diagnostic accuracy and the overall performance of CAD software were similar to the interpretation of digital CXR by a human reader, in

both the screening and triage contexts. Evaluations showed substantial variation in diagnostic accuracy across different contexts, implying that the use of CAD will require calibration for the purpose and setting in which it will be implemented.

C-reactive protein (CRP) is an indicator of inflammation that can be measured using point-of-care tests performed on capillary blood collected via finger prick. The accuracy of CRP to detect bacteriologically confirmed TB in people living with HIV was assessed with a meta-analysis of individual patient data of people screened in high- and medium-burden settings. CRP was found to have similar sensitivity and higher or similar specificity to symptom screening in all subpopulations tested. CRP offers a clinically significant improvement in accuracy over the WHO-recommended four-symptom screen among ambulant people living with HIV who are newly in care and not yet on antiretroviral treatment, a subpopulation for whom the accuracy of the four-symptom screen is low.

Molecular WHO-recommended rapid diagnostic tests for TB (mWRDs; eg Xpert MTB/RIF) were reviewed for use as TB screening tools among different populations at high risk of TB. Evidence shows improved accuracy and effectiveness in people living with HIV and in other high-risk populations. The evidence is strongest for hospitalized patients with HIV in settings with a high burden of TB, given the limited value of symptom screening and the grave consequences of missing the opportunity to initiate TB treatment promptly in this patient group.

Based on these updates, a set of 17 new and revised recommendations for screening for TB disease have been developed (Table 1). The main changes from the previous WHO guidance are summarized in Box 1. The new guidelines replace all previous WHO guidance on TB screening. The recommendations are accompanied by updated operational guidance, the *WHO operational handbook on tuberculosis. Module 2: screening – systematic screening for tuberculosis disease*, that includes further details on target populations and tools to use for systematic screening, including revised algorithms and modelled estimates of their performance.

Table 1. Recommendations in the WHO consolidated guidelines on tuberculosis. Module 2: screening – systematic screening for tuberculosis disease, 2021

Screening for TB in targeted populations	
1	Systematic screening for TB disease may be conducted among the general population in areas with an estimated TB prevalence of 0.5% or higher <i>(updated recommendation: conditional recommendation, low certainty of evidence).</i>
2	Systematic screening for TB disease may be conducted among subpopulations with structural risk factors for TB. These include urban poor communities, homeless communities, communities in remote or isolated areas, indigenous populations, migrants, refugees, internally displaced persons and other vulnerable or marginalized groups with limited access to health care <i>(existing recommendation: conditional recommendation, very low certainty of evidence).</i>
3	People living with HIV should be systematically screened for TB disease at each visit to a health facility <i>(existing recommendation: strong recommendation, very low certainty of evidence).</i>
4	Household contacts and other close contacts of individuals with TB disease should be systematically screened for TB disease <i>(updated recommendation: strong recommendation, moderate certainty of evidence).</i>
5	Systematic screening for TB disease should be conducted in prisons and penitentiary institutions <i>(updated recommendation: strong recommendation, very low certainty of evidence).</i>

Screening for TB in targeted populations

- 6 Current and former workers in workplaces with silica exposure should be systematically screened for TB disease
(*existing recommendation: strong recommendation, low certainty of evidence*).
- 7 In settings where the TB prevalence in the general population is 100/100 000 population or higher, systematic screening for TB disease may be conducted among people with a risk factor for TB who are either seeking health care or who are already in care
(*existing recommendation: conditional recommendation, very low certainty of evidence*).
- 8 People with an untreated fibrotic lesion seen on chest X-ray may be systematically screened for TB disease
(*existing recommendation: conditional recommendation, very low certainty of evidence*).

Tools for screening for TB

- 9 Among individuals aged 15 years and older in populations in which TB screening is recommended, systematic screening for TB disease may be conducted using a symptom screen, chest X-ray or molecular WHO-recommended rapid diagnostic tests, alone or in combination
(*new recommendation: conditional recommendation, very low certainty of evidence for test accuracy*).
- 10 Among individuals aged 15 years and older in populations in which TB screening is recommended, computer-aided detection software programmes may be used in place of human readers for interpreting digital chest X-rays for screening and triage for TB disease
(*new recommendation: conditional recommendation, low certainty of evidence*).
- 11 Among adults and adolescents living with HIV, systematic screening for TB disease should be conducted using the WHO-recommended four symptom screen and those who report any one of the symptoms of current cough, fever, weight loss or night sweats may have TB and should be evaluated for TB and other diseases
(*existing recommendation: strong recommendation, moderate certainty of evidence*).
- 12 Among adults and adolescents living with HIV, C-reactive protein using a cut-off of >5mg/L may be used to screen for TB disease
(*new recommendation: conditional recommendation, low certainty of evidence for test accuracy*).
- 13 Among adults and adolescents living with HIV, chest X-ray may be used to screen for TB disease
(*new recommendation: conditional recommendation, moderate certainty of evidence for test accuracy*).
- 14 Among adults and adolescents living with HIV, molecular WHO-recommended rapid diagnostic tests may be used to screen for TB disease
(*new recommendation: conditional recommendation, moderate certainty of evidence for test accuracy*).
- 15 Adult and adolescent inpatients with HIV in medical wards where the TB prevalence is > 10% should be tested systematically for TB disease with a molecular WHO-recommended rapid diagnostic test
(*new recommendation: strong recommendation, moderate certainty of evidence for test accuracy*).

-
- 16 Among individuals younger than 15 years who are close contacts of someone with TB, systematic screening for TB disease should be conducted using a symptom screen including any one of cough, fever or poor weight gain; or chest radiography; or both (*new recommendation: strong recommendation, moderate to low certainty of evidence for test accuracy*).
-
- 17 Among children younger than 10 years who are living with HIV, systematic screening for TB disease should be conducted using a symptom screen including any one of current cough, fever, poor weight gain or close contact with a TB patient (*new recommendation: strong recommendation, low certainty of evidence for test accuracy*).
-

TB: tuberculosis.

Box 1. Main changes to the guidance in the current update

- ➔ Community-wide systematic screening using an accurate screening and diagnostic algorithm may be used in settings with a TB prevalence of 0.5% and higher, based on new evidence of public health benefit.
- ➔ Computer-aided detection (CAD) is being recommended for the first time as an alternative to human interpretation of digital chest X-ray (CXR) for screening and triage for TB. Its use should be limited to the interpretation of plain CXRs for pulmonary TB in individuals aged 15 years or older.
- ➔ Molecular WHO-recommended rapid diagnostic tests (mWRDs) may be used to improve the accuracy of symptom screening in populations at high risk of TB.
- ➔ When scaling up CXR and innovations such as CAD, C-reactive protein and mWRDs for screening, due consideration should be given to avoid creating inequities. The use of mWRDs needs to be prioritized for diagnostic testing for people with presumptive TB. Implementation of CAD will require thorough consideration of its infrastructure requirements, including the availability of digital radiography equipment, computers and internet access, as well as the costs to licence and use different CAD products.
- ➔ In adults and adolescents living with HIV:
 - screening with CXR improves the sensitivity of the WHO-recommended four-symptom screen (cough, fever, weight loss or night sweats) for detecting TB, including in people who attend HIV care services for antiretroviral treatment;
 - C-reactive protein may be used for TB screening in addition to the WHO-recommended four-symptom screen in all people living with HIV in settings with a high TB burden;
 - mWRDs may be used to screen for TB in all people living with HIV as well as in other high-risk populations in settings with a high TB burden. They offer a distinct opportunity to improve timely diagnosis and treatment in hospitalized patients with HIV in medical wards in settings with a high TB burden.

1. Introduction

1.1 Background

Tuberculosis (TB) is a leading cause of death from a single infectious agent, despite being largely curable and preventable. In 2019, an estimated 2.9 million of the 10 million people who fell ill with TB were not diagnosed or reported to the World Health Organization (WHO) (7). WHO's End TB Strategy envisions a 90% reduction in TB incidence and a 95% reduction in TB deaths by 2035 (2, 3), and the Declaration adopted by the United Nations General Assembly in September 2018 at the High-Level Meeting on the Fight Against Tuberculosis commits to diagnosing and treating 40 million people with TB by 2022 (4). In order to achieve these ambitious targets, there is an urgent need to deploy strategies to improve the diagnosis and initiation of care for people with TB. One key strategy is systematic screening for TB disease, which is included in the End TB Strategy as a central component of its first pillar, aimed at ensuring early diagnosis for all with TB.

To help facilitate the implementation of TB screening at the country level, WHO published guidelines on systematic screening for TB in 2013 (5). Since then, there have been important new studies evaluating the impact of screening interventions on both individual-level and community-level outcomes related to TB, as well as new research evaluating innovative tools for screening for TB among important populations at high risk for TB disease. In light of these new developments and continual requests by Member States for guidance on how to conduct effective TB screening, in 2020 WHO convened a Guideline Development Group (GDG) to examine the evidence in order to update WHO's guidelines and recommendations for screening for TB disease.

1.2 Definition and objectives of systematic screening for TB disease

For the purpose of this guideline, systematic screening for TB disease is defined as **the systematic identification of people at risk for TB disease, in a predetermined target group, by assessing symptoms and using tests, examinations or other procedures that can be applied rapidly**. The screening tests, examinations and other procedures should efficiently distinguish people with a high probability of having TB disease from those who are unlikely to have TB disease. Among those whose screening is positive, the diagnosis needs to be established by diagnostic evaluation, consisting of one or more diagnostic tests, and clinical assessment, which together have high accuracy.

Systematic screening for TB disease is predominantly provider initiated. It may be conducted among people who do not seek health care because they do not have or recognize symptoms, they do not perceive that they have a health problem that warrants medical attention, there are barriers to accessing care, or for other reasons. It may also target people seeking health care who do or do not have symptoms or signs compatible with TB and who may not be identified by passive case-finding as possibly having TB. People seeking care who may be eligible for TB screening include people with medical conditions that constitute risk factors for TB (such as people living with HIV or diabetes mellitus) who may be seeking care for reasons other than symptoms compatible with TB.

There are two primary objectives of screening for TB disease: the first objective is to ensure that TB disease is detected early and treatment is initiated promptly, with the ultimate aim of reducing the risk of poor treatment outcomes, health sequelae and the adverse social and economic consequences of TB; the second objective is to reduce the community-level prevalence of TB disease, thus reducing transmission of *Mycobacterium tuberculosis* and averting future incident TB. Beyond TB disease, screening can also identify individuals who are eligible for and would benefit from TB preventive treatment (TPT) once TB disease is ruled out, thus further averting future incident TB.

Arising from these two primary objectives, there are two primary levels at which screening can augment standard TB care practices: that of the individual and the community. For individuals with TB disease, screening interventions can speed up and improve the probability of receiving a diagnosis and initiating treatment by bypassing many of the barriers to diagnosis and care that people suffering from TB disease can face, including (6):

- lacking knowledge about the signs and symptoms of TB;
- not recognizing symptoms;
- being unable to seek care or lacking sufficient resources for care-seeking;
- lacking access to TB diagnostic and treatment services;
- seeking care from providers who lack sufficient knowledge of TB or in facilities where TB diagnosis and treatment services are not available or of sufficient quality.

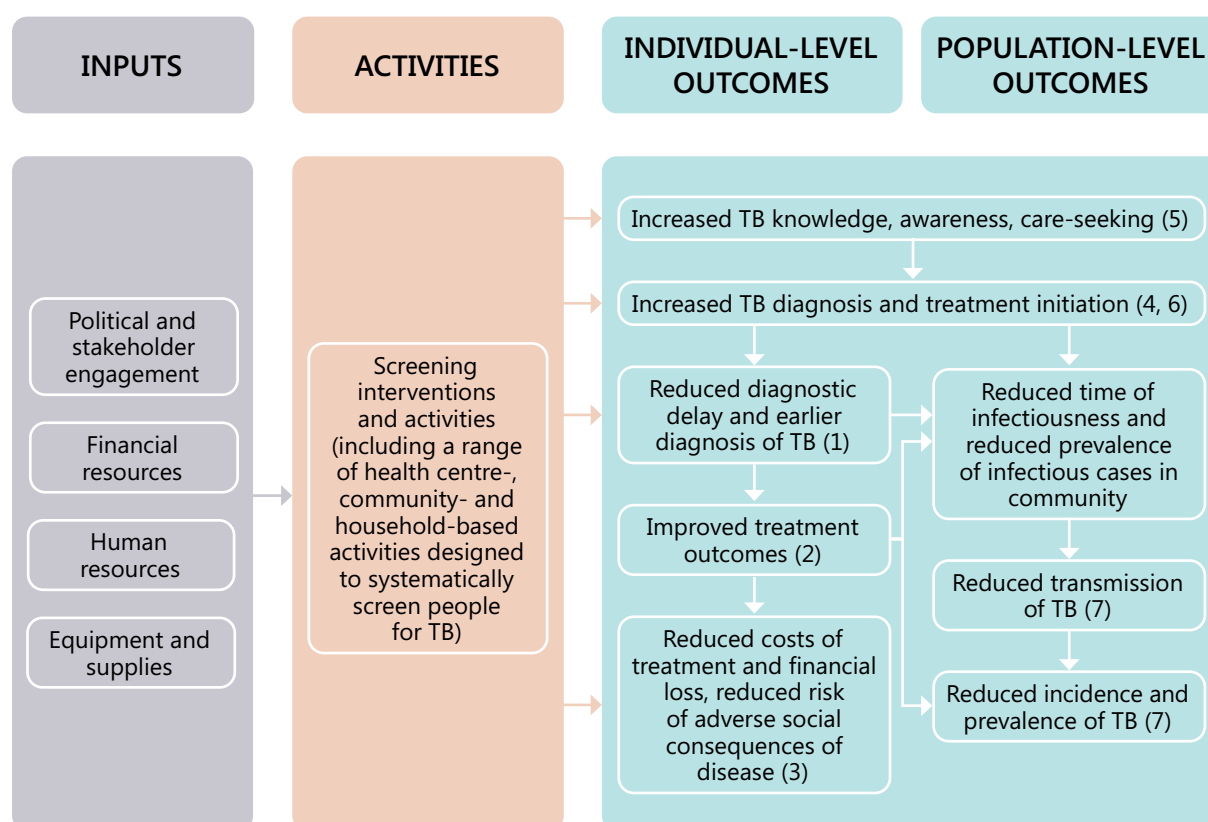
Beyond improving individual health and welfare, screening interventions for infectious diseases such as TB also address the epidemiology of the disease at the community level. By identifying and treating a proportion of the pool of individuals with TB disease in a given community, screening for TB seeks to reduce the prevalence, transmission and future incidence of the disease, with the long-term goal of eliminating the disease from the population.

The recommendations in this document apply to screening interventions that are conducted for the purpose of directly benefitting the individuals being screened by improving TB case detection and care and, thereby, benefitting the larger community by reducing the population burden of TB.

1.3 Scope of the 2021 update

Fig. 1 summarizes the potential contribution of TB screening to standard TB care practices based on the above frameworks for TB screening and shows the conceptual framework that guided the 2021 update to the TB screening guidelines.

Fig. 1. Conceptual framework for the 2021 WHO update to guidelines for systematic screening for TB. Numbers in parentheses refer to the PICO (population, intervention, comparator and outcome) questions that guided the evidence gathering



A series of PICO (population, intervention, comparator and outcome) questions was established and systematic reviews were sought or commissioned for each of the PICO questions and the priority background questions.

The first PICO questions aimed at summarizing the evidence on the effectiveness of systematic screening interventions for TB disease to address individual and community outcomes.

1. Among people with TB disease, does TB screening identify people at an earlier stage of disease compared with passive case-finding?
2. Is there a difference in TB treatment outcomes between people with TB identified through screening interventions compared with people with TB identified through passive case-finding?
3. For people with TB disease and their families, does receiving a diagnosis and undergoing a course of treatment after case detection through screening have a different cost of illness and risk of adverse social consequences compared with receiving a diagnosis and undergoing a course of treatment after diagnosis through passive case-finding?
4. For people being screened who do not have TB disease, what are the consequences of TB screening?
5. Does TB screening used in addition to passive case-finding affect subsequent health-seeking behaviour compared with passive case-finding alone?

6. Does TB screening initially increase the number of people with TB detected compared with passive case-finding alone?
7. Does TB screening affect the epidemiology of TB in a community, including the prevalence, incidence and transmission, compared with passive case-finding alone?

A series of questions focusing on the diagnostic accuracy and performance of screening tools was also prioritized.

8. Among people living with HIV, what is the performance of a range of screening tools compared with a microbiological reference standard?
9. Among the general population and high-risk groups eligible for TB screening, what is the performance of a range of screening tools compared with a microbiological reference standard?
10. Among children and adolescents eligible for TB screening, what is the performance of a range of screening tools compared with a composite or microbiological reference standard?
11. What is the performance of computer-aided detection (CAD) software for automated reading of digital chest radiographs (CXRs) for the detection of TB disease in the screening- and the triage-use cases?
12. Among the general population and high-risk groups eligible for TB screening, what is the performance of molecular WHO-recommended rapid diagnostic tests (mWRDs) for screening for TB disease compared with a microbiological reference standard?

Lastly, in addition to the PICO questions assessing the accuracy and effectiveness of screening interventions, there was a series of background questions prioritized to inform the implementation of TB screening interventions.

13. Among the general public and high-risk populations, what is the number needed to screen (NNS) to detect one person with TB disease?
14. Among contacts of people with TB disease, what is the NNS to detect one person with TB disease?
15. What are the estimated relative risks of TB associated with a range of risk factors?
16. What are the costs and cost-effectiveness of implementing TB screening interventions?
17. What are the perceptions and attitudes of communities towards TB screening programmes?

1.4 Rationale for the guideline update

Since the publication of the first WHO guidelines on systematic screening for TB in 2013 (5), there have been numerous studies, including reviews, randomized controlled trials, observational studies, modelling and cost-effectiveness research, evaluating the real or potential impacts of screening interventions on both individual-level and community-level outcomes related to TB. There have also been numerous prevalence surveys since 2013, which have shed new light on the magnitude of the burden of TB in several key countries. The majority of prevalence surveys have shown that the burden of TB is greater than previously thought.

1.5 Objectives of the guideline update

The specific objectives of the guideline update are:

1. to support Member States in implementing effective TB screening interventions by providing updated information about the expected impact of TB screening on patient-important outcomes and the epidemiology of TB, the expected yield of screening interventions and the expected performance of different screening tools and algorithms;

2. to contribute to finding more people with TB and finding them earlier in the course of disease in order to reduce disease burden, morbidity and mortality, and costs and financial hardship for people suffering from TB.

This update will allow policymakers in ministries of health to choose the best approach to planning and implementing screening and active case-finding activities, depending on the context. It will also provide a sound basis for developing or updating national guidelines for TB screening based on the epidemiology of TB and the health care delivery system in the country.

1.6 Target audience

This guideline is intended for personnel working in national TB programmes, national HIV/AIDS programmes or their equivalents, and other relevant national health programmes in ministries of health; other relevant ministries working in public health and screening; and for other health policymakers, clinicians and public health practitioners working on TB, HIV and infectious diseases in the public and private sectors. The recommendations provided here must be adapted to local settings. An accompanying operational handbook, *WHO operational handbook on tuberculosis. Module 2: screening – systematic screening for tuberculosis disease (7)* provides more detail and discussion of how screening recommendations can be adapted to address the conditions for and objectives of screening in specific contexts. National and subnational recommendations must be developed by national TB programmes and other national and subnational public health agencies and partners; recommendations developed by national and subnational agencies should guide health care staff working in specific settings.

2. Recommendations for systematic screening for TB disease in targeted populations

In this guideline, recommendations about systematic screening for TB disease are made for distinct populations for whom it is judged that the benefits and desirable effects of screening outweigh the potential harms.

Systematic screening for TB disease can be done for an entire population (community-wide screening) or it can be targeted at selected risk groups or subpopulations of people who may be at higher risk of being exposed to TB, developing TB disease or suffering poor outcomes from the disease, or some combination of these. It can target people who seek health care (with or without symptoms or signs compatible with TB) and people who do not seek care (because they do not perceive that they have a health problem that warrants medical attention, barriers make it difficult to access health care, or for other reasons). Furthermore, screening can help identify people who are at particularly high risk of developing TB disease and thus may require repeat screening, for example, people with an abnormal CXR compatible with TB but who are not diagnosed with TB disease at the time of screening or people for whom TPT is recommended. Combining screening for TB with screening for TB risk factors can also help map individual- or community-level risk factors, comorbidities and socioeconomic determinants that need to be addressed to more effectively prevent the disease.

Strong recommendations are made for those risk groups or subpopulations for which the desirable effects of adhering to the recommendation are judged to clearly outweigh the undesirable effects: for these recommendations, screening is judged to be feasible, acceptable and affordable in all settings. Conditional recommendations are made for those risk groups for which the desirable effects of TB screening probably outweigh the undesirable effects, but the trade-offs, cost-effectiveness, feasibility or affordability, or a combination of these, are uncertain. Reasons for uncertainty may include a lack of high-quality evidence to support the recommendation; limited evidence of benefit from implementing the recommendation; high costs or low feasibility or acceptability, or a combination of these.

It is important that TB is diagnosed promptly in groups that have a particularly high likelihood of undetected TB or a high risk of poor health outcomes, or both, in the absence of early diagnosis and treatment, even if direct evidence of benefit from screening is lacking. This is the rationale for making strong recommendations despite a lack of high-quality direct evidence for some risk groups. However, prioritization should also consider the risks associated with screening, including false-positive diagnoses, overtreatment and the opportunity costs of screening in terms of the resources required, both across risk groups and in relation to other interventions aimed at improving early diagnosis, treatment and prevention. Therefore, owing to the lack of high-quality direct evidence comparing benefits with harms and on the cost-effectiveness of screening in many of the populations considered, many of the recommendations made for screening in specific populations are conditional.

The degree of uncertainty about the trade-offs between the desirable effects and undesirable effects varies across settings, and it depends on the epidemiological situation and the health system. Therefore, a conditional recommendation in this guideline implies that the appropriateness of adhering to the

recommendation needs to be assessed in each setting, and there is a need to prioritize screening across risk groups in each setting.

For TB screening recommendations across all populations and in all settings, all people identified with presumptive TB disease should be offered HIV counselling and testing. In settings with a high prevalence of HIV, counselling and testing for HIV may be offered to all people screened for TB (8). People identified through screening who have symptoms or abnormalities suggestive of TB but who are not diagnosed with TB disease should be counselled and supported to seek medical care if their symptoms continue, emerge, re-emerge or worsen. If possible, repeat testing for TB should be offered to these individuals.

People who are diagnosed with TB disease through screening should have nutrition screening and assessment. If malnutrition is identified, it should be managed according to WHO's recommendations on nutritional care and support for people with TB. Concerns about weight loss or failure to gain weight should trigger further clinical assessment (e.g. to determine whether there is resistance to TB drugs, poor adherence, comorbid conditions) and nutrition assessment in order to determine the most appropriate interventions (9).

Screening population groups at higher risk for TB is ethically sound as long as it is designed and conducted to improve individual and public health and wellbeing; in this way screening contributes to the common good and upholds the ethical principle of accountability. As such, screening should always be done with the intent to provide care to those who need it, and never to exclude entry or employment or discriminate against individuals (10).

Nevertheless, screening population groups primarily based on risk raises a range of ethical issues (10). First, most of the individuals offered screening will not have TB disease and are not contagious. This makes the ethical obligations different from those associated with testing people who seek care because they are ill. For example, the absence of an immediate risk of transmission makes it unethical to restrict the movements of someone who has been offered screening. Refusal of screening should be respected and should not introduce discrimination of any sort. A screening programme should not create inequities in access to care for TB disease or any other reason.

Second, a positive screening test carries uncertainties, and confirmatory tests should be available to ensure an effective diagnostic pathway. Informed consent requires effective communication with each person about these uncertainties (e.g. false-positive results, risk of overtreatment). Appropriate mechanisms to obtain informed consent should comply with international human rights standards and account for different languages, literacy skills and legal status. Risk and uncertainty must be communicated in a way that is culturally and linguistically appropriate, including to those whose first language is foreign to the local setting, children, as well as to people in prison.

Third, TB disproportionately affects individuals and groups that are already disadvantaged due to disease, their socioeconomic situation or legal status, among other disadvantages. Therefore, efforts must be made to address existing inequities in access to services and to uphold human rights, so that the vulnerability of targeted groups does not impede their access to screening and treatment or violate their rights. Any intervention for vulnerable groups – including migrants, prisoners and children – should include measures to minimize the risk of stigmatization. Testing positive on a screening test – as with a confirmed diagnosis of TB – should not influence an immigration procedure or be used to force migration or deny entry to a country. Screening done for migration or employment reasons must uphold ethical principles; this should be reflected in laws or other policy regulations (10).

People should be offered screening in strict adherence to human rights and ethical considerations (11). Policies should be evaluated from an ethical perspective by those who use the guidelines, and the views and experiences of affected populations should be gathered after implementation, both to consider possible unexpected effects and to ensure that the evidence on which they are based remains current and relevant (12). Person-centred care entails, among other considerations, that it

is provided in an equitable fashion without placing marginalized and vulnerable populations at an added disadvantage; it focuses on the human rights aspects of screening so that there are appropriate safeguards in law, policy and practice to minimize any additional stigma, discrimination, violation of bodily integrity or restrictions on freedom of movement; and people who are offered testing and treatment should understand the associated uncertainties, which will help them make decisions about care options. Established human rights principles, such as consent, noncoercion, and confidentiality, should be respected.

2.1 Systematic screening for TB disease among the general population

1. Systematic screening for TB disease may be conducted among the general population in areas with an estimated TB prevalence of 0.5% or higher
(*updated recommendation: conditional recommendation, low certainty of evidence*).

2.1.1 Summary of the evidence and rationale

Systematic screening in the general population is conducted on the premise that it bears dual benefit: to the persons diagnosed with TB and to the community in which screening is conducted. Individuals found to have TB may benefit from less diagnostic delay, improved treatment outcomes, and lower costs and financial losses associated with the disease. It also benefits public health by reducing the population prevalence of TB, thereby reducing further transmission of TB.

There is limited direct evidence of individual benefit from improved treatment success or reduced mortality when TB screening is performed in the general population. There is some evidence that systematic screening helps reduce delay in TB diagnosis and that it detects patients at an earlier stage of their disease. Data also show that screening interventions result in a reduction in costs for patients who are detected through screening and, most critically, a reduction in the risk of catastrophic costs for patients detected through screening and their families. However, some data show that the proportion of people who do not start treatment is higher among those identified through screening than among those who present with illness.

With respect to the benefits of screening for the community, there is inconsistent evidence that systematic screening for TB improves detection and notification in the general population, with none coming from randomized trials. There is, however, evidence of an effect on TB prevalence and transmission. A trial in Viet Nam in a population with an estimated prevalence of 0.35% that used 3 years of annual door-to-door sputum collection and testing using the Xpert MTB/RIF assay showed that systematic screening reduced adult TB disease prevalence (73). An observational study in China conducted between 2013 and 2015 reported that three rounds of door-to-door symptom screening followed by CXR were associated with serial reductions in the absolute number of people with TB detected (74). In addition, two trials showed lower frequencies of TB infection among children in clusters where TB screening was done compared with others without the intervention (73, 75).

This is an updated recommendation: previously, systematic screening for TB disease in the general population was recommended in defined populations with extremely high levels of undetected TB, defined as a 1% prevalence or higher. Based on the updated evidence reviewed, the GDG concluded that the threshold of 1% recommended in the 2013 guidelines could be lowered, but considered that screening under programmatic conditions would not perform as well as was observed in the trial in Viet Nam and, therefore, proposed a 0.5% threshold to guide country implementation. Thus, the GDG recommended that general population screening may be considered in defined areas with a prevalence of undetected TB of 0.5% or more (see **Web Annex B**, Tables 1 and 2, and **Web Annex C**, Table 1).

2.1.2 Implementation considerations

The magnitude and balance of desirable and undesirable effects vary according to the epidemiological conditions (the prevalence of TB and of risk factors) and the intensity of the screening intervention being implemented (the coverage of the population and the sensitivity of the screening test and algorithm). There is currently no evidence that population-wide screening using less sensitive screening algorithms that begin with symptom screening are effective at reducing the population prevalence or transmission of TB. However, there is limited evidence that screening utilizing symptom-based screening tools may benefit individuals who are diagnosed with TB by leading to earlier diagnosis with less severe TB disease, lower costs to the patient and lower risk of catastrophic costs associated with the course of disease. The balance of potential benefits and harms of screening may tip in favour of the intervention depending on factors such as the prevalence of TB in the population to be screened (particularly undetected TB), the accuracy of the screening and diagnostic tests used, the degree of current TB case detection and the vulnerability of the population being screened.

Screening should be conducted using the most sensitive and specific screening algorithm possible, with a screening test that identifies those with a higher likelihood of having TB and a diagnostic test to confirm the diagnosis. Highly accurate screening tests such as CXR and mWRDs have high sensitivity, specificity, or both; however, the feasibility of their use is affected by resource and implementation requirements. Symptom screening is less accurate (reduced sensitivity, specificity, or both), but is generally considered much more feasible to implement. Recent evidence suggests that reductions in the population prevalence of TB can be achieved with repeated annual screenings using a sensitive screening algorithm (mWRD followed by clinical evaluation) (73). See **Section 3** for recommendations on tools for screening for more discussion on this topic.

While the implementation of a population-wide screening programme inevitably requires a significant investment of resources, overall there may be potential long-term savings to be made for health services through reduced future incidence. The costs of conducting community-wide screening will vary greatly, depending on the screening and diagnostic tests used and the population prevalence of TB. The cost-effectiveness of population-wide screening using a highly accurate algorithm is unknown, but the higher the prevalence, the more cost-effective the intervention will be. Using a highly accurate screening and diagnostic algorithm will inevitably require more resources than using symptom-based screening approaches, but it has a greater potential to reduce the population burden of TB.

The optimal frequency of screening in a general population to achieve individual- or community-level benefits remains unclear. There is also uncertainty about how people value the benefits expected from being screened. A review of qualitative studies of community perceptions of TB screening showed that, in general, screening was acceptable. However, resistance to screening was common, resulting from a combination of disparate factors, such as having a perception of being at low risk for TB among healthy participants, having little conviction in the effectiveness of screening, holding in poor regard TB services offered in the community, having concerns about confidentiality and the possible disclosure of diagnosis, and being anxious about the need to take TB medication.

Separate recommendations are made for select high-risk groups in the following sections. See the operational handbook for more details on the practicalities of implementing TB screening interventions using different tools and algorithms (7).

2.2 Systematic screening for TB disease among people with structural risk factors for TB

2. Systematic screening for TB disease may be conducted among subpopulations with structural risk factors for TB. These include urban poor communities, homeless communities, communities in remote or isolated areas, indigenous populations, migrants, refugees, internally displaced persons and other vulnerable or marginalized groups with limited access to health care (*existing recommendation: conditional recommendation, very low certainty of evidence*).

2.2.1 Summary of evidence and rationale

Populations with structural risk factors for TB are those that are at increased risks of TB and of poor health outcomes from TB due to structural determinants in their environment, defined as the conditions that generate or reinforce social stratification (e.g. socioeconomic inequalities, population growth, urbanization), and therefore give rise to an unequal distribution of key social determinants of TB epidemiology, such as poor housing, poverty and malnutrition, which in turn influence exposure to risk, vulnerability and ability to recover after developing the disease (16,17). Structural risk factors for TB include poverty; malnutrition; overcrowded and poorly ventilated living, working and gathering conditions; and limited or no access to health care. These risk factors are not mutually exclusive nor do they exclude the clinical risk factors described in other recommendations, and they often compound one another.

This recommendation has not been changed from the 2013 guidelines as, since the last GDG meeting, no new evidence was found to inform the question of the impact of systematic screening for TB disease compared with passive case-finding practices in these risk groups. Observational studies during 2013–2020 suggest that TB screening conducted among populations affected by structural risk factors may initially increase TB case notifications and decrease TB prevalence; however, all studies had a major risk of bias.

2.2.2 Implementation considerations

Due to the inherent lack of access to health care that defines the risk groups described in this recommendation, screening interventions would need to be conducted in and extended into the communities where members of these populations live and work in order to achieve effective reach and coverage.

The list of potential populations affected by structural risk factors included in this recommendation is not exhaustive, and this recommendation may apply to other groups with a high risk of TB and who have poor access to health care, including poor access to high-quality TB services.

There is no evidence about the effectiveness of different screening intervals; in the absence of such evidence, the choice of screening interval should be guided by feasibility. To the extent possible, community screening should be combined with screening for other diseases or risk factors and with health-promotion or social support activities. When screening is done in refugee camps and among displaced populations, consult *Tuberculosis care and control in refugee and displaced populations* (18) for recommendations on TB management and operational considerations.

2.3 Systematic screening for TB disease among people living with HIV

3. People living with HIV should be systematically screened for TB disease at each visit to a health facility

(existing recommendation: strong recommendation, very low certainty of evidence).

2.3.1 Summary of evidence and rationale

People living with HIV are approximately 19 times more likely to develop TB disease than those without HIV; in 2019, an estimated 44% of people living with HIV who also had TB disease did not reach care, and 30% of all HIV-related deaths were due to TB (7). Thus, ensuring early detection and treatment for TB among all people living with HIV is crucial for reducing morbidity and mortality in this group.

This recommendation, which applies to people of all ages, was first published in 2011 in WHO's *Guidelines for intensified tuberculosis case-finding and isoniazid preventive therapy for people living with HIV in resource-constrained settings* (19), and it remains in place. The GDG placed high value on ensuring that TB is diagnosed early in this risk group, which has a high likelihood of having undetected TB and a high risk of poor health outcomes in the absence of early diagnosis and treatment.

2.3.2 Implementation considerations

Recommendations on specific tools to be used for screening people living with HIV are described in detail in **Section 3**. Persons living with HIV who have a positive or abnormal screening test should go on to have primary diagnostic tests to confirm or rule out TB disease including an mWRD and lateral flow urine lipoarabinomannan assay (LF-LAM), if eligible (8). Once TB disease is ruled out, either through a normal screening test or after a negative diagnostic evaluation, people living with HIV should be evaluated for TPT as part of offering a comprehensive package of HIV care (20).

2.4 Systematic screening for TB disease among household and other close contacts of individuals with TB disease

4. Household contacts and other close contacts of individuals with TB disease should be systematically screened for TB disease

(updated recommendation: strong recommendation, moderate certainty of evidence).

2.4.1 Summary of evidence and rationale

Household and close contacts of individuals with TB disease are at high risk of TB infection and developing TB disease. A systematic review conducted for the guideline update found the weighted pooled prevalence of TB disease among all close contacts of TB patients was 3.6% (95% confidence interval [CI]: 3.3–4.0), with a median NNS of 35 (95% CI: 17–65). Systematic screening has been strongly recommended since 2012 for contacts of individuals with TB disease (27), given the high prevalence of disease in this population. At the GDG meeting, evidence was also presented on the individual- and community-level effects of screening of close contacts of TB patients. One trial of screening household contacts in Viet Nam showed a 2.5-fold increase in notification of contacts

diagnosed with TB resulting from the intervention (relative risk [RR]: 2.5; 95% CI: 2.0–3.2), along with a 40% reduction in all-cause mortality among household contacts (RR: 0.6; 95% CI: 0.4–0.8) (22). Another trial of contact screening in South Africa and Zambia showed evidence that screening household contacts may reduce TB prevalence in the larger community after 4 years during which contact investigation is conducted, although this did not reach statistical significance (prevalence ratio: 0.82; 95% CI: 0.64–1.04); there was also a suggestion that the intervention reduced the transmission of TB, measured through the incidence of TB infection in schoolchildren (RR: 0.45; 95% CI: 0.20–1.05) (see **Web Annex B**, Table 3, and **Web Annex C**, Table 2) (15).

Eligibility for TPT should be assessed among all TB contacts who are screened and found not to have indications of TB disease (20). Children aged < 5 years who are household and close contacts of people with bacteriologically confirmed pulmonary TB and who are found not to have TB disease after appropriate clinical evaluation or according to national guidelines should be given TPT even if testing for TB infection is unavailable. Children aged ≥ 5 years, adolescents and adults who are household and close contacts of people with bacteriologically confirmed pulmonary TB who are found not to have TB disease after appropriate clinical evaluation or according to national guidelines may also be given TPT. Contacts of TB patients who are living with HIV, no matter what their age, in whom TB disease has been ruled out, should receive TPT as part of a comprehensive package of HIV care (20).

2.4.2 Implementation considerations

Contact screening should always be done when a person with TB has any of the following characteristics: bacteriologically confirmed pulmonary TB, proven or presumed multidrug-resistant TB or extensively drug-resistant TB, is a person living with HIV or is a child younger than 5 years. Among contacts of patients with bacteriologically confirmed TB, the weighted pooled prevalence of TB was 3.4% (95% CI: 2.9–3.8). Among contacts of patients with multidrug-resistant or extensively drug-resistant TB the weighted pooled prevalence of TB was 3.7% (95% CI: 2.4–5.3). The weighted pooled prevalence of TB among HIV-positive contacts was 11.6% (95% CI: 8.2–15.4), with a median NNS of 9 (95% CI: 5–13). The weighted pooled prevalence among contacts younger than 5 years was 3.9% (95% CI: 2.5–5.4), with a median NNS of 30 (95% CI: 12–62), while the prevalence among contacts aged 5–14 years was 2.4% (95% CI: 1.6–3.4), with a median NNS of 36 (95% CI: 17–61). Contact investigation may also be performed for TB patients with all other forms of disease.

The definition of a household or close contact can vary greatly across settings, and in some contexts there is no clear distinction between households; therefore, decisions about screening household and close contacts should be based on local definitions and policies, prioritizing contacts for screening based on an assessment of their level of exposure.

2.4.3 Subgroup considerations

Children and adolescents are of particular importance in contact screening, given the high prevalence found in the youngest age groups and the importance of rapid diagnosis and treatment owing to the risk of rapid disease progression in children younger than 5 years (23), as well as the importance of initiating preventive treatment if TB disease has been ruled out (see **Section 2.4.2**).

Contacts who are persons living with HIV are also of particular importance in contact screening, given their extremely high prevalence of TB disease. Contacts living with HIV are also at increased risk for rapid progression to TB disease, and if they develop it, an increased risk of mortality. Contacts of TB patients who are also living with HIV are at a higher probability of themselves being HIV-positive, even they are unaware of their status. Thus, in settings with a high prevalence of HIV or when the index case is a person living with HIV, or both, all household contacts and close contacts should be offered counselling and testing for HIV. People living with HIV who are household contacts or close contacts of someone with TB disease and who, after an appropriate clinical evaluation, are found not to have TB disease should be evaluated for TPT.

2.5 Systematic screening for TB disease in prisons and other penitentiary institutions

5. Systematic screening for TB disease should be conducted in prisons and penitentiary institutions

(updated recommendation: strong recommendation, very low certainty of evidence).

2.5.1 Summary of evidence and rationale

People in prisons and other penitentiary institutions are at an increased risk for TB compared with the general population, and they often have limited access to health care services. The estimated incidence of TB among people residing in prisons is 23 times higher than that among the general population (24). Data reviewed for the GDG meeting suggest that screening in prisons may improve early case detection, increase overall case detection and reduce TB prevalence. One observational study showed that a smaller proportion of TB patients in prisons whose disease was detected through screening were at an advanced stage of disease compared with those diagnosed through passive case-finding services (25). Two observational studies showed that TB screening in prisons may reduce TB prevalence in the facilities (26, 27). One trial showed that screening in prisons may increase case detection by more than 50% (28).

Based on this new evidence and the high risk of TB in this population, the GDG felt that this risk group now merited a strong recommendation for TB screening (see **Web Annex B**, Table 4, and **Web Annex C**, Table 3). The GDG felt that implementing TB screening in prisons has the potential to increase equity in access to health care, particularly in settings where health services in prisons are suboptimal.

2.5.2 Implementation considerations

A prisoner is anyone held in a criminal justice facility or correctional facility during the investigation of a crime, anyone awaiting trial and anyone who has been sentenced. In addition, people residing in a correctional facility are almost always in close contact with several other inmates; thus, whenever a person residing in a prison is diagnosed with TB, prisoners who have been in close contact with that person should be investigated (see **Recommendation 4**). People who work in prisons and other penitentiary institutions are also at high risk of exposure to TB and should also be eligible for screening.

At a minimum, screening in prisons and other penitentiary institutions should always include screening when a person enters a detention facility, annual screening and screening upon release to prevent the reintroduction of TB into the broader community. Treatment and follow up after release should also be ensured.

Screening in prisons should be combined with efforts to improve living conditions and provide infection control measures. If possible, TB screening in prisons and other penitentiary institutions should be combined with screening for other diseases, including HIV, and health-promotion activities targeted towards people in these institutions. When starting screening, it is important to ensure that high-quality treatment and effective case management are in place, as well as effective mechanisms for continuing treatment after transfer or release. For recommendations on and operational aspects associated with TB care and prevention in prisons and other penitentiary institutions, see WHO's *Guidelines for the control of tuberculosis in prisons* (29). However, even if TB management practices are suboptimal, screening may be initiated to assess the burden of undetected TB among inmates to provide a rationale for strengthening general diagnostic and treatment services for TB, as well as implementing measures to improve infection control and living conditions. In addition to systematic screening, any inmate who develops symptoms suggestive of TB should have easy access to diagnostic

testing. People identified through screening and presumed to have TB disease but in whom active TB has not been diagnosed, should be informed about the importance of seeking medical care if TB symptoms continue, emerge, re-emerge or worsen.

2.6 Systematic screening for TB disease among miners and others exposed to silica dust

6. Current and former workers in workplaces with silica exposure should be systematically screened for TB disease

(existing recommendation: strong recommendation, low certainty of evidence).

2.6.1 Summary of evidence and rationale

Exposure to silica dust and silicosis are among the strongest risk factors for TB, with a relative risk of 2.8–39 for silicosis, depending on the severity of the disease (30). Silicosis is common in miners (31, 32), which is a primary reason for the high incidence of TB among them, and this is often compounded by a high prevalence of HIV. TB patients with silicosis have an increased risk of death (RR: 3.0; 95% CI: 1.4–6.3) (33). Thus, while the quality of the direct evidence is low for the benefit of TB screening in employees in workplaces where they are exposed to silica, the GDG placed high value on ensuring that TB is diagnosed early in this risk group, which has a high likelihood of having undetected TB as well as other pulmonary diseases that may be detected through screening. This recommendation has not been changed from the 2013 guidelines as no new robust evidence was found to inform the subject since the last guideline meeting.

2.6.2 Implementation considerations

There is limited evidence on the effectiveness of different screening intervals. The GDG suggests that the screening interval should be no longer than 12 months if possible, while an interval shorter than 12 months may be more beneficial. To the extent possible, TB screening should be combined with screening for other diseases and with health-promotion activities, as well as with efforts to improve working conditions (especially by reducing exposure to silica) and living conditions. During employment, screening should be considered to be the responsibility of the employer, and countries may have occupational health and safety legislation that addresses this.

2.7 Systematic screening for TB disease among people attending health care services who have clinical risk factors for TB

7. In settings where the TB prevalence in the general population is 100/100 000 population or higher, systematic screening for TB disease may be conducted among people with a risk factor for TB who are either seeking health care or who are already in care

(existing recommendation: conditional recommendation, very low certainty of evidence).

8. People with an untreated fibrotic lesion seen on chest X-ray may be systematically screened for TB disease

(existing recommendation: conditional recommendation, very low certainty of evidence).

2.7.1 Summary of evidence and rationale

Several clinical characteristics, conditions and comorbidities can indicate an increased risk for developing TB disease or suffering worse outcomes from the disease, or both. Individuals identified as having untreated fibrotic lesions on CXR and who are not diagnosed with TB disease are at increased risk of developing TB disease (34–37). These individuals are often identified through TB screening or clinical evaluation or during a clinical evaluation done for other reasons. Individuals with other risk factors for TB or risk factors for poor outcomes from TB often can be identified most easily in health care settings (38–66). **Table 2** summarizes the evidence about the primary risk factors for TB and for poor health outcomes related to TB. Groups with other risk factors – such as people with malignancies and other disorders that compromise their immune system and people receiving immunomodulatory therapies – may also be prioritized, depending on the local epidemiology and capacity of the health system.

This recommendation has not been changed from the 2013 guidelines because since the last GDG meeting, no new, robust evidence was found about the impact of systematic screening for TB disease compared with passive case-finding for screening of individuals with risk factors.

Table 2. Risk factors for tuberculosis (TB) to be considered when prioritizing TB screening among people attending for health care

Risk factor	Risk of TB or of poor outcomes
Fibrotic lesions on chest X-ray	Studies have shown an increased risk of developing TB disease among individuals with fibrotic lesions identified on chest X-ray but who are not diagnosed with TB disease. ^a
Diabetes mellitus (DM)	Systematic reviews have found that the risk of TB for patients with DM ranges from 1.5 to 2.0 to 3.1, with a decreasing risk in patients with well-controlled DM. Patients with DM were also at an increased risk of relapse, treatment failure and death. ^b
Previous TB	Patients with a history of TB are at increased risk of subsequent TB episodes, poor outcomes and developing drug-resistant TB. ^c
Chronic lung disease	Studies have found hazard ratios for developing TB among individuals with COPD ranging from 2.5 in China to 3.0 in Sweden. ^d
Smoking	Systematic reviews have found that people who smoke or who have a history of smoking have an increased risk of TB, with the RR ranging from 1.5 to 2.0 to 3.3; smokers are also at an increased risk of drug-resistant TB and poor outcomes from TB, including relapse and death. ^e
Alcohol use disorder	Systematic reviews have found a 1.35 to 1.9 RR associated with alcohol use and a 3 to 3.33 RR associated with alcohol use disorder, as well as increased risks of treatment failure and development of drug-resistant TB. An exposure–response analysis showed that for every 10–20 g of daily alcohol intake, there was a 12% increase in TB risk. ^f
Substance use disorder	People with substance use disorder are at increased risks of treatment failure, development of drug resistance, and mortality from TB due to low adherence and coincident clinical, socioeconomic and structural risk factors. ^g
Malnourishment	A systematic review found that lower BMI is associated with an increased risk of TB, with a reduction in TB incidence of 13.8% (95% confidence interval: 13.4–14.2) per unit increase in BMI within the range 18.5–30 kg/m ² . There are multiple pathways by which undernourishment can increase the risk of TB, including cell-mediated immunity and micronutrient deficiency, and other conditions can increase the risk of malnourishment and TB, including mental health and substance use disorders. ^h

Risk factor	Risk of TB or of poor outcomes
Pregnancy	A national registry study found IRRs for TB in pregnant women of 1.4 and of 1.9 for postpartum women compared with non-pregnant women. TB in pregnancy is associated with adverse outcomes and complications during birth. These outcomes include a roughly 2-fold increased risk of premature birth, low birthweight and intrauterine growth retardation, and a 6-fold increased risk of perinatal death. ⁱ
Immunocompromising conditions (organ transplant, renal failure, dialysis)	Systematic reviews have found that patients with immunosuppression for reasons other than HIV, including those undergoing organ transplantation or haemodialysis and those with renal failure, have a greatly increased risk of TB: one study found that the incidence of TB is 20–74 times higher in organ transplant patients compared with the general population; one study found that the incidence of TB is 10–25 times higher in dialyzed individuals compared with the general population; and one cohort study found that the incidence of TB is 100 times greater in persons undergoing dialysis compared with general population. ^j
Health care workers	A systematic review found the IRR of TB disease for health care workers was 2.94 compared with the general population. ^k

BMI: body mass index; COPD: chronic obstructive pulmonary disease; DM: diabetes mellitus; IRR: incidence rate ratio; RR: relative risk.

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^bHayashi S, Chandramohan D. Risk of active tuberculosis among people with diabetes mellitus: systematic review and meta-analysis. *Trop Med Int Health*. 2018;23:1058–70. doi:10.1111/tmi.13133. Al-Rifai RH, Pearson F, Critchley JA, Abu-Raddad LJ. Association between diabetes mellitus and active tuberculosis: a systematic review and meta-analysis. *PLOS ONE*. 2017;12:e0187967. doi:10.1371/journal.pone.0187967. Harries AD, Kumar AM, Satyanarayana S, Lin Y, Zachariah R, Lönnroth K, et al. Addressing diabetes mellitus as part of the strategy for ending TB. *Trans R Soc Trop Med Hyg*. 2016;110:173–9. doi:10.1093/trstmh/trv111. Baker MA, Harries AD, Jeon CY, Hart JE, Kapur A, Lönnroth K, et al. The impact of diabetes on tuberculosis treatment outcomes: a systematic review. *BMC Med*. 2011;9:81. doi:10.1186/1741-7015-9-81.

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^dInghammar M, Ekblom A, Engström G, Ljungberg B, Romanus V, Löfdahl CG, et al. COPD and the risk of tuberculosis—a population-based cohort study. *PLOS ONE*. 2010;5:e10138. doi:10.1371/journal.pone.0010138. Lee CH, Lee MC, Shu CC, Lim CS, Wang JY, Lee LN et al. Risk factors for pulmonary tuberculosis in patients with chronic obstructive airway disease in Taiwan: a nationwide cohort study. *BMC Infect Dis*. 2013;13:194. doi:10.1186/1471-2334-13-194.

^eJayes L, Haslam PL, Gratziau CG, Powell P, Britton J, Vardavas C, et al. SmokeHazz: systematic reviews and meta-analyses of the effects of smoking on respiratory health. *Chest*. 2016;150:164–79. doi:10.1016/j.chest.2016.03.060. Lin HH, Ezzati M, Murray M. Tobacco smoke, indoor air pollution and tuberculosis: a systematic review and meta-analysis. *PLOS Med*. 2007;4:e20. doi:10.1371/journal.pmed.0040020. Wang MG, Huang WW, Wang Y, Zhang YX, Zhang MM, Wu SQ, et al. Association between tobacco smoking and drug-resistant tuberculosis. *Infect Drug Resist*. 2018;11:873–87. doi:10.2147/idr.s164596. Alavi-Naini R, Sharifi-Mood B, Metanat M. Association between tuberculosis and smoking. *Int J High Risk Behav Addict*. 2012;1:71–4. doi:10.5812/ijhrba.5215.

^fImtiaz S, Shield KD, Roerecke M, Samokhvalov AV, Lönnroth K, Rehm J. Alcohol consumption as a risk factor for tuberculosis: meta-analyses and burden of disease. *Eur Respir J*. 2017;50:1700216. doi:10.1183/13993003.00216-2017. Simou E, Britton J, Leonardi-Bee J. Alcohol consumption and risk of tuberculosis: a systematic review and meta-analysis. *Int J Tuberc Lung Dis*. 2018;22:1277–85. doi:10.5588/ijtld.18.0092. Rehm J, Samokhvalov AV, Neuman MG, Room R, Parry C, Lönnroth K, et al. The association between alcohol use, alcohol use disorders and tuberculosis (TB). A systematic review. *BMC Public Health*. 2009;9:450. doi:10.1186/1471-2458-9-450.

^aDeiss RG, Rodwell TC, Garfein RS. Tuberculosis and illicit drug use: review and update. *Clin Infect Dis*. 2009;48:72–82. doi:10.1086/594126. Getahun H, Gunneberg C, Sculier D, Verster A, Ravigliione M. Tuberculosis and HIV in people who inject drugs: evidence for action for tuberculosis, HIV, prison and harm reduction services. *Curr Opin HIV AIDS*. 2012;7:345–53. doi:10.1097/COH.0b013e328354bd44. Silva DR, Muñoz-Torrico M, Duarte R, Galvão T, Bonini EH, Arbex FF, et al. Risk factors for tuberculosis: diabetes, smoking, alcohol use, and the use of other drugs. *J Bras Pneumol*. 2018;44:145–52. doi:10.1590/s1806–37562017000000443.

^bLönnroth K, Williams BG, Cegielski P, Dye C. A consistent log-linear relationship between tuberculosis incidence and body mass index. *Int J Epidemiol*. 2010;39:149–55. doi:10.1093/ije/dyp308. Cegielski JP, McMurray DN. The relationship between malnutrition and tuberculosis: evidence from studies in humans and experimental animals. *Int J Tuberc Lung Dis*. 2004;8:286–98.

^cJonsson J, Kühlmann-Berenzon S, Berggren I, Bruchfeld J. Increased risk of active tuberculosis during pregnancy and postpartum: a register-based cohort study in Sweden. *Eur Respir J*. 2020;55:1901886. doi:10.1183/13993003.01886–2019. Sugarman J, Colvin C, Moran AC, Oxlade O. Tuberculosis in pregnancy: an estimate of the global burden of disease. *Lancet Glob Health*. 2014;2:e710–6. doi:10.1016/s2214–109x(14)70330–4. doi:10.1016/S2214–109X(14)70330–4. Loto OM, Awowole I. Tuberculosis in pregnancy: a review. *J Pregnancy*. 2012;2012:379271. doi:10.1155/2012/379271.

^dMuñoz P, Rodríguez C, Bouza E. *Mycobacterium tuberculosis* infection in recipients of solid organ transplants. *Clin Infect Dis*. 2005;40:581–7. doi:10.1086/427692. Moore DA, Lightstone L, Javid B, Friedland JS. High rates of tuberculosis in end-stage renal failure: the impact of international migration. *Emerg Infect Dis*. 2002;8:77–8. doi:10.3201/eid0801.010017. Moran E, Baharani J, Dedicoat M, Robinson E, Smith G, Bhomra P, et al. Risk factors associated with the development of active tuberculosis among patients with advanced chronic kidney disease. *J Infect*. 2018;77:291–5. doi:10.1016/j.jinf.2018.06.003.

^eUden L, Barber E, Ford N, Cooke GS. Risk of tuberculosis infection and disease for health care workers: an updated meta-analysis. *Open Forum Infect Dis*. 2017;4:ofx137. doi:10.1093/ofid/ofx137.

2.7.2 Implementation considerations

This recommendation concerns interventions that should be undertaken in addition to passive case-finding practices, namely properly triaging and evaluating people seeking care who report signs or symptoms of TB, which should be done in all settings and is particularly important to implement rigorously among people who have risk factors for TB.

Groups should be prioritized based on their risk of TB, the risk of poor treatment outcomes if diagnosis is delayed and the size of the risk group in a given setting. People who are living with HIV, people who have had recent close contact with a person who has TB and people who have silicosis should always be screened for TB as described above (**Recommendations 3, 4 and 6**). Other risk factors that indicate an individual should be considered for screening are listed in **Table 2**.

For several of the clinical risk factors indicated above, TPT may be beneficial for the patient, subject to their exposure and eligibility (20).

Health care workers are a specific group that merits consideration for screening in health facilities, given the potentially high level of occupational exposure and the risk of further transmission to patients.

3. Recommendations for tools for systematic screening for TB disease

TB screening tools are designed to distinguish people with a higher probability of having TB disease from those with a low probability and can be assumed to be free of TB disease. They are not intended to provide a definitive diagnosis. In general, they need to be able to be implemented easily and relay results rapidly in order to be informative in a screening context. Screening tests need to be followed by a diagnostic test, offered as part of a comprehensive clinical evaluation, to confirm or rule out TB disease in individuals who screen positive.

The most desirable screening strategy would be one with a high total yield of true-positive cases of TB, few false positives, low NNS, low cost, a rapid and simple algorithm and high client acceptability. In practice, many of these factors tend to run in opposite directions, so a multifactorial analysis is needed to decide which screening tools and algorithms to use.

In 2014, WHO released a report summarizing the desirable characteristics, or target product profiles, of screening tests for detecting TB disease (67). The report highlighted that the minimal requirements for a target screening test would be an overall sensitivity of 90% and a specificity of 70% to detect pulmonary TB disease or rule it out in individuals being screened. Based on these benchmarks, an array of potential tools for screening for TB disease in different populations was considered by the GDG.

3.1 Tools for screening for TB disease among the general population and high-risk groups

9. Among individuals aged 15 years and older in populations in which TB screening is recommended, systematic screening for TB disease may be conducted using a symptom screen, chest X-ray or molecular WHO-recommended rapid diagnostic tests, alone or in combination (*new recommendation: conditional recommendation, very low certainty of evidence for test accuracy*).

3.1.1 Summary of the evidence and rationale

The data used to inform this recommendation came from a systematic review of the diagnostic accuracy of using symptoms and chest radiography to detect TB disease among individuals aged 15 years and older with negative or unknown HIV status. The review included studies of screening conducted in the general population (including several prevalence surveys conducted in African and Asian countries), as well as screening conducted in high-risk groups (including contacts of TB patients, prisoners and others). Data from a separate review of the diagnostic accuracy of mWRDs used as a screening tool in individuals from high-risk groups aged 15 years and older (including contacts of TB

patients, prisoners and miners, all with negative or unknown HIV status) were also used to inform this recommendation. All data were pooled to estimate the sensitivity and specificity of each screening tool, as there was insufficient evidence to make estimates of accuracy for the screening tools in each high-risk group (the performance of tools among people living with HIV is presented later, in **Recommendations 11–15**). The reference standard used throughout was bacteriologically confirmed pulmonary TB. **Table 3** summarizes the diagnostic accuracy of the screening tools considered (see **Web Annex B**, Tables 5–10 for more details).

Table 3. Diagnostic accuracy of symptoms, chest radiography, and molecular WHO-recommended rapid diagnostic tests for screening for TB disease

Screening test	No. of studies (no. of participants)	Sensitivity	No. of studies (no. of participants)	Specificity
WHO target product profile	NA	> 0.90	NA	> 0.70
Prolonged cough (≥ 2 weeks)	40 (6 737)	0.42	40 (1 284 181)	0.94
Any cough	21 (2 734)	0.51	21 (768 291)	0.88
Any TB symptom (cough, haemoptysis, fever, night sweats, weight loss)	28 (3 915)	0.71	28 (460 878)	0.64
Chest radiography (any abnormality)	22 (4 243)	0.94	22 (1 012 752)	0.89
Chest radiography (suggestive abnormality)	19 (2 152)	0.85	19 (464 818)	0.96
Molecular WHO-recommended rapid diagnostic test	5 (337)	0.69	5 (8 619)	0.99

NA: not applicable.

3.1.2 Implementation considerations for all tools

The GDG considered that all three approaches – symptom screening, CXR and mWRD – may have roles in screening for TB disease in the general population. The ranking of the tools according to accuracy proposed by the GDG is: first, CXR; second, mWRDs; third, screening for any TB symptom (higher sensitivity and lower specificity); and fourth, screening for any cough or cough lasting 2 weeks or longer (lower sensitivity and higher specificity). However, ease of implementation is highest by far for symptom screening, and less so for CXR and mWRDs with the requirement of additional equipment and resources. The GDG noted that factors related to patient selection, flow and timing may affect the measures of accuracy observed for CXR to detect TB. The inferences made about using mWRDs for screening in the general population relate primarily to pooled studies in high-risk populations and, therefore, there is uncertainty about whether the findings are directly applicable to a general population with a comparable burden of TB. See **Web Annex C**, Table 1, for more details of the recommendation.

The GDG noted that different symptom screening approaches have varying trade-offs in sensitivity and specificity. The feasibility of implementing symptom screening makes it a much more accessible option programmatically. Symptom screening is a standard part of a clinical workup and can be repeated as often as it is needed.

In contrast, additional resources are needed to undertake chest radiography and mWRDs. Radiography involves exposure to some ionizing radiation, which may increase the long-term risks of cancer. Innovations in radiography in recent years have substantially reduced radiation exposure levels. CXR is largely considered safe, with a radiation dose of 0.1 mSv, which corresponds to 1/30 of the average annual radiation dose from the environment (3 mSv) and 1/10 of the annual accepted dose of ionizing radiation for the general public (1 mSv). Pregnant women are especially vulnerable to ionizing radiation from radiography, and children have a longer life expectancy and, therefore, more time to develop radiation-induced health effects. However, for a pregnant woman and her fetus and for children, CXR has been deemed to not pose a significant risk, provided that good practices are observed, as the primary beam is targeted away from the pelvis (68).

When used for screening, mWRDs have different accuracy than when they are used for diagnosis (69), and different predictive values are associated with a positive test and a negative test due to differences in prevalence of TB in the populations being tested. Therefore, results should be interpreted appropriately, and those who screen positive for TB using an mWRD should receive a thorough clinical evaluation that may include further tests and procedures – such as CXR, repeat mWRD on additional sputum samples and other examinations – to establish a diagnosis of TB definitively prior to initiating treatment. These tests can be used on sputum only when a person can expectorate. Scaling up mWRDs for diagnosis should be prioritized (if full access has not yet been achieved) prior to scaling up mWRDs for screening. Using an mWRD as a screening tool requires significant resources, including increased capacity in and expansion of diagnostic and sample transportation networks. Depending on the feasibility and resources available, countries may choose to prioritize TB screening using mWRDs among certain subpopulations with a higher risk of TB.

Countries should position symptom screening, CXR and mWRDs within national TB screening and diagnostic algorithms, according to the goals and objectives of the screening, the populations being screened, the feasibility, resources available, and equity. A range of possible screening algorithms is presented in the operational handbook, including modelled performance of the expected yield.

3.2 Use of computer-aided detection software for automated reading of digital chest radiographs

10. Among individuals aged 15 years and older in populations in which TB screening is recommended, computer-aided detection software programmes may be used in place of human readers for interpreting digital chest X-rays for screening and triage for TB disease (*new recommendation: conditional recommendation, low certainty of evidence*).

3.2.1 Summary of the evidence and rationale

The use of CXR to screen for TB is a practice that goes back several decades. CXRs are also routinely used for triage of patients presenting to care who are displaying signs, symptoms or risk factors for TB to determine the most appropriate clinical pathway for proper evaluation. However, in many settings, the use of CXR for TB screening and triage for TB disease is limited by the unavailability of trained health personnel to interpret radiography images and by substantial intra- and inter-reader variability in its accuracy to detect abnormalities associated with TB (70–72).

Numerous specific software packages that provide CAD, or automated interpretation of digital CXR images for the express purpose of determining the likelihood of TB disease, have been developed and offer a potential technological answer to the numerous implementation challenges inherent in human interpretation of CXRs.

The GDG considered the performance of CAD software separately for the screening- and triage-use cases. For this guideline, triaging is defined as the process of deciding the diagnostic and care pathways for people based on their symptoms, signs, risk markers and test results. Triage involves assessing the likelihood of various differential diagnoses as a basis for making clinical decisions (73). It can follow more- or less-standardized protocols and algorithms, and it may be done in multiple steps (68). A triage test for TB is one that can be rapidly conducted among people presenting to a health facility to differentiate those who should have further diagnostic evaluation for TB (those whose TB triage test is positive or abnormal) from those who should undergo further investigation for non-TB diagnoses (for those whose TB triage test is negative or normal) (74). While there may be overlap between triaging and screening, there are several reasons to distinguish screening from triage when evaluating the performance of CAD software:

- The disease presentation may be different in screening populations in which one is more likely to encounter CXR findings of earlier TB than when compared with triage populations. Therefore, the same sensitivity and specificity point may not be achieved or may be achieved but with a different threshold score.
- TB prevalence will typically be much lower in screening populations (< 5%) than in triage populations (10–20%), which will impact a test's predictive values and the numbers of individuals correctly and incorrectly diagnosed.
- The ethical consequences of not detecting TB or other non-TB-related CXR findings (but clinically relevant abnormalities) that require follow-up examination are different for populations that do not seek care than for those that do (77).

A previous assessment of using CAD for automated interpretation of digital CXRs for TB by WHO determined that in order to adequately assess diagnostic accuracy, it was necessary to evaluate CAD software using a standard panel of CXR files with associated demographic and clinical data, including TB diagnosis, drawn from a representative population for the corresponding use case for the technology. It was deemed essential that such evaluations ensure that CXR libraries used in an evaluation not be made available for CAD software development, training or evaluation (68). For this GDG meeting, a scoping review for independent evaluations that met these criteria was conducted. Three independent evaluations for both the screening use case and the triage use case that assessed the performance of three distinct CAD programmes were identified and presented to the GDG, and they included all products that had received a CE mark (for *Conformité Européenne*, indicating a product's conformity with the European Economic Area's directives or standards) by January 2020.² The GDG was blinded to the brand names of the software programmes. A separate quality assessment of the evaluations was conducted and results presented to the GDG.

CAD programmes produce a numerical abnormality score for each digital image read that can then be compared to a threshold defined by the user to indicate if the patient is to be referred for further TB diagnostic evaluation. Because the abnormality scores produced are continuous, the sensitivity and specificity can vary from 0 to 100%, depending on where the threshold is set. For evaluation for the GDG, each software programme was set to a threshold that corresponded to 90% sensitivity for detecting pulmonary TB disease based on a microbiological reference standard. The resulting accompanying specificity for the software at that threshold was then reported and compared with the diagnostic accuracy of human readers interpreting CXRs in the same studies.

Due to specific methodological challenges, the estimates of CAD diagnostic accuracy were not able to be pooled across software programmes or across evaluations. Thus, the performances of CAD programmes and human readers from the included evaluations were presented as ranges (see **Table 4**). The three included evaluations assessed each programme's performance in different populations and in different settings (see **Web Annex B**, Tables 11 and 12, and **Web Annex C**, Tables 4 and 5).

² The three technologies that had received a CE mark by January 2020 and were included in all the evaluations are CAD4TB v6, Delft Imaging; Lunit Insight CXR, Lunit Insight; and qXR v2, Qure.ai.

Table 4. Sensitivity and specificity ranges of computer-aided detection software and human readers interpreting digital chest radiographs for detection of bacteriologically confirmed TB across three software programmes, from three independent evaluations of the software in a range of populations and settings

Type of case and type of reader	Accuracy estimate range	
	Sensitivity	Specificity
WHO target product profile	> 0.90	> 0.70
Screening use case		
CAD software	0.90–0.92	0.23–0.66
CXR with human reader	0.82–0.93	0.14–0.63
Triage use case		
CAD software	0.90–0.91	0.25–0.79
CXR with human reader	0.89–0.96	0.36–0.63

CAD: computer-aided detection; CXR: chest X-ray.

The results showed the variability of both human readers and CAD software programmes across different settings and populations. In comparing the range of accuracy of CAD to that of human readers interpreting CXRs and noting the variability of readers and the substantial overlap between the two ranges, the data suggested there is little difference between the two. Therefore, the GDG considered that CAD software programmes can be considered accurate when compared with human readers.

Other desirable effects beyond the accuracy of the technologies would likely include the possibility to scale up and thus increase the access to chest radiography, given the scarcity of radiologists in many settings. In addition, GDG members noted that in many settings, general practitioners or other providers without specific training in radiology are often tasked with interpreting chest radiographs, and they may not be as highly skilled as the readers used for comparison in the evaluations considered, thus indicating that the comparisons presented here may represent an underestimate of the true comparative accuracy of CAD software for detecting TB.

The drawback of using CAD interpretation in place of human readers for chest radiographs included the fact that it cannot detect other lung pathologies beyond TB. The capacity of CAD technologies to simultaneously screen for multiple pulmonary or thoracic conditions could be attractive for programmes, but no data on the performance of CAD for differential diagnosis were available to be assessed by the GDG.

CAD technologies have the potential to increase equity in the reach of TB screening interventions and in access to TB care if they facilitate the scale up of radiography for TB screening and triage and improve the interpretation of images.

The recommendation applies to software brands that upon external validation demonstrate a performance that is not inferior to the products reviewed by the GDG in 2020. The analysis for this recommendation was restricted to bacteriologically confirmed TB and, thus, the recommendation may not necessarily apply to other forms of TB (e.g. exclusively extrapulmonary TB, clinically diagnosed TB). This recommendation is specific to adults and adolescents aged 15 years and older. The recommendation applies only to the interpretation of anteroposterior or posteroanterior views of digital plain CXRs for pulmonary TB: it does not apply to the interpretation of lateral or oblique views, and its applicability to the interpretation of analogue CXRs is unknown.

3.2.2 Implementation considerations

The evaluations reviewed by the GDG demonstrated substantial variation in the diagnostic accuracy (sensitivity and specificity) of CAD programmes across settings, even when using the same technology set to the same threshold. Thus, it will be essential to calibrate the threshold to be used for any given software for each setting and population in which it will be used in order to ensure that the accuracy, predictive values, overall yield and requirements for further diagnostic testing are as expected. Further guidance on calibrating CAD in a new setting is provided in the operational handbook accompanying the guidelines, which links to a protocol for collecting the requisite data and a web-based tool to assist with the analysis of data and calculation of receiver operating characteristic curves and sensitivity and specificity values across a range of thresholds (7).

The feasibility of implementing CAD depends heavily on the setting, including ensuring access to the required equipment for conducting digital radiography, a stable internet connection and required maintenance for the hardware and software. The required resources and the cost-effectiveness will depend on the setting, including the availability and salaries of human readers.

3.3 Tools for screening for TB disease among people living with HIV

11. Among adults and adolescents living with HIV, systematic screening for TB disease should be conducted using the WHO-recommended four symptom screen and those who report any one of the symptoms of current cough, fever, weight loss or night sweats may have TB and should be evaluated for TB and other diseases

(existing recommendation: strong recommendation, moderate certainty of evidence).

12. Among adults and adolescents living with HIV, C-reactive protein with a cut-off of > 5 mg/L may be used to screen for TB disease

(new recommendation: conditional recommendation, low certainty of evidence for test accuracy).

13. Among adults and adolescents living with HIV, chest X-ray may be used to screen for TB disease

(new recommendation: conditional recommendation, moderate certainty of evidence for test accuracy).

14. Among adults and adolescents living with HIV, molecular WHO-recommended rapid diagnostic tests may be used to screen for TB disease

(new recommendation: conditional recommendation, moderate certainty of evidence for test accuracy).

15. Adult and adolescent inpatients with HIV in medical wards where the TB prevalence is > 10% should be tested systematically for TB disease with a molecular WHO-recommended rapid diagnostic test

(new recommendation: strong recommendation, moderate certainty of evidence for test accuracy).

3.3.1 Summary of the evidence and rationale

TB remains the primary cause of AIDS-related morbidity and mortality worldwide, despite impressive scale up of antiretroviral treatment (ART). In 2019, TB was associated with an estimated 208 000 (30%) AIDS-related deaths (7). Global estimates show a 44% gap in case detection among people with HIV-associated TB (7). A systematic review of postmortem studies of global AIDS-related deaths in adults found TB to be the primary cause of death in 37.2% of cases (95% CI: 25.7–48.7). TB was undiagnosed

prior to death in 45.8% of cases (95% CI: 32.6–59.1) (75). Ensuring early detection and timely treatment of TB among people living with HIV is of paramount importance for reducing mortality.

A key component of the HIV care cascade is the WHO-recommended four-symptom screen (W4SS) for adults and adolescents living with HIV, comprising screening for a current cough, fever, night sweats or weight loss. This has been the only WHO-recommended TB screening strategy for people living with HIV since its introduction in 2011, and it is recommended for use at every patient encounter with a health care worker (79). Designed primarily for ruling out active TB prior to the initiation of TPT, due to its high negative predictive value, the development of the W4SS was informed by a meta-analysis of individual patient data (IPD) from in observational studies conducted prior to 2010 and prior to the scale up of ART (76). However, a more recent systematic review found that the pooled sensitivity of the W4SS among people living with HIV who were on ART was 51% (95% CI: 28.4–73.2) and the specificity was 70.7% (95% CI: 47.7–86.4) (77). For people living with HIV who were not on ART, the sensitivity was 89.3% (95% CI: 82.6–93.9), and the specificity was 27.2% (95% CI: 17.3–40). In another study, the W4SS was found to have 42.9% sensitivity among pregnant women in Kenya attending services for prevention of mother to child transmission (78).

Therefore, a systematic literature review and meta-analysis of IPD was undertaken to assess further the accuracy of the W4SS among people living with HIV overall and within important subgroups and to identify other screening tools and strategies that would enhance TB detection among people living with HIV. Screening tools and strategies reviewed by the GDG included the use of C-reactive protein (CRP), CXR and mWRDs as stand-alone tests, as well as in combination with the W4SS. Culture was the reference standard for assessing the accuracy of the different screening strategies. These recommendations apply to adults and adolescents aged 10 years and older living with HIV; for recommendation on children under 10 years living with HIV see [section 3.4](#).

3.3.1.1 WHO-recommended four-symptom screen

The 2020 meta-analysis of IPD included 23 studies of 16 269 participants living with HIV, all of which reviewed the accuracy of the W4SS. The studies primarily focused on pulmonary TB disease. The unweighted average TB prevalence among participants within these studies was 9.2%, ranging from 1% to 26%; and 52% of people living with HIV screened positive on the W4SS. The sensitivity of the W4SS among all people living with HIV was 83% (95% CI: 74–89) and specificity was 38% (95% CI: 25–53). Estimates of the accuracy of the W4SS in different subgroups of people living with HIV are shown in [Table 5](#). When used alone, the W4SS was found to have its lowest sensitivity among outpatients on ART and among pregnant women, and it had markedly low specificity among medical inpatients.

Table 5. Diagnostic accuracy of the WHO-recommended four-symptom screen among different subpopulations of people living with HIV compared with culture as a reference standard

Population	No. of studies (no. of participants)	Sensitivity (95% CI)	Specificity (95% CI)
WHO target product profile	NA	> 0.90	> 0.70
All people living with HIV	23 (16 269)	0.83 (0.74–0.89)	0.38 (0.25–0.53)
Inpatients	4 (672)	0.96 (0.92–0.98)	0.11 (0.08–0.14)
Outpatients on ART	9 (4 309)	0.53 (0.36–0.69)	0.70 (0.50–0.85)
Outpatients not on ART	19 (11 159)	0.84 (0.75–0.90)	0.37 (0.25–0.50)
CD4 ≤ 200 cells/μL	22 (5 956)	0.86 (0.77–0.92)	0.30 (0.18–0.45)
Pregnant women living with HIV	8 (1 937)	0.61 (0.39–0.79)	0.58 (0.39–0.75)

ART: antiretroviral treatment; CI: confidence interval; NA: not applicable.

While there may be real-life limitations to the W4SS in terms of consistency and quality of delivery that might not be reflected in studies, it remains the simplest non-invasive tool to implement in any setting, requiring no infrastructure. However, the high proportion of W4SS positivity (94%) and very low specificity in medical inpatients living with HIV in settings where TB prevalence among study participants was > 10% gives it limited utility as a screen to rule in TB prior to diagnostic confirmation by mWRD in this very ill population.

The meta-analysis of IPD found no alternative screening tools or strategies that were significantly higher in both sensitivity and specificity than the W4SS or that met the WHO target product profile for a screening test on both parameters. In all cases, when sensitivity was higher and met the minimal requirements of the target product profile, specificity was compromised, and vice versa. Depending on a programme's decision to prioritize higher sensitivity or higher specificity, other tools or combinations of tools may be used to complement the W4SS.

3.3.1.2 C-reactive protein

CRP is an indicator of general inflammation that can be measured using point-of-care tests performed on capillary blood collected via finger prick. The evidence reviewed for the performance of CRP included 6 studies from Kenya, South Africa and Uganda with a total of 3 971 participants (see **Web Annex B**, Table 13, and **Web Annex C**, Table 6). The average unweighted prevalence of TB among participants in the studies was 14%, ranging from 1% to 26%.

Data on the accuracy of CRP using a cut-off value of > 5 mg/L and of > 10 mg/L as indicators of TB disease were reviewed and both cut-offs were considered to have similar or superior accuracy when compared with the W4SS. The cut-off of > 5 mg/L was recommended because it is the lowest threshold indicating abnormality in many clinical settings, and it has higher sensitivity than the cut-off of > 10 mg/L. The choice of cut-off will depend on the availability of CRP technology in a given setting, the prevalence of TB and of other conditions that may increase CRP and the preference for increased sensitivity or increased specificity.

The IPD meta-analysis on CRP using a cut-off of > 5 mg/L reported similar sensitivity to and higher or similar specificity to the W4SS in all subpopulations assessed (see **Table 6**). When combined with the W4SS and used in parallel, whereby a positive screen for either tool led to a diagnostic test, it was found to have similar or higher sensitivity and specificity to the W4SS for all populations, depending

on the cut-off threshold used and the subpopulation assessed. CRP was found to be most accurate among outpatients who were not on ART, compared with the W4SS alone, which had a sensitivity of 0.84 (95% CI: 0.75–0.90) and specificity of 0.37 (95% CI: 0.25–0.50) in this subpopulation. When performed sequentially after a positive W4SS among people living with HIV not on ART, CRP with a cut-off of > 5 mg/L was found to be as sensitive (0.84; 95% CI: 0.73–0.90) as the W4SS alone but to have significantly higher specificity (0.64; 95% CI: 0.55–0.72). Similar to the W4SS, the specificity of CRP for TB screening among inpatients living with HIV was found to be extremely low, likely due to competing comorbidities that would also result in raised CRP levels and the presence of symptoms (see **Web Annex C**, Table 6).

Table 6. Diagnostic accuracy of C-reactive protein using a cut-off > 5 mg/L among different subpopulations of people living with HIV compared with culture as a reference standard

Population	No. of studies (no. of participants)	Sensitivity (95% CI)	Specificity (95% CI)
WHO target product profile	NA	> 0.90	> 0.70
All people living with HIV	6 (3 971)	0.90 (0.78–0.96)	0.50 (0.29–0.71)
Inpatients	1 (400)	0.98 (0.93–1.00)	0.12 (0.09–0.17)
Outpatients on ART	1 (381)	0.40 (0.10–0.80)	0.80 (0.75–0.84)
Outpatients not on ART	4 (3 186)	0.89 (0.85–0.92)	0.54 (0.45–0.62)
CD4 ≤ 200 cells/μL	6 (1 829)	0.93 (0.87–0.97)	0.40 (0.22–0.62)
Pregnant women living with HIV	2 (62)	0.70 (0.12–0.97)	0.41 (0.12–0.78)

ART: antiretroviral treatment; CI: confidence interval; NA: not applicable.

As a point-of-care biomedical test, CRP represents an opportunity for enhancing TB screening among people living with HIV. Health staff and patients might be more motivated to pursue a confirmatory diagnostic test following a positive screen for CRP. The specificity and predictive value of the test for detecting TB, however, will likely be reduced in settings with a lower TB prevalence than in those included in the meta-analysis.

3.3.1.3 Chest radiography

CXR is recommended by WHO to be used in parallel with the W4SS where CXR is available to assist in ruling out active TB prior to initiating TPT among people living with HIV who are on ART. The GDG agreed that, due to the increased sensitivity, the evidence supported using CXR in addition to the W4SS as a parallel screening strategy in which a positive or abnormal result on either screen would indicate a referral for diagnostic evaluation. Data on “any abnormality” and an “abnormality suggestive of TB” detected by CXR were reviewed and either approach is recommended, depending on the context, the availability of radiological expertise, resources and preference towards higher sensitivity or higher specificity.

The evidence reviewed for the performance of CXR and the W4SS for all people living with HIV came from 8 studies conducted in Benin, Botswana, Brazil, Guinea, India, Kenya, Malawi, Myanmar, Peru, South Africa and Zimbabwe, with a total of 6 238 participants (see **Web Annex B**, Table 14, and **Web Annex C**, Table 7). The average prevalence of TB in all people living with HIV in the studies was 7%, ranging from 3% to 18%. Among outpatients on ART, the average prevalence was 2.6%.

CXR alone was found to have similar sensitivity to and similar or higher specificity than the W4SS across all subpopulations. When combined in a sequence whereby CXR followed a positive W4SS screen, CXR had a lower or similar sensitivity with higher or similar specificity. When combined and used in parallel with the W4SS, it had a higher or similar sensitivity and similar specificity (see **Table 7**). The IPD meta-analysis found this strategy to have the highest sensitivity (0.85; 95% CI: 0.69–0.94) compared with the W4SS (0.53; 95% CI: 0.36–0.69) and the other tools and strategies assessed for TB screening in outpatients on ART. While the data were limited for inpatients living with HIV, the combined strategy of CXR and the W4SS had a very low specificity (0.07; 95% CI: 0.03–0.19), similar to findings for using CRP or the W4SS alone (see **Web Annex C**, Table 7).

Table 7. Diagnostic accuracy among different subpopulations of people living with HIV of the WHO-recommended four-symptom screen combined with chest X-ray (any abnormality) compared with culture as the reference standard and using a positive or abnormal result on either screen or both

Population	No. of studies (no. of participants)	Sensitivity (95% CI)	Specificity (95% CI)
WHO target product profile	NA	> 0.90	> 0.70
All people living with HIV	8 (6 238)	0.93 (0.88–0.96)	0.20 (0.10–0.38)
Inpatients	1 (52)	0.90 (0.33–0.99)	0.07 (0.03–0.19)
Outpatients on ART	4 (2 670)	0.85 (0.69–0.94)	0.33 (0.15–0.58)
Outpatients not on ART	8 (3 516)	0.94 (0.89–0.96)	0.19 (0.09–0.34)
CD4 ≤ 200 cells/μL	8 (2 232)	0.94 (0.90–0.97)	0.14 (0.07–0.25)
Pregnant women living with HIV	1 (8)	0.75 (0.11–0.99)	0.56 (0.24–0.84)

ART: antiretroviral treatment; CI: confidence interval; NA: not applicable.

3.3.1.4 Molecular WHO-recommended rapid diagnostic tests for medical inpatients living with HIV in settings with a high TB burden

TB is the main cause of hospitalization and mortality among people living with HIV. The assessment of the performance of an mWRD used as a combined TB screening and diagnostic strategy for medical ward patients with HIV included 4 studies in Ghana, Myanmar and South Africa with a total of 639 participants (see **Web Annex B**, Table 15, and **Web Annex C**, Table 8). The prevalence of TB in the included studies was 23.8%, ranging from 7% to 26%. The mWRD test assessed in the IPD was primarily the Xpert MTB/RIF assay.

Using the W4SS alone had 96% sensitivity and 11% specificity in the IPD meta-analysis of medical ward inpatients living with HIV, 94% of whom were positive on the W4SS. Thus, the difference in accuracy was minimal between the full screening and diagnostic strategy of using W4SS followed by mWRD and using mWRD alone. Therefore, the value of the W4SS was judged to have limited utility in screening for TB in this population prior to an mWRD test, and the GDG recommended that medical inpatients should be screened and tested with an mWRD, irrespective of symptoms, to inform a decision about whether to treat for TB. A 10% threshold TB prevalence among hospital inpatients living with HIV is recommended, taking into account the TB prevalence among the participants studied and striking a balance between ensuring rapid diagnosis in this critically ill population and the need to avoid overtreatment. In lower prevalence settings, a screening and diagnostic strategy with mWRD alone would give rise to higher numbers of false positives, with overtreatment and the related social and

economic consequences, including potential delay in starting ART. This recommendation may not be applicable to settings with a lower pretest probability of TB.

3.3.1.5 Molecular WHO-recommended rapid diagnostic tests for all other people living with HIV

The systematic review of the performance of an mWRD used to screen for TB among people living with HIV included 14 studies with a total of 9 209 participants (see **Web Annex B**, Table 16, and **Web Annex C**, Table 9). The Xpert MTB/RIF assay was the primary mWRD used in these studies. The prevalence of TB in the studies ranged from 1% to 26%. The average TB prevalence among participants attending outpatient facilities was 8.6%.

Using an mWRD alone was found to have sensitivity of 0.69 (95% CI: 0.60–0.76) and specificity of 0.98 (95% CI: 0.97–0.99) compared with using the W4SS followed by an mWRD as a diagnostic test, which had sensitivity of 0.62 (95% CI: 0.56–0.69) and specificity of 0.99 (95% CI: 0.97–0.99) (see **Table 8**). There were no significant differences in the accuracy of the mWRD between the different subpopulations when compared with using the W4SS followed by the mWRD.

Due to the increased sensitivity of mWRDs, but also in consideration of the likely challenges relating to access, high costs and feasibility in many countries, mWRDs are recommended as an option for screening for TB disease among all adults and adolescents living with HIV who are not medical inpatients in settings where the TB prevalence exceeds 10%. As with all screening tools, the GDG emphasized the importance in all settings of following up an mWRD screen with a diagnostic assessment to prevent the potential harm of overtreatment. In addition, due consideration should be made to prioritizing mWRDs as a diagnostic test for all people with presumptive TB before scaling up mWRD as a screening test.

Table 8. Diagnostic accuracy of molecular WHO-recommended rapid diagnostic tests for screening for TB among different subpopulations of people living with HIV compared with culture as a reference standard

Population	No. of studies (no. of participants)	Sensitivity (95% CI)	Specificity (95% CI)
WHO target product profile	NA	> 0.90	> 0.70
All people living with HIV	14 (9 209)	0.69 (0.60–0.76)	0.98 (0.97–0.99)
Inpatients	4 (639)	0.77 (0.69–0.84)	0.93 (0.89–0.96)
Outpatients on ART	4 (2 645)	0.54 (0.20–0.84)	0.99 (0.97–1.00)
Outpatients not on ART	10 (5 796)	0.72 (0.64–0.79)	0.98 (0.98–0.99)
CD4 ≤ 200 cells/μL	12 (3 422)	0.76 (0.68–0.82)	0.97 (0.95–0.98)
Pregnant women living with HIV	4/473	0.55 (0.33–0.75)	0.99 (0.97–0.99)

ART: antiretroviral treatment; CI: confidence interval; NA: not applicable.

3.3.2 Implementation considerations for all tools for screening people living with HIV

Countries should position the W4SS, CRP, CXR and mWRD in combination with diagnostic evaluation using mWRDs and LF-LAM (8) within national TB screening and diagnostic algorithms according to their feasibility, the level of the health facility, resources and equity. Algorithms exploring the available

screening tools are presented in the operational handbook, including modelled performance of accuracy and yield (7). While all of the screening tools presented are recommended for all people living with HIV for ease of programming, evidence showed notable accuracy of CRP for TB screening in people not yet receiving ART and that CXR enhanced the sensitivity of the W4SS among people receiving ART, both of which might be considered when choosing algorithms.

Among inpatients on medical wards in settings with a high TB burden, evidence showed that the W4SS, CRP and CXR had limited accuracy, either due to extremely low specificity or suboptimal sensitivity. Therefore, using mWRD as an upfront screening and diagnostic test is warranted, particularly given the urgency of timely diagnosis in this population.

While there are minimal data on optimal frequency of screening across all tools, data presented to the GDG on results from the WHIP3TB trial (evaluating the effectiveness of weekly rifapentine plus isoniazid for 3 months compared with periodic rifapentine plus isoniazid for 3 months and 6 months of daily isoniazid in HIV-positive individuals) (79) underscored the need for more intensified routine screening in addition to the W4SS, even among those receiving ART who have received TPT. The GDG suggested that conducting more intensified screening in addition to the W4SS should occur at the time of an initial diagnosis of HIV or during the first antenatal care visit for pregnant women, and then annually thereafter. To reduce the burden on the patient, screening should be aligned with other routine HIV care visits, such as those for viral load monitoring or for ruling out TB disease prior to initiating TPT, depending on the setting and the national guidelines on HIV. Where applicable, the W4SS should also be conducted as part of a comprehensive clinical evaluation and to inform the need for increased infection control and for other diagnostic tests, such as LF-LAM. Otherwise screening with the W4SS alone should be carried out during all other interactions between patients and health care workers.

Consideration should also be given to the added benefit of including CRP for ruling out TB disease prior to initiating TPT among people living with HIV. In a setting of 1% TB prevalence, among 1 000 outpatients screened with the W4SS followed by CRP, 742 would be true negatives and eligible for TPT compared with only 416 found eligible by the W4SS. As is the case for using CXR for ruling out TB disease prior to initiating TPT, restricted access to CRP or CXR should not be a barrier to initiating TPT.

When considering using an mWRD as a TB screening tool among people living with HIV, it should be ensured that universal access to mWRD as a diagnostic test for everyone with presumptive TB is achieved first. The use of an mWRD as a screening tool requires significant resources for implementation, including increased capacity for diagnostic networks and expansion of sample transportation networks. Depending on the feasibility and resources available, countries may choose to prioritize TB screening using mWRDs among certain subpopulations, such as medical inpatients in settings where the TB prevalence is less than 10%, among other patients who are acutely unwell or among pregnant women living with HIV. Screening with an mWRD in lower prevalence settings than those included in the IPD meta-analysis may result in higher false positives should the diagnosis not be confirmed, with the associated overtreatment and related social and economic consequences, including potential delays in starting ART. Careful clinical assessment is recommended to ensure that TB is the primary cause of illness and that other conditions leading to the clinical presentation are also managed. A negative mWRD does not exclude TB. Patients in such settings who are mWRD negative but are manifestly sick may not be able to produce sputum of sufficient quality or may have extrapulmonary TB. For patients with a history of TB within the past 5 years, a positive result may be due to the presence of DNA detected from previously treated TB. Should the patient be unable to provide sputum, other biological specimens should be considered, as indicated (69). The TB prevalence among people living with HIV in medical wards may be calculated as the percentage of people with HIV-related admissions during a 6–12-month period who are diagnosed with TB.

To inform programming and resource planning, countries are encouraged to monitor and evaluate the yield of TB screening among people living with HIV, disaggregated by screening tool.

3.4 Tools for systematic screening for TB disease among children and adolescents

16. Among individuals younger than 15 years who are close contacts of someone with TB, systematic screening for TB disease should be conducted using a symptom screen including any one of cough, fever or poor weight gain; or chest radiography; or both
(*new recommendation: strong recommendation, moderate to low certainty of evidence for test accuracy*).

17. Among children younger than 10 years who are living with HIV, systematic screening for TB disease should be conducted using a symptom screen including any one of current cough, fever, poor weight gain or close contact with a TB patient
(*new recommendation: strong recommendation, low certainty of evidence for test accuracy*).

3.4.1 Summary of the evidence and rationale

Case detection is a crucial step in the cascade of care for children with TB; however, for most children who die from TB, the disease is never diagnosed (80). Children and adolescents who are younger than 15 years represented approximately 12% of incident cases but 16% of the estimated 1.4 million deaths from TB in 2019 (7). This relatively higher share of mortality in children highlights the urgent need for improved case detection and subsequent access to preventive and curative treatment in this age group, particularly for those at highest risk.

These recommendations relate to the two subpopulations of children for whom TB screening is strongly recommended but for whom there is as yet no standard screening approach: children and adolescents younger than 15 years who are close contacts of individuals with TB, and children younger than 10 years who are living with HIV. Adolescents living with HIV who are aged 10–19 years are covered in previous recommendations for screening people living with HIV (see **Recommendations 11–15**), and screening of contacts 15 years of age and older is covered in the previous recommendation on screening tools (see **Recommendations 9–10**).

Data from a systematic review of the diagnostic accuracy of multiple screening tools used to detect TB disease among children and adolescents, which were compared against a microbiological or composite reference standard, were used to inform this recommendation. Because bacteriological testing for TB is difficult in children, a composite reference standard is often used when evaluating diagnostic accuracy in this age group.

3.4.1.1 Close contacts younger than 15 years

The evidence reviewed about the performance of symptom screening among children and adolescents younger than 15 years who are close contacts of someone with TB included 4 studies with a total of 2 695 participants (see **Web Annex B**, Table 17, and **Web Annex C**, Table 10). In this population, a symptom screen involving any one of cough, fever or poor weight gain (where the presence of any symptom constitutes a positive screen), when compared against a composite reference standard, had a pooled sensitivity of 0.89 (95% CI: 0.52–0.98) and a pooled specificity of 0.69 (95% CI: 0.51–0.83).

The evidence reviewed about the performance of chest radiography among close contacts younger than 15 years of TB patients included 4 studies with a total of 2 550 participants (see **Web Annex B**, Table 18, and **Web Annex C**, Table 8). Compared with a composite reference standard, chest radiography using abnormalities suggestive of TB as a positive screen had a pooled sensitivity of 0.84 (95% CI: 0.70–0.92) and a pooled specificity of 0.91 (95% CI: 0.90–0.92).

Despite the absence of high-certainty evidence for test accuracy, the GDG felt that a strong recommendation was warranted for both of the tools being considered for close contacts younger than 15 years, given the high risk of disease and of mortality if the diagnosis is missed and TB is left untreated

3.4.1.2 Children younger than 10 years who are living with HIV

The evidence reviewed about the performance of symptom screening (any one of current cough, fever, poor weight gain or close contact with someone with TB) included 2 studies conducted in the outpatient setting with a total of 20 926 participants and including 20 3135 screens (see **Web Annex B**, Table 19, and **Web Annex C**, Table 10). In this setting, the combined symptom screen (where the presence of any symptom constitutes a positive screen) was found to have a pooled sensitivity of 0.61 (95% CI: 0.58–0.64) and a pooled specificity of 0.94 (95% CI: 0.86–0.98).

Again, despite the absence of high-certainty evidence, the GDG felt that a strong recommendation was warranted for symptom screening for children younger than 10 years who are living with HIV, given the high risk of disease and of mortality if the diagnosis is missed and TB is left untreated.

3.4.2 Considerations for screening children and adolescents

None of the screens investigated reached the target product profile of 90% minimum sensitivity in these high-risk subpopulations of children, although CXRs came the closest. Concerns were noted about the risk of incorporation bias when using a composite reference standard in this group, thus potentially inflating the estimates of accuracy observed. Concerns were also noted about the increased risk of false-positive diagnoses in children following a false-positive screening test compared with the risk in adults because children are more likely to be diagnosed using clinical evaluation rather than bacteriological confirmation, and the process of confirming a clinical diagnosis will weight the results of the screening test or tests. When screening high-risk groups of children, including close contacts and children living with HIV, the balance between the benefits of early case detection arising from true-positive screening results and the possible risk of overtreatment from false-positive screening results is in favour of screening. When screening populations of children with a lower risk of TB or in lower prevalence settings, the trade-off between early case detection and possible overdiagnosis will be different and should be carefully considered. In such situations, while a highly sensitive screening approach is important in order to maximize early case detection, health care workers must remain vigilant to possible false-positive diagnoses and monitor responses to treatment carefully, including evaluating children and adolescents for other potential diagnoses if symptoms or CXR abnormalities persist.

Given the high risk of TB and of mortality if TB is left untreated among children with HIV and among those in close contact with TB patients, there is an urgent need to utilize any and all available screening tests to increase timely diagnosis among these high-risk populations.

Children living with HIV represent an important group that should be considered for regular TB screening and the provision of preventive therapy, given their high risk of TB and of poor outcomes if not diagnosed in a timely manner. An essential minimum screening strategy for this group would be to ensure regular screening for TB symptoms at each visit to a health centre. While data were lacking to evaluate CXR as an initial screen for children living with HIV, CXR is a useful part of a diagnostic evaluation for TB in all children, including those living with HIV, especially younger children in whom bacteriological evaluation is commonly negative (68). Children and adolescents who are close contacts of someone with TB likewise represent an important group for screening for TB disease and for the initiation of preventive therapy, given their high risk for developing TB and rapid disease progression.

For children at high risk of TB, countries should position symptom screening and screening chest radiography within national TB screening and diagnostic algorithms, according to their feasibility, level of health facility, resources and equity. Algorithms exploring the different screening tools are presented in the operational handbook (7).

4. Monitoring and evaluation

In this section, we consider the monitoring and evaluation required to support the implementation of TB screening programmes. The operational handbook contains more details about the monitoring, surveillance and evaluation of TB screening interventions (7).

When considering implementing screening in the general population, it is important to know the epidemiology of TB and the prevalence of undetected TB in the community. This will help to estimate the impact of any screening programme. TB disease prevalence surveys are used to assess the baseline occurrence of TB in high-burden settings (81). However, these surveys represent a major undertaking and are not practical in all situations. Surveillance data based on TB notifications can be used as a starting point, but it must be carefully considered alongside other factors that affect the risk of TB and the likelihood of diagnosis, including the availability of health services and the prevalence of other risk factors for TB (e.g. HIV, diabetes mellitus, smoking, malnutrition, poverty, overcrowded living conditions, older age, substance use disorder, and other lung conditions, including silicosis or fibrotic chest lesions).

In settings that are generally considered to have a higher TB burden than is found in the overall population, such as prisons, careful monitoring and surveillance to assess TB prevalence and incidence should be done alongside TB screening. If the burden of TB is found to be low and sustained at a low level for a period of time, then TB screening can be discontinued. However, surveillance should be conducted to minimize the risks of TB disease and transmission.

4.1 Indicators

Continued monitoring can help programme managers assess the performance of the TB screening components that are within their purview. The following indicators should be considered for each targeted risk group:

1. the number of people eligible for screening;
2. the number of people screened (considering the first screening and second screening separately, if applicable);
3. the proportion of those eligible for screening who were screened;
4. the number of people with presumptive TB who were identified;
5. the number of people undergoing diagnostic investigation;
6. the number of people diagnosed with TB, stratified by type of TB;
7. the proportion of those undergoing diagnostic investigation who have TB;
8. the NNS to detect one person with TB;
9. the proportion of initial defaulters (that is, the number of people diagnosed with TB who do NOT start treatment divided by number of people diagnosed with TB);
10. the treatment success rate and the death rate (using standard cohort analyses).

Additional disaggregation may be done – for example, by age group and sex – but this requires that more detailed data are collected for each individual who is screened.

The uptake of screening in a risk group (the proportion of those eligible for screening who are actually screened) can be assessed only if the size of the target group has been well defined. It is normally possible to obtain the relevant information when screening is conducted within health facilities or closed settings (such as prisons) and through contact investigations. However, it is often difficult to obtain this information from outreach screening programmes – for example, when screening is done in the community – although the estimated population size of a targeted community may be used to obtain a rough estimate of the eligible population.

4.2 Routines for recording and reporting

In order to obtain the required information for the indicators described above, a TB recording and reporting system needs to include a minimal set of data elements. Although paper-based systems have been used to collect such data, it is now becoming increasingly feasible to collect data electronically, and this should be the standard aspired to for monitoring TB screening activities. The following strategies can be used to collect the necessary data:

- maintain a log of the number of people screened in each risk group. A special register with individual-level information for each person screened may be used to obtain more refined data about subcategories of persons within a risk group. Collecting these data is resource intensive, but it may be relevant when a screening programme is started as part of operational research. It may be feasible to implement this type of data collection on a continual basis for certain risk groups, such as people seeking care in medical facilities;
- maintain a register of all patients with presumptive TB who undergo further diagnostic evaluation (if a register is used to collect individual-level information for all people who are screened, then this information can be included in it);
- include a field in the laboratory register for noting whether the tested patient was identified through screening and to which risk group the patient belongs;
- include a field in the treatment register to note whether the patient was identified through screening and to which risk group the patient belongs.

4.3 Programmatic evaluations

A successful screening programme may lead to a diminishing yield, at least if the risk group is a fixed population. Over time, changes in the background burden of TB, as well as changes in the profile of TB patients in the community (e.g. a trend towards fewer patients with symptomatic TB) can lead to a reduction in the yield from screening, an increase in the NNS, a reduction in cost-effectiveness and a change in the ratio of benefits to harms. Trends in all of these indicators need to be monitored, and the prioritization of risk groups, choice of screening approach and screening interval should be regularly reassessed. Criteria for stopping screening should be established before a screening initiative is implemented.

Monitoring the indicators discussed above may point to the need for a special assessment, for example, to explore the reasons for a low uptake of screening, an unexpectedly low proportion of people with presumptive TB that was identified through screening, a low proportion of those with presumptive TB undergoing a diagnostic investigation and a higher than expected NNS. Additional quantitative and qualitative analyses may be needed to determine whether there are barriers to screening and to identify opportunities to improve the screening approach.

4.4 Initial calibration for computer-aided detection technologies

When CAD is implemented, local calibration is advised to customize the score thresholds to the requirements of the programme (see the operational handbook for more details about how this can be done) (7). After this initial calibration, ongoing monitoring and analysis of CAD performance should be carried out to assess the correlation with human readers' interpretations and with bacteriological confirmation, and with the proportion of images read as abnormal and requiring further investigation.

5. Research gaps

5.1 Screening for TB in targeted populations

5.1.1 The general population and high-risk groups

Well-designed trials and rigorous quasi-experimental studies in different settings are required to investigate the effects of population-wide systematic screening for TB on individual-level outcomes (diagnostic delay, treatment outcomes, costs to patients, social consequences) and population-level outcomes (TB prevalence, incidence, transmission), as well as to guide implementation choices (including the method of delivery, screening algorithms, the duration of screening intervals and frequency of screening, and the mode of delivering the intervention).

Research on the longer-term impacts of screening, including any potential evidence of averted morbidity or mortality, is not necessarily captured in the existing data. Research on the cost-effectiveness of screening is needed, using a longer time horizon to adequately capture all potential costs and longer-term effects, including potential reduced future prevalence and incidence.

Observational research and programmatic evaluations reporting the impact of community-wide screening on TB case notification rates – which provide an important source of evidence on the impacts of screening under programmatic conditions – must be carefully designed and analysed to minimize bias.

Studies of screening interventions should incorporate both qualitative and quantitative assessment of the indirect effects of screening, given the importance of health-seeking behaviour in TB care engagement and the potential impact of population-wide screening to change it and the importance of assessing any unintended mental, social or economic consequences of screening (including adverse effects, the burden of the test and downstream outcomes of clinical management that is guided by the outcomes of test results).

5.1.2 People living with HIV

Well-designed clinical trials are needed to strengthen the evidence about the accuracy, effectiveness (including the impact on patient-important outcomes such as mortality), feasibility and cost implications of using the W4SS, CRP, CXR and mWRD to screen for TB across all HIV subpopulations in settings with low, medium and high burdens of HIV and TB with and without high ART coverage.

Subpopulations of people living with HIV for whom further investigation is required would include, but not be limited to, inpatients, acute care service attendees, patients for whom ART has failed, patients newly diagnosed as HIV-positive enrolling in ART clinics, stable patients established on ART, pregnant women and children and adolescents living with HIV.

More data are needed on the effectiveness, cost-effectiveness, feasibility and acceptability, frequency and optimal periodicity of routine, regular screening with the W4SS, CRP, CXR and mWRD among people living with HIV. Specifically, more studies are needed that explore the optimal placement of mWRDs for screening in antenatal care settings versus within ART clinics. Lastly, more research is needed on the potential for screening people living with HIV with mWRDs using specimens other than sputum.

5.1.3 Children and adolescents

The GDG considered data on using mWRDs for screening children and adolescent outpatients accessing health care. They felt that the data, which included 2 studies with 787 participants and had results demonstrating substantial heterogeneity, represented insufficient evidence to establish an accurate and reliable estimate of the diagnostic accuracy of mWRDs, and, thus, the GDG decided not to issue a recommendation for or against their use as a screening tool for children and adolescents. This highlights as a research priority the need for more rigorous studies evaluating the use of mWRDs for screening children and adolescents.

Also highlighted is the urgent overall need for more research and the development of better screening tools and approaches for use in this population, including more data on screening approaches that target specific and distinct age ranges including infants younger than 12 months, children younger than 5 years, children up to the age of 10 years and those aged 10–19 years. More data are needed to determine the frequency with which screening should be conducted among the subpopulations of children at highest risk of TB, and further research is needed from well-designed clinical trials to provide evidence on patient-important outcomes for TB screening in children.

5.2 Tools for screening for TB

5.2.1 Computer-aided detection

Further evidence is needed about the performance of CAD software stratified according to the characteristics of the individual being evaluated (e.g. by smear status, HIV status, age cohort, history of TB, smoking status, sex) to allow for better setting-specific and patient-specific calibration of CAD programmes.

More research on users' perspectives is needed about CAD technologies in TB screening and triage, including their perceived acceptability to patients, providers and other stakeholders.

The development and evaluation of CAD programmes for automated detection of TB in children is urgently needed, as CXR is an important tool for detecting pulmonary TB in children and adolescents, given the difficulty of bacteriological testing and diagnosis.

5.2.2 C-reactive protein

For people living with HIV in settings with different TB prevalences, more research is needed to evaluate the accuracy and predictive value of measuring CRP above any cut-off higher than 5 mg/L for TB screening, when it is used either alone or in combination with other screening tests.

5.2.3 Screening algorithms

Across all populations and tools, more research is needed to evaluate the accuracy and effectiveness of complete screening and diagnostic algorithms, including symptom screening, CXR, CRP and mWRDs used in various combinations with diagnostic evaluation. Research into their effectiveness should include measures of the impacts on patient-important outcomes, such as mortality and treatment success.

5.3 Operational research

Standard monitoring and evaluation procedures may be complemented by operational research aimed at improving the performance of screening in the local setting as well as research aimed at improving the global evidence base for screening. Topics that may be explored include:

- assessing the accuracy and performance of different algorithms for screening and diagnosis;
- identifying operational challenges and solutions;
- identifying the best ways to improve acceptability and minimize the harms of screening;
- establishing the effectiveness and cost-effectiveness of screening in different risk groups and in different epidemiological situations;
- establishing local calibration of CAD software for the specific case of a programme.

There is a need for more, larger and better randomized trials to assess the short-term and long-term effectiveness and cost-effectiveness of screening. Implementing such studies requires careful planning and considerable resources.

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Supplementary Table

Summary of changes to existing recommendations

2013 Recommendation	2021 Recommendation
Household contacts and other close contacts should be systematically screened for active TB (<i>strong recommendation, very low-quality evidence</i>).	Household contacts and other close contacts of individuals with TB should be systematically screened for TB disease (<i>strong recommendation, moderate certainty of evidence</i>).
People living with HIV should be systematically screened for active TB at each visit to a health facility (<i>strong recommendation, very low-quality evidence</i>).	People living with HIV should be systematically screened for TB disease at each visit to the health facility (<i>strong recommendation, very low certainty of evidence</i>).
Current and former workers in workplaces with silica exposure should be systematically screened for active TB (<i>strong recommendation, low-quality evidence</i>).	Current and former workers in workplaces with silica exposure should be systematically screened for TB disease (<i>strong recommendation, low certainty of evidence</i>).
Systematic screening for active TB should be considered in prisons and other penitentiary institutions (<i>conditional recommendation, very low-quality evidence</i>).	Systematic screening for TB disease should be conducted in prisons and penitentiary institutions (<i>strong recommendation, very low certainty of evidence</i>).
Systematic screening for active TB should be considered in people with an untreated fibrotic chest X-ray lesion (<i>conditional recommendation, very low-quality evidence</i>).	People with an untreated fibrotic lesion seen on chest X-ray may be systematically screened for TB disease (<i>conditional recommendation, very low certainty of evidence</i>).
In settings where the TB prevalence in the general population is 100/100 000 population or higher, systematic screening for active TB should be considered among people who are seeking health care or who are in health care and who belong to selected risk groups (<i>conditional recommendation, very low-quality evidence</i>).	In settings where the TB prevalence in the general population is 100/100,000 population or higher, systematic screening for TB disease may be conducted among people with a risk factor for TB who are seeking health care or who are in health care (<i>conditional recommendation, very low certainty of evidence</i>).
Systematic screening for active TB may be considered for geographically defined subpopulations with extremely high levels of undetected TB (1% prevalence or higher) (<i>conditional recommendation, very low-quality evidence</i>).	Systematic screening for TB disease may be conducted among the general population in areas with an estimated TB prevalence of 0.5% or higher (<i>updated recommendation – conditional recommendation, low certainty of evidence</i>).

2013 Recommendation	2021 Recommendation
Systematic screening for active TB may be considered also for other subpopulations that have very poor access to health care, such as people living in urban slums, homeless people, people living in remote areas with poor access to health care, and other vulnerable or marginalized groups including some indigenous populations, migrants and refugees (<i>conditional recommendation, very low-quality evidence</i>).	Systematic screening for TB disease may be conducted for subpopulations with structural risk factors for TB. These include urban poor communities, homeless communities, communities in remote or isolated areas, indigenous populations, migrants, refugees, internally displaced persons , and other vulnerable or marginalized groups with limited access to health care (<i>conditional recommendation, very low certainty of evidence</i>).
Adults and adolescents living with HIV and screened with a clinical algorithm for TB, and who report any one of the symptoms of current cough, fever, weight loss or night sweats may have active TB and should be evaluated for TB and other diseases. (<i>strong recommendation, moderate quality of evidence</i>)	Among adults and adolescents living with HIV, systematic screening for TB disease should be conducted using the WHO-recommended four symptom screen and those who report any one of the symptoms of current cough, fever, weight loss or night sweats may have TB and should be evaluated for TB and other diseases (<i>strong recommendation, moderate certainty of evidence</i>).



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